Partisanship in Belgian Parliamentary Speech: Evidence from the Chamber of Representatives, 1831–2025*

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Abstract

What can nearly two centuries of parliamentary debate tell us about divisions between Belgian parties? We study speeches in the Chamber of Representatives from 1831 to 2025, using the measure of Gentzkow et al. (2019). It asks how easily a listener could identify a speaker's party from a single phrase. It measures differences in wording, not distances between policy positions. We remove routine parliamentary formulas and other phrases with little political content. Catholic and non-Catholic parties use similar language under census suffrage but become easier to distinguish in the decade after the Socialists enter parliament in 1894. Religious partisanship is modest for much of the twentieth century and never exceeds its random-label benchmark after the early 1990s. Despite the conflicts over the monarchy and schools, the 1950s are not a decade of unusually distinct wording once procedural language is removed. The left is most distinct from other parties in 1898–1940 and again in the short period 2024–2025, after several decades of relatively similar wording. Government and opposition are most distinct between the wars, but their partisanship also exceeds its random benchmark in more than half the periods since 1995. In these three series, the model implies at most a 0.63–0.66 probability of identifying a speaker's group after one minute of speech, below the corresponding peaks in the United States and Finland. A narrower left–right comparison peaks in 1904–1932; the right has become more distinct from all other parties since the 1970s, especially in 2024–2025. Since 1979, parties in opposition together have used more similar wording. A multiparty index over all pairs of party families peaks in 1936–1939, 1896–1898 and 2024–2025. Before the national parties split, Flemish and Walloon members differed chiefly in their regional interests, without a growing gap in wording; members of the same party divided in roll calls mainly over language legislation. Event studies find that coalition partners align their votes as soon as they enter government, while their wording changes little more than placebo vocabularies. After elections, the record shows more disorder, approval and laughter for about three months. As in studies of the United States, Finland and Norway, left–others and government–opposition partisanship have risen in recent years.

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*Preliminary draft; comments welcome. The series are estimated on the version of the dataset of 12 October 2026, with a vocabulary that leaves out phrases without content. Text in brown still reports earlier estimates of the series estimated last and will be updated.

1 Introduction

Political disagreement is often measured through votes. Roll-call scores place legislators on an ideological scale and track the distance between parties (Poole and Rosenthal, 1985; McCarty et al., 2006). Yet voting records reveal only the choices brought to the floor. When party discipline is strong, members may vote together even if their views differ. Speech is less constrained by party discipline than voting and provides another way to study political divisions (Proksch and Slapin, 2015; Schwarz et al., 2017). This is especially useful in Belgium, where parties have long controlled parliamentary careers and voting discipline has been high (Depauw, 2003; Deschouwer, 2012). We examine how the language of parties in the Belgian Chamber of Representatives changed between 1831 and 2025. Following Gentzkow et al. (2019), we call differences in wording between parties *speech partisanship*, or simply partisanship. The measure asks how easy it is to identify a speaker's party from what they say. It does not tell us how far apart the parties stand on policy: members may use different words because they debate different topics, perform different parliamentary duties or describe the same policy differently. These distinctions matter when interpreting the long-run series. Polarization in preferences or policy positions (DiMaggio et al., 1996) can impede decisions (Jones, 2001; Binder, 2003) or undermine democratic stability (Arbatli and Rosenberg, 2021), but a change in the language measure is not itself evidence of either effect.

Our source is the Belgian Chamber Debates dataset (Graziani, 2026), which covers the available official reports of the plenary sittings from the first sitting in 1831 through 2025. Coverage is incomplete in 1832–1843 and 1992–1995, and some other sittings may be missing (Section 3). Speeches are linked to members, parties, electoral districts and biographical information. We turn the [nSpeeches] speeches by Chamber members into stemmed two-word phrases, counted by speaker and roughly two-year period. We exclude procedural formulas, forms of address, titles, names of parties and people, numerical expressions, fragments and filler-word pairs. This leaves a vocabulary more closely related to the content of debate. Our measure, developed by Gentzkow et al. (2019) for the US Congress, is the expected probability that a neutral listener assigns to a speaker's actual group after hearing one phrase. We use their penalized estimator, random-label benchmark, magnitude calculations and phrase-level measure. Related work applies the same approach to Norway (Fiva et al., 2021) and Finland (Simola et al., 2025); supervised text classifiers have also been used to identify parties in the British and Swedish parliaments (Peterson and Spirling, 2018; Goet, 2019; Wising et al., 2026). Like the Norwegian and Finnish studies, we compare two groups at a time, and we combine these comparisons in a multiparty index.

Belgium offers an unusually long period of observation. The series begins under census suffrage, when only a small minority of adult men could vote, and follows plural voting in 1893, proportional representation in 1899, equal male suffrage in 1919 and women's suffrage in 1948. Its politics also developed around three major divisions: Catholics versus anticlericals, labour versus capital, and Dutch- versus French-speakers (Lorwin, 1966; Lipset and Rokkan, 1967; Lijphart, 1981). We study these divisions separately rather than treating Belgian politics as a single left–right scale. Language presents a further challenge. As the national parties split between 1968 and 1978, each successor party came to represent one linguistic community.

A combined French–Dutch vocabulary would conflate linguistic differences with partisan differences. Instead, we estimate separate series within each language and combine them using their shares of speech.

The first series, religious partisanship, compares the Catholic party family with all other parties. The two sides are difficult to tell apart under census suffrage: between the Liberals' organization as a party in 1846 and 1893, net partisanship averages less than 0.0001 and exceeds its random-label benchmark in just four of 23 periods. The distinction becomes clearer after plural voting brings Socialists into the Chamber. All five periods from 1894 to 1904 exceed the benchmark, with net values between 0.0007 and 0.0017. The series remains moderate from the First World War to the early 1990s, with periods above the benchmark between the wars, in 1944–1950, and in the 1960s and 1980s. It reaches its highest value, [relPeakNet], in [relPeakLabel], soon after the Christian democrats' linguistic split in 1968. That estimate comes mostly from French speeches and does not exceed the benchmark without covariates (Section 5.1). Since the beginning of the federal era, no period exceeds the benchmark.

The 1950s are a useful warning against equating political conflict with distinctive wording. Despite the end of the Royal Question and the second school war, no period between 1950 and 1963 exceeds the religious series' benchmark. Earlier estimates, which included every phrase, gave the 1950s a prominent peak. Much of it came from the routine language used by ministers and rapporteurs to introduce bills and reports (Section 5.1.1). The religious one-minute probability is highest in [relMinutePeakLabel]: the model predicts that a listener who hears one minute of speech would then identify the speaker's group with probability [relMinutePeak] per cent; in the latest period, the figure is [relMinuteLast] per cent. The corresponding figures reached 73 per cent for the US Congress in 2007 (Gentzkow et al., 2019) and just under 70 per cent for Finland in the 1970s (Simola et al., 2023).

The left–others series compares Socialist and Communist parties with everyone else. It is high from 1898 to 1940, exceeding the benchmark in 16 of 19 periods. The pre-war peaks fall in 1904–1908, followed by an interwar maximum during the economic crisis and special-powers governments of 1932–1934. After 1945 the gap narrows. The series is lowest between the mid-1980s and 2021, when only two of 15 periods exceed the benchmark. The series then rises in 2021–2024 and reaches [lrPeakNet] in the short period [lrPeakLabel]. Much of this last estimate comes from Dutch speeches. In earlier estimates, the recent values were also sensitive to the assignment of members of the joint green group to language communities. The unpenalized leave-out estimator ranks 2024–2025 above every period since 1914 but below most periods before that year. Unlike the Finnish result (Simola et al., 2025), the recent peak is not driven by the radical left. Excluding that family removes the increase in 2021–2024 and lowers the 1932–1934 estimate, but leaves 2024–2025 as the highest period.

We also compare members of governing parties with the opposition. Their wording differs little under census suffrage or plural voting. The divide is strongest between the wars, exceeding the benchmark in seven of the eight defined periods from 1921 to 1940. Socialists mostly opposed Catholic–Liberal governments until joining a national-union coalition in 1935, after which smaller parties made up the opposition. Between 1947 and 1994, just two periods exceed the benchmark. Since 1995, eight of 14 do, more than in any of the other main party groupings,

and the series rises again in 2021–2025. Across periods, government–opposition partisanship is correlated with the left–others series (0.55), but only weakly with religious partisanship (0.23). The left is more distinct on average when most Socialists are in opposition, although many periods of Socialist government also exceed the benchmark. We do not interpret these correlations as causal effects of government status.

The phrases behind the estimates show what each comparison picks up. Religious divisions appear in school vocabulary (*écoles communales*, *instruction publique*, *enseignement libre*); the left speaks distinctively about labour and voting rights (*classe ouvrière*, *suffrage universel*). The Royal Question brings in *consultation populaire*. Government–opposition phrases change with the issues of the day: in 1932–1934, Socialists in opposition discuss the cabinet's *pleins pouvoirs*; in 2024–2025, opposition speeches refer to *flexi-jobs* and *nieuwe belastingen*. We remove the routine formulas used by ministers and rapporteurs, since these reveal parliamentary roles rather than political arguments. Including them would make religious partisanship look highest in the 1950s (Section 5.12). We do not control for government status in the main estimates. Simola et al. (2025) do so in Finland, while Gentzkow et al. (2019) control for majority-party status in Congress. For much of Belgian history, however, one family held power alone, making the Catholic grouping and government status impossible to separate statistically.

A multiparty parliament lets us look beyond the main divisions. We compare Catholic, Liberal and Socialist families in pairs, and individual parties from 1979. Catholic and Liberal language is difficult to distinguish in most eras; their partisanship exceeds its benchmark mainly under census suffrage, when together they held nearly every seat. Catholic and Socialist wording differs more consistently: their series exceeds its benchmark in every period from 1894 to 1919 and all but one period between the wars. The largest differences between Liberals and Socialists occur before 1914, when both opposed Catholic governments. Since 1979, pairs of parties speaking the same language have used more similar wording when both are in opposition, but not when both govern. The wording of N-VA and Vlaams Belang shows no clear trend towards greater similarity. Pairwise comparisons are symmetric; they cannot tell us which party changed its language.

Our multiparty index measures expected partisanship between two members of different families. It is highest in 1936–1939, when the Flemish nationalists and Rex entered the Chamber in strength, followed by 1896–1898, after the Socialists' entry, and 2024–2025. The recent rise comes mainly from comparisons involving PVDA-PTB's radical left or the radical right. Before the national parties split, Flemish and Walloon members differed in their French speech mainly because of their regional agendas: canals, coal and the Walloon industrial basins on one side, and linen, sugar and ports on the other. Language policy appears among their distinctive phrases only in some periods. Their speech did not become easier to tell apart in the decades before the split. In roll-call votes, Flemish and Walloon members of the same party seldom diverge, except on some language bills before the Second World War; all Flemish and all Walloon members vote most differently when one region's members are mostly in government and the other's mostly in opposition, as before 1914 and in 2014–2019.

The label "left–others" is deliberate: its comparison group includes Christian democrats, Greens and regionalists as well as Liberals and the radical right. We therefore also compare the

left with a narrower right—Liberals, the populist radical right and N-VA—and the right with everyone else. The left–right series is high from 1904 to 1932, exceeding the benchmark in eight of 12 periods, but does so only in 2016–2019 after 1946. Right–others partisanship remains low until the 1970s and is higher thereafter. It exceeds the benchmark in seven of the 14 periods since 1995, with its highest value in 2024–2025.

Changes in voting rights are only loosely associated with shifts in partisan wording. Religious and government–opposition partisanship rise after the 1894 election under plural voting, when Socialists enter the Chamber, consistent with the Chamber–Senate comparison in Graziani and Vanschoonbeek (2026), a companion paper that studies these elections in much more detail. Following the first proportional election in 1900, government–opposition partisanship falls, whereas the other two series change by less than two standard errors. Religious and left–others partisanship change little around equal male suffrage. Government–opposition wording is more distinct after the national-union cabinets of 1918–1921 than before the war. Around the first election with women's suffrage in 1949, all three changes are smaller than two standard errors. These comparisons describe changes around reforms; they do not isolate the reforms' effects. Event studies compare each reform with the elections without a reform since 1831 (Section 8.4). With the penalized estimator, the rise in religious partisanship at the 1894 election is the largest at any election. Partisanship measured on numbers and dates rises too, though not on filler words, and no comparison of a reform with the other elections survives false-discovery-rate control.

Belgium's highest one-minute probabilities in the three main divisions, [relMinutePeak], [lrMinutePeak] and [govMinutePeak] per cent, are below those of the US Congress in 2007 and of Finland's peak in the 1970s (Section 5.11). Its religious divide is most visible in speech in the decade after 1894 and, after the Christian democrats split, around 1970 and in the early 1980s; it is not visible in the federal era. Left–others partisanship reaches its highest value in 2024–2025, and government–opposition partisanship its highest value since the 1930s in 2021–2024, as partisanship has also risen in recent decades in the United States, Finland and Norway. From 1981 to 1999, the left–others probability is close to that of the Norwegian Storting in the same years, and it is lower afterwards. As in other parliamentary systems, the government–opposition divide is clearly visible in speech, between the wars and again since 1995.

We examine whether parties differ in the subjects they discuss as well as in their choice of words (Section 6). Topic labels for each speech allow us to compare vocabulary within broad policy areas. Within policy areas, partisanship usually lies above its random series for the religious and left–others groupings and broadly follows partisanship over all phrases, more closely for left–others than for the religious divide. Differences in how much attention parties give to a topic exceed a random benchmark mainly for the left before 1979.

The parliamentary record also preserves reactions in the Chamber (Section 7). Recorded disorder peaks in the 1950s, even though parties are not unusually easy to distinguish by their vocabulary. Across decades, disorder is moderately associated with left–others partisanship but not with the religious or government–opposition measures. In every decade from the 1840s through the 2000s, opposition speakers face more disorder than government speakers. Members refer by name to politicians from other party families more than expected under random pairing;

from 1900, they are less likely to refer across the linguistic divide. The module's coding of emotional tone is still in progress.

To see what changes around specific events, we use a catalogue chosen before estimation (Section 8). When two parties start to govern together, their agreement in roll calls rises by almost half of its scale from the first period. Their politically meaningful vocabulary, however, changes little relative to placebo vocabularies made up of filler words and other low-content phrases. In the three months after elections, the parliamentary record contains more disorder, approval and laughter. The increases in disorder and approval are not confined to the sittings in which the Chamber installs itself or hears a new government's declaration. Yet government and opposition wording shows little change. These patterns concern voting and reactions more than partisan language. Roll calls also change around the financial crisis of 2008 and the refugee crisis and Paris attacks of 2015. Other events in the catalogue, from school wars and party splits to COVID-19, produce no robust pattern relative to placebo dates (Online Appendix F).

These results require several qualifications. Most importantly, wording is not policy distance or the intensity of political conflict. During the first school war, for example, both sides talked about education roughly in proportion to their numbers. Religious partisanship therefore rose much less than historical accounts of the confrontation might lead us to expect (Lamberts and Lory, 1986; Witte et al., 2009). Nor should levels be compared mechanically across eras. Speech volume and vocabulary size differ substantially: members used more than twice as many retained phrases per period under plural voting as in the federal era, and the Dutch vocabulary is roughly one-quarter the size of the French one. Parliamentary procedure matters too. In the earlier all-phrase estimates, which included routine formulas for submitting bills and reports and other phrases without content, the 1950s were the decade of highest religious partisanship and the second highest, after the incomplete 2020s, of left–others and government–opposition partisanship; the filtered vocabulary removes these phrases, and we also remove the translations printed inside speech blocks (Sections 3 and 5.12). Finally, the intervals have 80 per cent nominal coverage, are not adjusted for multiple comparisons, and each main random benchmark relies on a single permutation. We put more weight on sustained changes than on isolated periods above the benchmark.

Our approach builds on a growing literature that studies party divisions in parliamentary speech outside the US Congress. Peterson and Spirling (2018) and Goet (2019) examine British debates, the former using classifier accuracy and the latter tracing the Commons back to 1811. In the Irish Dáil, Lauderdale and Herzog (2016) recover positions from speeches and find that the government–opposition divide is the main dimension. Fiva et al. (2021) use the measure of Gentzkow et al. (2019) for Norway since 1981, and Simola et al. (2025) for Finland over a century. Wising et al. (2026) analyze Swedish parliamentary blocs; Hirst et al. (2014) show that Canadian text classifiers often learn the difference between governing and opposition roles rather than ideology. Who speaks, and what they are free to say, also depend on parliamentary rules and party control (Proksch and Slapin, 2015; Bäck et al., 2021). For Congress, Jensen et al. (2012) and Gentzkow et al. (2019) document long-run changes in partisan language.

We add to this literature in four ways. First, the series begins at the Chamber's first sitting in 1831, earlier than those of the measure for the US Congress, from 1873, and for Finland, from

1907, and it follows a single parliament from census suffrage to federalization. It measures the religious and socioeconomic cleavages separately within each of the two languages of debate, and the linguistic cleavage in French speech before the national parties split.

Second, we combine two-group comparisons in a multiparty index. Applications of the measure of Gentzkow et al. (2019) to multiparty parliaments compare two groups at a time: left and right blocs, government and opposition, or single pairs of parties (Fiva et al., 2021; Simola et al., 2025). Wising et al. (2026) adapt supervised classifiers to the multiparty Riksdag. Our index averages the two-group measure over every pair of party families with at least 15 speakers in a period, weighting each pair by the product of the two families' numbers of speakers, so that it measures the expected partisanship between two randomly drawn members of different families. Because each pair keeps the baseline of one half, the index does not fall mechanically when more parties enter, as the probability of identifying one party among many would. To our knowledge, it is the first index that summarizes the measure of Gentzkow et al. (2019) over all the party families of a parliament.

Third, we use changes of coalition and placebo vocabularies to help separate parties from their roles in parliament. Whether differences in parliamentary speech reflect parties or the roles of government and opposition remains an open question (Hirst et al., 2014; Lauderdale and Herzog, 2016). In Belgium, government status follows party for long periods: when one family governs alone, a control for government status leaves the religious series unidentified (Section 4). Changes of coalition since 1979 move parties between government and opposition while the parties stay the same. When two parties start to govern together, their roll-call agreement rises at once, but their wording changes little more than phrases without political content. Being on the same side of the government–opposition divide goes with more similar wording, but also with a fall in partisanship on pairs of filler words that is even larger with the penalized estimator and of similar size with the leave-out estimator. That association may therefore reflect, at least in part, the kinds of speech that go with the two roles.

Fourth, we show how much the measure depends on phrases without political content, and we use them as a diagnostic. The formulas with which ministers and rapporteurs submit bills and reports, which only those who govern use, account for much of the 1950s peak in the earlier estimates on all phrases. The phrases that we leave out of the vocabulary, such as numbers, amounts and dates and pairs of filler words, form placebo vocabularies with little political content, which we estimate in the same way as the main vocabulary. Some changes in the measure also appear in them: at the first election under plural voting, in 1894, partisanship measured on numbers and dates rises too, and the difference from the main vocabulary is not unusual (Section 8.4).

We also relate wording to roll-call votes and to the reactions recorded in the sittings. For 1880–1912, Graziani and Vanschoonbeek (2026) use the same data and the leave-out estimator of Gentzkow et al. (2019) to compare the Chamber and the Senate around the reforms of 1894 and 1900. Here, we study the Chamber's full history and use the penalized estimator as our main measure.

The paper proceeds as follows. Section 2 introduces the historical setting, Section 3 describes the corpus, and Section 4 sets out the measure and estimation. Section 5 presents the principal

series, party comparisons, institutional associations, the comparison with other parliaments and the comparison of speech with roll-call votes; Section 5.7 examines regional speech and roll-call voting within those results. Section 6 separates topics from wording, while Section 7 examines recorded reactions and references to other members. Section 8 studies coalitions, elections, electoral reforms and other events. Section 9 concludes. Online Appendix B gives the party-family definitions and government histories.

2 Background

2.1 Belgium, 1830–2025

Belgium gained independence in 1830 and adopted its Constitution on 7 February 1831. It was the first country on the European continent to industrialize. Coal and iron production developed in Wallonia and textile manufacturing around Ghent, while most of Flanders remained agricultural (Witte et al., 2009). Figure 10 plots GDP per capita and its growth from 1846, when the annual series begins (Bolt and van Zanden, 2024, 2025). Growth continued through much of the nineteenth century, was interrupted by sharp falls during both German occupations, accelerated after 1945 and slowed after the crises of the 1970s. After the Second World War, economic activity increasingly shifted from Wallonia to Flanders, adding to tensions between the language communities (Deschouwer, 2012).

Three divisions shaped Belgian party politics (Lorwin, 1966; Lijphart, 1981). Religion separated Catholics from anticlerical Liberals, above all over schooling and the Church's place in public life. The Liberal education law of 1879 sparked the first school war, which ended after the Catholic victory of 1884 and a law allowing communes to replace their own schools with Catholic ones (Lamberts and Lory, 1986; Witte et al., 2009). The second school war followed the education measures of the Socialist–Liberal government of 1954–1958 and ended with the School Pact of 1958. Class became a separate political divide after the Belgian Workers' Party was founded in 1885 and major strikes broke out in 1886. Socialists first entered the Chamber in 1894. Language divided the Flemish movement from the French-speaking state. Key episodes included the Equality Law of 1898, the language legislation of the 1930s, the split of national parties in 1968–1978 and successive state reforms in 1970–1993, which turned Belgium into a federation (Deschouwer, 2012). Around these divisions grew separate networks, or pillars, of parties, unions, mutual societies, schools and newspapers (Lijphart, 1981).

Other conflicts crossed these divisions. The Royal Question concerned King Leopold III's return after the Second World War. It divided Catholics from Socialists and Liberals, as well as Flanders from Wallonia. It was settled after the popular consultation of 1950 and the King's abdication in 1951 (Witte et al., 2009; Conway, 2012).

2.2 The Chamber of Representatives, 1831–2025

The Constitution of 1831 established two houses. The Chamber was elected under census suffrage, a direct-tax qualification, in multi-member districts (*arrondissements*), with half its seats renewed every two years (Graziani, 2026). The qualification fell in 1848, but the electorate

remained small until 1893. Following a general strike, the revision of 1893 introduced universal male suffrage with plural votes for some men, alongside compulsory voting. The law of 1899 made Belgium the first country to adopt list proportional representation in national parliamentary elections; it was applied in 1900. In 1919, equal male suffrage replaced plural voting and elections began renewing the entire Chamber. Women first voted in national elections in 1949 (Ahmed, 2013; Witte et al., 2009; Graziani, 2026).

Until the state reform of 1993, the Chamber and Senate shared legislative power on equal terms. We do not study the Senate's proceedings. The Chamber grew with the population, reaching 212 members in 1949, and has had 150 members since 1995. It did not sit during most of the German occupations, from August 1914 to November 1918 and from May 1940 to September 1944.

For long periods, governments came from a single party family. After the early unionist cabinets, Liberal and Catholic governments alternated from 1847. Catholics then governed alone from 1884 to the First World War. Coalitions became the norm after 1918. Governments of national union included all three traditional families in 1918–1921, 1926–1927 and most of 1935–1939. The Christian democrats again governed alone in 1950–1954. Under a right established in the Constitution of 1831, ministers may address the Chamber whether or not they are members (Graziani, 2026).

The Chamber has kept a verbatim record of its sittings throughout the period. The record appeared in the official gazette, the *Moniteur belge*, until 1844 and then in the *Annales parlementaires*. For more than a century and a half, it was prepared from shorthand notes. Audio recordings replaced shorthand in 2000. Since 1999, each intervention has been numbered and accompanied by a summary in the other national language (Model, 2021; Graziani, 2026).

Sittings have been public since 1831. For most of the period, voters encountered the debates through newspapers belonging to the same pillars as the parties. French dominated nineteenth-century debates. Dutch speeches were almost absent before the mid-1890s, accounted for about a quarter of speeches by the late 1930s, and have been the majority since the 1980s (Graziani, 2026).

3 Data

The Belgian Chamber Debates dataset (Graziani, 2026) contains the available official plenary reports from 1831 to 2025. The version used here, dated 12 October 2026, has [nBlocks] debate blocks, including 2,370,355 speeches. These blocks come from OCR of the *Moniteur belge* and *Annales parlementaires*, and from the Chamber's PDF verbatim reports from 1999. The updated dataset recovers sittings previously missed on days with multiple sittings. For 533 of the 535 days between 1999 and 2018 that the earlier version drew from the Chamber's website, we instead use PDF reports: the website interleaves each speech with its summary in the other language. We also repair misplaced accents in reports from 1 July 1999 to 14 July 2000, where accents appear after their letters (*pre´sident*). Against hand-annotated excerpts, segmentation achieves an F1 score of 89.1 per cent for block boundaries. Overall, 98.5 per cent of non-chair plenary speeches are linked to a member or minister. We record the speech language and the

speaker's party, district, gender and ministerial office on the day. Online Appendix B gives further details on selection, preprocessing and group definitions.

Sample. We retain French and Dutch speeches in plenary sittings by members serving in the Chamber on that date. Like Simola et al. (2023), who omit the Speaker and Deputy Speakers, we exclude the chair's turns. We also remove collective interjections, clerks' text, documents read into the record, translations and summaries in the other language, and committee proceedings. Ministers who were senators or held no parliamentary seat are excluded because they had no Chamber district. The starting sample has [nSpeechesSelected] plenary speeches linked to a member or minister; [nSpeeches] were given by Chamber members and enter the analysis. Of these, [shareDutch] per cent are in Dutch.

Periods. We work with periods of roughly two years. Until 1919, when half the Chamber was renewed at a time, a legislature in the register typically covers two years and forms one observation period. From 1919, elections renewed the whole Chamber for terms of up to four years, extended to five from 2014. We divide any legislature lasting at least three years at its midpoint. This produces periods close to the two-year US Congresses of Gentzkow et al. (2019) and the parliamentary years used by Simola et al. (2025). The two wars interrupt the terms elected before them. In total, we observe [nPeriods] periods, from [firstPeriod] through [lastPeriod].

Phrases. We clean each speech by removing parenthetical interruptions, replacing punctuation and symbols with spaces, converting the text to lowercase and removing stopwords (Gentzkow et al., 2019). Words are reduced to stems with the appropriate Snowball stemmer (Porter, 1980, 2001), and adjacent stems form two-word phrases (bigrams). We exclude bigrams that contain names or acronyms of parties, members' surnames, months or addresses to the chair, as well as listed procedural expressions. We also filter out phrases without content: formulas, courtesies, titles, other personal names, ministry names used as titles, numerical expressions and dates, word fragments and pairs of filler words. References to a law and its year or to a person's age can remain, as in *67 ans*. The rules match those used on the project's website (Online Appendix B). Filtering matters: formulas used by ministers and rapporteurs for introducing bills and reports were responsible for much of the apparent religious peak in the 1950s in the earlier all-phrase estimates (Section 5.12). Following Gentzkow et al. (2019), a phrase enters the vocabulary only if it appears at least 100 times overall, at least 10 times in one period and in at least 10 separate speaker-periods. The French and Dutch vocabularies contain [vocabFr] and [vocabNl] phrases, accounting for [phrasesFr] and [phrasesNl] occurrences. The content filter removes 7,488 French and 4,195 Dutch phrases; it retains 85.8 per cent of French and 71.7 per cent of Dutch occurrences. We count retained phrases by member, language and period. The resulting sample has [nSpeakerPeriods] speaker-periods for [nPersons] members.

Parties, regions and government. We use the party labels listed for each legislature and group parties into families: Catholic and Christian democrat, Liberal, Socialist, Communist and radical left, Flemish nationalist, Francophone regionalist, Green, and other. Where noted, the populist

radical right—Rex, Vlaams Blok and Vlaams Belang, Front National and Parti Populaire—forms a separate group. After a family splits, each party belongs to one language community. Members of the joint parliamentary group of the two Green parties belong to the community of their district or, in Brussels, of the language they speak. Families group parties over time: a family's predecessors and successors form a grouping for comparison, not a claim of organizational continuity. Members who switch parties within a period are assigned to their party at the start of that period. Speakers whose party is unknown, 2.0 per cent of member-periods, are left out of every comparison. Online Appendix Table B2 lists every party label of the register with its family, its speakers and speeches, its side in every comparison and its years in government.

We reconstruct each government's composition from the register of ministers. A party is in government on a given day if one of its members holds ministerial office. Before the national parties split, the party is the national party; afterwards, it is each community's party. Where a family has several parties at the same time—such as the Volksunie and the Vlaams Blok, which the register places in the same family, FDF and the Rassemblement Wallon, or Agalev and Ecolo—each party counts separately, and ministers without a party place no party in government. Online Appendix Table B3 lists the parties in government. We classify a member-period as governing if at least half the member's speeches were delivered while the party was in government. The province of the electoral district serves the same purpose as state in Gentzkow et al. (2019) and dialect region in Simola et al. (2023): it captures regional language differences and members' regional agendas.

Quality of the record. The source has three limitations. First, only part of the proceedings survives in our digitized material for 1832–1843, when we rely on the *Moniteur*, and in 1992–1995, when much plenary work moved to a Central Commission not covered here. Other isolated sittings may also be missing. We shade the two incomplete stretches in figures and leave them out of summary tables, since missing sittings can change which debates enter a period. Second, speaker linkage is weaker in the 1830s: 74 per cent of speeches are linked to a member, compared with over 92 per cent in every later decade (Online Appendix Figure D3).

Third, the printed record sometimes includes translations and procedural text inside a speaker's block. Between 1936 and 1958, blocks often append translations of remarks by rapporteurs and ministers submitting documents (*Traduction :*, *Vertaling :*). In 1993–1995 and 1997–2007, speeches often end with roll-call formulas (*Résultat du vote*, *Uitslag van de stemming*). Language labels are also occasionally wrong, and members sometimes switch languages mid-speech. We remove inserted translations and trailing roll-call formulas, correct language labels when function words clearly show an error, and count sentences in the other language as speech in that language (Online Appendix B). After these corrections, phrases containing a function word from the other language make up at most 0.25 per cent of French phrases in any period and, from 1919 onward, at most 0.21 per cent of Dutch phrases. Section 5.12 shows which results depend on these choices.

4 Model and estimation

4.1 Partisanship measure

For each comparison, we divide members into groups A and B: for example, the Catholic family and all remaining parties. Let c_{ijt} count speaker i's uses of phrase j in period t, and let $m_{it} = \sum_j c_{ijt}$ be the number of phrases spoken. The share accounted for by phrase j in group P is $q_{jt}^P = \sum_{i \in P_t} c_{ijt} / \sum_{i \in P_t} m_{it}$. A listener starts with equal probabilities for the two groups. After hearing phrase j, the probability assigned to group A is (Gentzkow et al., 2019):

$$\rho_{jt} = \frac{q_{jt}^A}{q_{jt}^A + q_{jt}^B}. \tag{1}$$

We model speaker i's phrase counts as multinomial:

$$\boldsymbol{c}_{it} \sim MN\left(m_{it}, \boldsymbol{q}_t^{P_i}(\boldsymbol{x}_{it})\right), \tag{2}$$

with phrase-choice probabilities

$$q_{jt}^{P_i}(\boldsymbol{x}_{it}) = \frac{e^{u_{ijt}}}{\sum_k e^{u_{ikt}}}, \qquad u_{ijt} = \alpha_{jt} + \boldsymbol{x}_{it}'\gamma_j + \varphi_{jt}\mathbf{1}_{i \in A_t}, \tag{3}$$

Here, α_{jt} captures the phrase's popularity in that period and φ_{jt} is the party effect. The controls in $\boldsymbol{x}_{it}$ are gender and the province of the electoral district, which may affect speech independently of partisanship. Their effects are constant over time, as are most covariate effects in Gentzkow et al. (2019).

Gentzkow et al. (2019) control for majority-party status, while Simola et al. (2025) control for government status. Our main specification omits a control for whether the speaker's party is in government. In Belgium, government status closely follows party membership for long periods. Catholics governed alone in 1846–1847, 1870–1878, 1884–1914 and 1950–1954, and Liberals in 1840–1841, 1847–1855, 1857–1870 and 1878–1884. A nearly unpenalized government coefficient that is constant over time cannot then be separately identified from the penalized party effects.

The estimates confirm this problem. Adding government status produces religious partisanship values of 0.64 to 0.72 in nine periods between 1868 and 1898, all under governments of a single family. With gender and province also controlled for, as in Simola et al. (2025), the values are 0.53 to 0.70 in eight periods between 1868 and 1902, also under governments of a single family. These values lie far outside the range of the other penalized specifications (Online Appendix Figure A2). In the other periods, net partisanship averages near zero. We instead measure the government–opposition divide directly in Section 5.4.[1]

For characteristics $\boldsymbol{x}$ in period t, partisanship is the expected posterior probability assigned

[1] Allowing the covariate effects to vary across six institutional periods leaves the real religious series almost unchanged (correlation 0.98) and changes average net religious partisanship little (0.00024, against 0.00028 with constant effects). Individual periods move more, because each specification draws its own random benchmark (Online Appendix Figure A11).

to the speaker's actual group after one phrase, starting from equal prior probabilities:

$$\pi_t(\boldsymbol{x}) = \tfrac{1}{2}\, \boldsymbol{q}_t^A(\boldsymbol{x}) \cdot \boldsymbol{\rho}_t(\boldsymbol{x}) + \tfrac{1}{2}\, \boldsymbol{q}_t^B(\boldsymbol{x}) \cdot (1 - \boldsymbol{\rho}_t(\boldsymbol{x}))\,, \tag{4}$$

We average this measure across speakers:

$$\pi_t = \frac{1}{|A_t \cup B_t|} \sum_{i \in A_t \cup B_t} \pi_t(\boldsymbol{x}_{it}). \tag{5}$$

The measure cannot fall below 0.5. It equals 0.5 when the two groups use every phrase at the same rate, and rises as their wording becomes easier to distinguish. Switching the labels of the groups leaves it unchanged. A higher value does not imply a larger ideological or policy distance.

4.2 Estimation

The vocabulary is large and individual phrases are often rare. Raw frequencies would therefore exaggerate partisanship: a phrase used once would look perfectly associated with its speaker's group (Gentzkow et al., 2019). We follow Gentzkow et al. (2019) and Taddy (2015), fitting multinomial logit phrase probabilities with a lasso penalty on group effects. Estimation uses the Poisson approximation:

$$c_{ijt} \sim \text{Poisson}\left(\exp(\mu_{it} + u_{ijt})\right), \qquad \mu_{it} = \log m_{it}. \tag{6}$$

This approximation allows us to estimate a separate regression for each phrase. For phrase j, we minimize

$$\frac{1}{N} \sum_t \sum_i \left[m_{it} e^{u_{ijt}} - c_{ijt} u_{ijt} \right] + \psi \Big(\sum_t |\alpha_{jt}| + \|\gamma_j\|_1 \Big) + \lambda_j \sum_t |\varphi_{jt}|, \tag{7}$$

where N is the number of speaker-periods. The small penalty $\psi = 10^{-5}$ is the one Gentzkow et al. (2019) use to ensure convergence. As in their paper, we choose λ_j using the Bayesian Information Criterion (Schwarz, 1978). We evaluate 100 penalty values, starting at the smallest value that sets all φ_{jt} to zero.

All covariates are indicators. The likelihood therefore depends only on counts and exposures aggregated by period, group and covariate pattern, which simplifies estimation. Given γ_j, define group P's exposure as $e_t^P = \sum_{i \in P_t} m_{it} e^{\boldsymbol{x}_{it}' \gamma_j}$ and its phrase count as c_{jt}^P. Also define $s_{jt} = c_{jt}^A + c_{jt}^B$, $\pi_t^0 = e_t^A / (e_t^A + e_t^B)$, $\nu = N\psi$ and $\kappa_j = N\lambda_j$. The fitted count for group A is

$$\hat{\mu}_{jt}^A = \begin{cases} c_{jt}^A - \kappa_j & \text{if } c_{jt}^A - \pi_t^0 (s_{jt} + \nu) > \kappa_j, \\ c_{jt}^A + \kappa_j & \text{if } c_{jt}^A - \pi_t^0 (s_{jt} + \nu) < -\kappa_j, \\ \pi_t^0 (s_{jt} + \nu) & \text{otherwise,} \end{cases} \tag{8}$$

The fitted count for group B is $\hat{\mu}_{jt}^B = s_{jt} + \nu - \hat{\mu}_{jt}^A$. The corresponding coefficients are $\hat{\alpha}_{jt} = \log(\hat{\mu}_{jt}^B / e_t^B)$ and $\hat{\varphi}_{jt} = \log(\hat{\mu}_{jt}^A / e_t^A) - \hat{\alpha}_{jt}$. The lasso moves group A's count towards its nonpartisan value by $N\lambda_j$. The penalty on α_{jt} adds $N\psi$ to the fitted total. We can therefore compute the BIC in closed form at every point on the penalty path. We estimate γ_j with proximal Newton

steps, replacing $(\alpha_{jt}, \varphi_{jt})$ in the objective with their optimal values. Next, we choose λ_j again and repeat until the selected point on the path stops changing.

Two languages. We build separate French and Dutch vocabularies and estimate the model for each language. A speaker-period contributes its phrases in the relevant language. After the Catholic (1968), Liberal (1972) and Socialist (1978) families split, members of these families enter only the estimation for their own language community. From 1968, the same rule applies to the parties that belong to a single language community, such as the Flemish nationalists, the Francophone regionalists and the Greens. This ensures that we compare parties speaking the same language. Before the splits, the province control accounts for differences between Flemish and Walloon members within a party. Period t's partisanship is the weighted average of both languages:

$$\pi_t = \sum_{\ell \in \{\text{fr,nl}\}} w_{\ell t}\, \pi_{\ell t}, \tag{9}$$

The weights $w_{\ell t}$ are each language's share of the phrases spoken by the two groups. We include a language only if each group has at least 10 speakers in it. We include a period only if each group has at least 20 speakers overall, or 15 in pairwise comparisons. Simola et al. (2023) use a threshold of 30. Our lower threshold reflects the Chamber's size: it had 102 to 152 members before 1912, the Socialists entered with 28 members in 1894, and each party belongs to only one language group after the national parties split.

Levels are not comparable across languages. The Dutch vocabulary is about a quarter the size of the French vocabulary. With fewer phrases, each phrase conveys more information about the speaker. We therefore compare the timing of the two series rather than their levels.

4.3 Inference

We construct confidence intervals by subsampling (Politis et al., 1999; Gentzkow et al., 2019). For each period, we draw 100 subsamples, each containing 20 per cent of speakers selected without replacement. We estimate the series in each subsample k. The intervals are

$$\left[0.5 + \exp\left(\log(\hat{\pi}_t - 0.5) - Q_t^{(90)}/\sqrt{N_t}\right),\ 0.5 + \exp\left(\log(\hat{\pi}_t - 0.5) - Q_t^{(10)}/\sqrt{N_t}\right)\right], \tag{10}$$

where $Q_t^{(p)}$ is the p-th percentile of $Q_{kt} = \sqrt{N_{kt}}\,[\log(\hat{\pi}_{t,k} - 0.5) - \log(\bar{\pi}_t - 0.5)]$. Here, $\bar{\pi}_t$ is the subsample average of $\hat{\pi}_{t,k}$ and N_t is the number of speakers. The intervals have 80 per cent nominal coverage.

For comparison, we re-estimate each series after randomly permuting group labels within periods, holding group sizes fixed (Gentzkow et al., 2019). This reveals how much apparent partisanship the estimator produces even when labels contain no information about speech. The random series does not stay exactly at 0.5: with only a few dozen speakers, it usually ranges from 0.500 to 0.503 and rises when one group is small. It also varies with the particular permutation. For 1884–1886, for instance, the real religious estimates are 0.5029 with covariates and 0.5030 without, while the respective random estimates are 0.5019 and 0.5044. We define *net partisanship* as $\hat{\pi}_t - \hat{\pi}_t^{\text{random}}$. We say that a period exceeds its random benchmark, and the tables

mark it as significant, when the lower 80 per cent confidence bound of the real series exceeds the upper bound of the random series. This is a descriptive convention, not a formal test of their difference. We do not correct the intervals for the many comparisons, and the random series uses only one draw. Individual periods should therefore be interpreted cautiously; patterns persisting across periods are more informative.

4.4 Phrase partisanship

The partisanship of phrase j at $\boldsymbol{x}$ measures how removing it changes the expected posterior probability assigned to group A (Gentzkow et al., 2019):

$$\zeta_{jt}(\boldsymbol{x}) = 0.5 - 0.5 \sum_{k \neq j} \left(\frac{q^A_{kt}(\boldsymbol{x})}{1 - q^A_{jt}(\boldsymbol{x})} + \frac{q^B_{kt}(\boldsymbol{x})}{1 - q^B_{jt}(\boldsymbol{x})} \right) \rho_{kt}(\boldsymbol{x}). \tag{11}$$

Positive values identify phrases associated with group A; negative values identify phrases associated with group B. We average across the period's speakers to obtain ζ_{jt}.

4.5 OCR noise

OCR errors turn some phrase occurrences into strings outside the vocabulary. Since these errors affect all parties, they tend to blur party differences in older volumes (Hill and Hengchen, 2019; van Strien et al., 2020). We assess this effect by adding simulated OCR errors to the born-digital reports of 2019–2025: common character confusions, deletions, insertions, and split or merged words. We process the altered text with the same pipeline and vocabulary, then re-estimate partisanship. The ratio $(\pi_t^{\text{noisy}} - 0.5)/(\pi_t^{\text{clean}} - 0.5)$ measures attenuation at each character error rate.

5 Results

We begin with the religious and socioeconomic divides: Catholic versus non-Catholic parties (Section 5.1) and the left versus the remaining parties (Section 5.2). Following the broad classification in Simola et al. (2023), the left includes Socialist parties—the Belgian Workers' Party and its successors—together with Communist and radical-left parties. Its comparison group includes every other party, not just the right; we therefore call the series *left–others*. Section 5.3 instead compares the left with a narrower definition of the right and the right with all other parties. Section 5.4 turns to government and opposition, and Section 5.5 to individual party pairs, party families, gender and today's parties. Sections 5.6 and 5.7 examine Flemish and Walloon members, their speech topics and their voting records. For the three principal series, Figures 1, 2 and 4 plot estimates against the random-label benchmark.

Tables 1 and 2 give period and decade summaries. None of the principal series peaks in the 1950s. The highest net values occur in [relPeakLabel] for the religious divide, [lrPeakLabel] for left–others and [govPeakLabel] for government–opposition. Applied to one minute of speech, the model gives maximum classification probabilities of [relMinutePeak], [lrMinutePeak] and

Table 1: The main series

			All periods		Highest period			Last
Series	Years	Periods	Signif. (%)	Mean	Period	Net	One minute	Net
Catholic vs other parties	1843–2025	81	28	0.28	1969–1971	2.00	0.60	0.00
Left vs other parties	1894–2025	58	47	0.83	2024–2025	5.24	0.62	5.24
Left vs right	1900–2025	52	21	0.93	1929–1932	2.96		2.15
Right vs other parties	1900–2025	52	23	0.44	2024–2025	4.30		4.30
Government vs opposition	1845–2025	77	30	0.35	1934–1936	3.53	0.66	2.24
Catholic vs Liberal	1843–2025	81	15	0.04	1974–1977	1.89		-0.27
Catholic vs Socialist	1894–2025	58	47	0.53	1898–1900	4.13		0.25
Liberal vs Socialist	1894–2025	56	30	0.83	1912–1914	4.63		0.25
Multiparty index	1843–2025	83	35	0.60	1936–1939	4.82		3.96
Women vs men	1974–2025	18	56	1.63	2021–2024	3.72		1.11

Notes: Net partisanship: real minus random series, multiplied by 1,000. Periods: periods when the series is defined; periods with low coverage of the sittings (1832–1843, 1992–1995) are excluded. Signif.: share of the periods when the lower bound of the 80 per cent interval of the real series lies above the upper bound of the interval of the random series (a descriptive rule, not a test; Section 4). Highest period: among the periods above the benchmark, the one with the highest net partisanship. One minute: model-based probability of guessing a member's group correctly after one minute of speech (33 phrases, treated as independent) in that period, computed from the real series (Section 5.10). The multiparty index is estimated with the language rule for every party (Section 5.5). Produced by `code/python/20_main_tables.py`.

[govMinutePeak] per cent, respectively, in [relMinutePeakLabel], [lrMinutePeakLabel] and [govMinutePeakLabel].

Left–others partisanship exceeds the benchmark in 47 per cent of periods, compared with 30 per cent for government–opposition and 28 per cent for religion. The narrower left–right and right–others series do so in 21 and 23 per cent. From the 1990s onward, religious net partisanship averages between -0.9 and 0.4 per thousand by decade, and no period after 1991 exceeds the benchmark. Left–others decade averages from the 1990s through the 2010s lie between -0.1 and 0.3, and government–opposition averages between 0.1 and 0.4. In the incomplete 2020s they reach 3.0 and 2.1, respectively. The same tables also cover the family pairs, multiparty index and gender comparison (Section 5.5).

5.1 Religious partisanship

Figure 1 tracks Catholic versus non-Catholic wording, with controls for gender and province. Under census suffrage, the difference is generally small. Before 1846, net partisanship ranges from -0.0011 to 0.0006; only 1833–1835 exceeds the random benchmark. Six of these eight periods have incomplete source coverage, and neither well-covered period, 1843–1845 or 1845–1848, exceeds the benchmark. This fits a Chamber dominated by loose alliances of notables and early unionist cabinets. After the Liberals form a party in 1846, net partisanship still averages less than 0.0001. Only four of the 23 periods through 1893 exceed the benchmark. The largest census-era values occur in 1880–1882 and 1884–1886, during and after the first school war (0.0011 in both cases), but only when province is controlled for. Without covariates, they are no larger than in the 1870s (Online Appendix Figure A2).

After the first plural-suffrage election, the distinction becomes clearer. Net partisanship is between 0.0007 and 0.0017 in each period from 1894 to 1904, and every one exceeds the benchmark. Socialists have now entered the Chamber, alongside Liberals in the non-Catholic

Table 2: Net partisanship by decade

Decade	Religious	Left–others	Left–right	Right–others	Gov.–opp.	Cath.–Lib.	Cath.–Soc.	Lib.–Soc.	Multiparty	Women–men
1840s	0.3				-0.2	0.7			0.2	
1850s	-0.1				0.2	0.2			-0.0	
1860s	-0.3				-0.3	0.1			-0.3	
1870s	-0.0				-0.1	0.3			0.2	
1880s	0.4				-0.1	0.6†			0.5†	
1890s	0.9†	1.1			0.3	1.0	3.6†	3.2	2.3	
1900s	0.5	1.4†	2.0†	0.6	0.3	-0.2	2.4†	2.9†	1.2†	
1910s	0.4	0.8†	1.9†	0.4	-0.3	0.1	1.8†	3.2†	1.3†	
1920s	0.5	1.3†	1.6	0.1	1.0†	-0.1	1.3†	0.9	0.8†	
1930s	0.6	1.7†	1.4	-0.2	2.0†	0.3	1.0†	1.3	2.1†	
1940s	0.6†	0.2	0.9†	-0.4	0.4†	0.1	0.7†	0.7	0.2	
1950s	0.3	0.4	0.6	-0.1	0.1	-0.9	0.0	-0.1	0.2	
1960s	0.5†	1.0†	0.0	0.2	-0.3	-1.1	0.2	-0.0	-0.1	
1970s	0.3	0.3	0.6	0.4	0.4	-0.3	0.7†	1.4†	0.1	0.5
1980s	0.8	0.5	0.6	0.8	0.3	0.7	0.4	0.7	0.3	-0.4
1990s	0.4	0.1	-0.3	0.5	0.1	-0.4	-0.5	-0.6	-0.1	2.1
2000s	-0.0	-0.1	0.7	0.3	0.4†	0.2	-2.0	-0.1	1.0	1.9†
2010s	-0.9	0.3	0.5	0.9†	0.3	-0.4	-0.9	-0.6	-0.5	1.5†
2020s	0.0	3.0†	1.0	2.4†	2.1†	-0.1	-0.8	0.7	1.9†	2.6†

Notes: Average net partisanship (real minus random series, multiplied by 1,000) of the periods whose middle year falls in the decade; periods with low coverage of the sittings are excluded. †: the real series is above its random benchmark (the 80 per cent intervals do not overlap) in at least half of the periods of the decade. Religious: Catholic family vs other parties; left–others: Socialist and Communist families vs other parties; left–right: Socialist and Communist families vs the Liberal family, the populist radical right and N-VA; right–others: the Liberal family, the populist radical right and N-VA vs other parties; government–opposition: members of the parties in government vs the others; Cath.–Lib., Cath.–Soc. and Lib.–Soc.: pairs of the Catholic, Liberal and Socialist families, without covariates; multiparty: average partisanship of the pairs of all party families, weighted by the product of their numbers of speakers, without covariates; women–men: women vs men of the same family (Section 5.5). The 2020s are not a complete decade: they include the periods to 2025.

group. When we compare Catholics with Liberals alone, no period between 1894 and 1914 exceeds the benchmark (Section 5.5). The broader religious series falls back to 0.0001–0.0003 in 1904–1912 before exceeding the benchmark again in 1912–1914. After low values under the national-union governments of 1918–1921, it is moderate, exceeding the benchmark in 1921–1923, 1927–1934, 1944–1946 and 1947–1950. None of the 1950–1963 periods, including the second school war and the School Pact, exceeds it: net values in 1950–1958 range from -0.0001 to 0.0007 (Section 5.1.1). Net partisanship reaches its maximum in [relPeakLabel], when the real estimate is [relPeakReal].

After the Christian democrats split in 1968, the series is noisy. It exceeds its benchmark in five of the 23 periods with adequate coverage: 0.0008 in 1968–1969, 0.0020 in 1969–1971, 0.0019 in 1981–1983, 0.0013 in 1983–1985 and 0.0008 in 1989–1991. The value of 1969–1971, the highest of the series, rests mainly on French speech, in which the Christian democrats have 19 speakers. Their most partisan French phrases in that period concern development cooperation and the European Parliament (Table 11). In 1981–1985, under centre-right governments, special powers (*pouvoirs spéciaux*) are among the remaining parties' most partisan French phrases (Online Appendix C). The last period includes the abortion law, which passed against the votes of the Christian democrats despite their being in government; *ongeboren leven* is among their most partisan Dutch phrases.

In the federal era, the real series never exceeds the random benchmark. Net partisanship ranges from [*relLowNet*] in [relLowLabel] to 0.0004 and averages [*relNetP6*]. In 2019–2024, the Christian democrats have fewer than 20 speakers across both languages, so the series is

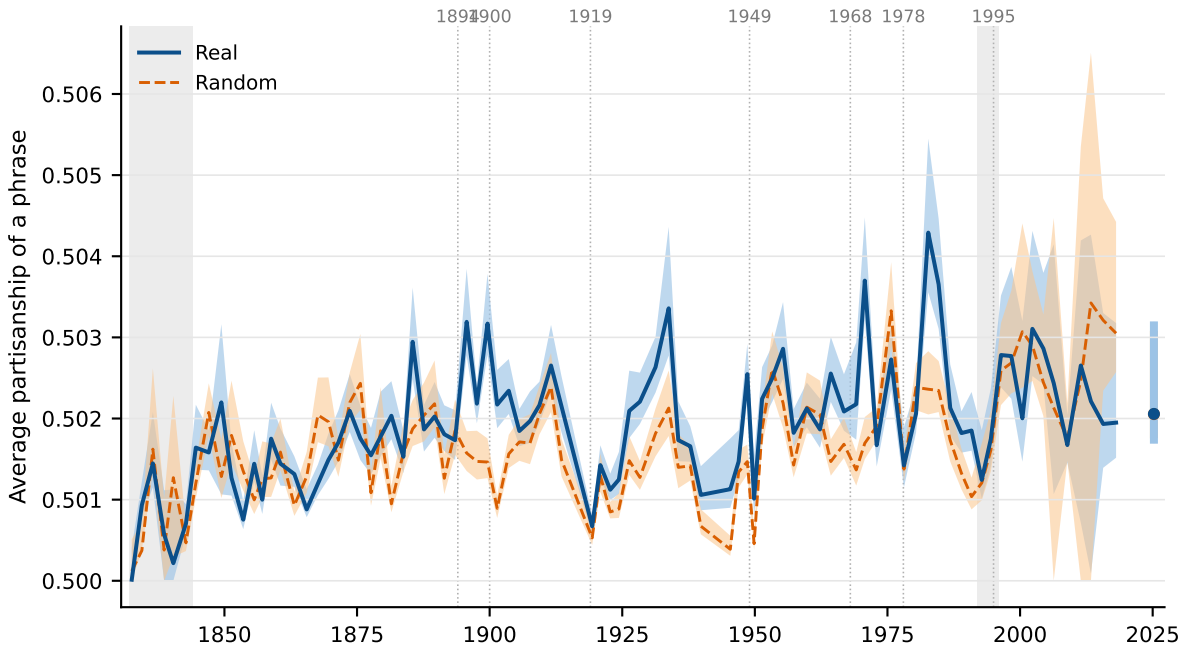


Figure 1: Religious partisanship: Catholic family vs other parties, controls for gender and province

Notes: Average partisanship of a phrase, $\hat{\pi}_t$, by period of about two years; French and Dutch averaged by their share of speech. Controls for gender and the province of the district. Real: actual party labels; random: labels assigned at random among the speakers of each period, holding group shares fixed. Shaded areas: subsampling confidence intervals with 80 per cent nominal coverage, centred at the estimate. Dotted lines: the reforms of 1894, 1900, 1919, 1949, 1968, 1978 and 1995. Grey bands: years when the source covers the sittings only in part (1832–1843, before the *Annales parlementaires*, and 1992–1995, when much of the plenary business went to a Central Commission). The series breaks when a group has fewer than 20 speakers in total; a language enters the average when each group has at least 10 speakers in it.

undefined. Net partisanship is 0.0000 in 2024–2025. Across the full sample, the real series' lower confidence bound exceeds the random series' upper bound in [relShareAboveRandom] per cent of all periods, and in 28 per cent of the periods with adequate coverage (Table 1).

The timing is only partly robust across specifications (Online Appendix A). Without covariates, the series also exceeds its benchmark in most periods of 1894–1908, in 1932–1936 and in 1981–1985, and in only one period of the 1950s (1950–1952). Its highest value is then in 1894–1896, and 1969–1971 does not exceed the benchmark. Allowing covariate effects to vary by institutional period leaves the real series almost unchanged, but the specification draws its own random benchmark: with it, [relPeakLabel] falls to 0.0009 and no longer exceeds the benchmark, and the highest value is 0.0019, in 1932–1934 (Figure A11). Without covariates, changing ψ to 10^{-4} or 10^{-6} alters levels, but the highest value remains under plural voting (Figure A7). Without the speeches that members delivered as ministers, none of the five periods after 1968 that exceed the benchmark in the main specification does so, including [relPeakLabel] (0.0005), while both periods of 1950–1954 do (Figure A10). Excluding the populist radical right changes the real series by less than 0.0001 before 1985 and by at most 0.0005 afterwards, but with its own random benchmark the net value of [relPeakLabel] falls to 0.0007 (Figure A4).

The series based on empirical phrase-choice probabilities, the plug-in estimator of Gentzkow et al. (2019), moves with its random benchmark (Figure A5). This suggests that finite-sample bias dominates without penalization. The leave-out estimator of Gentzkow et al. (2019), which does not penalize phrase effects, is positive in every period of 1843–1893 but exceeds its benchmark in only three of them. It exceeds its benchmark in all ten periods of 1894–1914, is highest in 1894–1896 and exceeds it in no period of the 1950s (Figure A9). The Dutch component exceeds its benchmark in six periods, among them 1954–1956 (0.0029), during the second school war. The French component exceeds its benchmark in no period of the 1950s and is highest in 1969–1971 (0.0028; Figure A8). Levels are not comparable between the languages, whose vocabularies differ in size (Section 4).

5.1.1 Schools, the Royal Question and the language of governing

Simola et al. (2025) link Finland's high partisanship in the 1970s to the extreme left's phrases about the Soviet Union. In Belgium, schools and the Royal Question are the natural candidates for such a role. Figure 19 compares the share of phrases about schools and education with religious partisanship. School phrases peak during the conflicts identified by historians: 3.7 per cent of all phrases in 1878–1880, when the Liberal school law was debated; 4.5 per cent in 1884–1886, when the Catholic majority replaced it; 3.3 per cent in 1912–1914, during the compulsory education law; and 2.6 per cent in 1954–1956, during the second school war. Religious partisanship does not follow these peaks. It exceeds its benchmark in 1884–1886 and 1912–1914, but not in 1878–1880 or 1954–1956.

The attention paid to schools suggests why (Figure 20). In the first school war, both camps discussed education roughly in proportion to their representation. Catholics accounted for 52 per cent of school phrases in 1878–1880, compared with 45 per cent of speakers and 43 per cent of seats. In 1880–1884, their share of school phrases was 49 per cent, close to their 46–49 per cent share of speakers. The difference lay more in the terms each side preferred. The

Liberal government spoke of *écoles communales*, the communal schools covered by the 1879 law. Catholic opponents used *instruction publique* and *enseignement libre*. After the 1884 election, Liberals in opposition used phrases such as *écoles communales*, *écoles adoptées* and *pères de famille* (Online Appendix C). Much of the remaining vocabulary was common to both sides, keeping the aggregate measure relatively low.

The Royal Question leaves a clearer trace in the attention the parties gave to it. Related phrases reach their highest share in 1949–1950. Catholic members used only 19 per cent of them, despite representing 46 per cent of speakers and half the seats (Online Appendix Figure A15). In that period, *consultation populaire* and *question royale* are the remaining parties' most partisan French phrases (Table 11), and religious partisanship exceeds its benchmark at a modest level (0.0006).

Table 11 reports the five French phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group, selecting one period every ten years, as Gentzkow et al. (2019) do for selected Congresses. It also reports predicted counts per 100,000 phrases. Online Appendix C lists phrases for every period and both languages. Figure A13 tracks the most distinctive phrases of each decade.

Three kinds of phrases recur. The first concerns the cleavages themselves: secondary and religious education for Catholics in 1848–1850 (*enseignement moyen*, *enseignement religieux*); *classe ouvrière* and *suffrage universel* for other parties after 1894; family allowances for Catholics in the 1930s (*allocations familiales*); *consultation populaire* for their opponents in 1949–1950; public, technical, secondary and free education for Catholics in the 1950s (*enseignement technique*, *enseignement libre*; *technisch onderwijs* in Dutch); and development aid and cooperation for Christian democrats from the 1960s (*assistance technique*, *coopération au développement*).

The second kind reflects governing and opposition roles. Opposition members speak of the special or full powers of the government (*pouvoirs spéciaux*, *pleins pouvoirs*), and members of both sides of the previous government (*vorige regering*). In earlier versions of the vocabulary, the clearest examples were procedural formulas and titles. Rapporteurs and ministers submit reports and bills with fixed formulas (*j'ai l'honneur de déposer sur le bureau de la Chambre*, *section centrale*; *ter tafel neerleggen* in Dutch), which only those who govern use, and members address ministers by their titles (*ministre de l'intérieur*, *premier ministre*). These phrases switch sides when governments change. In 1950–1958, one bloc governed and the other opposed throughout: the Christian democrats in 1950–1954 and the Socialists and Liberals in 1954–1958. In estimates on all phrases, the submission formulas account for 57 per cent of the religious partisanship of 1952–1954 in French and 75 per cent in Dutch, and the religious series is highest in 1950–1958, at 0.0026–0.0031. The vocabulary used here leaves out procedural formulas, forms of address and titles, and other phrases without content (Section 3). Without them, no period of 1950–1958 exceeds the benchmark. The leading Dutch phrases of these years concern technical and public education, social insurance and pensions on the Catholic side, and the length of military service on the other (*technisch onderwijs*, *sociale voorzorg*, *twintig maanden*). Section 5.12 discusses how the vocabulary affects the other series.

Before the corrections described in Section 3, Dutch versions of the same formulas (*legt ter tafel het verslag over het wetsontwerp*) also led the French series in these years, because the record prints them within the rapporteur's or minister's French speech. After the translations

are removed, phrases containing a Dutch function word account for at most 0.026 per cent of the French phrases of Catholics, Liberals, Socialists and Communists in 1950–1958 (Online Appendix B). The third kind of phrase reflects noise in the record, discussed in Section 5.12.

5.2 Left–others partisanship

Figure 2 compares the left with other parties from 1894, when the Belgian Workers' Party entered the Chamber. The series exceeds its random benchmark in 47 per cent of periods. Net partisanship is 0.0006 in 1894–1896 and 0.0008 in 1896–1898, neither above the benchmark, and averages 0.0013 from 1898 to 1914, with its highest pre-war values in 1904–1908 (0.0020–0.0022). From 1894 to 1914, 29 to 40 Socialist speakers opposed Catholic-only governments. Their distinctive phrases concerned the working class, universal suffrage, working hours, the eight-hour day and labour courts (*classe ouvrière*, *suffrage universel*, *heures de travail*, *huit heures*, *conseils de prud'hommes*; Table 12).

Partisanship declines under the national-union governments of 1918–1921, then rises through the 1920s to 0.0039 in 1932–1934, and it exceeds its benchmark in every period from 1921 to 1940. The early 1930s were the years of strikes, special powers for Catholic–Liberal governments and the Plan of Labour. The Socialists were in opposition, with *pleins pouvoirs* and *classe ouvrière* among their most partisan phrases. Partisanship falls to 0.0010 in 1936–1939, when the Socialists join tripartite governments, and is near zero under the governments of the Liberation.

After 1945 the series is lower. From 1944 to 1958, it exceeds its benchmark only in 1954–1956, and net partisanship is at most 0.0011. From 1958 to 1985, it exceeds its benchmark in six of 12 periods, with values of up to 0.0017 in 1979–1981. The value of 1974–1977 (0.0014), under Christian-democratic and Liberal governments that excluded the Socialists, comes mainly from Dutch speech and does not exceed the benchmark. From 1985 to 2021, net partisanship averages 0.0001 and exceeds its benchmark in two of 15 periods, the longest stretch of low values in the series. It then rises to 0.0030 in 2021–2024 and reaches its highest value, [lrPeakNet], in [lrPeakLabel]. This last value comes mainly from Dutch speech (0.0105, against 0.0009 in French, which does not exceed its benchmark) and rests on 65 sittings. In Dutch, the left's most partisan phrases in 2021–2025 include *werkende klas* and *sterke overheid*, and the remaining parties' include *gegarandeerde dienstverlening*, guaranteed service (Online Appendix C). The recent values are also sensitive to how the members of the joint green group are assigned to language communities: in earlier estimates on all phrases, correcting that assignment raised the values of 2021–2024 and 2024–2025 from 0.0017 and 0.0029 to 0.0039 and 0.0062. The leave-out estimator, which does not penalize phrase effects, places 2024–2025 (0.023) above every other period after 1914 but below most periods before 1914 (Figure 3).

Simola et al. (2025) show that the extreme left drives Finland's 1970s peak. We perform the same check by excluding the Communist and radical left family in Figure 18. The family first has speakers in 1925; before then, the series changes only because penalties and covariate effects are estimated jointly across periods. Excluding it lowers the 1932–1934 value from 0.0039 to 0.0024 and removes the rise of 2021–2024 (0.0001 rather than 0.0030): in that period, the radical left, PVDA-PTB, has 13 speakers. The highest value remains that of [lrNoCommPeakLabel] ([lrNoCommPeakNet], against [lrPeakNet]). In 1950–1958, the series exceeds its benchmark only

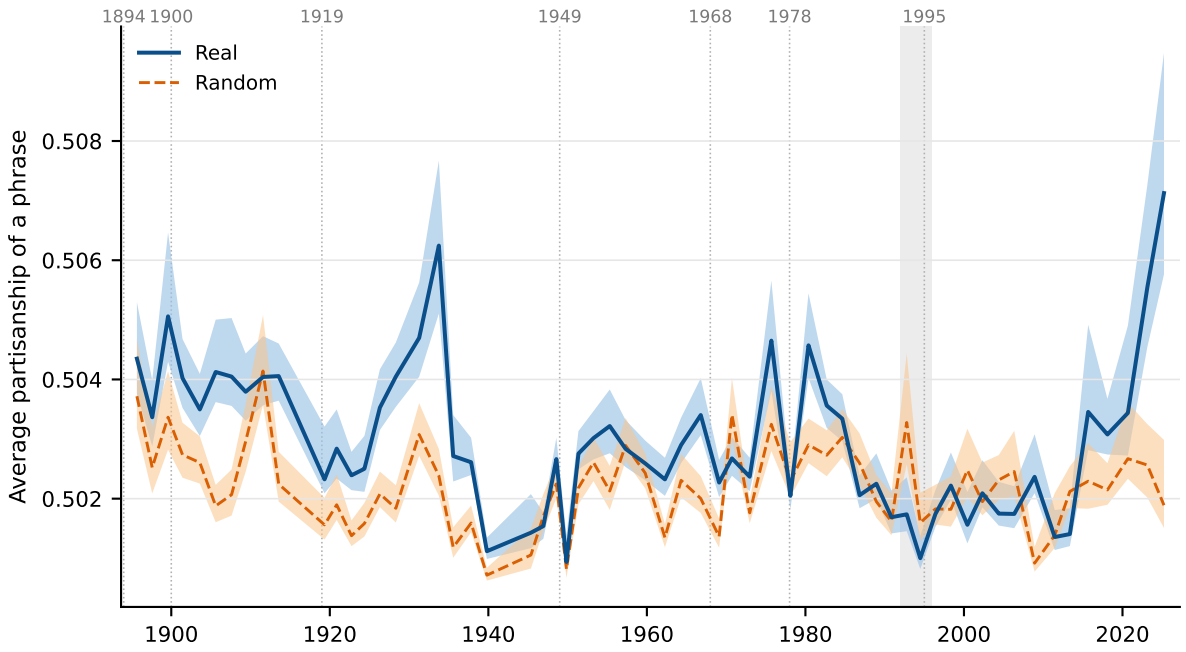


Figure 2: Left–others partisanship: Socialist and Communist families vs other parties, controls for gender and province

Notes: As Figure 1, from the first period with Socialist members (1894).

in 1954–1956 with the radical left, and in no period without it.

Online Appendix Figure A4 excludes the populist radical right: Rex, Vlaams Blok and Vlaams Belang, Front National and Parti Populaire. This follows Simola et al. (2025)'s analysis excluding Finland's Rural Party. The three highest values remain those of 2024–2025, 1932–1934 and 2021–2024. Adding government status to the main controls, as in Simola et al. (2025), lowers the real series in 36 of 58 periods, including 1932–1934 and 2024–2025 (Online Appendix Figure A3). Allowing covariate effects to vary by institutional period lowers the real series most in 2019–2025: net partisanship in 2024–2025 is 0.0029 rather than 0.0052, still above its benchmark, and 1932–1934 (0.0045) becomes the highest value (Online Appendix Figure A11).

5.3 Left–right and right–others partisanship

The left–others grouping places Christian democrats, Greens and regionalists in the same group as Liberals and the radical right. We therefore also compare the left with the right in a narrower sense: the Liberal family, the populist radical right and N-VA. Other parties are left out of this comparison. We also compare the right with all other parties. Both series use the penalized estimator with controls for gender and province, as the principal series do, with subsampling intervals and random benchmarks. They are undefined in 1894–1900, 1946–1949 and 1952–1954, when the right has fewer than 20 speakers. The leave-out estimator of Gentzkow et al. (2019) has subsampling intervals with 80 per cent nominal coverage. Table 3 reports averages by institutional period, and Figure 3 shows the series. Values in this subsection are net partisanship multiplied by 1,000.

Left–right partisanship is highest from 1904 to 1932. It exceeds its benchmark in 8 of the 12

Table 3: Left–others, left–right and right–others partisanship by institutional period

	Penalized: mean net (×1,000)			Leave-out: significant periods		
Institutional period	Left–others	Left–right	Right–others	Left–others	Left–right	Right–others
Plural voting	1.17	1.98	0.54	10/11	7/8	0/8
Equal male suffrage	1.39	1.45	−0.08	6/11	8/11	2/11
Post-war unitary state	0.44	0.43	−0.06	1/13	1/10	2/10
State reforms	0.49	0.43	0.72	2/9	3/9	2/9
Federal Chamber of 150	0.72	0.62	0.97	6/14	4/14	5/14

Notes: Penalized estimator: mean net partisanship (real minus random series, multiplied by 1,000), controls for gender and province. Leave-out estimator (Gentzkow et al., 2019), without covariates: periods in which the lower bound of the 80 per cent interval of the real series lies above the upper bound of the interval of the random series, out of the periods in which the series is defined. Left: Socialist and Communist families; right: Liberal family, populist radical right and N-VA; left–right leaves the other parties out. Averages are over the periods in which each series is defined: left–right and right–others are undefined in 1894–1900, 1946–1949 and 1952–1954, when the right has fewer than 20 speakers. Periods with low coverage of the sittings are excluded. Produced by `code/python/27_paper_extras.py`.

periods of these years, and its averages under plural voting and equal male suffrage, 1.98 and 1.45, are higher than those of left–others partisanship (1.17 and 1.39). Its highest net value, 3.81 in 1927–1929, does not exceed the benchmark, because the right has only 23 speakers and the random interval is wide; the highest value above the benchmark is 2.96 in 1929–1932 (Table 1). In both periods, Liberals governed with Catholics and the Socialists were mostly in opposition. After 1945, left–right partisanship is low. It averages 0.43 in the post-war unitary state and during the state reforms and 0.62 in the federal era, and after 1946 it exceeds its benchmark only in 2016–2019. In these decades, the left's main opponents, in government and in opposition, were often the Christian democrats, who are left out of this comparison. Across periods, left–others and left–right partisanship are moderately related (correlation 0.41 over 52 periods). The leave-out series exceeds its random benchmark in 7 of 8 periods under plural voting and 8 of 11 under equal male suffrage, but in only 1 of 10 in the post-war unitary state.

Right–others partisanship is low until the 1970s. Its averages are at most 0.54 before 1971, and it exceeds its benchmark in 3 of the 29 periods before 1971. The leave-out series exceeds its benchmark in 6 of 38 periods before 1995. Its averages are higher during the state reforms and in the federal era (0.72 and 0.97), and it exceeds its benchmark in 7 of the 14 periods since 1995, in 2003–2005 partly because of text of the voting procedure (Section 5.12). Its highest values are 4.30 in 2024–2025, 2.50 in 1983–1985 and 1.90 in 1974–1977; the second and third come when the Liberals governed with the Christian democrats. The leave-out series exceeds its benchmark in 4 of the 5 periods since 2014. The recent rise of left–others partisanship therefore coincides with a period in which the wording of the right and that of all other parties have also become easier to tell apart. In a variant in which the Greens join the left, left–right partisanship exceeds its benchmark in 2021–2024 and 2024–2025 (2.42 and 4.32), and in only one other period after 1946 (1995–1997). In 2024–2025, the two estimators rank the left's comparisons differently: the penalized estimator places left–others partisanship (5.24) above left–right partisanship (2.15), while the leave-out estimator places left–right (37.0) above left–others (23.3).

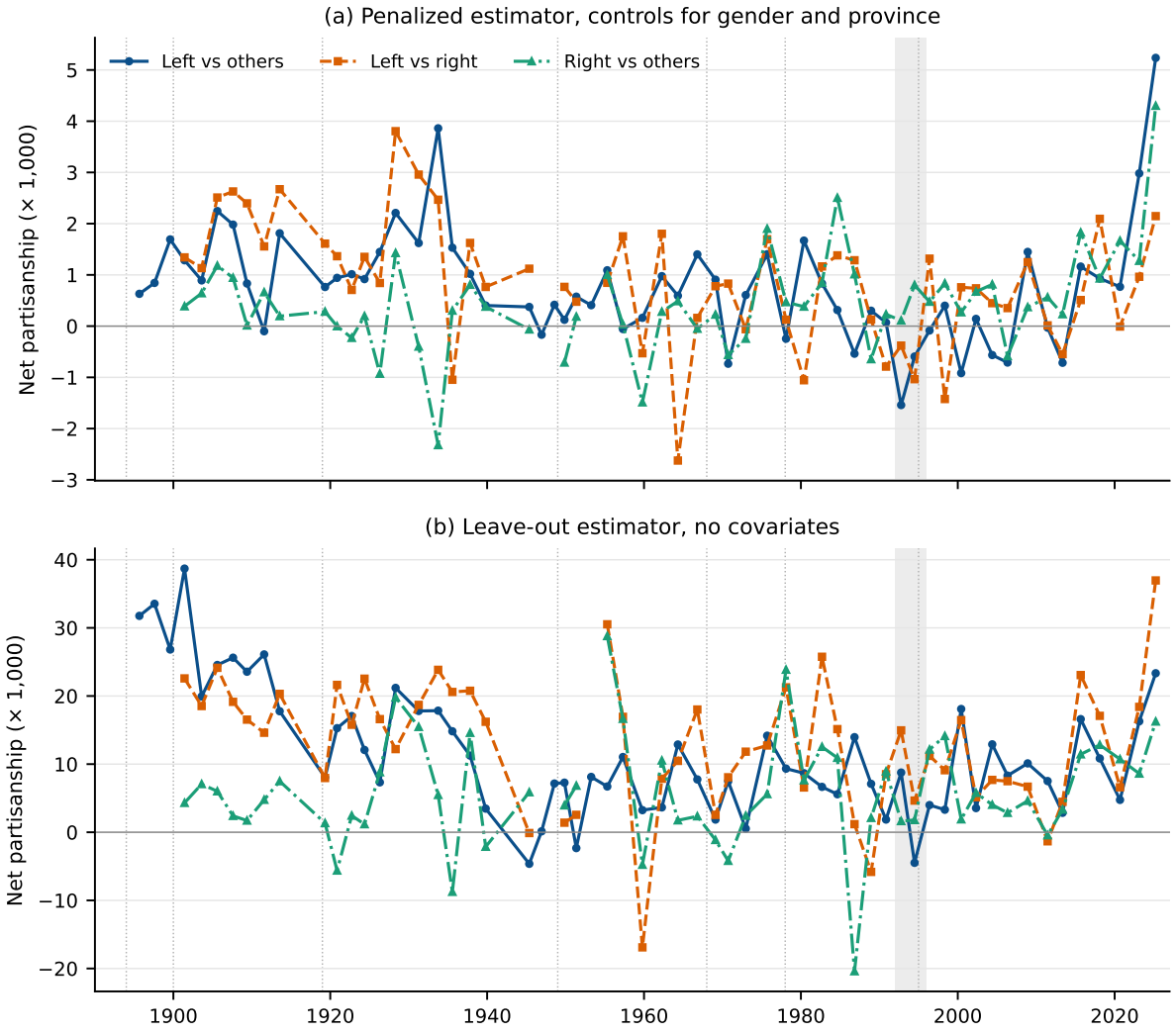


Figure 3: Left–others, left–right and right–others partisanship

Notes: Net partisanship (real minus random series), multiplied by 1,000. (a) Penalized estimator with controls for gender and province. (b) Leave-out estimator without covariates. Left: Socialist and Communist families; right: Liberal family, populist radical right and N-VA; left–right leaves the other parties out. Gaps: periods when the right has fewer than 20 speakers. Shaded: low coverage of the sittings; dotted lines: reforms. Produced by `code/python/27_paper_extras.py`.

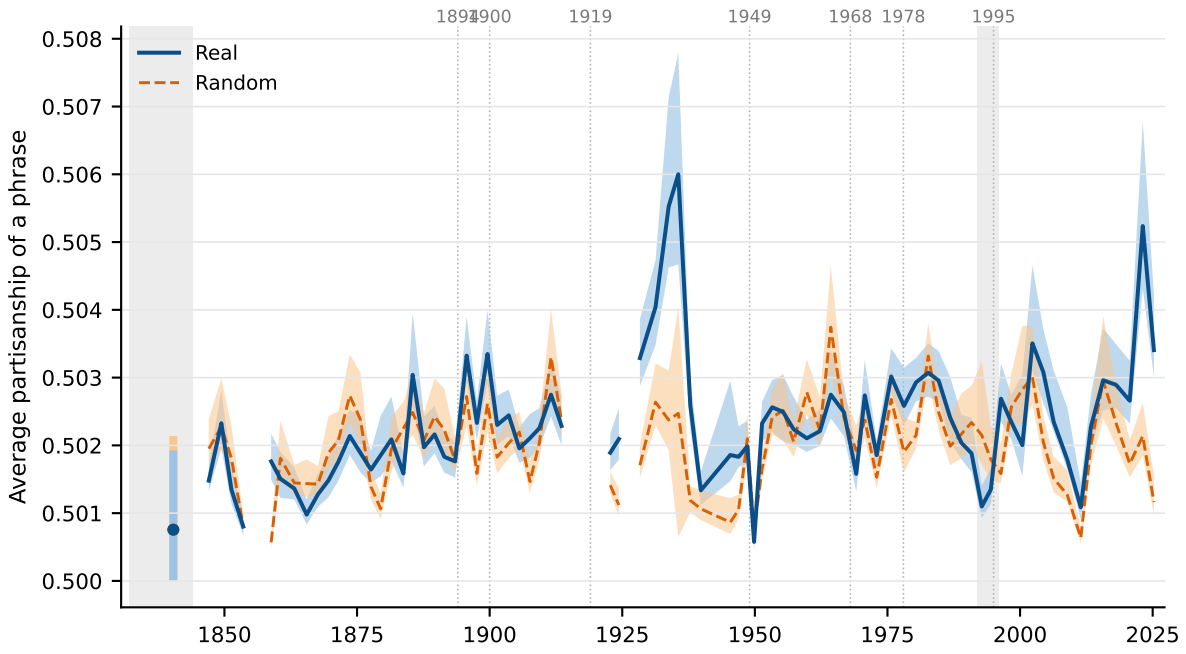


Figure 4: Government–opposition partisanship, controls for gender and province

Notes: Group A: members of the parties in government; group B: members of the parties in opposition. A member-period is in government if at least half of the member's speeches were delivered while the party was in government. Gaps: periods when the opposition had fewer than 20 speakers (unionist and national-union governments). Otherwise as Figure 1.

5.4 Government and opposition

Figure 4 compares the language of governing and opposition parties, controlling for gender and province. In Belgium, parliamentary governments date from 1831; Finland did not establish full responsibility to parliament until after 1917. The Belgian series is therefore available almost throughout the period. It is undefined in eleven periods with too few opposition speakers: the unionist or mixed cabinets of 1831–1845, except 1839–1841, and of 1854–1857, the national-union governments of 1918–1921 and 1926–1927, and 1925–1926, when the ministerial register includes an extra-parliamentary Liberal minister in a Catholic–Socialist cabinet.

The series exceeds its random benchmark in 30 per cent of periods (Table 1), less often than the left–others series (47 per cent) and about as often as the religious series (28 per cent). Its institutional-period averages (Table 4) are −0.0001 under census suffrage and 0.0003 under plural voting. They are highest under equal male suffrage, at 0.0015, near zero in the post-war unitary state and during the state reforms (0.0000 and 0.0002), and 0.0008 in the federal era.

The series is highest between the wars. It exceeds its benchmark in seven of the eight periods from 1921 to 1940 in which it is defined, and in 1944–1947. In 1927–1934, Socialists opposed Catholic–Liberal governments, and all three main series exceed their benchmarks. Net partisanship reaches 0.0032 in 1932–1934 and its highest value, [govPeakNet], in [govPeakLabel]. In that period, the Socialists joined the government of national union formed in March 1935, and the opposition was small: 27 of its members enter the series. The opposition's most partisan phrases include the decree-laws and the national-union government in French (*arrêtés-lois*, *union nationale*) and the Flemish people in Dutch (*Vlaamsche volk*). Before 1919, the series is low. Under

census suffrage, it exceeds its benchmark only in 1857–1859 and during the first school war, in 1878–1880; under plural voting, only in 1896–1898 and 1906–1908.

Under governments formed by a single family, the government–opposition grouping is the same as the religious grouping. This is the case in 32 periods with adequate coverage between 1848 and 1914 and in 1950–1954. In these periods, the real series of the two groupings differ by at most 0.0002, but their random benchmarks, each from a single permutation, differ by up to 0.0012, and in 13 of the 32 periods only one of the two series exceeds its benchmark. Single periods above or below the benchmark should therefore be read with caution (Section 4).

After 1945, the series is low. From 1947 to 1994, it exceeds its benchmark only in 1950–1952 and 1979–1981. Under the centre-right governments of 1981–1987, which used special powers, it does not exceed its benchmark, while religious partisanship does in 1981–1985. In the federal era, government–opposition partisanship exceeds its benchmark in 8 of 14 periods, more often than the right–others series (7), the left–others series (4), the left–right series (1) and the religious series (none). Its values in 2021–2024 (0.0031) and 2024–2025 (0.0022) are its highest since the 1930s. In 2021–2024, the government included Liberals, Socialists, Greens and Christian democrats and was opposed mainly by Flemish nationalists, the radical right and the radical left. As Lauderdale and Herzog (2016) find for the Irish Dáil, government and opposition form a recurring division in speech, even where other cleavages structure the party system.

The most distinctive phrases reflect the policies of the government of the day and the opposition's objections to them. In 1932–1934, the Socialist opposition speaks of the working class, the full powers and unemployment benefits (*classe ouvrière*, *pleins pouvoirs*, *allocations de chômage*). In 2024–2025, the governing side speaks of the government agreement and the energy transition (*accord de gouvernement*, *transition énergétique*), and the opposition of flexi-jobs and new taxes (*flexi-jobs*, *nieuwe belastingen*; Online Appendix C). The formulas with which the governing side submits bills and reports, and the titles with which members address ministers, are not in the vocabulary (Section 3). In estimates on all phrases, the series exceeds its benchmark in 12 of the 14 periods of the federal era (Section 5.12). The unpenalized leave-out estimator depends differently on speech volume (Online Appendix Figure A9). It exceeds its benchmark in 11 of the 14 periods of the federal era, and its federal-era average (0.016) is as high as under plural voting and equal male suffrage (0.015 and 0.016). Its highest value, 0.035, is that of 2024–2025, a short period.

5.5 Other comparisons between parties

Pairwise comparisons. Figure 5 compares the Catholic, Liberal and Socialist families in pairs, without covariates. This follows Simola et al. (2025), who compare four parties present throughout their sample. Levels and timing differ across the three pairs (Tables 1 and 2).

Catholics and Liberals, separated only by the religious cleavage, speak similarly for most of the period. Net partisanship averages less than 0.0001 and exceeds its random benchmark in 12 of 81 periods. Nine of these fall under census suffrage, when the two families held almost all the seats, among them three of the four periods from 1878 to 1886, during and after the first school war (0.0007–0.0010). After 1894, the pair exceeds its benchmark only in 1929–1932, 1974–1977 and 2003–2005. Its highest value, 0.0019 in 1974–1977, comes when the two families governed

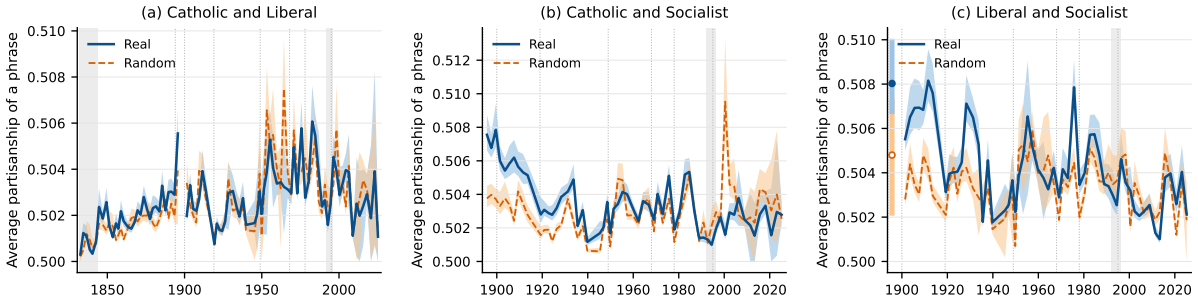


Figure 5: Partisanship, pairwise comparisons

Notes: Pairs of the Catholic, Liberal and Socialist families, without covariates; real and random series. Gaps: periods when one of the two families had fewer than 15 speakers. Otherwise as Figure 1.

together. During the second school war, net partisanship is 0.0002 in 1954–1956 and −0.0014 in 1956–1958, and the pair's decade averages are negative in the 1950s and 1960s.

Catholics and Socialists, divided by religion and class, are the pair whose partisanship most often exceeds its random benchmark: in 27 of 58 periods, including every period from 1894 to 1919 and all but one between the wars. Net partisanship averages 0.0026 under plural voting, 0.0012 under equal male suffrage and 0.0002 in the post-war unitary state. Its highest values come soon after the Socialists entered the Chamber: 0.0041 in 1898–1900 and 0.0038 in 1894–1896. The interwar maximum is 0.0025 in 1932–1934. In 1950–1958, the pair exceeds its benchmark only in 1956–1958 (0.0013), and it never does so after 1978. Since 1985, when the two families often governed together, its net partisanship is negative in 12 of 17 periods; the lowest value, −0.0079 in 1999–2001, comes from a random benchmark far above the real series.

Liberal and Socialist wording is easiest to tell apart before 1914, when both families were in opposition to Catholic governments. The pair exceeds its benchmark in every period from 1900 to 1919, and net partisanship averages 0.0030 under plural voting. It peaks at 0.0046 in 1912–1914, after their electoral cartel's defeat. Between the wars, it exceeds its benchmark in 1927–1934, when the Liberals governed with the Catholics and the Socialists were in opposition. In the 1950s and 1960s, it does so only in 1968–1969. It exceeds its benchmark again in the three periods of 1974–1981 (0.0034 in 1974–1977), when one of the two families governed and the other was in opposition, but not under the centre-right governments of 1981–1987, which the Socialists also opposed. Since 1995, it exceeds its benchmark in two of 14 periods, 1999–2001 and 2019–2021.

The pairs are not consistently related to government status. Here a family governs in a period when at least half of its speakers are in government. Catholic–Socialist partisanship is higher, on average, in periods when one of the two families governs and the other is in opposition (0.0008 against 0.0002). This difference comes from the decades before 1945, when the Socialists opposed Catholic-led governments; after 1945 it is reversed. Catholic–Liberal and Liberal–Socialist partisanship is lower in such periods (near zero against 0.0002, and 0.0007 against 0.0009). These are associations: parties that enter or leave government also change their agendas and their members. Because each comparison is symmetric, these series show when two families' wording became more or less similar, but not which family changed.

Multiparty partisanship. With more parties, the probability of identifying the right one falls mechanically. We therefore combine binary comparisons between all families present in a period: Catholics, Liberals, Socialists, Communists and the radical left, Flemish nationalists, the populist radical right, Francophone regionalists and Greens. The multiparty index averages pairwise partisanship, weighting each pair by the product of the two families' speaker counts. It measures expected partisanship between two randomly selected members of different families. Each comparison has a baseline of one half, so the index retains the other series' scale without a mechanical dependence on party count. The pairs have no covariates, and each has its own random benchmark. A pair enters a period when each family has at least 15 speakers in total; a language enters the pair's average when each family has at least 10 speakers in it.

The pairs are estimated with the language rule of Section 4. From 1968, comparisons involving Flemish nationalists or the radical right therefore use Dutch speech, and those involving Francophone regionalists French speech; the other pairs average the two languages (Online Appendix B).

Figure 6 shows point estimates of the index and the contributions of the pairs among the three traditional families and of the pairs involving other families. It also reports an index based on comparisons that control for gender. To compare a pair with its random benchmark, we use the pair's own subsampling intervals. Until 1968, the index consists mostly of pairs among the traditional families: pairs involving other families enter it only in 1936–1940 and 1946–1949. Under census suffrage, it reduces to the Catholic–Liberal comparison. It rises when the Socialists enter the Chamber, to between 0.0030 and 0.0044 in 1894–1900; in 1896–1900, when the Liberals have fewer than 15 speakers, only the Catholic–Socialist comparison remains.

The index is highest in 1936–1939, at 0.0048, when the Flemish National League and Rex entered the Chamber in strength. Pairs involving the Flemish nationalists contribute 0.0032 and the other pairs involving Rex 0.0008, against 0.0007 for the pairs among the traditional families. All four pairs involving the Flemish nationalists exceed their random benchmarks, and none of the other three involving Rex does. From 1944 to 2001, the index lies between -0.0011 and 0.0010 in every period with adequate coverage, and between -0.0005 and 0.0010 in 1950–1958.

The index is higher in 2001–2010 (0.0010 to 0.0016), when pairs involving Vlaams Blok and Vlaams Belang contribute 0.0010 to 0.0020, and slightly negative in 2010–2019 (-0.0010 to -0.0002), when Vlaams Belang has too few speakers to enter it. It rises to 0.0004 in 2019–2021, 0.0014 in 2021–2024 and 0.0040 in 2024–2025, a short period. This is the third-highest value of the series, after 1936–1939 and 1896–1898. No pair exceeds its random benchmark in 2021–2024.

In 2024–2025, pairs involving PVDA-PTB's radical left contribute 0.0022 and the other pairs involving the radical right 0.0012. Six of the 13 pairs exceed their benchmarks, and five of these six involve the radical left or the radical right. Pairs among the three traditional families, which have governed together in various combinations since the 1980s, contribute between -0.0005 and 0.0001 in every period since 1999. As with the pairwise comparisons above, these contributions show which comparisons became easier or harder to tell apart, not which family changed its wording. Controlling for gender gives similar results: across the periods with adequate coverage, the two indexes are correlated at 0.84, and their three highest values fall in the same periods.

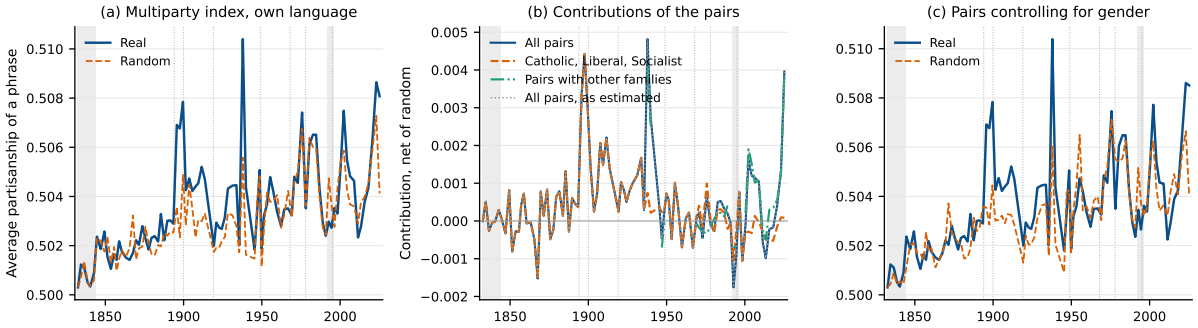


Figure 6: Multiparty partisanship

Notes: (a) Average pairwise partisanship of the party families present in each period, weighted by the product of their numbers of speakers; real and random series. Under the language rule, the pairs that involve the Flemish nationalists or the populist radical right are read in Dutch from 1968 and those that involve the Francophone regionalists in French; the other pairs average the two languages. (b) Contribution of the pairs to the index, net of the random series: all pairs, pairs among the Catholic, Liberal and Socialist families, and pairs that involve the other families; dotted grey: the same index computed from each pair's average of the two languages, which coincides with it because the language rule leaves a single language in the pairs concerned. (a) and (b) without covariates. For the three pairs of Figure 5, the real series are the same, but the random series are drawn anew. (c) the same index from pairs that control for gender. The province is not controlled for, because it is close to collinear with some of the smaller parties. Point estimates (`code/python/18_multiparty_ownlang.py`); subsampling intervals are not shown.

Parties since 1979. After the national parties split, we also compare parties within each language, following the comparisons between families above. For every pair of parties with at least 8 speakers in a language and period, we estimate partisanship without covariates, with its own random benchmark. Each comparison is symmetric: it shows when two parties' wording becomes more or less similar, not which party changed. Across 21 Dutch-language pairs (210 pair-periods), a regression with pair and period fixed effects associates being on the same side of the government–opposition divide with a difference in net partisanship of -0.82 per thousand (standard error 0.28, clustered by pair). The difference is larger when both parties are in opposition (-1.76, standard error 0.48) than when both govern (0.04, standard error 0.21). In French, pairs in which both parties are in opposition are also more similar (-1.68, standard error 0.88), while pairs in which both govern are, if anything, less similar (0.21, standard error 0.11). This is a descriptive relationship, not an estimated effect of government status: parties that enter or leave government also change their agendas, their roles and their members. In the period series of all pairs of parties of one language, a similar association appears and is larger on pairs of filler words, and governing together goes with much higher agreement in roll calls but with little change in wording beyond what phrases without political content also show (Section 8.2).

Figure 7 follows N-VA, with its predecessor the Volksunie, and Vlaams Belang. Their net partisanship shows no clear trend. Net of period effects estimated on the other pairs, its trend since 2003 is 1.32 per thousand per decade, with an 80 per cent interval from -0.45 to 3.05 over 6 periods, and 0.24 over all 8 periods. In 2024–2025, the largest differences in Dutch speech are those between VB–PVDA (21.53 per thousand) and N-VA–PVDA (17.70), while N-VA is closest to CD&V (0.32). The pair comparisons cannot identify which of the two parties changed, so they cannot tell whether either party has moved towards the other.

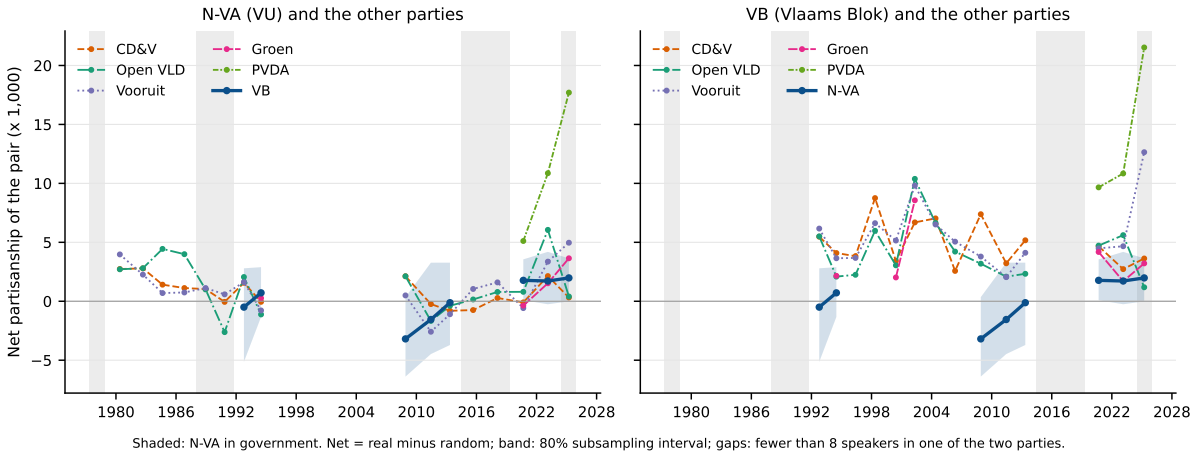


Figure 7: N-VA and Vlaams Belang compared with the other Dutch-speaking parties

Notes: Net partisanship of each pair of parties in Dutch speech, without covariates (real minus random series, multiplied by 1,000); band: 80 per cent subsampling interval; shaded: N-VA, or the Volksunie, in government; gaps: fewer than 8 speakers in one of the two parties. N-VA includes its predecessor, the Volksunie; predecessors are grouped for comparison and need not be the same organization. Produced by `code/python/24_dyads_analysis.py`.

Women and men. Women entered the Chamber in 1929 but remained below a tenth of members until 1995. Their share has exceeded a third since the parity law of 2002 and two fifths since 2019 (Online Appendix Table D2). Figure 23 measures how easily women and men can be told apart by their wording. We control for party family, allowing family effects to vary across eras beginning in 1929, 1945, 1971, 1995 and 2007. This compares women with men of the same family. Each group must have at least 15 speakers in total and 10 in a language. The series therefore begins in 1974–1977 and, until 1999, rests on Dutch speech alone.

Since 1995, net partisanship between women and men averages 0.0021 and exceeds its random benchmark in 10 of 14 periods: in every period from 1999 except 2012–2016. The highest of these values is 0.0037, in 2021–2024. Before 1995, the series is defined in four periods, with 15 to 18 women speakers, and never exceeds its benchmark. Without controls, the average since 1995 is 0.0016, and again 10 of 14 periods exceed the benchmark. Women and men are therefore about as easy to tell apart within families as across them. In the federal era, they are also easier to tell apart than the groups of the principal series, whose net partisanship averages [*relNetP6*] (religious), [lrNetP6] (left–others) and [govNetP6] (government–opposition; Table 4).

Today's parties. Figure 24 traces each party elected in 2024 back to its foundation or its predecessors; the predecessors are grouped for comparison and need not be the same organization. It compares the party's speech with that of other parties in its language group. Dutch-speaking parties are compared with Dutch-speaking parties, and French-speaking parties with French-speaking parties; before the split, each series follows the members of the predecessor party who spoke that language. PVDA-PTB, a national party, is compared across both languages. Table 8 reports legislature averages since 1995. The series have no covariates and require at least 8 speakers in the party's language. The random benchmark is therefore especially important for smaller parties.

Before 1995, the Flemish nationalists never exceed their random benchmark. The predeces-

sors of the traditional parties exceed it more often in French than in Dutch; the French-speaking Socialists, for example, do so in every period from 1894 to 1914. Since 1995, the series exceed their benchmarks in only 10 of 117 party-periods, all of Dutch-speaking parties: Vooruit in 2003–2010 and 2024–2025, Open VLD in 2003–2007 and 2021–2024, Vlaams Belang in 1997–1999 and 2001–2003, and N-VA, the largest party since 2010, in 2021–2024.

In Table 8, 62 of the 69 legislature averages lie within 0.003 of the random benchmark. The highest are those of Vooruit (0.0050) and PVDA-PTB (0.0049) in the legislature elected in 2024, which so far covers only 2024–2025, and of Vlaams Belang in the legislature elected in 1995 (0.0042). No PVDA-PTB period exceeds its benchmark: since 2019 the party has 12 to 15 speakers, and its intervals are wide. Ecolo's averages are negative in all five legislatures with enough speakers: a randomly selected group of French-speaking members of the same size is easier to tell apart from the others than Ecolo's members are. Levels cannot be compared across languages (Section 4).

5.6 The linguistic cleavage before the split

Before the national parties split, Flemish and Walloon members of the same party shared a parliamentary group. Figure 22 compares their French speech up to 1967, before the first split, controlling for party family and gender. The grouping is Flanders versus Wallonia, based on the electoral district, rather than party. Net partisanship averages 0.0008 and exceeds its random benchmark in 28 of the 58 periods with adequate coverage. Higher values occur in 1850–1861 (up to 0.0027 in 1859–1861), 1900–1910 (up to [regionPeakNet] in [regionPeakLabel]) and between the wars (0.0027 in 1927–1929). Values are lower after 1945: three of the eleven periods from 1946–1947 exceed the benchmark. There is no upward trend before the split. The last period above the benchmark is 1958–1961 (0.0015). The three following periods are below zero, and the last, 1965–1968, has the series' lowest value ([*regionLowNet*]). These last values depend on the specification: estimated together with the later periods, in which only the families that had not yet split are compared, the values of 1961–1968 lie between 0.0000 and 0.0004. In that extended comparison, net partisanship lies between −0.0003 and 0.0010 in 1968–1974 and never exceeds its benchmark; later periods have too few Flemish members speaking French (Figure 9, panel a).

The phrases show what distinguished the regions (Online Appendix C). Most reflect regional economic interests. Flemish members discuss beet sugar, linen, the port of Ostend and West Flanders. Walloon members discuss the Société Générale, the coal basins, the Charleroi canal, the canalization of the Meuse and, before 1914, the councils of industry and labour and work accidents (*industrie travail*, *accidents travail*). In 1936–1939, paid holidays and the forty-hour week distinguish Walloon members (*congés payés*, *quarante heures*), and in the 1950s the coal industry does (*industrie charbonnière*, *national charbonnages*). In 1936–1939 and 1958–1961, the Congo distinguishes Flemish members (*congo belge*, *ruanda urundi*).

Language policy appears among the distinctive phrases of Flemish members in several periods: the Flemish language in 1856–1864 (*langue flamande*), the Dutch-language university of Ghent in the 1920s (*université gand*, *université flamande*), cultural councils in 1939–1940 (*conseil culturel*) and the language question in 1961–1963, when the language border was fixed by law (*problème linguistique*). Walloon members refer to the Walloon question in 1944–1946 (*problème*

wallon). Within families, Flemish and Walloon Catholics speak similarly: their partisanship exceeds its random benchmark in only 4 of 42 periods (Online Appendix Figure A12). Socialists differ more, exceeding their benchmark in four periods between the wars and in 1952–1954. Before the split, the linguistic cleavage therefore appears in speech mainly through regional interests.

5.7 Flanders and Wallonia, 1831–2025

Wording can be compared only within one language, so the comparison above ends when the national parties split. Votes can be followed to 2025. We compare the roll-call votes of members elected in Flemish and in Walloon districts with the party-unlikeness index of Rice (1925): the absolute difference between the shares of the two groups voting yes. From each roll call's index we subtract its expected value when the region labels are assigned at random among the members voting, and we average over the roll calls of each period. Members elected in Brussels are left out, as in the speech comparison. The index measures votes, not language. Figure 8 shows the periods with at least 20 usable roll calls, which begin in 1870; Online Appendix A gives the definitions.

Panel (a) compares all Flemish with all Walloon members. Their votes differ most before 1914. The net division averages 0.17 in 1870–1893 and 0.23 under plural voting, and it peaks at 0.34 in 1898–1900, when the majorities of the two regions vote on opposite sides in 62 per cent of roll calls. It averages 0.06 between the wars, 0.07 in 1945–1970 and 0.04 in 1971–1994, when the two majorities are on opposite sides in 2 per cent of roll calls. In the federal era it rises again, to 0.13, with opposite majorities in 13 per cent of roll calls.

These divisions move with the composition of the government. Panel (c) counts the seats of the parties in government, those of all their members whether or not they speak: the seats elected in Flanders minus those elected in Wallonia, as a share of both, averaged over the sittings of each period. Before 1919, when half of the Chamber was renewed every two years, the members sitting on each day are reconstructed from their speeches and roll-call votes (Online Appendix B). Under the Catholic governments of 1870–1878 and 1884–1900, the measure is 0.47 to 0.67: members elected in Flanders hold 74 to 83 per cent of the government's seats from the two regions. Under the Liberal government of 1878–1884, it is -0.62 to -0.69 (16 to 19 per cent), and after proportional representation, in 1900–1914, 0.32 to 0.39 (66 to 70 per cent). Because Flanders elects more members, the measure is positive for a government that holds the same share of the seats of each region: a government of all the members of the two regions would score 0.05 in 1870–1872 and 0.30 in 2024–2025. Across periods, the absolute value of the measure is correlated with the net division in votes (0.57 over 66 periods); measured from the value of a government of all members, the correlation is 0.76. In the federal era, the majorities of the two regions are most often on opposite sides in 2014–2016 and 2016–2019 (51 and 39 per cent of roll calls), when MR was the only French-speaking party in government: the Michel government held 65 of the 87 seats of Flanders and 16 of the 48 seats of Wallonia. In 2024–2025, the measure is 0.21; the government formed in February 2025 holds 47 of the 87 seats of Flanders and 28 of the 47 seats of Wallonia. Like panels (a) and (b), the measure follows the districts and leaves out the members elected in Brussels. Counted by language community, with Brussels,

some governments have the opposite balance: the De Croo government (2020–2024) held 41 seats elected in Flanders, 37 in Wallonia and 10 in Brussels, where nearly all its members were French-speaking, so that its French-speaking parties held more seats than its Dutch-speaking ones.

Panel (b) compares Flemish and Walloon members of the same family: of the same national party before its split, and of the family's two parties afterwards (Christian democrats, Liberals, Socialists and Greens). Their votes differ much less. The net division averages 0.01 to 0.05 in each institutional period, and it is unrelated to the composition of the government in panel (c) (correlation with its absolute value 0.00). The highest values are in 2024–2025 (0.20), 1921–1923 (0.15) and 2007–2010 (0.12). Since 1995, the largest divisions within families occur when the two parties of a family sat on opposite sides of the government–opposition divide: the Socialists from December 2007 to December 2011 (PS in government, sp.a in opposition), the Christian democrats in 2014–2019 (CD&V in government, cdH in opposition), and the Liberals and Socialists in 2025 (MR and Vooruit in the government formed in February, Open Vld and PS in opposition). Parties of the same family also divide while governing together: in a roll call of February 2004 on the bill on the right of foreigners to vote in municipal elections, VLD voted against and MR for. In 1921–1923, the Chamber voted on the language of teaching at the University of Ghent: in a roll call of 22 December 1922 on the bill on the use of languages at the university, 95 per cent of Flemish and 3 per cent of Walloon Socialists voted yes.

Online Appendix Figure A20 classifies the roll calls by keywords in the title of the bill. Among all members, many subjects divide the regions in each institutional period, with values of up to 0.35 for subjects other than language. Within families, language is the only subject with large divisions: 0.33 under plural voting and 0.10 between the wars (10 and 12 roll calls), against at most 0.08 for any other subject in any institutional period. In the final vote on the bill on the use of Flemish in official publications, in March 1898, all Flemish and 8 per cent of Walloon Catholics voted yes; in a roll call of May 1913 on the bill on the use of languages in the army, 8 per cent of Flemish and 91 per cent of Walloon Liberals did. In 1970, Flemish and Walloon Liberals, still one party, voted on opposite sides in a roll call on the bill on the use of languages in the administration of the communes around Brussels, and so did the Christian democrats, by then two parties, on the bill on agglomerations and federations of communes. After 1945, the division within families on language bills falls to 0.04, and after the splits to 0.00 in 1971–1994 and 0.03 in the federal era, although language bills are now frequent (247 and 258 roll calls).

The subjects of speeches give a second comparison, which does not depend on wording. Panel (b) of Figure 9 assigns each French speech its most likely topic (Section 6) and asks how well the topic of one speech reveals the region of the speaker: the probability of guessing the region correctly with equal priors, net of its mean when the region labels are permuted at random among the speakers. This probability lies above the 90th percentile of its permutations in 26 of 55 periods until 1961–1963, after which fewer than 20 Flemish members speak French. It does so in 9 of the 10 periods of plural voting, when Walloon members devote 16 per cent of their substantive speeches to labour and Flemish members 9 per cent, while Flemish members speak more often of transport (12 against 8 per cent), agriculture (8 against 6) and language policy (4.4 against 2.7). The differences are small: the probability is at most 3.76 percentage

Table 4: Partisanship by institutional period

Institutional period	Catholic vs other	Left vs other	Government vs opposition
Census suffrage, 1831–1893	0.00004		-0.00010
	(0.00010)		(0.00010)
Plural voting, 1894–1918	0.00070	0.00117	0.00029
	(0.00024)	(0.00017)	(0.00017)
Equal male suffrage, 1919–1944	0.00055	0.00139	0.00153
	(0.00012)	(0.00036)	(0.00043)
Post-war unitary state, 1945–1970	0.00055	0.00044	-0.00001
	(0.00017)	(0.00014)	(0.00013)
State reforms, 1971–1994	0.00043	0.00049	0.00024
	(0.00032)	(0.00023)	(0.00014)
Federal Chamber of 150, 1995–2025	-0.00028	0.00072	0.00075
	(0.00024)	(0.00056)	(0.00033)

Notes: Mean partisanship net of the random series, by institutional period; Newey–West standard errors (two lags) in parentheses. Periods with low coverage of the sittings (1832–1843, 1992–1995) are excluded.

points above the mean of its permutations, in 1868–1870. From the late 1930s, fewer Flemish members speak French (24 to 43 in the periods measured after 1944), so the comparison rests on a smaller group.

After the split, Flemish members speak Dutch and Walloon members French, and the topics of the two languages are labelled by different classifiers. On all speech, the topic shares of Flemish and Walloon members differ by 16 per cent in the federal era (half the sum of the absolute differences), but Dutch and French speeches in the same sittings already differ by 10 per cent. We therefore do not compare the subjects of the two regions' speech after the split.

Taken together, Flemish and Walloon members of the same party differed before the split in the subjects they raised and, in votes, mainly on language bills; their wording did not become easier to tell apart before the split. The votes of all Flemish and all Walloon members tend to differ more when the government draws most of its seats from one region and leaves most members of the other in opposition, most of all before 1914; in the federal era, the highest value is in 2014–2016. These comparisons are descriptive, and the Rice index measures votes, not language.

5.8 Partisanship and institutional change

Table 4 reports average net partisanship by institutional period. Table 9 reports changes around each reform: plural voting, first used in 1894; proportional representation in 1900; equal male suffrage in 1919; women's suffrage in 1949; party splits in 1968 and 1978; and the federal Chamber in 1995. Figure 25 plots the series alongside institutional-period averages. These comparisons describe changes around each reform; they do not isolate its effect, because other events occurred at the same time.

Religious partisanship averages 0.00004 under census suffrage, 0.00070 under plural voting and 0.00055 under equal male suffrage and in the post-war unitary state. It is 0.00043 during the state-reform decades and -0.00028 in the federal era. Left–others partisanship averages 0.00117 under plural voting, 0.00139 under equal male suffrage and 0.00044 to 0.00072 afterwards; the federal-era mean has a large standard error because of the high values of 2021–2025.

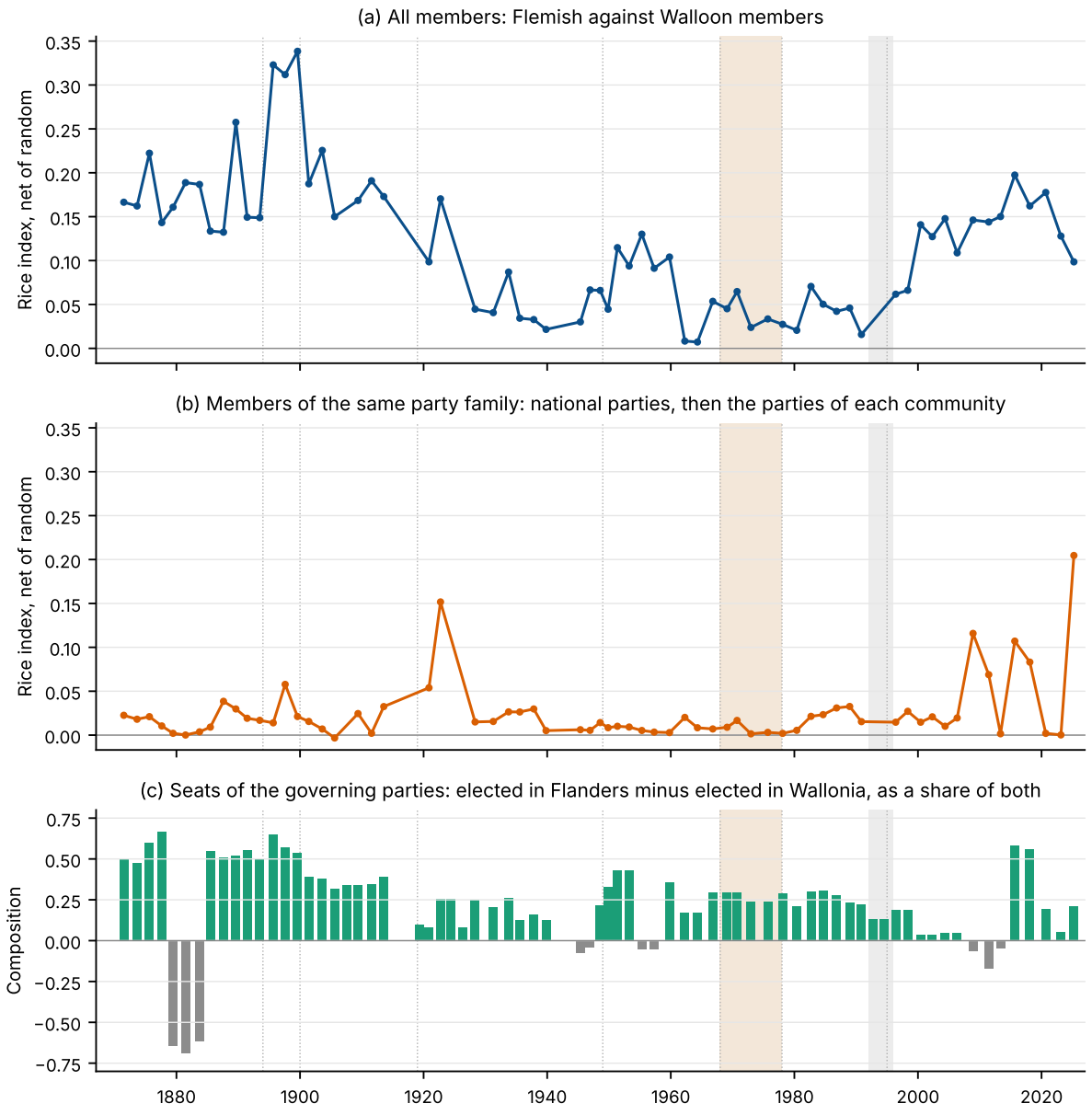


Figure 8: Flemish and Walloon members in roll-call votes, 1870–2025

Notes: (a) Members elected in Flemish against members elected in Walloon districts: Rice index of party unlikeness (Rice, 1925), the absolute difference between the two groups' shares of yes among their yes and no votes, minus its expected value when the region labels are assigned at random among the members voting; mean over the roll calls of the period with at least 10 yes or no votes in each region. (b) The same within each party family, with at least 5 yes or no votes in each region: the national party before its split (Catholics 1968, Liberals 1972, Socialists 1978), afterwards the family's parties in the two communities (CD&V and cdH/Les Engagés, Open Vld and MR, sp.a/Vooruit and PS, Groen and Ecolo); mean over the families, weighted by their votes. (c) Composition of the government: the seats of the parties in government elected in Flemish districts minus those elected in Walloon districts, as a share of both, mean over the sitting days of the period; all members count, whether or not they speak (Online Appendix B, "Seats of the governing parties"). Members elected in Brussels are left out, and so are the roll calls in which a member appears more than once (2,402 of 31,615). Periods with fewer than 20 roll calls and periods with low coverage of the sittings are not shown in panels (a) and (b). In panel (c), green: positive; grey: negative. Shaded: the years of the party splits, 1968–1978 (beige), and low coverage (grey); dotted lines: start of the institutional periods of Table 4. Produced by `code/python/29_regions.py`.

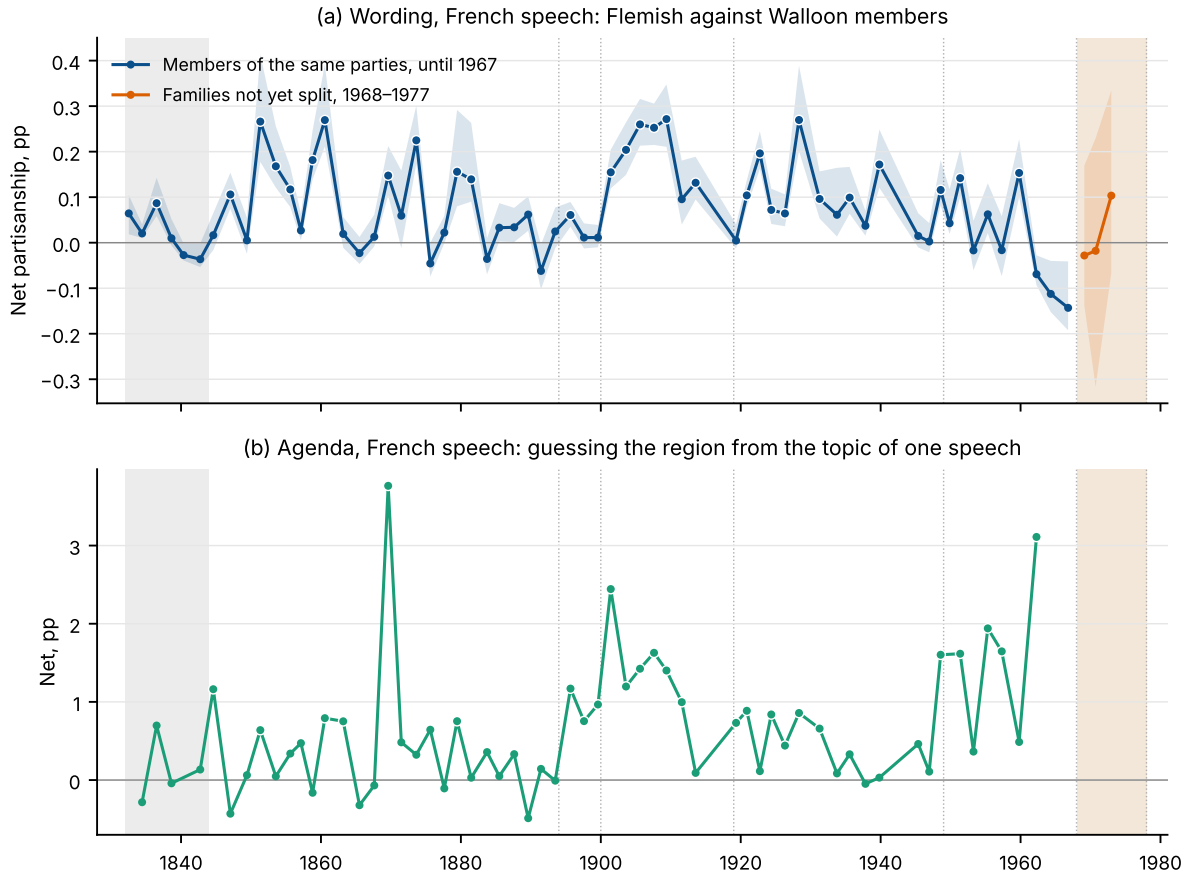


Figure 9: Flemish and Walloon members in French speech: wording and topics, 1831–1977

Notes: (a) Net partisanship between members elected in Flemish and in Walloon districts, French speech, controls for party family and gender, in percentage points (0.27 points: 0.0027), with 80 per cent intervals; dots: periods above the random benchmark. Blue: members of the national parties, until 1967 (Figure 22); orange: from 1968, members of the families that had not yet split (Liberals until 1971, Socialists until 1977), in a series estimated with all periods to 1977. (b) Probability of guessing the region of the speaker from the topic of one French speech, with equal priors, minus its mean when the region labels are permuted among the speakers of the period (1,000 permutations), in percentage points; speakers with at least five speeches on a substantive topic, and at least 20 speakers in each region; dots: periods above the 90th percentile of the permutations. Shaded: low coverage of the sittings (grey) and the years of the party splits, 1968–1978 (beige); dotted lines: start of the institutional periods. Produced by `code/python/29_regions.py`.

Government–opposition partisanship is −0.00010 under census suffrage and 0.00029 under plural voting. It is highest under equal male suffrage, at 0.00153, near zero in the post-war unitary state and during the state reforms (−0.00001 and 0.00024), and 0.00075 in the federal era.

The 1894 elections are followed by increases of 0.00126 in religious partisanship (standard error 0.00025) and 0.00092 in government–opposition partisanship (standard error 0.00011). This agrees with Graziani and Vanschoonbeek (2026), who find that the partisan gap widened in the Chamber relative to the Senate after these elections. As discussed above, the rise coincides with Socialists joining group *B*; the Catholic–Liberal pair exceeds its random benchmark in no period of 1894–1914 (Section 5.5). Section 8.4 compares the change of 1894 with the changes at the other elections since 1831 and on phrases without political content. After proportional representation in 1900, government–opposition partisanship is 0.00048 lower (standard error 0.00021), while the changes in religious and left–others partisanship (−0.00061 and 0.00042) lie within two standard errors. Around equal male suffrage, religious and left–others partisanship change little (−0.00010 and 0.00013), while government–opposition partisanship is 0.00104 higher (standard error 0.00024), in a comparison of four periods that leaves out the national-union governments of 1918–1921.

Around women's suffrage, first used in 1949, the three series change by less than two standard errors (−0.00028, 0.00034 and −0.00040), although the later periods of this comparison include the Royal Question and the beginning of the second school war. Women's voting rights changed the Chamber's composition little. After the Christian democrats split in 1968, left–others partisanship is 0.00074 lower (standard error 0.00031), mainly because of its low value in 1969–1971. The other changes after the party splits of 1968 and 1978 lie within two standard errors, including the higher religious partisanship after 1978 (0.00125, standard error 0.00070).

Partisanship can change through member turnover or through changes in sitting members' speech. Figure 26 and Table 10 decompose changes in leave-out partisanship between consecutive periods. The decomposition uses each member's leave-out score, pooled over the two languages, so from the 1890s its changes differ somewhat from those of the leave-out series of Online Appendix Figure A9, which combines the estimates of the two languages. One component measures changes among members present in both periods; the other measures entry and exit. For religious partisanship, the components have opposite signs in five of six institutional periods. Across the full sample, changes among continuing members add to partisanship, while turnover reduces it. This is consistent with members becoming more distinctive as their careers progress and entrants speaking less distinctively than those they replace. We treat this as a career pattern rather than a response to specific reforms and avoid interpreting individual periods. Neither component measures a change in political positions: sitting members' speech can become more distinct because they take up different subjects or roles.

5.9 Partisanship and the economy

Most variation in partisanship occurs within institutional periods (Figure 25). Figure 10 compares the three series with GDP-per-capita growth averaged over each period. Across all periods, correlations are 0.03 for religious partisanship, −0.17 for left–others partisanship and −0.13 for government–opposition partisanship. Within the institutional periods in Table 4, which account

for long-run changes, the correlations are −0.12, −0.25 and −0.15 (panels c–e). The slopes per percentage point of growth are −0.00004, −0.00010 and −0.00006, with Newey–West standard errors of 0.00003, 0.00003 and 0.00007. Only the slope for left–others partisanship is more than two standard errors from zero.

Within institutional periods, partisanship therefore tends to be somewhat higher when growth is slower than usual, most clearly between the left and other parties. The clearest case is the depression of 1932–1934, when all three series are above their institutional-period averages, left–others partisanship by 0.0022. Reconstruction in 1919–1923 combines the fastest growth with low partisanship in every series. Slow growth also coincides with high religious partisanship in 1981–1983 and with high left–others partisanship in 2007–2010. Other slow periods do not follow the pattern. In 1929–1932, the slowest growth relative to its institutional period, partisanship is close to its institutional-period average in all three series, and in 1936–1939 it is below average in all three. In the slowdowns of 1952–1954 and 1956–1958, religious and left–others partisanship are at or below their averages.

This relationship is descriptive. Growth and partisanship may share causes, and partisan conflict may itself slow growth. Reconstruction occurred under national-union governments. Excluding the post-war unitary state, whose slowdowns of the 1950s coincide with average or low partisanship, gives within-period correlations of −0.22, −0.28 and −0.17; leaving out any one institutional period gives within-period correlations between −0.29 and −0.04 for the three series. Evidence from elsewhere links economic hardship to political extremism in interwar Europe (de Bromhead et al., 2013), and shows increased polarization after financial crises but not ordinary recessions (Funke et al., 2016).

5.10 Magnitudes

For comparisons across parliaments, we follow Gentzkow et al. (2019) and express partisanship as the probability of identifying a member's group after one minute of speech, roughly 33 phrases. This number is implied by the estimated model, which treats the phrases as independent draws from the phrase-choice probabilities. It is not the observed accuracy of a classifier applied to real speeches. Online Appendix Figure A1 shows the projection as the number of phrases grows; Figure 11 plots the one-minute values by period with estimates from other parliaments. In the US Congress, the estimated probability rose from 57 per cent in 1989 to 73 per cent in 2007 (Gentzkow et al., 2019). Finland reached slightly under 70 per cent in the 1970s and was near 55 per cent at the end of its sample (Simola et al., 2023). For Norway, the reported values are 57 per cent in 1982–2000 and 61 per cent in 2001–2020 (Fiva et al., 2021). Section 5.11 discusses these comparisons.

In Belgium, the religious-group probability ranges from 52 to 60 per cent in 1846–1914, reaches [relMinutePeak] per cent in [relMinutePeakLabel] and is [relMinuteLast] per cent in the last period. The left–others probability is 61 per cent in 1894–1896, peaks at [lrMinutePeak] per cent in [lrMinutePeakLabel], and is [lrMinuteLast] per cent in the last period. The government–opposition probability reaches [govMinutePeak] per cent in [govMinutePeakLabel], below the US Congress in 2007, and is [govMinuteLast] per cent in the last period.

These projected probabilities come from the real series, with no correction for finite-sample

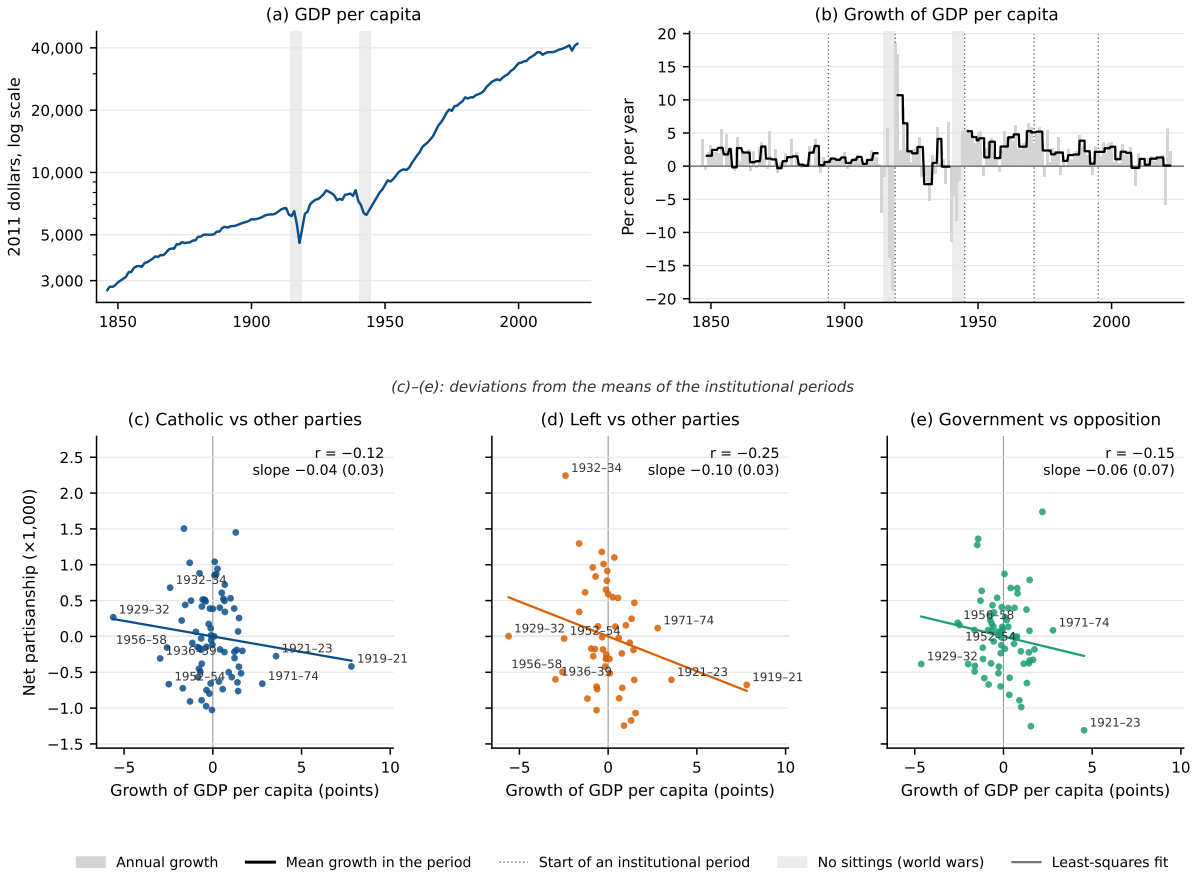


Figure 10: The Belgian economy and partisanship, 1846–2022

Notes: (a) GDP per capita in 2011 dollars, log scale. (b) Annual growth of GDP per capita, and its mean in each period of the partisanship series, with the years weighted by the days of the period that fall in them (periods that include a year of war or occupation left out); dotted lines: start of the institutional periods of Table 4 (1894, 1919, 1945, 1971, 1995). (c)–(e) Net partisanship (Figures 1, 2 and 4), multiplied by 1,000, against the mean growth of GDP per capita in the period, both as deviations from the means of the institutional period among the periods shown in the panel; line: least-squares fit; r: correlation; slope: change in partisanship (×1,000) per percentage point of growth, with fixed effects for the institutional periods, and its Newey–West standard error (two lags). Periods with low coverage of the sittings and the four periods that include a year of war or occupation (1912–1914, 1918–1919, 1939–1940, 1944–1946) are excluded; labels mark the periods whose growth is at least 2.3 points away from the mean of their institutional period. Shaded: no sittings of the Chamber (August 1914 to November 1918, May 1940 to September 1944). Source: Maddison Project Database 2023 (Bolt and van Zanden, 2024, 2025), from Gadisseur (1973) for 1846–1913, Carbonnelle (1959) for 1913–1950 and the Total Economy Database of the Conference Board for 2008–2022 (`code/python/22_economy.py`).

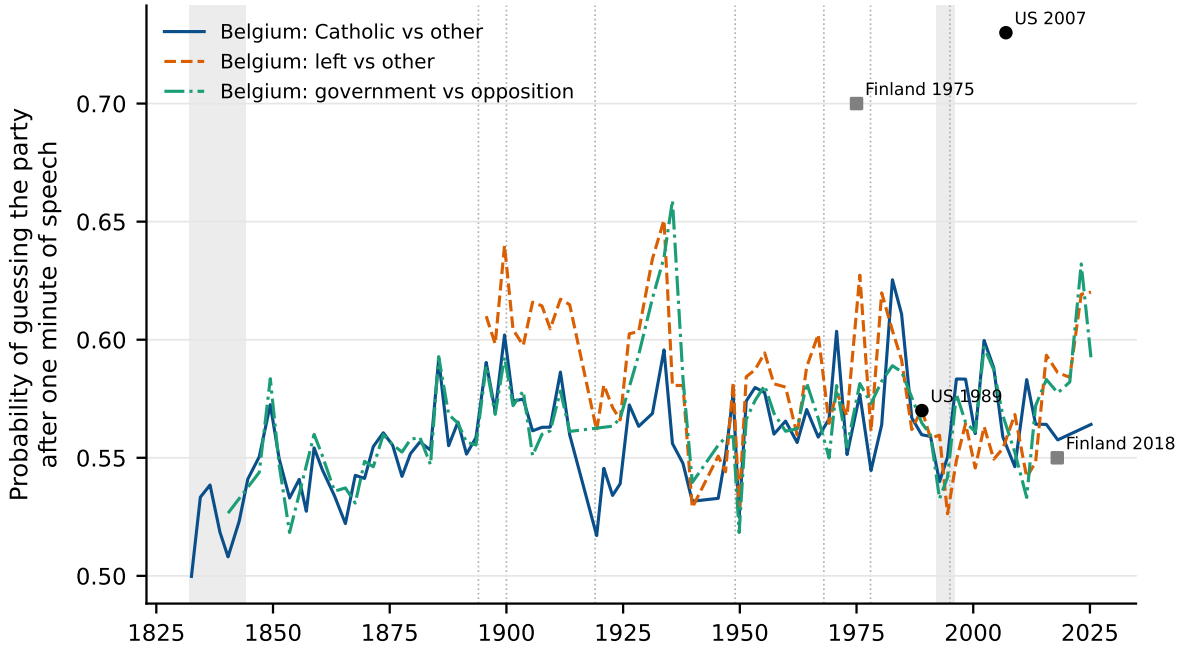


Figure 11: Probability of guessing the party after one minute of speech

Notes: Expected posterior probability assigned to the actual group after hearing 33 phrases (Gentzkow et al., 2019), by period, computed from the real series under the model, with the phrases treated as independent. Markers: values reported by Gentzkow et al. (2019) for the US Congress and by Simola et al. (2023) for Finland.

bias. Even a random-series estimate of 0.502 implies a one-minute probability of about 56 per cent when group labels contain no information. With that qualification, Belgium's highest projections remain below Finland's peak in the 1970s. In the latest Belgian period, probabilities between 56 and 62 per cent are higher than Finland's at the end of its sample and, for left–others, close to Norway's 2001–2020 figure.

Gentzkow et al. (2019) compare their speech measure with roll-call voting in the US Congress. We compare differences in wording with divisions in roll-call votes in three ways. The division between two groups in a roll call is the party-unlikeness index of Rice (1925): the absolute difference between the groups' shares voting yes, averaged over the roll calls of the period (Online Appendix A). The index measures votes, not language.

First, for the three main divides, the correlation across periods between the division in votes and net partisanship in speech is 0.19 for Catholics and other parties (64 periods), 0.18 for the left and other parties (54) and 0.05 for government and opposition (65; Online Appendix Figure A17 and Table A1). Within institutional periods, the correlations change sign. Votes divide Catholics from other parties most under plural voting (a mean of 0.65). Within that period, the two measures move in opposite directions (correlation −0.63): religious partisanship in speech is highest in 1894–1902, the division in votes in 1904–1914. Votes divide the left from other parties mainly when the Socialists are in opposition.

Second, since 1979, two parties of the same language that sit on the same side of the government–opposition divide vote alike (a mean division of 0.11) and are harder to tell apart in speech (a mean net partisanship of 0.94 per thousand) than two parties on opposite sides (0.57 and 2.67; Online Appendix Figure A18). Across the 278 pair-periods of 34 pairs, the correlation

between the two measures is 0.30; within pairs over time, it is 0.16. Third, before the splits, Flemish and Walloon members of the same national party seldom vote differently (Section 5.7), and the correlation between their division in votes and their partisanship in speech is 0.17 (49 periods; Online Appendix Figure A19).

Divisions in votes and differences in wording therefore move only loosely together. Votes record the side a group takes on a bill; wording records how its members speak, including on subjects that are not put to a vote. Figure 21 compares religious partisanship with five other features of the Chamber, following Simola et al. (2023). Online Appendix Figure A21 reports the OCR experiment. In the born-digital reports of 2019–2025, a character error rate of 5 per cent reduces left–others partisanship above 0.5 to 81 per cent of its clean-text value with covariates and 72 per cent without them. At 10 per cent error, these shares fall to 69 and 56 per cent. The religious series is undefined in most of these years. Historical volumes' error rates have not yet been measured, so we cannot use these results to correct nineteenth-century levels. If historical error rates were about 5 per cent, those levels would be understated by 19 to 28 per cent. A correction of that size would not change the rank of census suffrage among the institutional periods of Table 4.

5.11 Comparison with other parliaments

Table 5 places the Belgian estimates alongside studies of other parliaments. The comparisons are necessarily approximate: studies differ in the groups they compare, the units of text, the controls they use and how they address small-sample bias. Even so, three differences stand out.

First, the peaks occur at different times. Belgian religious partisanship is most visible in the decade after 1894 and, after the Christian democrats split, around 1970 and in the early 1980s; it never exceeds its random benchmark in the federal era. Left–others partisanship is highest in 2024–2025, a short period, and government–opposition partisanship between the wars. Elsewhere, peaks are recent or occur in the late twentieth century. US congressional partisanship is low and stable from 1873 to the early 1990s, then rises sharply (Gentzkow et al., 2019). Jensen et al. (2012) date an earlier increase in partisan language to the late 1960s. Finnish partisanship peaks in the mid-1970s and rises again from the 1990s to its 1930s level (Simola et al., 2025). In Norway's Storting, it stays at 0.502–0.503 per word until about 2000, then rises to a peak in 2019 (Fiva et al., 2021).

In the British House of Commons, party-classifier accuracy falls to its lowest level during the Second World War, rises sharply from 1979 and peaks around 1987. It then declines, faster after Tony Blair becomes Labour leader in 1994, returning to mid-1960s levels (Peterson and Spirling, 2018). In the religious and left–others series, Belgium shares Britain's lower partisanship under broad coalitions, such as the national-union governments of 1918–1921, and its decline in the 1990s. Like Finland, Norway and the United States, Belgium shows a recent rise: left–others partisanship reaches its highest value in 2024–2025, and government–opposition partisanship its highest value since the 1930s in 2021–2024.

Second, peak levels differ. Belgium's highest one-minute probabilities, [relMinutePeak] per cent for the religious grouping in [relMinutePeakLabel], [lrMinutePeak] per cent for the left–others grouping in [lrMinutePeakLabel] and [govMinutePeak] per cent for government and

opposition in [govMinutePeakLabel], are below Finland's 1970s peak of slightly below 70 per cent and the US Congress's 73 per cent in 2007. Belgium's left–others probability averages 57 per cent across the periods from 1981–1983 to 1997–1999, close to Norway's bloc-identification probability of 57 per cent in 1982–2000, and 56 per cent in those from 2001–2003 to 2016–2019, below Norway's 61 per cent in 2001–2020 (Fiva et al., 2021). These comparisons are approximate: Fiva et al. (2021) use single words rather than bigrams and compare two blocs, and none of these probabilities subtract finite-sample bias.

Third, the groups behind the peaks differ. Finland's extreme left drives its 1970s peak (Simola et al., 2025). Belgium's radical left contributes to the 1932–1934 value, to the rise of 2021–2024 and to part of the 2024–2025 value ([lrNoCommPeakNet] without it, against [lrPeakNet]), but it does not change which period is highest. The highest left–others value, in 2024–2025, comes mainly from Dutch speech, and the highest religious value, in 1969–1971, mainly from French speech (Sections 5.1 and 5.2).

The government–opposition divide recurs across studies of parliamentary systems. In the Irish Dáil, it is the main dimension of speech, with ministers more extreme than backbenchers (Lauderdale and Herzog, 2016). Classifiers trained on Canadian debates learn governing and opposition roles rather than ideology and fail when parties exchange roles (Hirst et al., 2014). In the House of Commons, the second dimension of party embeddings separates government from opposition (Rheault and Cochrane, 2020). Swedish parties within the same bloc spoke most similarly during the formal alliances of 2006–2014 (Wising et al., 2026). Government–opposition interests also explain roll-call voting better than policy positions in most parliamentary systems (Hix and Noury, 2016). Belgium extends this evidence over the long run. Government–opposition differences can be measured from the 1840s. They are clearest between the wars, and since 1995 they exceed their random benchmark more often than the religious, left–others, left–right and right–others series (Section 5.4).

5.12 Limits of the measure

Four limitations matter when comparing levels across eras.

The amount of speech. The penalized estimator retains a party coefficient only when the data support it. Speech volume has fallen: members use about 4,100 vocabulary phrases per period in 1894–1918 and 1,800 in 1995–2025. Phrases with a non-zero religious coefficient fall from about 300 to about 130 per period (Online Appendix Figures A6 and D4). Part of the low religious and left–others partisanship from the mid-1980s to the 2010s may therefore reflect less speech per member rather than smaller differences in speech.

The choice of estimator affects how the recent decades compare with the past. In the unpenalized leave-out series, left–others partisanship in the 2000s and 2010s exceeds its level in the 1950s and 1960s, and 2024–2025 is higher than any other period after 1914 (Online Appendix Figure A9). Leave-out government–opposition partisanship is as high on average in the federal era as under plural voting or equal male suffrage, and it peaks in 2024–2025. The trade-off is precision: the leave-out confidence intervals are more than ten times wider than those of the penalized estimator.

Table 5: Partisanship of parliamentary speech: Belgium and other parliaments

Study	Parliament	Measure	Highest values	Recent decades
This paper	Belgium, Chamber of Representatives, 1831–2025	GST, pairs of words; party families	Religious: 1969–1971 and 1981–1983, one minute 0.60 and 0.63; left–others: 1932–1934 and 2024–2025, 0.65 and 0.62; government–opposition: 1934–1936, 0.66	Religious never above its random benchmark since 1995; left–others highest in 2024–2025; government–opposition above its benchmark in 8 of 14 periods since 1995; one minute 0.56–0.57 (left–others) and 0.57 (government–opposition) in 1981–2019
Simola et al. (2025)	Finland, Eduskunta, 1907–2018	GST, pairs of words; left vs other parties	Mid-1970s, driven by the extreme left; one minute somewhat below 0.70	Rising since the 1990s, to the level of the 1930s; one minute about 0.55
Gentzkow et al. (2019)	United States, Congress, 1873–2016	GST, pairs of words; Democrats vs Republicans	2000s; one minute 0.73 in 2007	Sharp rise from the early 1990s
Fiva et al. (2021)	Norway, Storting, 1981–2020	GST, single words; left vs right bloc	2019, 0.506 per word	Rising from about 2000; one minute 0.57 (1982–2000) and 0.61 (2001–2020)
Peterson and Spirling (2018)	United Kingdom, House of Commons, 1935–2013	Accuracy of party classifiers (about 0.65–0.76)	Around 1987	Falling after 1994, to the level of the mid-1960s
Wising et al. (2026)	Sweden, Riksdag, 1991–2023	Party classifiers	Similarity within the blocs highest in 2006–2014	Declining after the formal alliances

Notes: GST: the measure of Gentzkow et al. (2019), the expected posterior probability assigned to a speaker's actual group after one phrase. One minute: model-based probability of guessing a speaker's group after one minute of speech (about 33 phrases, treated as independent), computed from the real series; for Belgium, Table 1 and Figure 11; for the United States, Gentzkow et al. (2019); for Finland, Simola et al. (2023); for Norway, Fiva et al. (2021). Levels are comparable only roughly across studies that differ in the unit of text, the groups compared, the covariates and the treatment of finite-sample bias.

Role and substance. Different words need not imply different policies. Ministers introduce legislation; opposition members question and criticize them. These duties give rise to different expressions even when the underlying issues are shared. Our vocabulary excludes routine titles and formulas, but retains substantive references to government decisions, such as the opposition's criticism of *pouvoirs spéciaux* (Section 5.1.1). Left–others partisanship tends to be higher when Socialists are in opposition, whereas religious partisanship is only weakly associated with the government–opposition divide. Conversely, two sides may fight over a policy while using many of the same words, as in the first school war. The separate government–opposition comparison, party pairs and dictionaries of education, monarchy, language and labour terms help interpret these differences. They do not remove the effect of institutional roles from the series.

The record. Language labels, procedural text and the conversion of the reports affect several results. Before counting phrases, we remove the translations printed within speech blocks and the roll-call formulas, relabel blocks whose function words belong to the other language, and count sentences in the other language as speech in that language (Section 3). From 1968, we also apply the language rule to every party that belongs to one language community. These corrections change several results of earlier estimates. Roll-call formulas appended to Christian-democratic members' French speeches no longer produce a religious peak in 2001–2003. French passages in Dutch blocks no longer drive the left–others estimate for 1974–1977, which now comes mainly from Dutch speech. Dutch blocks labeled French no longer inflate comparisons involving the Flemish nationalists and the radical right after 2010. Translations of the submission formulas no longer enter the French series of the 1950s (Section 5.1.1).

Some problems remain. Dutch words still occur in some French blocks: *ik citeer* and *handelingen plenaire* are among the most partisan French phrases of other parties in the left–others comparison of 1969–1971 and 1989–1991 (Table 12). Text of the voting procedure also remains in some Dutch blocks: vote tallies printed in both languages in 2001–2003 (*oui nee*, *abstentions totaal*) and the formula that applies the result of a previous vote in 2003–2005 (*uitslag vorige*, *stemming gelden*) are among the most partisan Dutch phrases of the Christian democrats in the religious comparison (Online Appendix C) and of the remaining parties in the right–others comparison, and the answers of a roll call (*non neen*, *neen geantwoord*) among those of the right in 1985–1989 (phrase tables of the replication package). The right–others value of 2003–2005, which exceeds its benchmark, rests on its Dutch component, in which these phrases are four of the five most partisan phrases of the remaining parties. In earlier estimates, the recent left–others values were sensitive to how the members of the joint green group are assigned to language communities (Section 5.2).

The vocabulary. We leave phrases without content out of the vocabulary (Section 3). Some of these phrases distinguish the groups only through their roles: only those who govern use the formulas with which ministers and rapporteurs submit bills and reports. In an earlier estimation on all phrases, with the version of the dataset of 11 October 2026, the three main series were high in the 1950s, when one bloc governed and the other opposed. The religious series was highest in 1950–1958, at 0.0026 to 0.0031, and exceeded its random benchmark in all four periods

of these years, as did the left–others and government–opposition series. The religious values of these years came largely from the submission formulas (Section 5.1.1). With the vocabulary used here, no period of 1950–1958 exceeds its benchmark in the religious series, and only one does in each of the other two (1954–1956 and 1950–1952).

Most other patterns depend little on the vocabulary. Period by period, net partisanship in the two estimations has a correlation of 0.75 in the religious series, 0.90 in the left–others series and 0.81 in the government–opposition series, and of 0.90, 0.97 and 0.89 when 1950–1958 is left out. Every period that exceeds its benchmark in the religious and left–others series also does so in the estimation on all phrases, and so do all but one in the government–opposition series. With the vocabulary used here, net partisanship is lower in 77 of 81, 49 of 58 and 75 of 77 periods, and fewer periods exceed their benchmarks, especially for government and opposition after 1945: 2 rather than 8 of the 13 periods of the post-war unitary state, 1 rather than 6 of the 9 periods of the state reforms, and 8 rather than 12 of the 14 periods of the federal era. Several results hold in both estimations. Religious partisanship is higher under plural voting than under census suffrage and never exceeds its benchmark in the federal era. Left–others partisanship exceeds its benchmark in 9 of the 11 periods under equal male suffrage and is highest in 2024–2025. Government–opposition partisanship exceeds its benchmark in most periods between the wars and since 1995.

6 Topics: what the parties talk about and how

Two parties can sound different because they discuss different issues or because they describe the same issue in different words. Following Gentzkow et al. (2019), we distinguish the two. *Between-topic partisanship* measures how much groups differ in the attention they allocate to topics. *Within-topic partisanship* measures how distinguishable their wording is within each topic, averaged across topics according to their share of speech.

The examples in Section 5.1.1 suggest that both mechanisms matter. During the first school war, Catholics and Liberals devoted comparable attention to schooling but chose different school vocabulary. By contrast, the Royal Question and labour issues produced clear differences in attention. In 1949–1950, Catholics accounted for 19 per cent of Royal Question phrases despite making up 46 per cent of speakers. Between 1894 and 1914, Socialists made up 18–22 per cent of speakers yet used 45–59 per cent of labour phrases, which themselves accounted for 1.4–2.7 per cent of all phrases (Online Appendix Figure A16). Language policy, however, was not primarily a Flemish agenda within the national parties: between the wars, Flemish members used 37–51 per cent of its phrases, close to their 45–48 per cent share of speakers.

Topic labels. A companion module assigns a topic to every speech in five steps, without fixing the topic list in advance. First, a language model freely labels agenda items from a sample of sittings drawn by half-legislature. It also converts keywords and EuroVoc descriptors from legislative files into labels. Second, these labels are grouped into 23 categories and a residual category for procedure. Third, the language model applies the resulting codebook to samples of passages of up to 300 words. Fourth, we train CamemBERT for French (Martin et al., 2020) and

the 2023 version of RobBERT for Dutch (Delobelle et al., 2020) on 5,868 French and 5,212 Dutch passages labelled in this way. On 612 French and 588 Dutch passages held out from training, the classifiers assign the language model's category to 74 per cent of French and 74 per cent of Dutch passages, with macro-averaged F1 scores of 0.62 and 0.63. A check of 150 passages per language by two human coders is planned. Fifth, the classifiers label every passage, and each speech takes the category with the highest mean probability across its passages, weighted by passage length. The residual category contains about half of the classified speeches but typically only 1 to 6 per cent of the phrases in a legislature, so its speeches are short or consist mostly of phrases outside the vocabulary. We group the 24 categories into nine policy areas (Table 6).

Estimation. For the religious and left–others groupings, we estimate partisanship without covariates over all phrases of the classified speeches, between topics, and within each topic and policy area. We use legislatures rather than periods as the unit of time, merging the periods of each legislature, so that each topic has more speakers. A topic or area enters a legislature when each group has at least 15 speakers in a language, and only those languages enter its average; estimates over all phrases and between topics require 20. As for the principal series, we repeat each estimate after permuting group labels among the legislature's speakers and report net partisanship. We do not compute subsampling intervals for these series, so values for individual legislatures are only indicative.

Within and between topics. Figure 12 and Table 6 report the results. Within-topic partisanship lies above its random series in 35 of 49 legislatures (the topic series have no intervals) for the religious grouping and in 32 of 36 for the left–others grouping. Partisanship within policy areas is lower than partisanship over all phrases and moves with it, closely for the left–others grouping and loosely for the religious grouping, with correlations of 0.65 and 0.33 across legislatures. Its median is 0.25 per thousand for the religious grouping, against 0.37 over all phrases, and 0.59 against 1.42 for the left–others grouping. The two levels are not directly comparable: each within-topic estimate uses fewer phrases, and the penalized estimator retains a party coefficient only when the data support it (Section 5.12).

Between topics, the random benchmark is high (Figure 12c–d). With 24 topics as the choices and thousands of phrases per member, a member's division of attention across topics distinguishes even groups drawn at random, plausibly because members specialize and random groups differ in their mix of specialists. For the religious grouping, the random series lies between 0.506 and 0.574 and the real series between 0.507 and 0.535, while partisanship over all phrases peaks at 0.506. For the left–others grouping, the real series lies above its random series in 21 of the 24 legislatures that begin before 1979, including all 13 that begin before 1946: until the late 1970s, the left divided its attention across subjects differently from the remaining parties. For the religious grouping, it does so in 20 of 39 legislatures before 1979. From 1979, the real series lies below its benchmark in 10 of 10 legislatures for the religious grouping and 9 of 12 for the left–others grouping. One possible reason is that parties divide the agenda among their members, so that each party covers subjects more evenly than a random group of the same size. Each between-topic benchmark comes from a single permutation, so we avoid interpreting individual legislatures of these series.

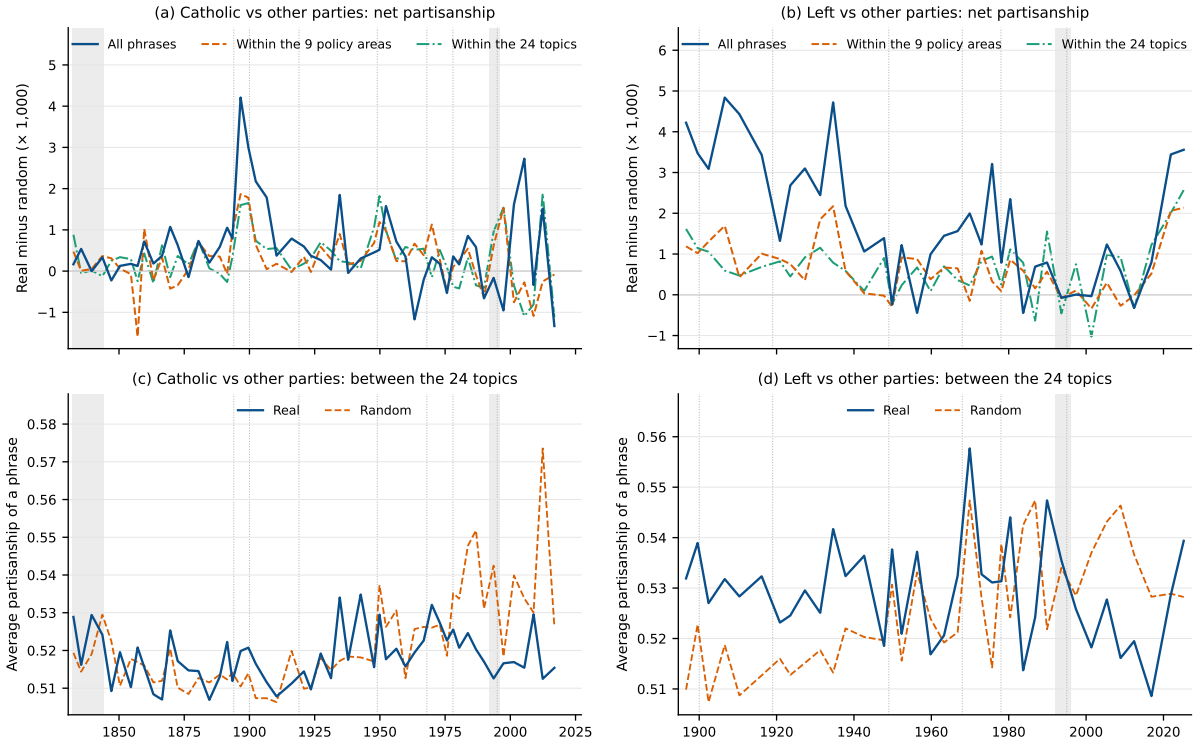


Figure 12: Partisanship within and between topics

Notes: (a)–(b) Partisanship net of the random series, multiplied by 1,000, of all the phrases of the classified speeches, and within the nine policy areas and the 24 topics (averages of the area or topic series, weighted by their phrases). (c)–(d) Partisanship between the 24 topics, which treats the topics as the choices: real series and series with randomly permuted labels. Religious and left–others groupings without covariates; time unit: legislature; points: legislatures whose neighbours have no value. Shaded: low coverage of the sittings; dotted lines: reforms. Produced by `code/python/26_topic_figures.py` from the series of `code/python/23_topic_areas.py`.

Policy areas. For the religious grouping, the most partisan areas over the whole period are public finance and money, institutions, and school, religion and culture, with medians of 0.54, 0.53 and 0.31 per thousand across legislatures; justice, rights and public order is the least partisan (0.00). Partisanship in school, religion and culture rises during both school wars: to 0.6 and 1.1 in the legislatures of 1878–1886 and to 1.9 and 1.5 in those of 1950–1958. It is the most partisan of the eight substantive areas in 1882–1886 and in both legislatures of 1950–1958, and the fourth in 1878–1882. During the first school war, this agrees with the dictionaries, which show Catholic and Liberal members using different school phrases. For the left–others grouping, public finance and money, and labour and welfare are the most partisan areas, with medians of 0.94 and 0.89, while justice, rights and public order, and school, religion and culture are the least partisan (-0.10 and -0.07). Both leading areas are most partisan under plural voting and lower in every later institutional period.

The procedural area is the most volatile area for the religious grouping. In 1950–1954, when one bloc governed and the other opposed, its net partisanship is 2.3 per thousand for the religious grouping and 1.3 for the left–others grouping, although these estimates leave out the formulas with which the governing side submits bills and reports, from which much of the religious partisanship of these years came in the estimates on all phrases (Section 5.12). Its highest values are 24.5 in 2003–2007 for the religious grouping, during the roll-call-formula years, after which it stays above 10 until 2014, and 7.3 in 2003–2007 for the left–others grouping. Removing the area changes within-area partisanship by more than 0.5 per thousand in only two legislatures, both for the religious grouping: 1999–2003 and 2003–2007.

Three questions. The dictionaries leave three questions open. Do the religious and left–others divides reflect different agendas, or different language about the same subjects? Does low religious partisanship in the federal era reflect a shared agenda or a shared vocabulary? How much of the government–opposition difference is procedural?

For both groupings, differences in language about the same subjects are more common than differences in agendas. Between-topic partisanship lies above its random series in 20 of 49 legislatures for the religious grouping and in 24 of 36 for the left–others grouping, less often than within-topic partisanship. Within-area religious partisanship is highest in 1894–1898, the first legislature with Socialist members, at 1.87 per thousand; between-topic partisanship also lies above its random series in that legislature. In the 1950s, within-area religious partisanship is 0.98 in 1950–1954 and 0.25 in 1954–1958. It is positive in 8 of 8 substantive areas in 1950–1954 but in only 4 of 8 in 1954–1958, and between topics the real series lies below its benchmark in both legislatures. Procedural text adds little: without the procedural area, within-area religious partisanship in 1950–1954 is 0.93 rather than 0.98. For the left–others grouping, within-area partisanship is highest in 1932–1936, at 2.17; in 1950–1958, 7 and 6 of the eight substantive areas are positive, and between-topic partisanship exceeds its benchmark in both legislatures.

In the federal era, religious partisanship within areas and within topics is near zero or negative, as during the state reforms, and about as low as under census suffrage (Table 6). On average, it is negative in economy and infrastructure, labour and welfare, justice, rights and public order, and school, religion and culture, while public finance and money, institutions,

Table 6: Net partisanship by policy area and institutional period

	1831–1893	1894–1918	1919–1944	1945–1970	1971–1994	1995–2025
A. Catholic vs other parties						
Legislatures	15	6	7	8	7	6
All phrases	0.4	2.1	0.5	0.3	0.1	0.5
Within the 9 policy areas	0.1	0.7	0.4	0.7	0.0	−0.2
Within the 24 topics	0.2	0.9	0.3	0.7	−0.1	0.0
Public finance and money	0.4	0.2	0.5	0.9	0.4	1.3
Economy and infrastructure	0.3	0.3	0.2	0.6	−0.3	−1.3
Labour and welfare	0.1	1.1	0.7	0.2	−0.6	−0.9
Institutions	0.0	0.8	0.8	1.0	−0.1	1.9
Language and territory	−0.6	2.0	0.4	0.5	−1.3	0.8
Justice, rights and public order	0.6	0.5	0.2	−0.3	−0.1	−1.8
Foreign affairs, defence and colonies	0.2	1.0	−0.5	1.1	0.2	0.4
School, religion and culture	−0.4	0.4	0.1	0.7	0.8	−1.0
Procedure and residual	0.2	1.1	−0.4	0.6	−0.1	7.5
B. Left vs other parties						
Legislatures		6	7	8	7	8
All phrases		3.9	2.5	1.0	1.2	1.2
Within the 9 policy areas		1.1	0.9	0.4	0.5	0.6
Within the 24 topics		0.9	0.7	0.4	0.7	0.9
Public finance and money		2.3	1.3	0.9	0.8	1.2
Economy and infrastructure		1.6	0.9	−0.3	0.4	1.0
Labour and welfare		2.2	1.4	0.5	0.7	0.5
Institutions		0.5	0.2	0.7	0.7	0.7
Language and territory		1.6	1.3	−0.5	1.4	
Justice, rights and public order		0.4	0.8	−0.2	−0.2	−0.2
Foreign affairs, defence and colonies		−0.8	0.4	0.2	0.0	0.9
School, religion and culture		−0.8	0.4	0.3	−0.2	0.1
Procedure and residual		1.1	0.5	0.6	0.9	2.0

Notes: Mean over the legislatures that begin in each institutional period (census suffrage, plural voting, equal male suffrage, post-war unitary state, state reforms and federal Chamber, as in Table 4) of partisanship net of the random series, multiplied by 1,000; no covariates. Legislatures with low coverage of the sittings, and legislatures in which partisanship over all phrases is undefined, are excluded. Within: average of the area or topic series, weighted by their phrases. Blank: fewer than two legislatures with a value. Single legislatures with small groups can dominate a cell: in 1898–1900, language and territory reaches 7.8 for the religious grouping and 13.3 for the left–others grouping. Areas and their topics: public finance and money (public finance and taxation; money and financial systems); economy and infrastructure (economic activity; agriculture and food; transport and maritime affairs; housing and public works; environment and energy); labour and welfare (employment and labour; social protection; health and medicine; family and personal status); institutions (constitutional government; public administration); language and territory (language policy; territorial government); justice, rights and public order (justice and law; rights and citizenship; public safety and emergency response); foreign affairs, defence and colonies (foreign relations and colonial governance; national defence); school, religion and culture (education; religion and church–state relations; communications, media and culture); procedure and residual (residual and procedural). Produced by `code/python/26_topic_figures.py`.

language and territory, and foreign affairs remain positive. Since the between-topic real series also lies below its benchmark from 1979, the low federal-era values reflect both a shared agenda and, in several areas, shared wording. In these estimates over all phrases, by contrast, federal-era religious partisanship is not low, owing to high values in 1999–2007 and 2010–2014 that do not appear within the policy areas. For the left–others grouping, the federal era is not uniformly low: within areas, partisanship is near zero from 1999 to 2014 and rises to 2.06 and 2.14 per thousand in 2019–2024 and 2024–2025, close to its highest value.

The figure and the table do not show the government–opposition grouping. For the groupings they show, the procedural area is the most partisan area in 10 of 49 legislatures for the religious grouping and in 4 of 36 for the left–others grouping; for the religious grouping, these include both legislatures of 1950–1958 and the four from 2003 to 2019. Estimates for government and opposition from the same module give a first answer to the third question. There, the procedural area is the most partisan area in 7 of 49 legislatures, including both of 1950–1958 and the four from 1999 to 2014. But it holds less than 7 per cent of the phrases of any legislature, so leaving it out changes within-area government–opposition partisanship by at most 0.17 per thousand. This does not separate the wording of roles within substantive speeches.

7 Reactions, references and emotions

Partisanship measures differences in speech rather than how members treat each other. A second companion module examines the affective side of debates across the entire period. It follows Peeters (2026), who study interactions between Dutch- and French-speaking federal parliament members since 1961. The module records three features. First, it detects stenographers' descriptions of disorder (interruptions, noise, protests and uproar), approval and laughter, plus the chair's calls to order and censures. Second, it detects references by name between members and whether they cross party families or language communities. Third, it will classify speakers' emotions using the eight categories of Widmann and Wich (2023), and the tone towards referenced members as no attack, civil attack or uncivil attack (Muddiman, 2017). A language model codes a sample, and a multilingual classifier (Conneau et al., 2020) extends these labels to the corpus. The first two features require no model and cover the entire period; the third is in preparation: the language model has labelled the samples, and two human coders will label part of the test samples.

The results below come from the module's run of 14 October 2026 on the version of the dataset used in this paper. Online Appendix E explains the measures. Four patterns stand out. Disorder is highest in the 1950s and, before then, from the 1870s to the 1910s. Opposition speakers receive more disorder than governing speakers. Across decades, disorder is moderately correlated with left–others partisanship, but not with religious or government–opposition partisanship. Members refer to other party families more often than a random benchmark predicts but, from 1900, to the other language community less often.

The temperature of the Chamber. Recorded disorder rises from 6 occurrences per 100,000 words spoken by members in the 1830s to 82 in the 1870s. It stays between 70 and 80 until the

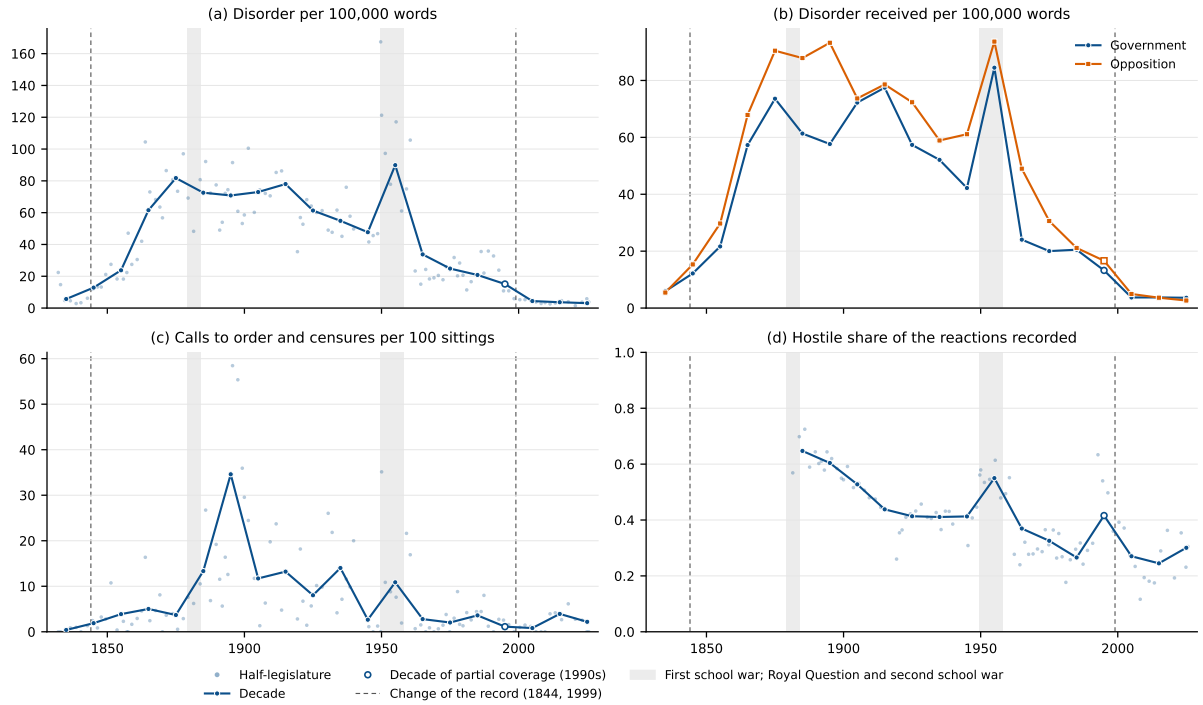


Figure 13: The temperature of the Chamber

Notes: (a) Stage directions recording disorder (interruptions, noise, protests, uproar) per 100,000 words spoken by members, the chair excluded. (b) The same, in the speeches of members of the governing parties, ministers included, and of the opposition. (c) Calls to order and censures by the chair per 100 sittings. (d) Hostile share of the reactions recorded by the stenographers, disorder / (disorder + approval), from 1880, since approval is rarely recorded between the 1840s and 1878. Points (panels a, c and d): half-legislatures; lines: decades; open markers: the 1990s, when much of the plenary business went to a Central Commission whose proceedings are not included (Section 3). Dashed lines: the record moves from the *Moniteur belge* to the *Annales parlementaires* (1844); interventions are numbered from 1999 and transcribed from audio recordings from October 2000, and the levels after 1999 are not comparable with the earlier ones. Shaded: first school war (1879–1884); Royal Question and second school war (1949–1958). Emotions module (`code/emotions`), run of 14 October 2026; produced by `code/python/21_emotions_link.py`.

1910s and falls to 48 in the 1940s (Figure 13a). It peaks at 90 in the 1950s. Around the turn of the decade, it reaches 167 in the first half of the short Royal Question legislature (1949–1950), which contains only seven sittings, and 121 in its second half. It reaches 117 in 1954–1956 during the second school war. After the School Pact, disorder rises again to 106 in 1959–1961, then falls to 34 in the 1960s and 15 in the 1990s. Levels after 1999 are not comparable because interventions became numbered and transcription later switched to recordings.

Calls to order and censures peak after the Socialists enter, at 55–58 per 100 sittings in 1894–1898. This is twice the earlier maximum of 27 in the 1884–1886 half-legislature. They rise again in the 2010s, to 4 per 100 sittings, and to 6 in 2016–2019 (Figure 13c). We calculate the hostile share as disorder divided by disorder plus approval. We begin in 1880 because stenographers rarely recorded approval from the 1840s to 1878. The share falls from 0.65 in the 1880s to 0.41 in the 1920s–1940s, rises to 0.55 in the 1950s, and stays between 0.24 and 0.37 from the 1960s, apart from 0.42 in the 1990s, a decade of partial coverage (Figure 13d).

Who received the disorder. Opposition speakers receive more disorder per word than governing-party speakers, including ministers, in every decade from the 1840s to the 2000s (Figure 13b). The ratio is 1.4–1.6 in the 1880s, 1890s, 1940s and 1970s, and about two in the 1960s: 49 versus

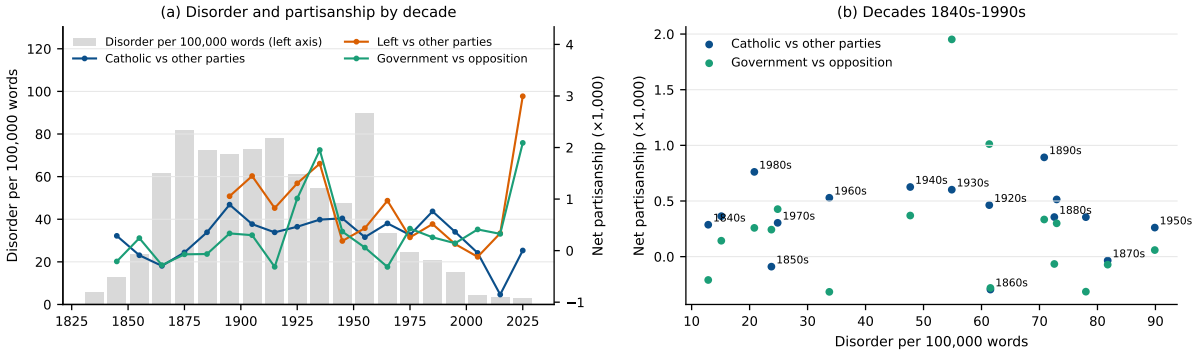


Figure 14: Disorder and partisanship

Notes: (a) Bars: stage directions recording disorder per 100,000 words spoken by members, by decade (left axis); lines: net partisanship of the religious, left–others and government–opposition series, multiplied by 1,000, averaged over the periods whose middle year falls in the decade (right axis). (b) The religious and government–opposition averages against disorder, decades 1840s–1990s. Periods with low coverage of the sittings are excluded. The comparison is made by decade because the emotions module divides the legislatures at different dates. After 1999 the levels of disorder are not comparable with the earlier ones (Figure 13). Produced by `code/python/21_emotions_link.py`, which writes the correlations to `output/series/emotions_corr.csv`.

24 occurrences per 100,000 words. The gap is small in the 1900s, 1910s and 1980s and reversed in the 2010s and the incomplete 2020s. A study of question time in 2010–2020 also finds that opposition members make most attacks (Poljak, 2023). Dutch speeches receive less disorder than French speeches until the 1980s, including 23 versus 42 per 100,000 words in the 1960s, and more from the 1990s.

Disorder and partisanship. Disorder and partisanship follow different long-run patterns (Figure 14). Across decades from the 1840s to the 1990s, disorder per 100,000 words is not correlated with net religious partisanship (-0.06) or with government–opposition partisanship (-0.01). From the 1890s, it is moderately correlated with left–others partisanship (0.44 over 11 decades): both are high from the 1890s to the 1930s and low from the 1970s. Excluding the 1950s gives correlations of -0.02, 0.03 and 0.66. Disorder is also correlated with Catholic–Socialist partisanship (0.55 over 11 decades, 0.85 without the 1950s; Section 5.5).

The 1950s show the difference most clearly. The Royal Question and the second school war produce more disorder than any other decade, but in all three series net partisanship in the 1950s is below its average over the decades compared (Figure 14a). Debates were loud, yet the parties' vocabularies were not unusually easy to tell apart.

References. References to members by name follow party and language lines (Figure 15). The module detects about 1.2 million references by name and links more than three quarters to a member or minister. In every decade, members refer to other party families more often than random speaker–target pairing within the half-legislature would predict. The gap reaches 15 percentage points in the 1870s (66 versus 51 per cent) and 12 in the 1950s (77 versus 65 per cent). From the 1860s to the 1890s, references to the other language community—or, before party splits, the other region—also exceed the benchmark by 6–7 points. From 1900, they fall below it by 6–13 points: 41 versus 49 per cent in the 1900s and 32 versus 41 per cent in the 2000s. Before proportional representation, most districts elected members of one party, so cross-region

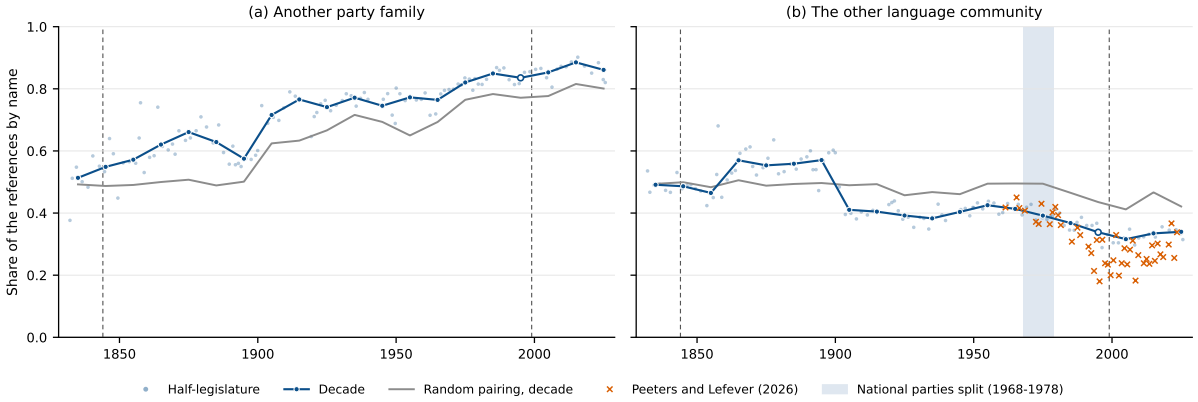


Figure 15: References between groups

Notes: Share of the references by name from one member to another, attributed to a member or minister, that point to (a) a member of another party family and (b) a member of the other language community, among the references whose speaker and target groups are known. Points: half-legislatures; lines: decades; grey lines: the share expected if, within each half-legislature, every reference paired its speaker with a target drawn at random from all the targets referred to. Crosses: the same share in the investiture and State of the Union debates of 1961–2023 studied by Peeters and Lefever (2026, Appendix 1), pooled by year. Language community: that of the party after the national parties split (1968–1978, shaded), otherwise the region of the constituency, so before the 1960s it separates Flemish from Walloon and Brussels constituencies rather than languages spoken. Dashed lines and open markers as in Figure 13. Emotions module (`code/emotions`), run of 14 October 2026; produced by `code/python/21_emotions_link.py`.

references were largely cross-party references.

We compare the module with the investiture and State of the Union debates of 1961–2023 studied by Peeters and Lefever (2026). Its intervention counts are within 15 per cent of theirs in 37 of 49 debates; the median ratio is 1.01 across the 20 debates of 1961–1991 and 0.97 across the 29 of 1993–2023. The largest shortfall is in 2006, when only one of the debate days is in the dataset. Cross-language reference shares track theirs, with a correlation of 0.80 across 49 debates (0.85 without 2006) and a mean absolute difference of 0.05 (crosses in Figure 15b). They find that only 31 per cent of references cross the language border. In their data, the share declines from 1979, after the last national-party split, and rises again from 2007.

Coding emotions and reference tone will show whether parties with more distinct speech also address each other with more anger and incivility; stage directions suggest so only for left–others partisanship. It will also show whether low partisanship in the federal era coexists with greater hostility.

8 Events: coalitions, elections and electoral reforms

Long-run series show when party language changed, but not what happened at particular political turning points. We therefore compare outcomes around events with outcomes around comparable dates. Our event catalogue was fixed before estimation and includes the electoral reforms from plural voting to the gender quotas, coalition changes since 1979, national-union cabinets, national party splits, elections and government changes, state reforms, domestic conflicts and major external shocks from the Great Depression to Russia's invasion of Ukraine (Online Appendix Table F1). Two findings survive false-discovery-rate control at 10 per cent. Coalition partners align their roll-call votes quickly when they govern together, but their wording

does not become easier or harder to tell apart beyond what placebo phrases with little political content also show. Roll-call agreement also shifts after the 2008 financial crisis and around the refugee crisis and Paris attacks in 2015. After elections, recorded disorder, approval and laughter rise for roughly three months, with no comparable shift in government–opposition wording; the rise does not appear in the months before elections or after changes of government that do not follow an election within four months. The electoral reforms leave a weaker trace. With the penalized estimator, the first election under plural voting, in 1894, is followed by the largest rise in religious partisanship at any election since 1831. Partisanship measured on numbers and dates rises too, though not on filler words, and no comparison of a reform with the other elections survives false-discovery-rate control. We do not find similarly robust changes around the other events (Online Appendix F). The appendix reports all 6,531 estimates. These comparisons remain observational: events may coincide with other developments.

8.1 Design

Outcomes. The results of this section use four outcomes. The first is the net partisanship of pairs of parties of the same language since 1979 (34 pairs with enough speakers by period), estimated by period without covariates, with the penalized estimator of Section 4 and with the leave-out estimator of Gentzkow et al. (2019). Its random benchmark is the mean of 10 (penalized) or 20 (leave-out) permutations of the party labels among the speakers of each period, rather than the single permutation of the principal series, so that the noise of one draw does not drive single periods. By half-year, the same comparison uses leave-out scores of single speeches (Online Appendix F). The second is agreement in roll calls between two parties: one minus the absolute difference between the shares of their members voting yes, averaged over roll calls (25 pairs). The third is the reactions recorded in the sittings (Section 7), per 100 speeches. The fourth is the government–opposition partisanship of the speeches of each month, from leave-out scores of single speeches. For the electoral reforms, we use the series by period of seven groupings (religious, left–others, government–opposition, left–right, right–others, Flemish–Walloon and women–men), estimated in the same way as those of the pairs.

Placebo vocabularies. The phrases left out of the main vocabulary (Section 3) provide two placebo vocabularies with little political content: numbers, amounts and dates (2,189 French and 852 Dutch phrases) and pairs of filler words (1,326 and 1,392). Their partisanship responds to who speaks and how much, and to the speech habits that a party's members share, but should not respond to a change in the political content of speech. The comparisons of wording between pairs of parties and the changes at elections are repeated on both placebo vocabularies; the monthly partisanship of government and opposition used in Section 8.3 has no placebo version.

Comparisons and inference. Each event study compares the change at an event with units that the event does not touch, with the same change at other dates, or both. The designs differ by event (Online Appendix Table F1).

- *Coalitions* (Section 8.2). The units are pairs of parties of the same language since 1979 (for roll-call agreement around single changes and crises, all pairs, including pairs of different

languages). The treated pairs are those whose two parties start or stop governing together. We regress the outcome on an indicator that both parties govern, with pair and period fixed effects, so that the pair-periods in which the two parties do not govern together serve as comparison, and report a wild cluster bootstrap by pair with the null hypothesis imposed (Cameron et al., 2008; Webb, 2023) and a placebo distribution from 499 random circular shifts of each pair's government path in time. Because parties enter and leave government at different dates, we also report the estimator of de Chaisemartin and D'Haultfœuille (2020, 2026), which uses the first change of each pair, and stacked event studies with clean controls (Cengiz et al., 2019): around each change, the pairs that change against the pairs with no change from two periods before to two periods after, with the period before the change as reference. A single change of coalition (N-VA entering government in 2014, the purple coalition of 1999, the Vivaldi coalition of 2020) is a difference-in-differences by half-year: the pairs whose status changes against the pairs whose status is stable, over four half-years before the change and four from the half-year of the change. Its p-value compares the estimate with the same estimate at every other half-year whose window does not overlap the event (62 to 81 placebo dates). For the financial crisis of 2008, the treated pairs are those on opposite sides of the government–opposition divide; for the refugee crisis and the Paris attacks of 2015, those that include Vlaams Belang or N-VA; the comparison pairs are all the others. For wording, the same estimates on the two placebo vocabularies, and on the difference between the paper's vocabulary and each placebo vocabulary, give a second comparison.

- *The Chamber after elections* (Section 8.3). The comparison is within each event: the sittings of months 0 to 2 after an election against those of months 12 to 10 before it. We stack the 73 elections since 1831, including the partial elections that renewed half of the Chamber before 1919, in bins of three months from twelve months before to eleven months after, with election and month-of-year fixed effects, and report a wild cluster bootstrap by election. Two comparisons check the result: the months before the election, and the changes of government that do not follow an election within four months, stacked in the same way. A further version leaves out the sittings of the 30 days after each election and of the 14 days after the start of the first government formed after it.

- *Electoral reforms* (Section 8.4). The comparison is a set of other dates: the elections without a reform. For each reform, we take the change in a series from the period before the first election under the new rules to the period after it, and compare it with the changes at the N elections without a reform since 1831 at which the series is defined ($N = 62$ for the religious grouping), relative to their mean. The p-value is the share of these $N + 1$ changes, the reform's included, that are at least as far from that mean, so it cannot fall below $1/(N + 1)$. The comparison is repeated with the elections within 40 years of the reform and with the elections without another shock. Three further controls follow, as in Online Appendix F, and a fourth check uses the members who stay. Placebo groupings: the mean change of the groupings that a reform should change minus that of the groupings that it should not; for 1894, the religious and government–opposition groupings against the Flemish–Walloon comparison, overall and within the Catholic and Liberal families.

Placebo vocabularies: the change on numbers and dates and on filler pairs, and the change on the paper's vocabulary minus the change on each of them. Exposure: whether families, or pairs of families, whose share of speeches changes more at the reform also change more in partisanship, compared with the same slope at the other elections. Stayers: the change among the members who speak in the same group in the periods before and after the election.

- *Splits of the national parties* (Online Appendix F). The units are the Catholic, Liberal and Socialist families by year from 1945 to 2000, with family and year fixed effects; the families that have not yet split serve as comparisons. Because there are only three splits, the p-values come from the six ways of assigning the three split years to the families and from moving the three years together by up to ten years in either direction.

- *Equality law of 1898*. The Flemish–Walloon partisanship of speech in French, by half-year: the mean of the two (or four) half-years after the law minus the mean of the two (or four) before, leaving out the half-year of the law, compared with the same change at every other half-year from 1870 to 1913 whose window does not overlap the law. Numbers and dates give the placebo vocabulary.

- *Policy areas*. The area (or the topics) that an event touches against the other areas (or topics), with area and time fixed effects, twelve years on each side unless the catalogue sets another window; by period, the period that contains the event is left out. The p-values come from the same estimate at dates more than one window away, from giving the treatment to each other area in turn (so that, when one area is treated, the smallest p-value is one eighth) and, when two areas or more are treated, from a wild cluster bootstrap by area. For the independence of the Congo, synthetic difference-in-differences (Arkhangelsky et al., 2021) weights the seven other substantive areas as donors over the three periods of 1952–1958 and compares them with foreign affairs, defence and colonies over the four periods of 1961–1969.

- *Members and single events in the sittings*. The members whose party enters or leaves government are compared with the other members, and members appointed minister for the first time with members not yet or never appointed. Single events in the sittings compare the mean of the three or six months after an event with the months before, against the same contrast at every other month of the surrounding twenty years, at least twice the window away from the event and at least three months from an election.

Because we report many tests, we compute the q-values of Benjamini and Hochberg (1995) within families of tests that answer the same question (Online Appendix Table F12). The results on coalitions and on the Chamber after elections keep a false discovery rate below 10 per cent in their families; those on the electoral reforms do not.

8.2 Governing together: votes and wording

Since 1979, parties within each language community have repeatedly entered and left governing coalitions together. Their voting records move sharply when coalition status changes. Their

wording, by contrast, moves little more than vocabulary with no clear political content (Figure 16, Table 7).

Agreement in roll calls between two parties, on a scale from 0 to 1, is 0.46 higher when they govern together ($p < 0.001$ with the wild bootstrap; no placebo date gives a larger coefficient). The change comes at once. In the stacked event study, agreement changes by 0.44 in the first period of a coalition and by -0.48 in the first period after a coalition ends; two periods before a coalition starts, the coefficient is small but significant (-0.04, $p = 0.031$). The estimator of de Chaisemartin and D'Haultfœuille (2020), which uses only the first change of each pair, gives 0.43, with a placebo of -0.02 in the period before. Single changes of coalition, compared with estimates around other dates, show the same pattern (Online Appendix Table F3). When N-VA entered government in 2014, the agreement of the new partners rose by 0.57 ($p = 0.015$), while that of the pairs that stopped governing together fell (-0.49, $p = 0.014$). After the purple coalition of 1999 and the Vivaldi coalition of 2020 were formed, the agreement of the new partners rose by 0.39 and 0.34 ($p = 0.014$ and 0.029). Votes also moved after two crises. After the financial crisis of 2008, the agreement of pairs of parties on opposite sides of the government–opposition divide fell (-0.32, $p = 0.029$); after the refugee crisis and the Paris attacks of 2015, which fall in the same half-year, that of pairs including Vlaams Belang or N-VA rose (0.20, $p = 0.014$).

Wording does not show the same change. With the penalized estimator, the partisanship of the two parties' wording does not change when they govern together (0.1 per thousand, $p = 0.471$ with the wild bootstrap, 0.560 against placebo dates), while the placebo vocabularies move in opposite directions: pairs of filler words become harder to tell apart (-1.5 per thousand, $p = 0.012$), numbers and dates easier (1.2, $p = 0.032$). The leave-out estimator gives -10.8 per thousand on the paper's vocabulary, but also -8.3 on filler pairs and -5.9 on numbers and dates; by half-year, it gives almost the same estimate on the paper's vocabulary and on filler pairs (-10.0 and -10.1). With both estimators, the partners' wording does not change significantly more than pairs of filler words; against numbers and dates, it converges somewhat more (by half-year with the leave-out estimator, -5.8 per thousand, $p = 0.101$ with the wild bootstrap and 0.024 against placebo dates). In single changes of coalition, where the wording of a pair changes, filler pairs change in the same direction. When N-VA entered government in 2014, the pairs that stopped governing together became easier to tell apart (31.9 per thousand with leave-out scores by half-year, $p = 0.014$ against placebo dates), and so did their wording on pairs of filler words (20.0, $p = 0.027$). When the Vivaldi coalition was formed in 2020, its new partners became harder to tell apart (-18.9, $p = 0.027$), and so did their wording on pairs of filler words (-14.1, $p = 0.041$). In neither case does the difference between the two vocabularies stand out from placebo dates ($p = 0.123$ and 0.479). Being on the same side of the government–opposition divide, the association reported in Section 5.5, behaves in a similar way: it goes with more similar wording on the paper's vocabulary (-0.5 per thousand) and with an even larger fall in partisanship on filler pairs (-2.2). That association may therefore reflect, at least in part, the kinds of speech that go with the roles of government and opposition rather than political vocabulary.

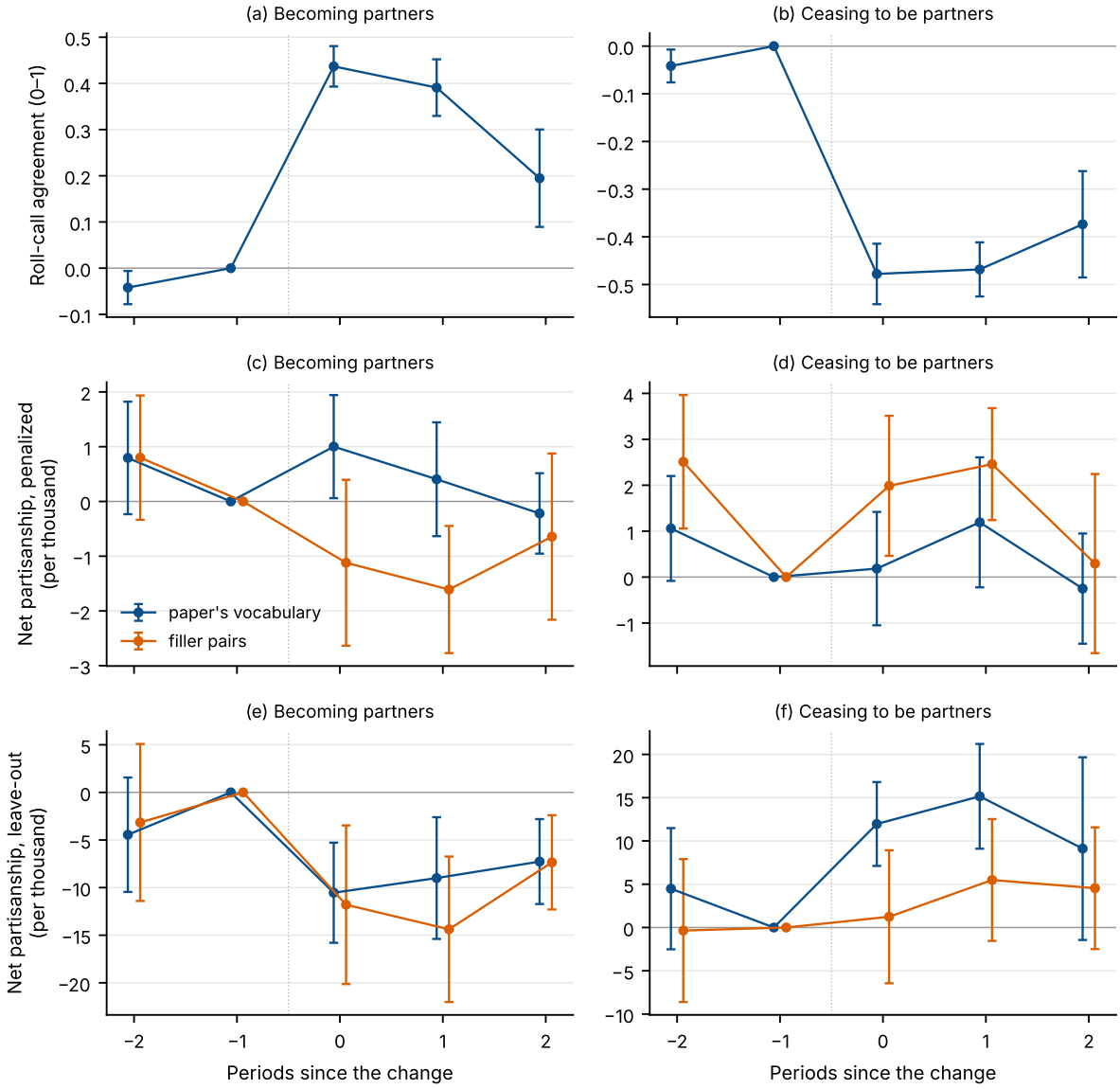


Figure 16: Roll-call agreement and wording around changes of coalition, pairs of parties, 1979–2025

Notes: Stacked event studies (Cengiz et al., 2019): pairs of parties of the same language whose government status changes (left: both parties come to govern together; right: they stop governing together), against pairs whose status does not change in the window; pair-by-event and period-by-event fixed effects; the period before the change is the reference. (a)–(b) agreement in roll calls, one minus the absolute difference between the two parties' shares of yes votes. (c)–(f) net partisanship of the pair, multiplied by 1,000: penalized (c, d) and leave-out (e, f) estimators, on the paper's vocabulary and on pairs of filler words. Bars: 95 per cent intervals, standard errors clustered by pair. Produced by `code/events/e95_paper.py`.

Table 7: Main estimates of the event studies

	Estimate	p	p, placebo
A. Coalitions, pairs of parties since 1979: both parties govern			
Roll-call agreement (0–1)	0.46	<0.001	0.002
Penalized, paper's vocabulary (per thousand)	0.12	0.471	0.560
filler pairs	−1.47	0.012	0.002
numbers and dates	1.21	0.032	0.018
Leave-out, paper's vocabulary (per thousand)	−10.81	<0.001	0.002
filler pairs	−8.27	0.011	0.002
numbers and dates	−5.94	0.168	0.070
B. Sittings after an election, months 0–2 (per 100 speeches)			
Disorder	3.42	<0.001	–
without installation and declaration sittings	2.86	0.004	–
Approval	3.04	<0.001	–
Laughter	0.83	0.004	–
C. Electoral reforms: change at the first election, penalized (per thousand)			
Plural voting (1894): religious	2.06	–	0.016
government–opposition	2.02	–	0.058
Flemish–Walloon	2.22	–	0.037
religious, filler pairs	0.34	–	0.302
religious, numbers and dates	3.71	–	0.111
Proportional representation (1900): religious	−0.73	–	0.175
Equal male suffrage (1919): religious	0.42	–	0.397
Women's suffrage (1949): religious	−0.41	–	0.397

Notes: Panel A: pairs of parties of the same language, 1979–2025, by period; coefficient on both parties governing, with pair and period fixed effects. Roll-call agreement on a scale from 0 to 1; penalized and leave-out: net partisanship of the pair without covariates (real minus the mean of the permuted series), multiplied by 1,000, on the paper's vocabulary and on the two placebo vocabularies. p: wild cluster bootstrap by pair (Cameron et al., 2008; Webb, 2023); placebo p: the same coefficient after shifting each pair's government path in time (499 shifts). Panel B: sittings stacked around the elections since 1831, months 0–2 after the election against months 12–10 before, with event and month-of-year fixed effects; reactions per 100 speeches; p: wild cluster bootstrap by election. Panel C: change in net partisanship (penalized, without covariates) from the period before the first election under each reform to the period after it, minus the mean change at the elections without a reform at which the series is defined, multiplied by 1,000; filler pairs and numbers and dates: the same change on the placebo vocabularies; placebo p: share of those elections, counting the reform, whose change is at least as far from their mean. Online Appendix F reports all estimates.

8.3 The Chamber after elections

The Chamber reacts differently for about three months after an election. Stacking the 73 elections since 1831 with sittings in the window (partial elections of half of the Chamber included), the sittings of the three months after an election record 3.4 more occurrences of disorder per 100 speeches than a year before (mean 6.1), 3.0 more of approval (mean 6.4) and 0.8 more of laughter (mean 3.9), with no change in the three months before the election (Table 7; Online Appendix Table F4). From the fourth month on, the differences are small. The rise is not confined to the sittings in which the new Chamber installs itself and debates the government's declaration: without the first 30 days after the election and the 14 days after the first government formed after it, disorder still rises by 2.9 ($p = 0.004$). Changes of government that do not follow an election within four months are not followed by a similar change. The wording of government and opposition does not move with the reactions: the government–opposition partisanship of the speeches of each month, measured with leave-out scores, changes by 0.62 per thousand in the three months after an election ($p = 0.770$).

8.4 Electoral reforms

A companion paper studies the elections of 1894 in much more detail: for 1880–1912, it compares the Chamber with the Senate around the reforms of 1894 and 1900 (Graziani and Vanschoonbeek, 2026). Here we ask only whether the change at each reform stands out among the changes at the other elections since 1831. Figure 17 plots the change in net partisanship between the period before and the period after each election, for the religious and government–opposition groupings, on the paper's vocabulary and on the two placebo vocabularies. At the first election under plural voting, in 1894, penalized religious partisanship rises by 2.1 per thousand relative to the mean change at the elections without a reform, the largest change at any election since 1831 ($p = 0.016$, the smallest possible value; Table 7). Net partisanship moves from 0.3 per thousand in 1892–1894 to 2.3 in 1894–1896, the highest value of this series without covariates. It does not move in the two periods before the election (-0.1 and 0.1 per thousand) and stays high in the next two (1.7 and 1.3). Government–opposition partisanship rises by 2.0 per thousand ($p = 0.058$) and Flemish–Walloon partisanship by 2.2 ($p = 0.037$). With the leave-out estimator, the religious change of 1894 is larger than at 98 per cent of the elections without a reform ($p = 0.032$). With 174 comparisons of reforms with the other elections, however, none of these changes keeps a false discovery rate below 10 per cent.

Phrases without political content give a mixed picture. With the penalized estimator, partisanship measured on pairs of filler words hardly moves in 1894 (0.3 per thousand), and the difference between the two vocabularies is unusual ($p = 0.016$). Partisanship measured on numbers and dates, however, rises by 3.7 per thousand, more than on the paper's vocabulary. Changes of this size are not rare in this smaller and noisier vocabulary ($p = 0.111$ against the other elections), and the difference from the paper's vocabulary is not unusual ($p = 0.397$). With the leave-out estimator, partisanship on numbers and dates rises less than on the paper's vocabulary (13.3 against 16.2 per thousand). The rise is also not confined to the groupings that the reform should have changed most: the Flemish–Walloon comparison, which we had listed among those that should not move, rises as much as the religious and government–opposition

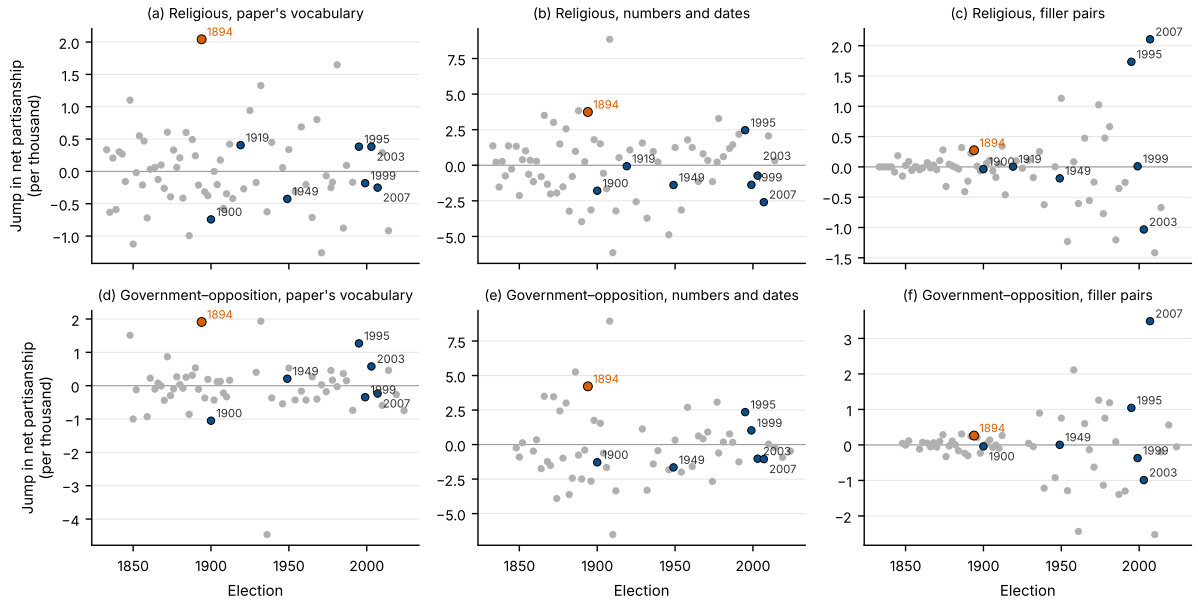


Figure 17: Change in net partisanship at every election, 1831–2025

Notes: Change in net partisanship (penalized series without covariates; real minus the mean of 10 random permutations of the group labels; multiplied by 1,000) from the period before each election to the period after it. Grey: elections without reform; blue: the first elections under a reform, labelled by year; orange: plural voting (1894). (a)–(c) religious grouping; (d)–(f) government and opposition. (a) and (d): the paper's vocabulary; (b) and (e): numbers, amounts and dates; (c) and (f): pairs of filler words (Section 8.1). Produced by `code/events/e95_paper.py`.

comparisons. This series, however, compares French speech without the party control of Section 5.6, and the 1894 elections changed the parties of each region's members: the Socialists won their seats in Wallonia, and the Catholics nearly all Flemish seats. Among the 74 members who spoke in the same religious group before and after 1894, religious partisanship rises by 3.6 per thousand ($p = 0.062$) with cross-fitted scores. With leave-out scores, their change has the opposite sign and is not unusual (-4.1, $p = 0.538$), so these members do not settle whether the rise reflects turnover.

With the penalized estimator, no other reform is followed by a change that stands out among the elections: in the religious, left–others, government–opposition and Flemish–Walloon groupings, 2 of the 23 changes after the other seven reforms have $p \leq 0.10$, about as many as chance would produce. With the leave-out estimator, 4 of 23 do, and three changes are larger than at every election without a reform: Flemish–Walloon and left–others partisanship after the first election with women's suffrage, in 1949 (19.0 and 13.6 per thousand, $p = 0.019$ and 0.057), and left–others partisanship after the gender quotas of 1999 (16.5, $p = 0.029$; Online Appendix Table F5). With the penalized estimator, religious and government–opposition partisanship fall after proportional representation in 1900; the first fall is not unusual ($p = 0.175$) and the second is borderline ($p = 0.096$). The groupings that each reform should have affected most do not move more than the others, and families more exposed to a reform, measured by the absolute change in their share of speeches, do not change more than the others at the reforms where this can be estimated, which exclude 1894 (Online Appendix Table F6). Pooled over the 7 reforms at which the series is defined, left–others partisanship measured by the leave-out estimator rises by 8.5 per thousand ($p = 0.001$), the only one of the 8 pooled comparisons on the paper's

vocabulary with $p < 0.05$. On numbers and dates, however, it rises by 11.6 ($p = 0.030$). These results qualify those of Section 5.8. The rise in religious partisanship after 1894 is the largest at any Belgian election with the penalized estimator and the second largest with the leave-out estimator. Partisanship on numbers and dates rises too, by more with the penalized estimator and by less with the leave-out estimator, so part of the rise may reflect changes in who speaks and how much rather than in political vocabulary.

8.5 What changes at events

The events that leave a visible trace in the Chamber go with changes in how members vote and react, not in how easily their parties can be told apart by their wording. This fits what the measure captures: differences in wording, not in positions or in conduct (Section 4). Of the electoral reforms, only plural voting in 1894 stands out among the elections, and partisanship on numbers and dates rises with it (Section 8.4). The other event studies find no change that stands out from comparable dates once placebo dates, placebo vocabularies and the number of tests are taken into account (Online Appendix F). They concern the speech of members whose party enters or leaves government, which cannot be separated from the party composition of the two sides; first appointments as minister; the splits of the national parties; the governments of national union; the equality law of 1898, after which only one of two estimators shows a change in Flemish–Walloon partisanship that stands out from placebo dates; the cordon sanitaire around the Vlaams Blok and the fall of the government over Brussels-Halle-Vilvoorde in 2010; the partisanship of the policy areas touched by the state reforms, conflicts and shocks, where 27 of 611 tests have $p \leq 0.05$ against placebo dates, about as many as chance alone would produce (31); and single events in the sittings, from the Royal Question to COVID-19.

9 Conclusions

What changed most in Belgian parliamentary language was not always what made politics most contentious. We have traced differences in the wording of parties in the Chamber from 1831 to 2025, across religious, left–others, left–right, right–others and government–opposition divisions, as well as comparisons of party pairs and regions. Under census suffrage, Catholic and Liberal politicians generally sounded alike. After Socialists entered in 1894, religious partisanship became more visible, exceeding its benchmark throughout 1894–1904. Left–others differences were pronounced from the turn of the century to 1940, while government and opposition were easiest to distinguish between the wars. Once procedural formulas are removed, the 1950s, the decade of the Royal Question and the second school war, show no unusual peak. Religious partisanship remains moderate until the early 1990s, is highest in the main specification in 1969–1971, soon after the Christian democrats split, and never exceeds its benchmark in the federal era. Left–others differences stay low from the mid-1980s to 2021 before reaching their highest value in the short 2024–2025 period. Government–opposition differences, meanwhile, exceed their benchmark in more than half of the periods since 1995. In the three main divisions, the model implies that one minute of speech would identify a speaker's group correctly with a probability of at most 63–66 per cent, below the corresponding figures for the United States in

2007 and Finland in the 1970s.

Separating political divisions matters. The left–others series is more closely related to the government–opposition divide than the religious series is: the left and the other parties tend to be easier to tell apart when most Socialists sit in opposition, although the relationship is far from mechanical. Among family pairs, Catholic–Socialist partisanship exceeds its benchmark most often and is higher on average when one family governs and the other opposes; the Catholic–Liberal and Liberal–Socialist pairs do not show the same pattern. Since 1979, two parties of the same language tend to sound more alike when both are in opposition. The multiparty index, which combines all pairs of families, is highest in 1936–1939, mainly because of comparisons involving the Flemish nationalists, in 1896–1898, when it reduces to the Catholic–Socialist comparison, and in 2024–2025, mainly because of comparisons involving the radical left and right. The linguistic divide follows a different path. Before the national parties split, Flemish and Walloon members of the same party mainly differed in the regional interests they discussed. Their wording did not become more distinct before the splits, despite the later centrality of language to party organization. Section 5.7 also documents how regional voting patterns changed across the whole period.

Around the reforms of voting rights, the clearest changes in partisan wording coincided with changes in the Chamber's composition, especially when Socialists gained seats in 1894. The religious divide is most visible in the decade after 1894 and, after the Christian democrats split, around 1970 and in the early 1980s, rather than during the second school war or the Royal Question. With the penalized estimator, the rise in religious partisanship at the 1894 election is the largest at any election since 1831. Partisanship measured on numbers and dates rises too, though not on filler words, so part of the rise may reflect who speaks rather than political vocabulary (Section 8.4). Graziani and Vanschoonbeek (2026) study these elections in much more detail. The more recent rise in left–others and government–opposition partisanship resembles the upward movement reported for the United States, Finland and Norway, although the series are not identical; the wording of the right and that of all other parties has also become easier to tell apart. Roll-call voting tells a partly different story. Across groups, party pairs and regions, differences in votes move only loosely with differences in wording. When parties join a coalition, they quickly vote more alike but show little change in politically meaningful vocabulary relative to placebo phrases. Parties on the same side of the government–opposition divide also use more similar wording, but their filler words converge by a similar amount, or by more with the penalized estimator, so part of this association may reflect how governing and opposition members speak rather than their political vocabulary. Following elections, the record shows more disorder, approval and laughter, again without a clear change in government–opposition wording (Section 8). Across decades, disorder is related to left–others partisanship but not to the religious measure, and opposition speakers usually face more recorded disorder.

The measure has clear limits. It records whether groups use distinguishable words, not whether they disagree on policy. A common shift towards harsher rhetoric would be invisible if all parties adopted it. Nor can binary comparisons, even combined in a multiparty index, fully describe Belgium's multiparty system (Simola et al., 2025). Results depend on speech volume, document quality, vocabulary selection and parliamentary roles. Indeed, including

the formulas used by governing parties to introduce legislation creates a misleading peak in religious partisanship in the 1950s. Placebo vocabularies of dates, numbers and filler words suggest that some differences, and some of their changes, may also reflect who speaks and in what institutional capacity, rather than substantive political language (Section 8).

Further work can separate these mechanisms more fully. Coding the emotions and tone of interventions will test whether groups that sound more distinct also address one another more aggressively; the stage directions so far suggest such a link mainly for left–others partisanship (Section 7). Identifying procedural speech within substantive debates would clarify how much of government–opposition language reflects parliamentary duties. Our vocabulary filter already reduces this component, especially after 1945, and topic labels offer another way to isolate it (Sections 5.12 and 6). Finally, the Senate offers a useful comparison: until 1993 it shared legislative power with the Chamber but was elected under different rules. Its debates could show whether those institutional differences also shaped partisan speech.

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Additional figures and tables

This section collects supplementary figures and tables for the main text. The accompanying paragraphs explain what each display shows and where it is discussed. The last two tables report the most partisan French phrases for the religious and left–others comparisons, in one period per decade.

Figure 18 repeats Figure 2 excluding the Communist and radical left family, following Simola et al. (2025)'s treatment of Finland's extreme left. Before 1925, when this family first has speakers, the real series changes only because penalties and covariate effects are estimated jointly across periods, by at most 0.0007 (in 1898–1900). Net values change more, by up to 0.0021 in 1910–1912, mainly because each specification draws its own random benchmark. Excluding the family lowers net partisanship in 1932–1934 from 0.0039 to 0.0024, although it has only three speakers, and in 2021–2024 from 0.0030 to 0.0001. The highest value remains that of [lrNoCommPeakLabel] ([lrNoCommPeakNet], against [lrPeakNet]), followed by 1898–1900 and 1906–1908. The radical left therefore contributes to the 1932–1934 and 2021–2024 values, but does not determine the highest one (Section 5.2).

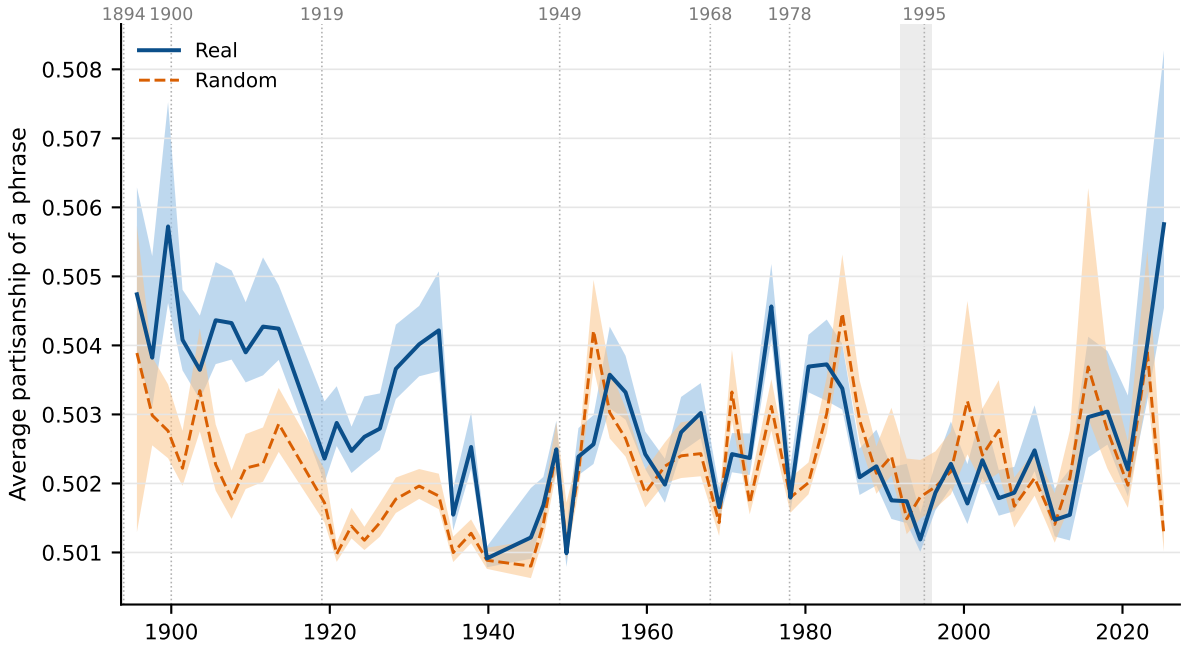


Figure 18: Left–others partisanship without the Communist and radical left family

Notes: As Figure 2, without the members of the Communist and radical left family.

Figure 19 plots school- and education-related phrases as a share of all phrases, alongside religious partisanship. The share peaks during each school conflict. It reaches its maximum of 4.5 per cent in 1884–1886, when the new Catholic majority replaces the Liberal school law, 3.3 per cent during the compulsory-education debate in 1912–1914, and 2.6 per cent during the second school war in 1954–1956. Religious partisanship does not follow these peaks: it exceeds its benchmark in 1884–1886 (0.0011) and 1912–1914 (0.0007), but not in 1954–1956 (0.0007) or in 1878–1880 (−0.0001), when the share is 3.7 per cent. School phrases almost disappear after education passes to the Communities in 1989. Across periods, the two have a correlation of 0.21:

topic attention and partisanship need not move together. Both sides discuss schools extensively during the first school war (Figure 20; Section 5.1.1).

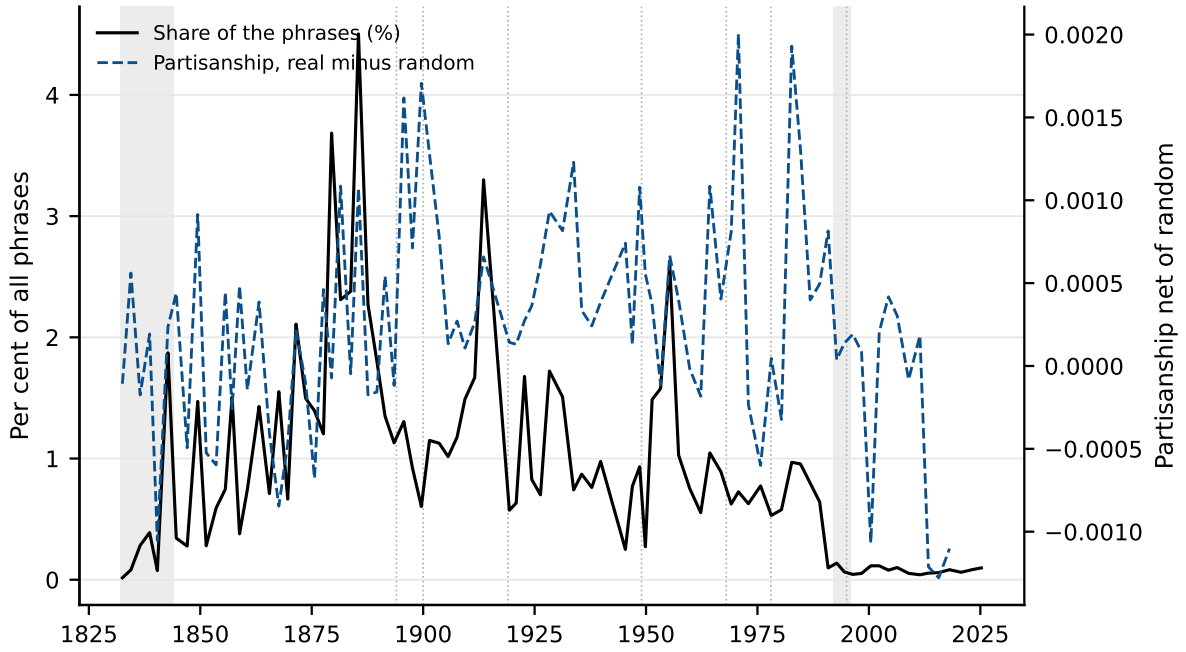


Figure 19: Phrases related to schools and education, and religious partisanship

Notes: Share of all phrases (both languages) that contain a stem related to schools and education (Online Appendix B), and religious partisanship net of the random series (Figure 1).

Figure 20 breaks down Figure 19's school phrases by speaker. Panel (a) counts Catholic members' school phrases; panel (b) shows their share of all school phrases. Counts are highest in 1884–1886 and 1878–1880. In the second school war, they are a little over half their 1878–1884 level, although the Chamber uses only about a fifth fewer phrases. The Catholic share of school phrases generally tracks its share of speakers (correlation 0.75). In 1878–1884, Catholics use 49–52 per cent of school phrases and account for 45–49 per cent of speakers. The 1950s differ. Catholics use 36–37 per cent in 1950–1954, when they govern alone and represent 52–53 per cent of speakers. In opposition in 1954–1956, they use 53 per cent while representing 46 per cent of speakers. After 1989, shares rely on at most about 400 phrases per period. During the first school war, both sides discuss schools; their choice of phrases distinguishes them (Section 5.1.1).

Figure 21 compares religious partisanship with six features of the Chamber. The roll-call comparison in panel (a) follows Gentzkow et al. (2019)'s US Congress analysis and is discussed in Section 5.10; the other five follow Simola et al. (2023)'s Finnish analysis. These indicators follow trends of their own. Roll calls rise from fewer than twenty per period before 1870 to several hundred in most periods after 1945, and more than a thousand in some periods from 1974. The effective number of parties rises from about two before 1894 to seven to ten in the federal era, when 40–61 per cent of members are new in each legislature. Very long speeches are most common in the 1960s–1980s and, less so, the 1850s–1880s, and they are only weakly correlated with religious partisanship (0.10 over 78 periods since 1850).

Figure 22 groups members by region rather than party. It compares national-party members elected in Flemish and Walloon districts, using French speech and controlling for party family

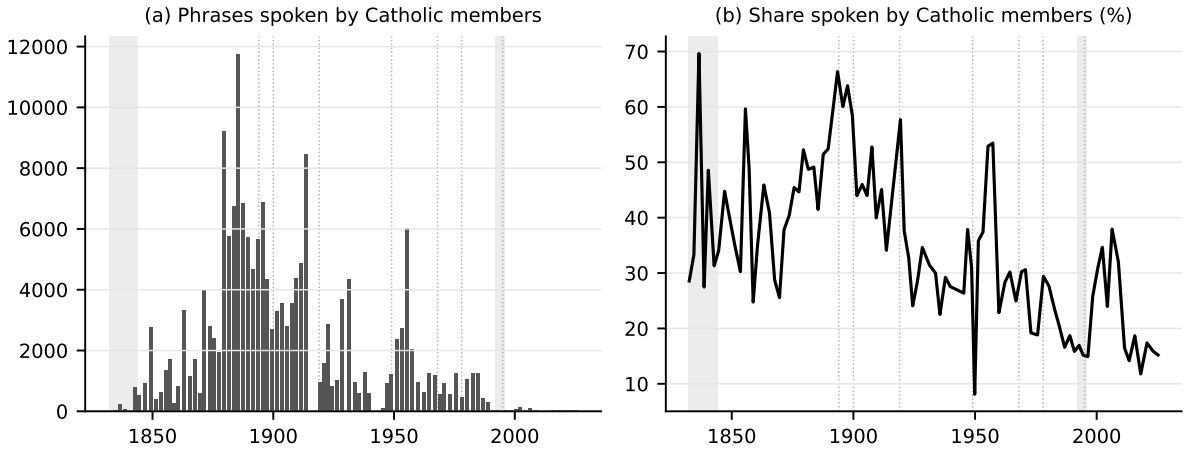


Figure 20: Catholic members and the school phrases

Notes: (a) Number of phrases related to schools and education spoken by Catholic members; (b) share of these phrases spoken by Catholic members.

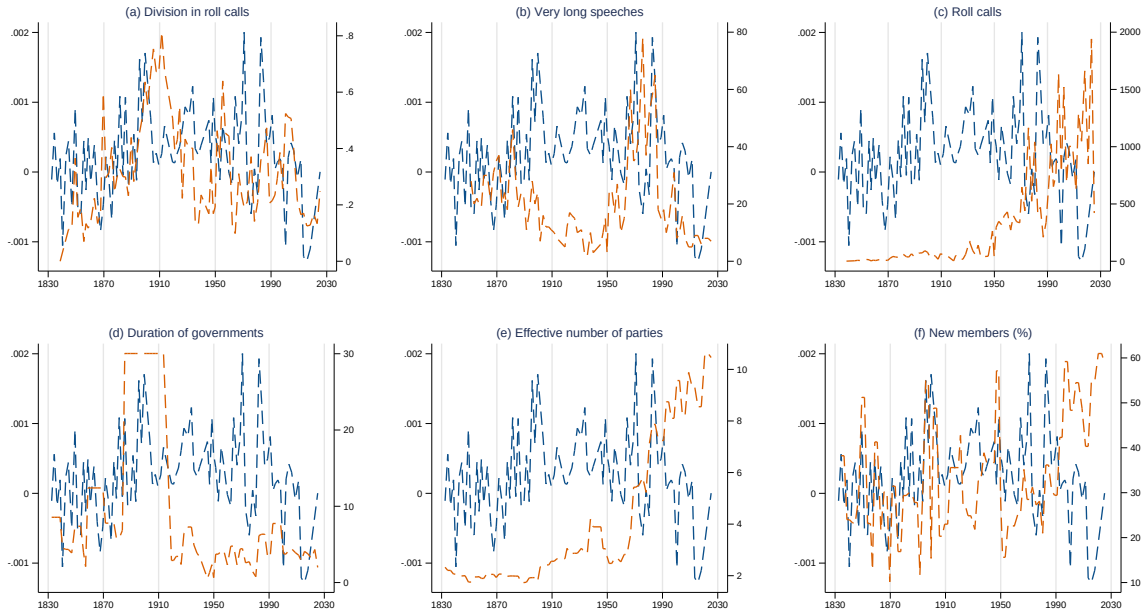


Figure 21: Religious partisanship and other features of the Chamber

Notes: Blue (left axis): religious partisanship net of the random series. Orange (right axis): (a) division between Catholics and other parties in roll calls (Rice, 1925), as in Online Appendix Figure A17; (b) number of very long speeches (top 0.1 per cent of the speeches since 1850); (c) number of roll calls; (d) mean duration of the governments in office; (e) effective number of parties among the members; (f) share of the members of a legislature who did not sit in the previous legislature.

and gender. Net partisanship averages 0.0008 and exceeds its random benchmark in 28 of the 58 periods with adequate coverage. Values are highest in the 1900s, 1850s and 1920s, peaking at [regionPeakNet] in [regionPeakLabel]. The 1960s are the lowest decade, with no period above the benchmark, and the final period, 1965–1968, has the lowest value of the series ([*regionLowNet*]). The regions' speech therefore does not become easier to tell apart before the party splits. As Section 5.6 argues, linguistic divisions appear mainly through regional interests (Online Appendix Table C94).

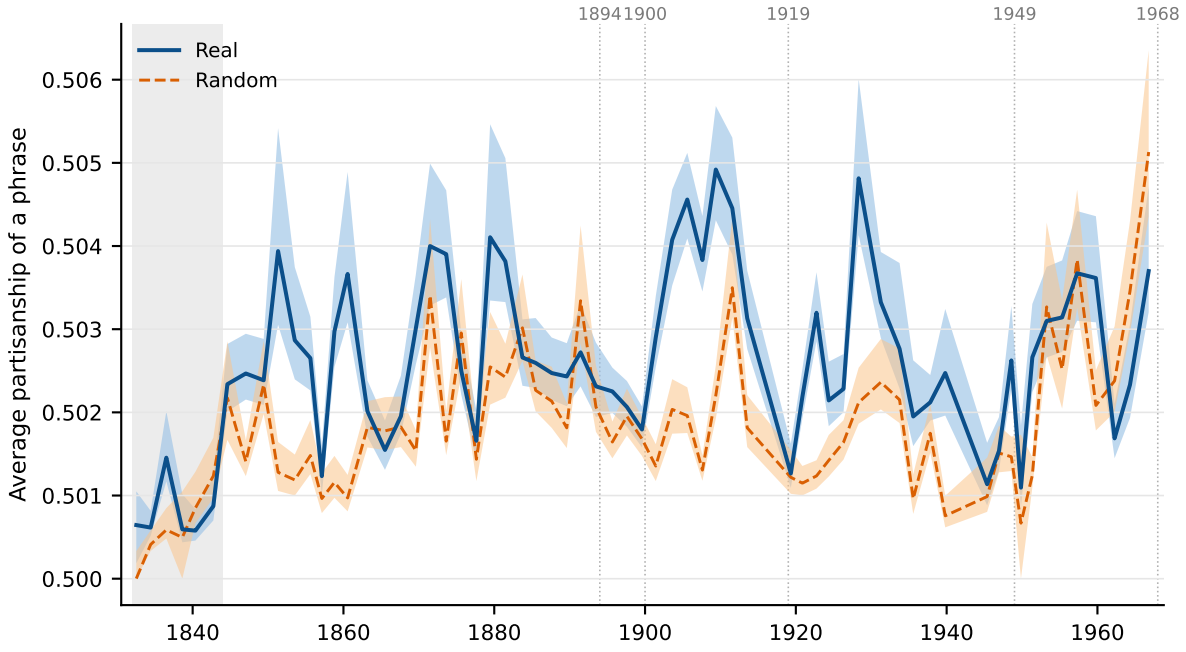


Figure 22: Flemish vs Walloon members of the national parties, before the split (French speech)

Notes: Group A: members elected in Flemish districts; group B: members elected in Walloon districts; members from Brussels are excluded. French speech, periods that begin before 1968, when the first of the national parties split. Controls for the party family and gender. Otherwise as Figure 1.

Figure 23 measures how easily women and men can be told apart by their wording. Before 1995, the series is defined in only four periods, each with 15 to 18 women speakers; intervals are wide and no period exceeds its random benchmark. Since 1995, as women's share of members rises to two fifths, the real series exceeds its random benchmark in 10 of 14 periods, both with and without party-family controls. Net partisanship averages 0.0021 with controls and 0.0016 without them. The leave-out estimator (panel c) exceeds its random benchmark in 5 of these 14 periods. Women and men are therefore about as easy to tell apart within families as across them. In the federal era, they are easier to tell apart than the religious, left–others and government–opposition groups (Section 5.5).

Figure 24 traces parties elected in 2024 through their predecessors. Without covariates, it compares each with other parties in its language group. Before 1995, the predecessors of the traditional parties exceed their random benchmarks more often in French than in Dutch: the Socialists in 20 of 44 periods in French, among them every period from 1894 to 1914, against 7 of 39 in Dutch; the Catholics and Christian democrats in 24 of 69 against 2 of 42; and the Liberals in 13 of 69 against 5 of 40. In Dutch, the Christian democrats and the Socialists both exceed their

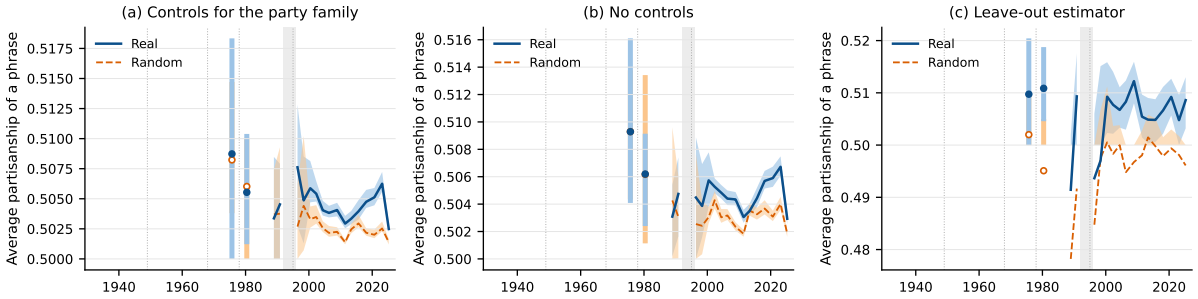


Figure 23: Women vs men

Notes: Group A: women; group B: men. (a) Controls for the party family, with effects that differ by era; (b) no controls; (c) leave-out estimator, no controls. Periods that begin in 1929 or later; a period enters the series when each group has at least 15 speakers in total and 10 in a language. Otherwise as Figure 1.

benchmarks in 1952–1954 (0.0017). Levels cannot be compared across languages (Section 4).

Smaller parties rarely exceed the benchmark of a random group of equal size. The Flemish nationalists never do so before 1995, and afterwards only in 2021–2024. The series of DéFI, which groups the FDF with the Rassemblement wallon, has enough speakers only in eight periods of 1968–1985; its net partisanship is negative in six of them and exceeds the benchmark only in 1983–1985, with eight speakers. The Greens and the radical left never exceed their benchmarks. Since 1995, 10 of 117 party-periods do so, all of Dutch-speaking parties (Table 8; Section 5.5).

Table 8 averages Figure 24's series by legislature from 1995, multiplying values by 1,000. Of 69 entries, 62 lie between -3 and 3. Seven of the 11 parties with entries have both positive and negative values, so rankings change across legislatures; Vlaams Belang and PVDA-PTB have only positive values, and Ecolo and Groen only negative ones. The largest positive values belong to Vooruit and PVDA-PTB in the legislature elected in 2024, which so far covers only 2024–2025, and to Vlaams Belang in 1995. No PVDA-PTB period exceeds its benchmark. The largest negative values are Ecolo's, down to -8.95 in 2010, from a single period with eight speakers in which the random estimate lies far above the real one. Vlaams Belang's averages are positive in all seven legislatures with entries, but only two of its periods exceed the benchmark; no other federal-era party consistently stands apart from the remaining parties of its language group. This agrees with low religious and left–others partisanship in most of these years; the high values of Vooruit and PVDA-PTB in 2024–2025 coincide with the highest left–others value (Sections 5.2 and 5.5).

Figure 25 plots religious and left–others partisanship alongside the institutional-period means in Table 4. Religious means do not follow successive expansions of voting rights. They rise from 0.00004 under census suffrage to 0.00070 under plural voting, are 0.00055 under equal male suffrage and in the post-war unitary state and 0.00043 during the state reforms, and are negative in the federal era (-0.00028). Left–others means are 0.00117 and 0.00139 under plural voting and equal male suffrage and 0.00044 to 0.00072 afterwards. Most variation occurs within these periods. Period means explain about a quarter of religious-series variance and 12 per cent of left–others variance. For example, the post-war mean combines values of -0.00011 to 0.00067 in 1950–1958 with the series' highest value, 0.00200 in 1969–1971. These shifts coincide more closely with changes in Chamber and government composition than with voting rights (Section 5.8).

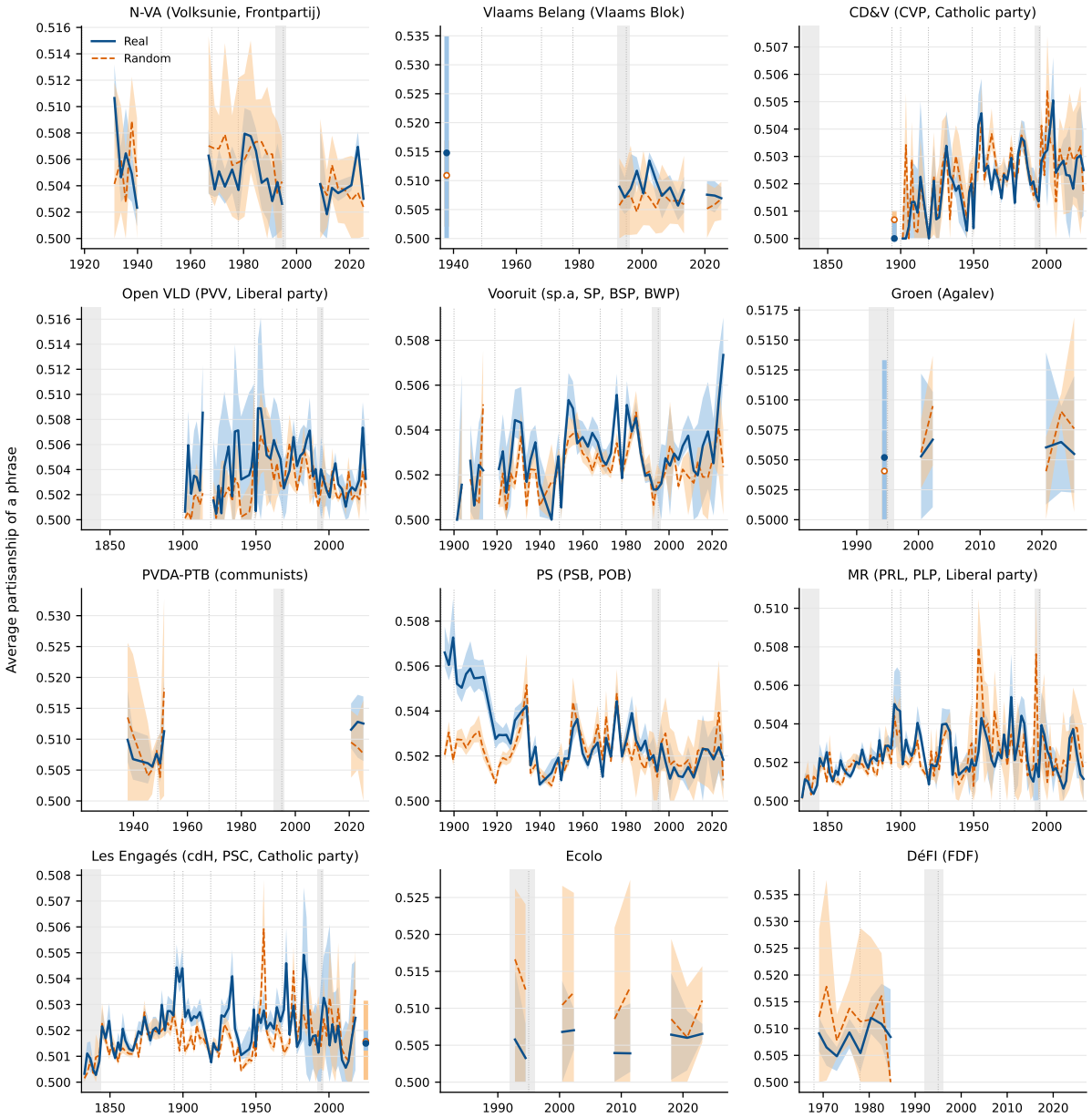


Figure 24: The parties of today's Chamber: partisanship against all the other parties

Notes: Each panel: group A is the party named in the title, followed back through its predecessors; group B, all the other parties of the same language group. Dutch series for the Dutch-speaking parties, French series for the French-speaking ones, both languages for PVDA-PTB. No covariates; a period enters the series when the party has at least 8 speakers. Otherwise as Figure 1.

Table 8: Partisanship of today's parties by legislature, 1995–2025

	Legislature elected in							
Party	1995	1999	2003	2007	2010	2014	2019	2024
N-VA				-0.05	-1.58	-0.19	2.31	0.64
Vlaams Belang	4.17	2.96	2.16	2.35	0.81		1.98	0.11
CD&V	-0.26	-0.83	1.34	-0.60	-0.13	-1.04	-0.22	0.87
Open VLD	-0.15	-0.43	1.76	0.24	-1.10	0.63	2.51	0.74
Vooruit	0.28	-0.41	1.11	1.82	-0.02	1.71	0.45	5.00
Groen		-1.56					-0.28	-2.11
PVDA-PTB							3.10	4.87
PS	-0.66	-0.44	-0.48	-0.31	-0.34	0.04	-0.86	0.92
MR	0.35	0.54	-0.12	0.18	-1.12	0.52	-1.21	-0.47
Les Engagés	0.94	-1.06	-0.34	-0.40	-0.21	-0.54		-0.08
Ecolo		-4.39		-4.61	-8.95	-2.10	-2.28	
DéFI								

Notes: Partisanship of each party against all the other parties of its language group, real minus random series, multiplied by 1,000; mean of the periods of each legislature. Blank: fewer than 8 speakers.

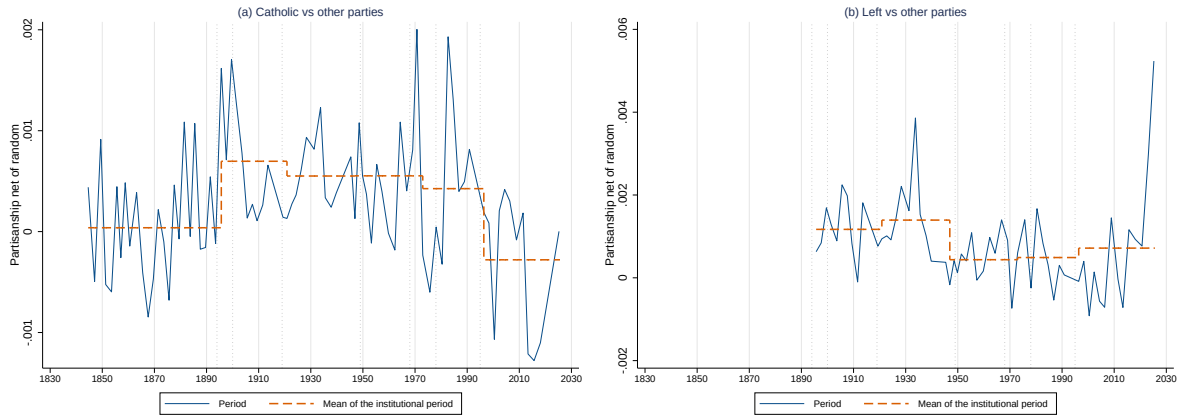


Figure 25: Partisanship and the institutional periods

Notes: Partisanship net of the random series, and its mean in each institutional period: census suffrage (to 1893), plural voting (1894–1918), equal male suffrage (1919–1944), the post-war unitary state (1945–1970), the state reforms (1971–1994) and the federal Chamber of 150 members (from 1995).

Figure 26 decomposes consecutive-period changes in leave-out religious partisanship. The within component measures changes among members present in both periods; renewal measures entry and exit. Both are cumulated from 1845. Over the whole period, the within component adds 0.036 and renewal subtracts 0.027. Renewal mainly reduces partisanship because departing members spoke more distinctly than those who remained: exits subtract 0.068, while entries add 0.041. Entry contributes most in 1894–1896, the first period under plural voting (0.023), partly offset by exit (−0.012). Individual changes are noisy, and the components have opposite signs in just over half of them (45 of 81). We therefore interpret careers and long swings rather than effects of specific reforms (Section 5.8). Online Appendix Figure A22 reports the left–others decomposition.

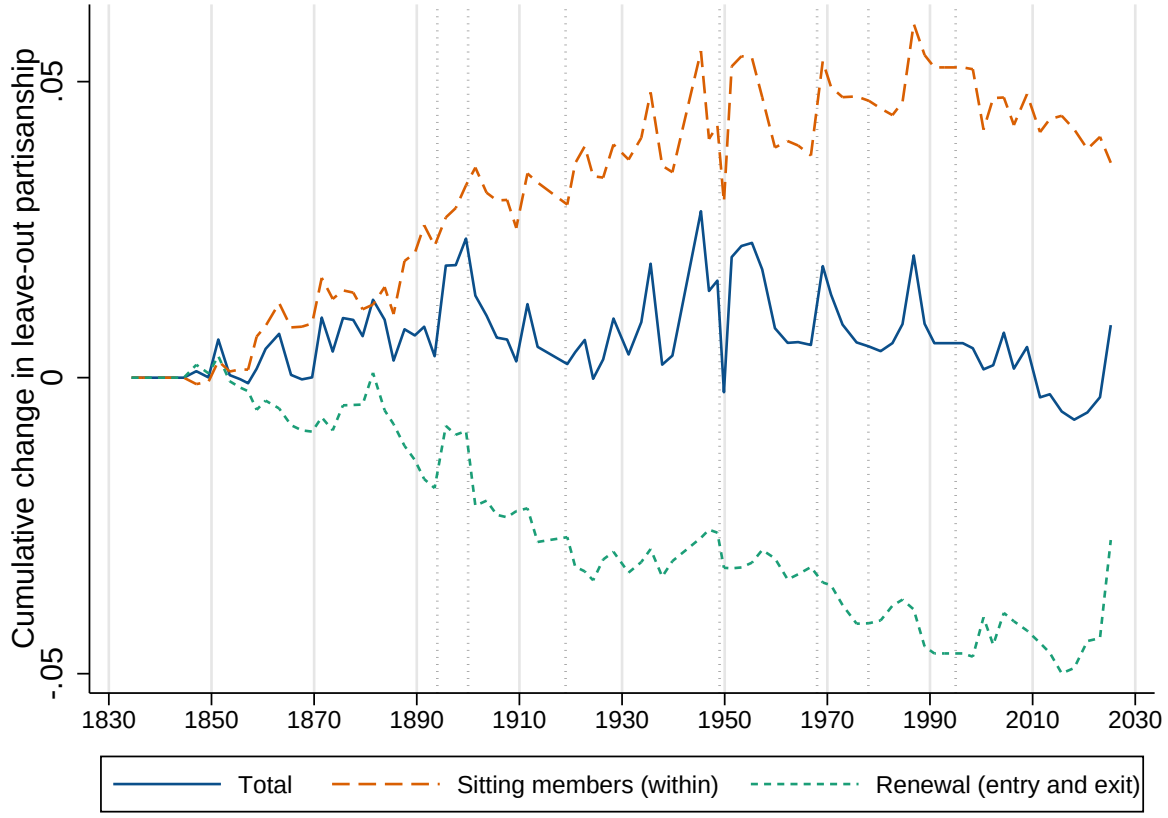


Figure 26: Renewal of the Chamber and adaptation of sitting members: Catholic vs other parties

Notes: Cumulative change in leave-out partisanship (Gentzkow et al., 2019) between consecutive periods, decomposed into the change among members who sat in both periods (within) and the contribution of entry and exit (renewal). Changes into or out of a period with low coverage of the sittings are not counted.

Table 9 compares periods beginning within five years after each reform with those beginning within six years before it. Each comparison uses four to seven periods, so standard errors are only indicative. The clearest changes occur after 1894 and, for government and opposition, around 1919. Following the Socialists' entry in 1894, religious and government–opposition partisanship rise by 0.00126 and 0.00092. After proportional representation, government–opposition partisanship is 0.00048 lower (standard error 0.00021), while the changes in the other two series lie within two standard errors. Around equal male suffrage, religious and left–others partisanship change little, while government–opposition partisanship is 0.00104 higher, in a

comparison of four periods. Around 1949, all three series change by less than two standard errors. After the 1968 split, left–others partisanship is 0.00074 lower (standard error 0.00031); the other changes after both party splits lie within two standard errors. Too few periods around 1995 have adequate sitting coverage (Section 5.8).

Table 9: Partisanship around the reforms

Reform	Catholic vs other	Left vs other	Government vs opposition
1894	0.00126		0.00092
	(0.00025)		(0.00011)
1900	-0.00061	0.00042	-0.00048
	(0.00033)	(0.00047)	(0.00021)
1919	-0.00010	0.00013	0.00104
	(0.00009)	(0.00038)	(0.00024)
1949	-0.00028	0.00034	-0.00040
	(0.00020)	(0.00024)	(0.00031)
1968	0.00012	-0.00074	0.00057
	(0.00052)	(0.00031)	(0.00049)
1978	0.00125	0.00036	-0.00017
	(0.00070)	(0.00079)	(0.00018)
1995			

Notes: Difference in partisanship net of the random series between the periods that begin up to five years after the reform year and those that begin up to six years before (for 1919, from 1910, because the Chamber did not sit during the war); Newey–West standard errors (one lag) in parentheses. Periods with low coverage of the sittings are excluded, which leaves too few periods around 1995.

Table 10 sums the changes in Figure 26 within each institutional period, alongside the left–others changes from Online Appendix Figure A22. For religious partisanship, the within component is positive in the three periods before 1945 and negative in the post-war unitary state (-0.006). The components have opposite signs in five of six periods, partly offsetting each other. The largest total, 0.026 under equal male suffrage, comes almost entirely from sitting members. In the federal era, sitting members become less distinct (-0.016), while renewal adds 0.019. For left–others partisanship, both components have the same sign under plural voting and equal male suffrage. The federal-era rise of 0.026 comes from sitting members (0.058), partly offset by renewal (-0.033). These totals accumulate noisy changes and describe careers rather than reform effects (Section 5.8).

Table 10: Renewal and adaptation, by institutional period

	Catholic vs other			Left vs other		
Institutional period	Total	Within	Renewal	Total	Within	Renewal
Census suffrage, 1831–1893	0.00362	0.02228	-0.01866			
Plural voting, 1894–1918	-0.00132	0.00696	-0.00828	-0.02830	-0.01327	-0.01503
Equal male suffrage, 1919–1944	0.02581	0.02593	-0.00012	0.00493	0.00401	0.00093
Post-war unitary state, 1945–1970	-0.01409	-0.00608	-0.00801	-0.00054	0.00279	-0.00332
State reforms, 1971–1994	-0.00819	0.00332	-0.01152	-0.00086	-0.00358	0.00271
Federal Chamber of 150, 1995–2025	0.00305	-0.01610	0.01915	0.02569	0.05823	-0.03254

Notes: Sum over the periods of each institutional period of the change in leave-out partisanship (total), and of its components: change among sitting members (within) and entry and exit (renewal). Changes into or out of a period with low coverage of the sittings (1832–1843, 1992–1995) are not counted.

Table 11 lists the five French phrases most typical of each group in one period per decade, following Gentzkow et al. (2019, Table I)'s comparison of selected Congresses. Most concern the cleavages or the policies of the day. Role-related phrases are fewer than in earlier versions of the vocabulary, which kept the formulas with which the governing side submits bills and reports and the titles with which members address ministers. The clearest remaining example concerns the special powers of governments (*pouvoirs spéciaux*): in the full-period lists (Online Appendix C), it belongs to the group with the smaller share of governing members in 7 of the 8 periods in which it appears.

Cleavage-related phrases include religious education for Catholics in 1848–1850, *instruction publique* versus *écoles communales* during the first school war, working-class and universal-suffrage phrases for other parties after 1894, family allowances for Catholics in 1929–1932 and 1939–1940, the Royal Question for other parties in 1949–1950, the School Pact for other parties in 1958–1961, and development cooperation for Christian democrats from 1969–1971.

Table 11: Most partisan phrases, Catholic and Other parties, French speech: one period every ten years

Catholic			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1839–1841					
société générale	256	135	sucre exotique	105	185
adjudication publique	45	41	tribunal commerce	51	104
chemin fer	513	513	chemins vicinaux	49	73
travaux publics	266	266	caisses prévoyance	9	20
plus grande	145	144			
1848–1850					
chemin fer	611	307	garde civique	49	84
enseignement religieux	131	49	société générale	23	57
enseignement moyen	159	86	banque nationale	18	45
conseil mines	89	28	lettre voiture	8	35
canal charleroy	93	42	code pénal	18	43
1859–1861					
vol domestique	130	24	garde civique	10	88
passe port	99	20	bruxelles louvain	43	85
fonds communal	147	83	sucre betterave	18	48
cours légal	85	44	cour appel	42	72
communes octroi	119	78	cours certificat	10	39
1868–1870					
listes électorales	100	41	sociétés coopératives	4	76
conseil communal	109	75	assemblée générale	42	97
service militaire	90	56	sociétés anonymes	85	137
visites domiciliaires	50	19	écoles normales	11	57
police judiciaire	35	9	travail enfants	3	32
1878–1880					
instruction publique	279	119	chemin fer	349	455
corps électoral	117	42	écoles communales	81	133
valeur locative	76	18	marques fabrique	19	50
enseignement libre	80	28	département travaux	35	64
contribution personnelle	60	18	voies navigables	26	52
1888–1890					
chemin fer	400	299	sûreté publique	21	105
prud hommes	245	161	agents provocateurs	4	51
conseils prud	172	106	ville gand	17	63
pays flamand	90	46	écoles spéciales	63	103
programme candidature	23	3	conseil communal	44	82
1898–1900					
juge paix	106	43	banque nationale	137	358
chemin fer	308	270	classe ouvrière	37	132
fer vicinaux	58	25	tramways bruxellois	32	118
cour militaire	41	9	suffrage universel	52	130
conseil guerre	89	61	conseil communal	48	96
1908–1910					
comité spécial	113	29	chemin fer	333	420
loi 1902	84	23	classe ouvrière	35	105

Table 11: Most partisan phrases, Catholic and Other parties, French speech: one period every ten years (continued)

Catholic			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
temps service	93	50	fer postes	34	93
effectif paix	70	36	écoles normales	40	94
union minière	48	18	postes télégraphes	37	90
1918–1919					
dommages guerre	204	142	classe ouvrière	61	115
oeuvre nationale	43	8	comité national	50	93
tribunaux dommages	95	62	avant guerre	143	174
écoles normales	76	45	chemin fer	149	177
orphelins guerre	39	12	pendant guerre	159	179
1929–1932					
pensions vieillesse	95	56	société nationale	114	166
allocations familiales	73	41	classe ouvrière	75	127
main oeuvre	82	52	sciences arts	61	102
produits agricoles	41	14	écoles moyennes	19	56
responsabilité limitée	31	5	écoles communales	19	47
1939–1940					
chemin fer	157	80	personnel enseignant	26	69
pommes terre	85	18	nationalistes flamands	12	46
assurance chômage	53	18	politique indépendance	11	37
allocations familiales	91	58	sous officiers	25	48
caisses chômage	42	9	politique étrangère	12	25
1949–1950					
assurés libres	73	17	consultation populaire	201	406
travailleurs indépendants	66	15	question royale	105	148
conseil entreprise	65	20	stations épuration	9	31
pouvoirs représentants	53	9	sociétés habitations	14	31
central provincial	45	8	eaux usées	27	43
1958–1961					
politique scientifique	53	3	classe ouvrière	26	74
budget ordinaire	59	26	loi unique	62	109
gouverneur général	61	31	haute autorité	27	72
force publique	62	31	personnel enseignant	9	54
économique régional	39	10	pacte scolaire	33	78
1969–1971					
pays voie	282	26	police judiciaire	13	94
voie développement	262	27	gaz naturel	11	54
coopération développement	136	15	santé publique	72	112
parlement européen	127	40	travailleurs indépendants	16	50
gouvernement belge	121	40	personnes âgées	12	45
1979–1981					
gouvernement belge	128	46	organisations syndicales	26	56
communauté germanophone	97	23	travailleurs indépendants	15	44
coopération développement	84	29	région bruxelles	32	56
parlement européen	68	17	communauté flamande	26	46
education nationale	113	72	communautés régions	41	61
1989–1991					
juge instruction	162	72	culturelles nationales	16	69
services police	161	80	scientifiques culturelles	17	69
précompte mobilier	103	36	loi financement	26	72
agents change	69	7	publics institutions	24	65
communauté germanophone	64	12	etat major	10	33
1999–2001					
coopération développement	151	43	services police	32	92
arc ciel	76	31	criminalité organisée	14	55
principe précaution	77	35	union européenne	129	169
nations unies	66	29	réduction temps	18	55
soumis sanction	53	17	police locale	36	71
2007–2010					
partenaires sociaux	243	117	intérêts notionnels	32	73
demandeurs emploi	111	29	fraude fiscale	69	102
plan relance	113	57	demandeurs asile	20	52
titres services	107	54	aide financière	12	34
coopération développement	109	56	médecins généralistes	53	69
2016–2019					
juge instruction	125	45	sécurité sociale	153	201
secret professionnel	84	29	saut index	10	53
aide juridique	68	16	hommes femmes	21	49

Table 11: Most partisan phrases, Catholic and Other parties, French speech: one period every ten years (continued)

Catholic			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
toute série	125	74	droits femmes	21	45
arabie saoudite	88	44	droits humains	17	41
2024–2025					
union européenne	459	178	flexi jobs	40	115
coopération développement	127	33	malades longue	17	86
norme croissance	119	48	travailleurs travailleuses	14	73
consommation drogues	47	6	travailleuses travailleurs	11	62
spf justice	39	4	tableaux budgétaires	10	56

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Catholic: positive; Other parties: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 1; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table 12 presents the same comparison for the left and other parties from 1898–1900. Before the Second World War, social issues distinguish the left. *classe ouvrière* appears among its five most partisan phrases in every displayed period before 1945, alongside universal suffrage and the eight-hour day. The other side's phrases concern the courts, the army and the colony before the First World War (*juge de paix*, *conseil de guerre*, *union minière*), the Flemish language between the wars (*langue flamande*) and, from the late 1960s, the communities and regions (*conseil culturel*, *région wallonne*). After the Second World War, the left's phrases concern the School Pact and the *loi unique* in 1958–1961, then public health and trade unions; in 2007–2021, they concern health care, social security, the financial crisis and tax havens (*paradis fiscaux*), and in 2024–2025 working women and men, the pension age and purchasing power (*travailleuses et travailleurs*, *67 ans*, *pouvoir d'achat*).

As in the religious comparison, *pouvoirs spéciaux* marks the side out of government. In the full-period lists, it belongs to the left in the four periods when French-speaking Socialists are mostly out of government and to other parties in the five when they govern.

Table 12: Most partisan phrases, Left and Other parties, French speech: one period every ten years

Left			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1898–1900					
banque nationale	414	139	juge paix	27	113
classe ouvrière	186	31	conseil guerre	42	96
suffrage universel	143	57	députation permanente	39	88
caisse épargne	155	71	cour militaire	5	50
tramways bruxellois	111	36	cour appel	15	51
1908–1910					
classe ouvrière	180	34	comité spécial	6	85
chemin fer	477	342	union minière	4	45
prud hommes	299	168	loi 1902	26	61
conseils prud	200	121	budget extraordinaire	33	63
huit heures	120	46	effectif paix	31	60
1918–1919					
classe ouvrière	258	46	dommages guerre	43	214
comité national	133	55	tribunaux dommages	13	99
huit heures	50	10	écoles normales	22	73
suffrage universel	90	50	juge unique	40	77
suffrage femmes	69	37	langue flamande	10	44
1929–1932					
industrie travail	180	77	sociétés privées	6	33
classe ouvrière	164	69	langue flamande	14	42
société nationale	191	116	libre échange	24	52
huit heures	63	21	responsabilité limitée	2	27
assurances sociales	61	24	deux langues	20	43
1939–1940					
personnel enseignant	102	23	santé publique	80	116
instruction publique	156	105	budget extraordinaire	21	51
classe ouvrière	81	47	classes moyennes	33	63
union soviétique	29	2	défense nationale	239	269
masses laborieuses	25	1	fournitures bureau	3	32
1949–1950					
membres conseil	149	70	eaux usées	10	66
consultation populaire	400	323	dommages guerre	29	63
libre belgique	62	16	société nationale	53	83
question royale	162	122	collège électoral	18	47
vrai oui	46	11	organisations ouvrières	15	44
1958–1961					
classe ouvrière	94	20	congo belge	70	172
loi unique	120	58	table ronde	99	141
pacte scolaire	89	33	politique scientifique	3	40
services publics	103	52	ruanda urundi	73	107
personnel enseignant	60	12	marché commun	71	100

Table 12: Most partisan phrases, Left and Other parties, French speech: one period every ten years (continued)

Left			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1969–1971					
santé publique	153	75	conseil culturel	34	106
distribution eau	70	9	police judiciaire	42	99
organisations syndicales	114	59	ik citeer	9	62
etats unis	119	75	conseil économique	41	89
marché commun	105	67	six communes	19	66
1979–1981					
loi programme	184	103	région bruxelles	15	66
organisations syndicales	96	28	communauté germanophone	5	56
etats unis	94	37	finances publiques	37	86
défense nationale	126	74	région wallonne	106	149
sécurité sociale	194	144	matières personnalisables	14	50
1989–1991					
défense nationale	191	92	handelingen plenaire	250	352
etat major	90	10	loi financement	27	76
office contrôle	88	31	communauté européenne	73	110
santé publique	84	33	cour arbitrage	14	46
sécurité sociale	154	105	précompte professionnel	13	35
1999–2001					
conséquence proposition	57	4	organisations syndicales	33	49
proposition modification	54	8	poser question	98	114
parole déclaration	75	33	non seulement	169	170
temps travail	95	55	union européenne	160	161
défense européenne	43	10	celle ci	154	155
2007–2010					
santé publique	182	77	titres services	29	75
soins santé	183	82	coopération développement	48	76
sécurité sociale	193	95	jeux hasard	27	51
crise financière	159	84	droits enfant	20	40
paradis fiscaux	82	24	état fédéral	31	49
2019–2021					
sécurité sociale	236	61	droit passerelle	62	181
paradis fiscaux	105	19	entités fédérées	61	117
soins santé	237	152	niveaux pouvoir	35	76
services publics	136	63	état islamique	4	44
conditions travail	123	56	gouvernement fédéral	40	77
2024–2025					
travailleuses travailleurs	119	10	énergies fossiles	8	58
67 ans	106	6	transition énergétique	6	32
travailleurs travailleuses	125	28	aidants proches	12	34
pouvoir achat	205	123	santé publique	36	56
services publics	161	85	travailleurs sociaux	29	49

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Left: positive; Other parties: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 2; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Online Appendix A: Partisanship results

This appendix adds results for Section 5. It examines identification after multiple phrases and robustness to covariates, estimators, the penalty ψ, language and sample selection. It also compares regions within families, studies individual phrases and phrase groups, compares speech with roll calls, tests OCR noise and decomposes left–others changes through Chamber renewal. Each specification draws its own random benchmark (Section 4). Differences in net partisanship for a single period may therefore reflect the random draw. We mainly compare real series. Most checks follow Gentzkow et al. (2019): magnitude, estimates without covariates, the penalty ψ, plug-in and leave-out estimators, and roll-call comparisons. Government-status controls and checks excluding the radical left or right follow Simola et al. (2023)'s Finnish analysis.

Figure A1 shows the expected posterior probability assigned to a speaker's actual group after up to 100 phrases, following Gentzkow et al. (2019, Figure 3). It selects eight religious-series periods from 1859–1861 to 2024–2025 and seven left–others periods from 1894–1896: the periods closest to 1860, 1880, 1900, 1925, 1950, 1975, 2000 and 2025 in which each series is defined. The probability rises slowly. After one minute, or 33 phrases, the model implies probabilities of 0.53 to 0.59 for the religious grouping and of 0.53 to 0.63 for the left–others grouping. The steepest religious curve is that of 1898–1900, under plural voting; the four steepest left–others curves (1894–1896, 1898–1900, 1974–1977, 2024–2025) lie close together, well above the other three. These curves use real estimates and include finite-sample bias. For example, the religious curve for 1974–1977 is the second steepest even though its real estimate lies below its random benchmark. The selected periods do not include the religious maximum of [relPeakLabel]. Figure 11 reports every period's one-minute probability (Section 5.10); it comes from separate, smaller simulations, and for the selected periods its values differ from those of the curves by less than 0.01.

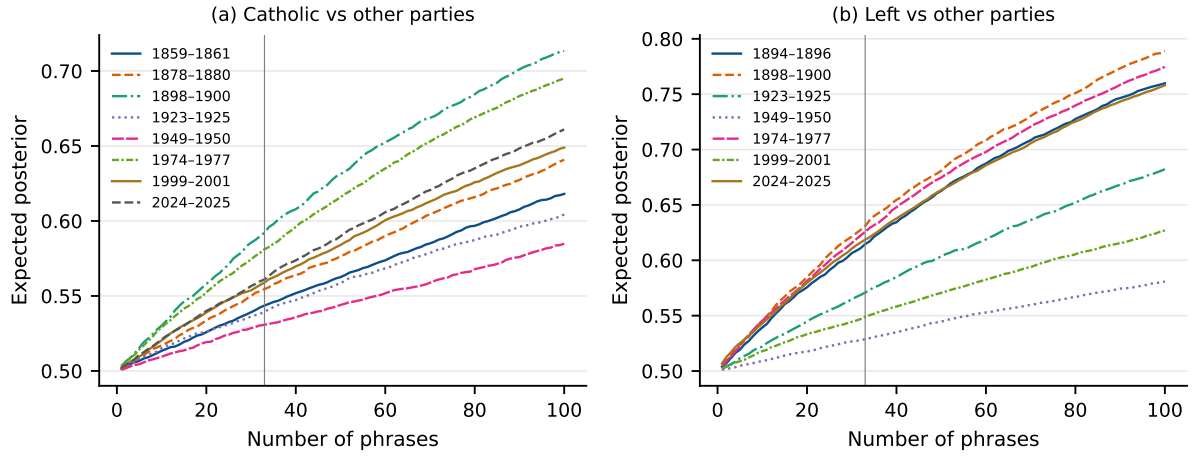


Figure A1: Magnitude of partisanship when more than one phrase is heard

Notes: Expected posterior on the true group after hearing up to 100 phrases, for selected periods; computed as in Gentzkow et al. (2019). Vertical line: one minute of speech (33 phrases).

Figure A2 re-estimates religious partisanship with alternative controls. Without covariates (panel a), the real series closely matches the main specification, with a correlation of 0.93. Most

differences in net estimates come from the random benchmarks. Government status, whether alone or with gender and province as in Simola et al. (2025) (panels b and c), nearly coincides with the religious grouping in most periods of 1848–1914 and in 1950–1958. Catholics then either govern alone or form almost the whole opposition. Party and government effects cannot be separately identified in these periods. The real series jumps to 0.64–0.72 in nine periods of 1868–1898 with government status alone, and to 0.53–0.70 in eight periods of 1868–1902 with gender and province as well. In other periods, including the 1950s, it stays close to its random benchmark (average net partisanship −0.0001 in panel b and 0.0001 in panel c). We therefore omit government status from the main specification (Section 4) and measure government–opposition differences directly in Section 5.4.

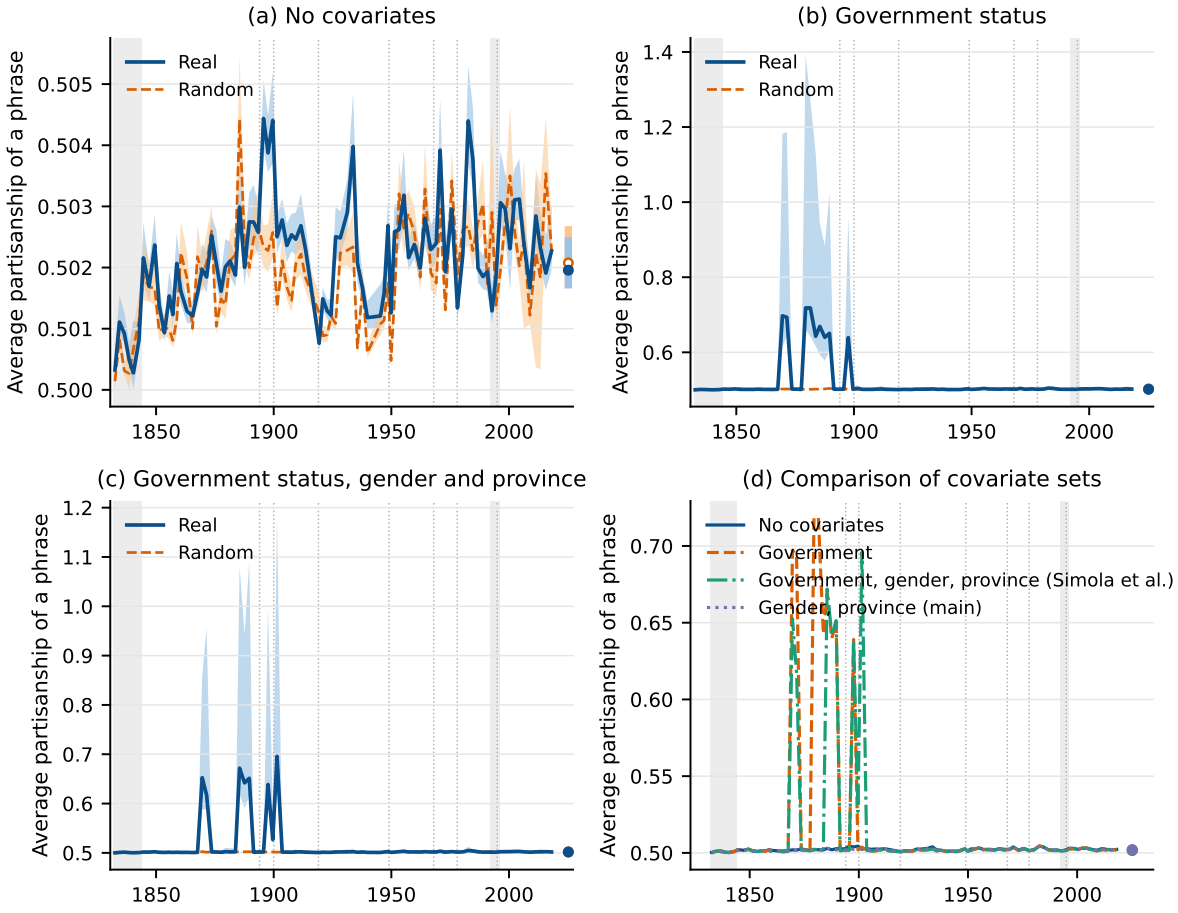


Figure A2: Religious partisanship: impact of covariates

Notes: (a)–(c) Real and random series for different sets of covariates; (d) comparison of the covariate sets: the main specification controls for gender and province; Simola et al. (2025) also control for government status, as Gentzkow et al. (2019) do for the majority status of the party. With government status, the series is not identified in the periods when a single family held office (values between 0.53 and 0.72 in 1868–1902).

Figure A3 repeats this comparison for left–others partisanship. Government status produces no jumps like those of the religious series. With government status alone (panel b), the real series is lower than without covariates in 41 of 58 periods; with government status, gender and province (panel c), it is lower than in the main specification in 36 of 58. Before 1960, the reductions are larger in the eleven periods when the left almost coincides with the opposition, including 1896–1900, 1927–1934 and 1949–1954: with government status alone, the real series

falls by 0.0013 on average in these periods and by 0.0006 in the other periods. In the eleven periods, average net partisanship is 0.0019 without covariates and 0.0004 with government status alone (0.0012 in the main specification and 0.0006 in panel c), and the 1932–1934 value largely disappears (0.0011 in panel b, against 0.0053 without covariates and 0.0039 in the main specification). The highest value of the main specification, [lrPeakNet] in [lrPeakLabel], also falls, to 0.0028 in panel (c). Without covariates, the real series is higher than in the main specification in all 30 periods before 1960, and its highest net value is that of 1932–1934 rather than 2024–2025. As with religion, government status absorbs much of partisanship when the cleavage aligns with government versus opposition (Section 5.4).

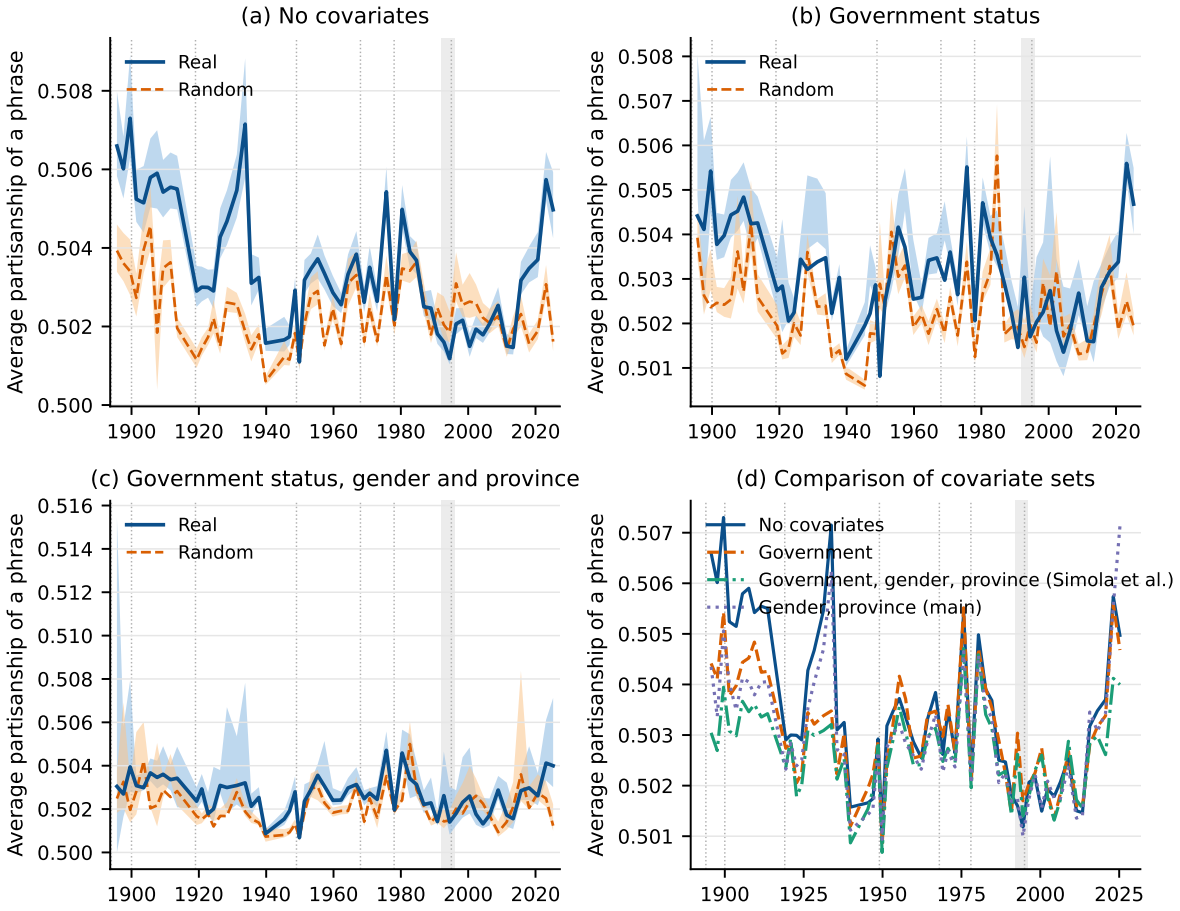


Figure A3: Left–others partisanship: impact of covariates

Notes: As Figure A2.

Figure A4 excludes the populist radical right: Rex from 1936, Vlaams Blok and Vlaams Belang from 1978, Front National from 1991 and Parti Populaire from 2010. This follows Simola et al. (2023)'s exclusion of Finland's Rural Party. Real series change little: before 1985, religious and left–others estimates move by less than 0.0001, and afterwards by at most 0.0005. They change even before 1936, when no radical-right member sits, because phrase penalties and covariate effects are estimated jointly across periods. The three highest left–others values remain those of 2024–2025, 1932–1934 and 2021–2024.

Net estimates change more because each specification draws its own random benchmark. In [relPeakLabel], the religious maximum of the principal series, net partisanship falls from

[relPeakNet] to 0.0007 while the real estimate is almost unchanged ([relPeakReal] and 0.5038). Without the radical right, the highest religious value is that of 1932–1934 (0.0022). The radical right does not determine the real series, but the ranking of single periods by net partisanship depends on the random draw.

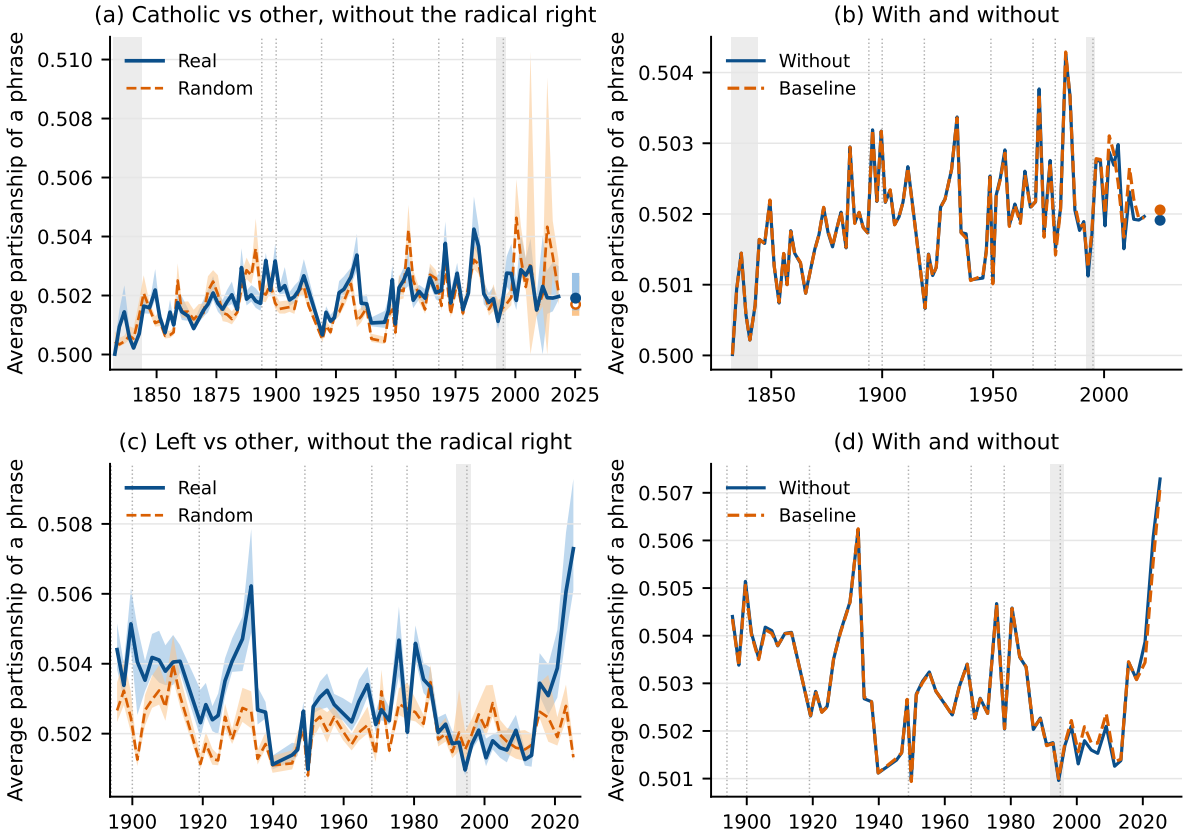


Figure A4: Partisanship without the populist radical right

Notes: Members of Rex, the Vlaams Blok and Vlaams Belang, the Front National and the Parti Populaire excluded.

Figure A5 calculates partisanship from observed phrase-choice probabilities without penalization. This is the plug-in estimator used by Gentzkow et al. (2019, Figure 1) to demonstrate finite-sample bias. In the periods with adequate coverage, real estimates range from 0.56 to 0.75, far above penalized estimates. However, random benchmarks follow them closely and account for more than nine tenths of their distance from 0.5 on average. Levels also track speech volume: the correlation between real estimates and log phrases per speaker is -0.94 for the religious grouping and -0.90 for the left–others grouping. Peaks occur in short or poorly covered periods, including 1831–1833, 1949–1950 and 1991–1995. The unpenalized empirical estimator therefore mainly captures finite-sample bias, motivating our use of penalization (Section 4).

Figure A6 shows the information available for each religious-series period. Panel (a) counts French and Dutch vocabulary phrases used at least once. Panel (b) counts phrases whose party coefficient $\hat{\varphi}_{jt}$ remains non-zero after penalization. The first count averages about 69,000 per period in 1894–1970 and 51,000 in the federal era, about a quarter fewer. The second falls more, from about 300 per period under plural voting to about 130 in the federal era. It follows speech volume (correlation 0.90 with the number of phrases spoken) and drops in short or poorly

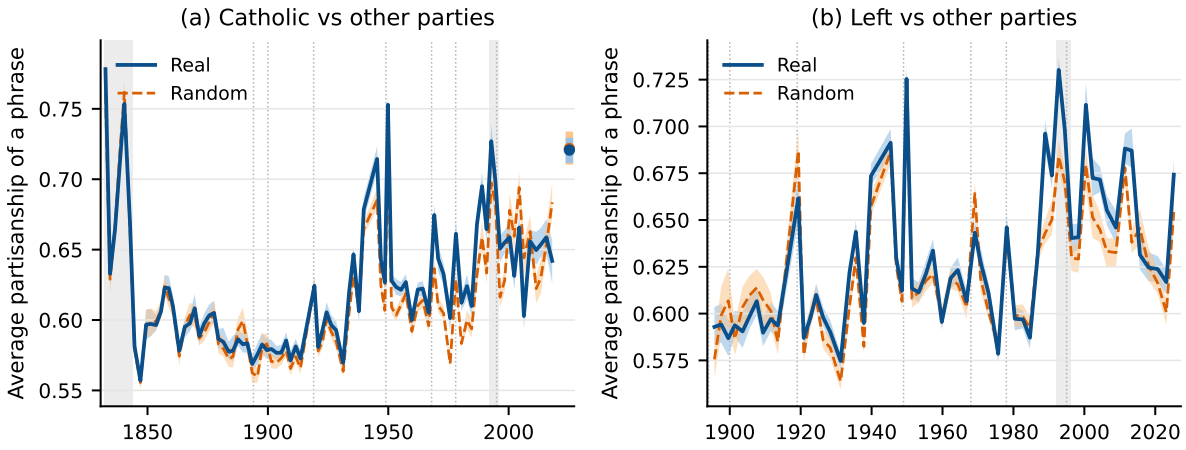


Figure A5: Partisanship computed from empirical phrase choice probabilities

Notes: Choice probabilities $\hat{q}_{jt} = c_{jt} / \sum_k c_{kt}$ by group, without penalization (the plug-in estimator of Gentzkow et al., 2019).

covered periods such as 1944–1946, 1949–1950, 1991–1995 and 2024–2025. Since partisanship depends on non-zero coefficients, lower recent estimates may partly reflect less speech rather than smaller differences in speech (Section 5.12).

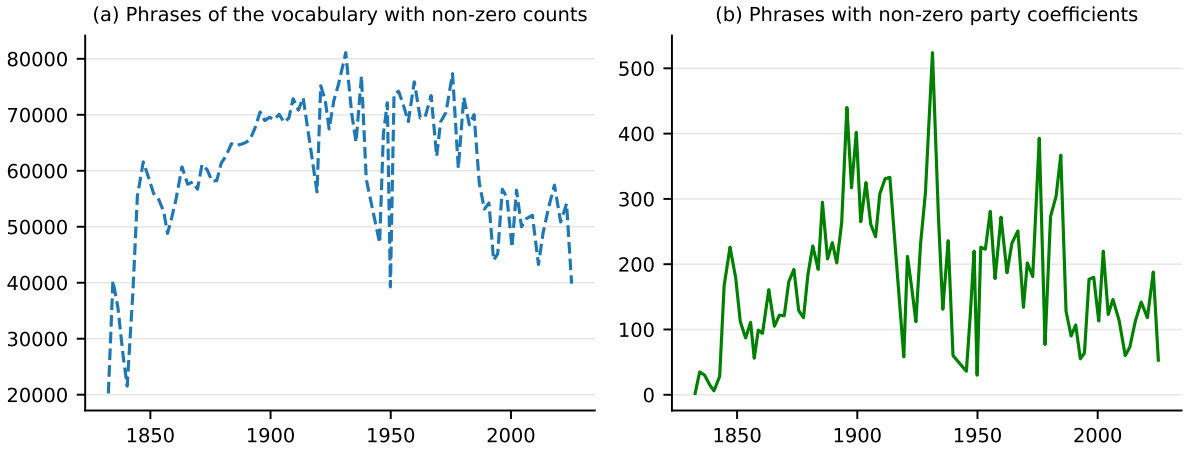


Figure A6: Phrase usage and partisanship

Notes: (a) Phrases of the vocabulary with non-zero counts; (b) phrases with non-zero party coefficients $\hat{\varphi}_{jt}$, religious split, main specification.

Figure A7 re-estimates the series without covariates, increasing or decreasing the penalty ψ tenfold, as in Gentzkow et al. (2019)'s robustness checks. The reference estimates are panel (a) of Figures A2 and A3, with $\psi = 10^{-5}$. At $\psi = 10^{-4}$, every real estimate moves towards 0.5. The median reduction is about a fifth of its distance from 0.5 for the religious series and a sixth for the left–others series. Random estimates fall by about the same amount, leaving average net partisanship almost unchanged: 0.0003 for the religious series and 0.0012 for the left–others series, as with $\psi = 10^{-5}$. At $\psi = 10^{-6}$, real estimates are almost identical to those at $\psi = 10^{-5}$, with correlation above 0.999. At every value of ψ, the highest religious value falls under plural voting, in 1894–1896 or 1898–1900, and the highest left–others value is that of 1932–1934. The timing of the highest values without covariates is therefore robust to the penalty.

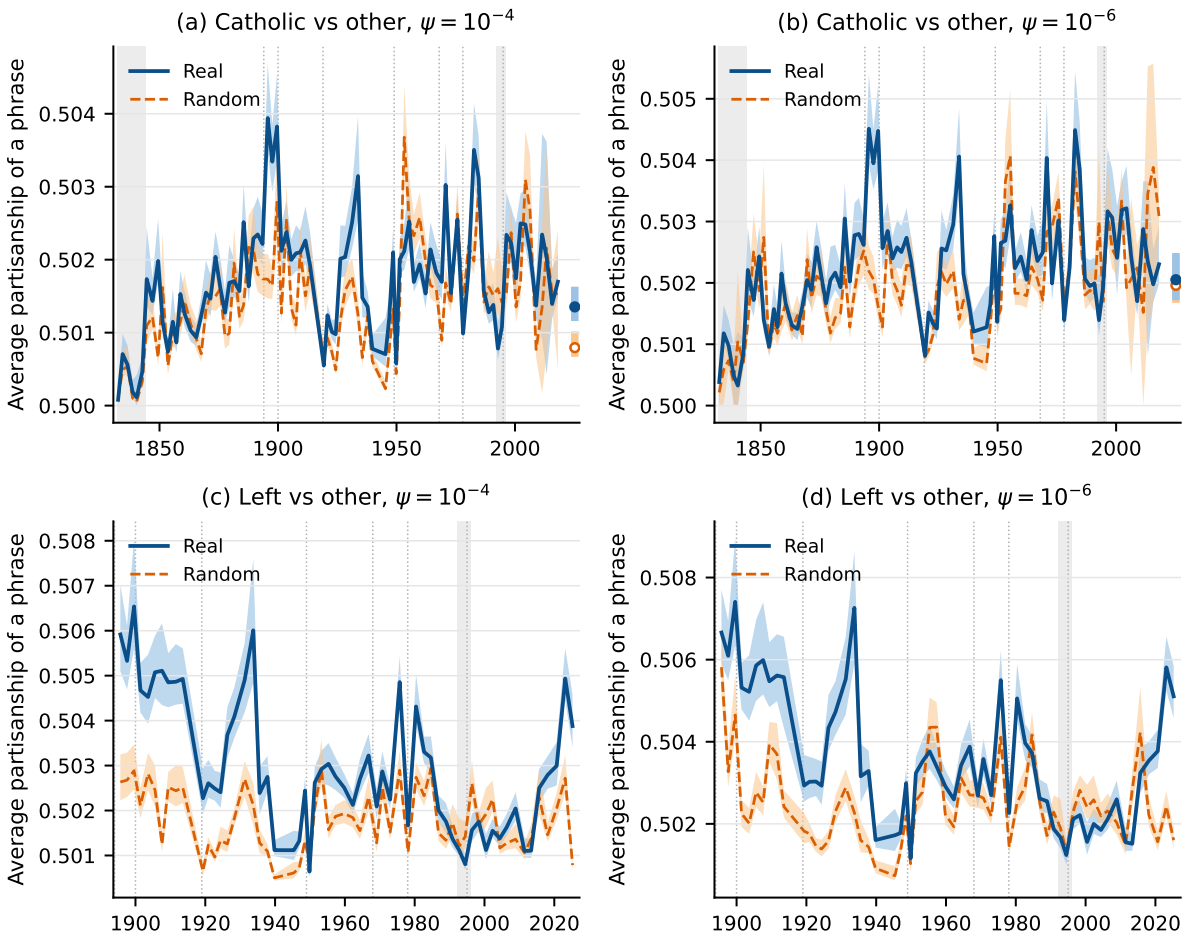


Figure A7: The impact of ψ

Notes: Series without covariates, with $\psi = 10^{-4}$ and $\psi = 10^{-6}$ (the main specifications use $\psi = 10^{-5}$).

Figure A8 separates the French and Dutch components of the principal series, which are weighted by their phrase shares. Dutch enters in 1900 but carries little weight before 1919. It accounts for less than a fifth of interwar phrases and for 17–26 per cent in 1950–1958. The Dutch religious series is highest in 1954–1956 (0.0029), during the second school war; this is the only period of 1950–1958 in which either Dutch series exceeds its benchmark. The Dutch left–others series is highest in 2024–2025 (0.0105). French religious estimates in 1950–1958 (0.0000–0.0004) are below the plural-voting maximum (0.0017 in 1898–1900) and the 1969–1971 value (0.0028), the highest of the French series, when 19 Christian democrats spoke French. The French left–others series peaks in 1932–1934 (0.0041). French speech contributes 96 per cent of the religious maximum of [relPeakLabel], and Dutch speech 91 per cent of the left–others maximum of [lrPeakLabel]. Levels cannot be compared across languages because the Dutch vocabulary is about a quarter the size of the French vocabulary (Section 4).

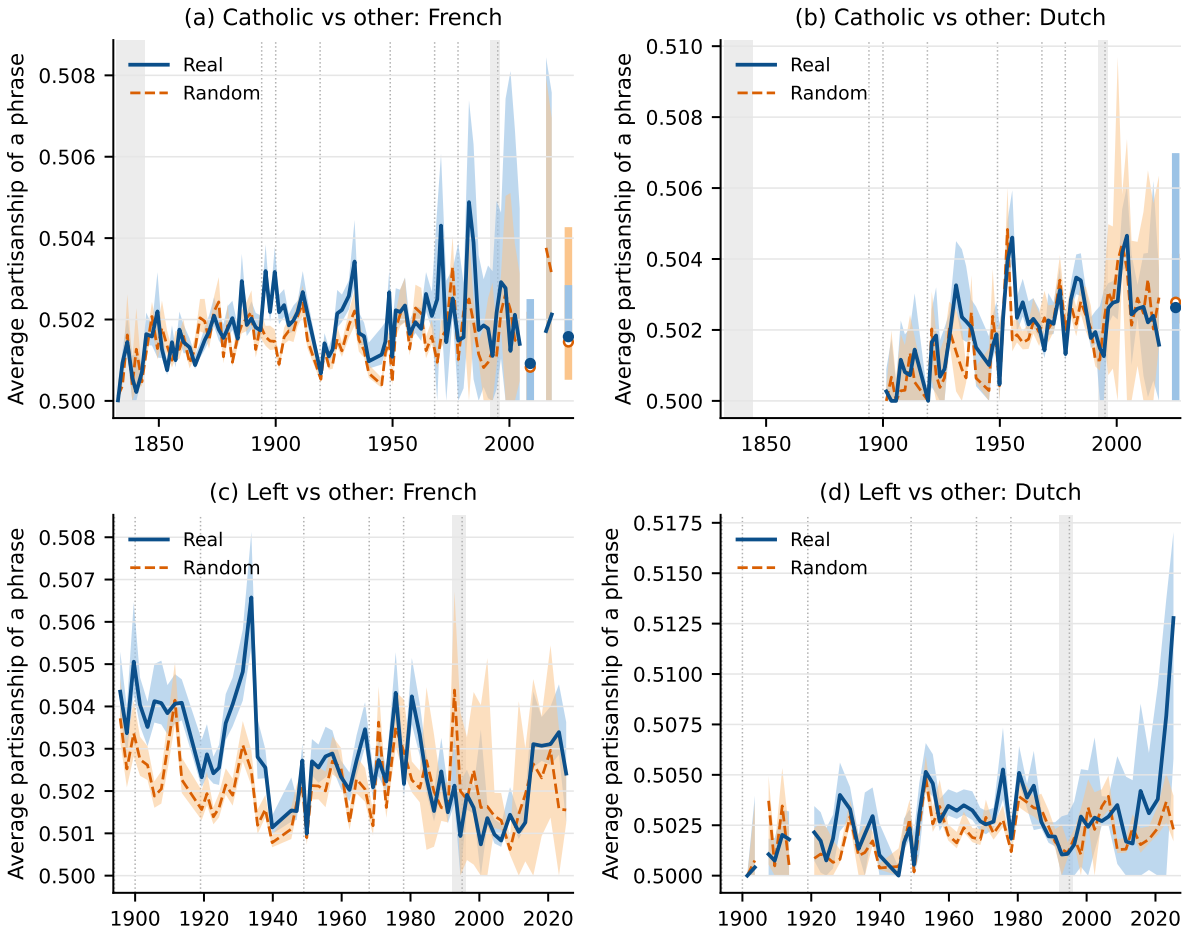


Figure A8: Partisanship by language

Notes: The French and Dutch components of Figures 1 and 2. Levels are not comparable across languages: the Dutch vocabulary is about a quarter of the French one.

Figure A9 reports the unpenalized leave-out estimator of Gentzkow et al. (2019) for the three main groupings, without covariates. Its random benchmark averages close to 0.5, but estimates are noisy. Some fall below 0.5, and intervals are more than ten times wider than penalized intervals. Religious estimates are positive in every period of 1843–1893 but exceed their benchmark in only three. They exceed it in all ten periods of 1894–1914 and are highest in

1894–1896; they exceed it in no period of 1950–1961 and are lowest, and negative, in 1974–1978. Left–others estimates are highest before 1914, when they exceed their benchmark in all ten periods, and lower after 1945. Government–opposition estimates exceed their benchmark in most periods under plural voting and equal male suffrage and in the federal era (9 of 10, 7 of 9 and 11 of 14 periods).

The leave-out series rank recent periods higher than the penalized series do. In 2024–2025, a short period, left–others leave-out partisanship (0.023) exceeds every value after 1914 but remains below most values of 1894–1914, and government–opposition leave-out partisanship (0.035) is the highest of its series. Averaged over the federal era, the latter (0.016) is as high as under plural voting and equal male suffrage (0.015 and 0.016), whereas the penalized government–opposition series averages half as much in the federal era as under equal male suffrage. The religious leave-out series exceeds its benchmark in 2024–2025, unlike the penalized series, which never does so in the federal era (Section 5.12).

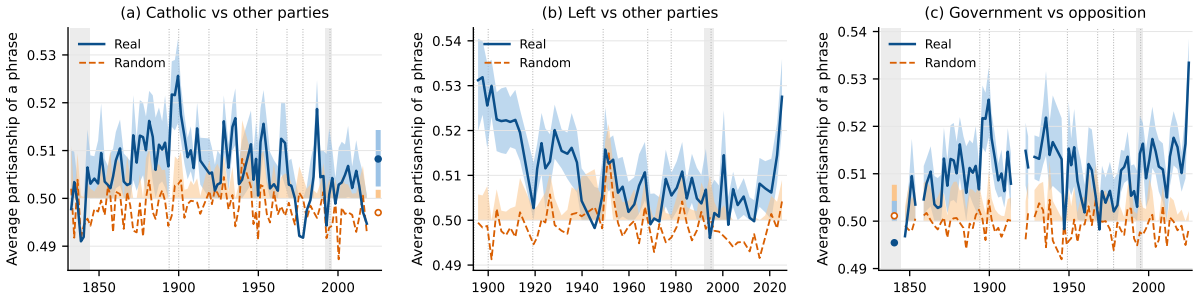


Figure A9: Leave-out estimator

Notes: Leave-out estimator of Gentzkow et al. (2019), as used by Graziani and Vanschoonbeek (2026); no covariates.

Figure A10 compares the main real series with estimates excluding speeches delivered by members as ministers, which account for about a sixth of phrases. It omits random benchmarks and confidence intervals. Before 1968, religious estimates are higher without ministerial speeches in 52 of 58 periods, but after the Christian-democratic split in only 11 of 23. None of the five periods after 1968 in which the main religious series exceeds its random benchmark does so without ministerial speeches. In [relPeakLabel], net partisanship falls from [relPeakNet] to 0.0005. Conversely, 1950–1952 and 1952–1954, when the Christian democrats governed alone, exceed the benchmark only without ministerial speeches (0.0012 and 0.0011). The left–others series is higher without ministerial speeches in 41 of 58 periods and exceeds its benchmark in 33 periods rather than 27. Its highest value is then that of 1932–1934 (0.0041), and the value of [lrPeakLabel] falls from [lrPeakNet] to 0.0028 but still exceeds its benchmark. The religious values above the benchmark after 1968 therefore depend on speeches delivered by ministers, while the left–others series keeps its high values of 1932–1934 and 2024–2025.

Figure A11 compares constant gender and province effects with effects that vary by institutional period: six periods for the religious series and five for the left–others series, which begins in 1894. The real series are close, with correlations of 0.98 and 0.94 over the periods with adequate coverage. They differ most in the federal era for the left–others grouping: with varying effects, its real series is 0.0028 lower in 2024–2025 and 0.0013 lower in 2021–2024. Average net religious partisanship is 0.00028 with constant effects and 0.00024 with varying effects; average

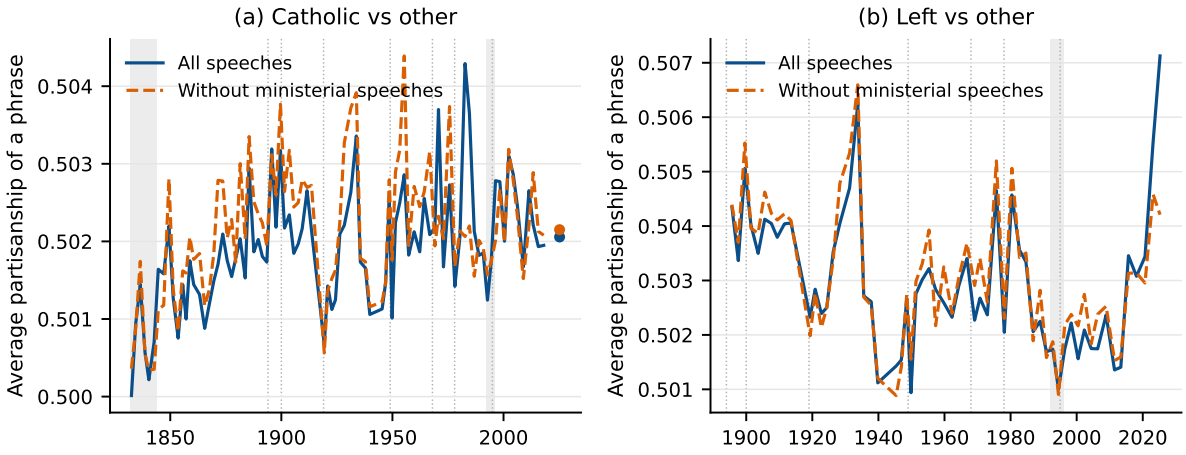


Figure A10: Speeches delivered in a ministerial capacity left out

net left–others partisanship is 0.00083 and 0.00091. The figure shows the real series only.

Individual net estimates vary more, because each specification draws its own random benchmark. With varying effects, religious partisanship in [relPeakLabel], the highest value of the principal series, falls from [relPeakNet] to 0.0009 and no longer exceeds its benchmark, mainly because the random estimate rises. For the same reason, 1894–1896 falls from 0.0016 to 0.0001. The highest religious value is then 0.0019, in 1932–1934, where the random estimate falls. Left–others partisanship in [lrPeakLabel] falls from [lrPeakNet] to 0.0029, because its real series is lower, and still exceeds its benchmark; the highest left–others value is then 0.0045, in 1932–1934.

With varying effects, the religious series exceeds its benchmark in 20 rather than 23 of 81 periods, and the left–others series in 30 rather than 27 of 58. Institutional-period averages keep the ranking of Table 4: religious partisanship is highest under plural voting and lowest in the federal era, and left–others partisanship is highest under equal male suffrage and lowest in the post-war unitary state.

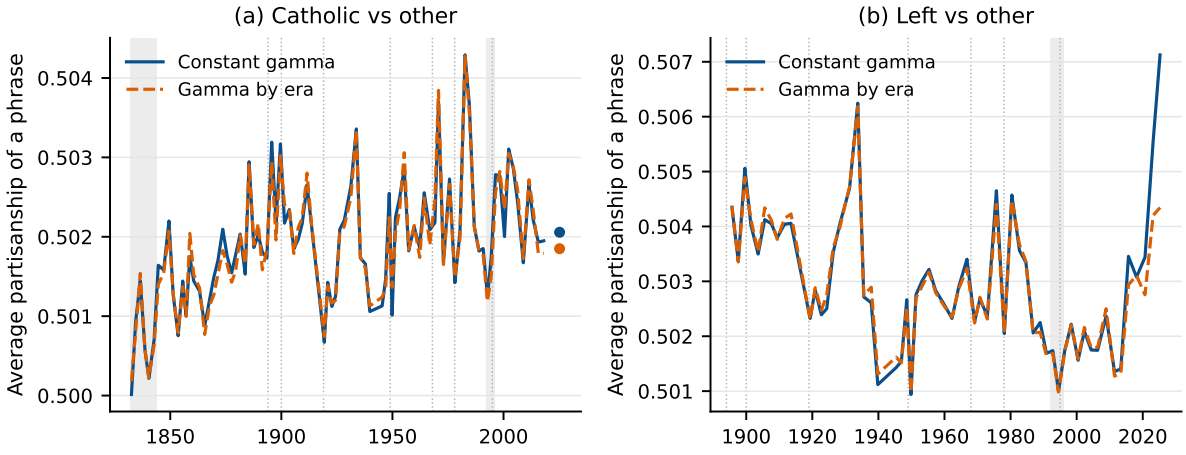


Figure A11: Covariate effects that vary by institutional period

Figure A12 compares Flemish and Walloon members within national parties before each party's split, using French speech without covariates. Each region needs 15 speakers. This creates

Catholic-series gaps before 1884, when few Walloon Catholics speak. The Liberal series covers only the years to 1864 and 1904–1914, and the Socialist series begins in 1919. Catholics speak similarly across regions: net partisanship averages 0.0004 and exceeds its random benchmark in 4 of 42 periods, all before 1914. Socialists differ more, averaging 0.0022, and exceed their benchmark in 5 of 23 periods: four between the wars and 1952–1954. Their intervals are wide, however, with only 15 to 25 Flemish Socialists speaking French. Liberal differences average 0.0017 and exceed the benchmark only in 1850–1852 and 1857–1859. Regional differences within national parties therefore appear mainly among Socialists between the wars and among Liberals in the 1850s, and rarely among Catholics (Section 5.6).

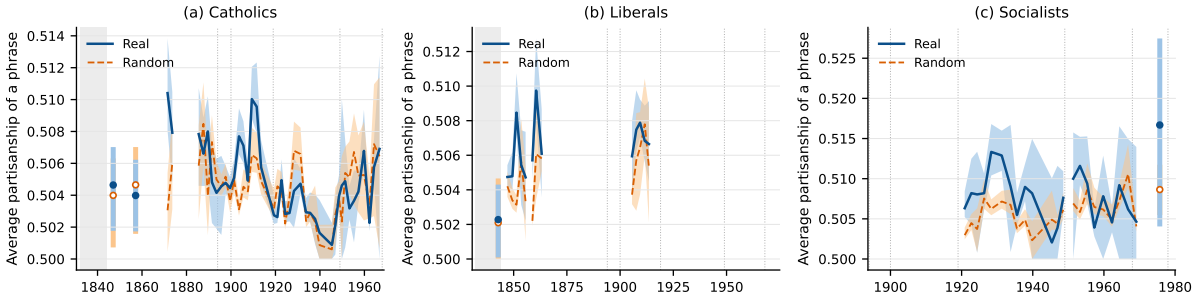


Figure A12: Flemish vs Walloon members within each party family, before its split

Notes: French speech; no covariates.

Figure A13 tracks French phrase partisanship $\hat{\zeta}_{jt}$ (Gentzkow et al., 2019), multiplied by 1,000. Eligible phrases were among a decade's three most partisan. Each panel selects phrases first identified in its half-century and displays the five with the largest absolute value in any period. Cleavage-related phrases mark particular episodes: *instruction publique* for Catholics in 1878–1884 and *écoles communales* for Liberals in 1884–1886, during and after the first school war; *suffrage universel* and *classe ouvrière* for other parties from the 1890s; *consultation populaire* for other parties in 1949–1950; and *pays en voie de développement* for Christian democrats in 1969–1971 (Section 5.1.1). Fewer phrases reflect institutional roles than in earlier versions of the vocabulary, which kept procedural formulas and titles. The clearest is *pouvoirs spéciaux*, which identifies other parties in 1981–1983 and 1995–1997, under governments that included the Christian democrats. The largest values come from the first decades: *los renten* for Catholics in 1835–1837 (1.00), a period with low coverage of the sittings, and *chemin de fer* for Catholics in 1843–1845 (0.99). Since 1945, the largest is that of *union européenne* for the Catholic family in 2024–2025 (0.70).

Figure A14 repeats this exercise for left–others partisanship. The first panel is empty because the series begins in 1894. Before 1914, *banque nationale*, *suffrage universel* and *classe ouvrière* identify the left, at about 0.7 in 1898–1902 and 1912–1914. Later panels mix policy and role vocabulary. *pleins pouvoirs* identifies the Socialist opposition in 1932–1934 (0.41), and *sécurité sociale* the left in 1995–1997 (0.60), when the Socialists governed. *pouvoirs spéciaux* changes sides with government status: it identifies the left when the Socialists are in opposition, as in 1985–1987 (0.43), and other parties when they govern, as in 1995–1997 (-0.63).

In earlier versions of the vocabulary, two name sequences ranked first and third among the left's most partisan phrases in 1894–1896: *Cousin Coiseau* and *MM. Cousin*, the names of the

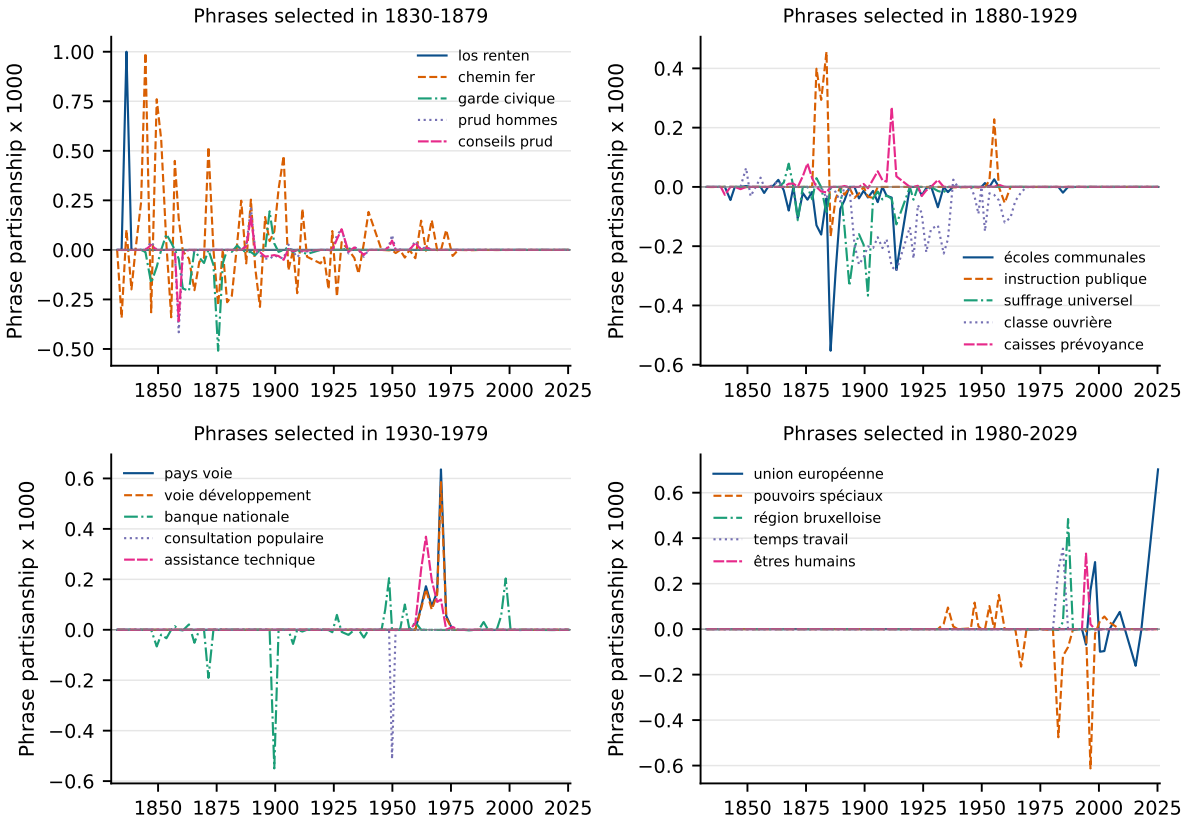


Figure A13: Religious partisanship ζ_{jt} of selected phrases

Notes: Phrase partisanship multiplied by 1000. The phrases are selected among the three most partisan of each decade (French speech); each panel shows, among those first selected in the decades of its half-century, the five with the largest absolute value in any period. Positive values: Catholic phrases.

Zeebrugge port contractors whose concession was debated in August 1895. The vocabulary now leaves out names of persons (Section 3). In the lists of Online Appendix C, the left's most partisan phrase in 1894–1896 is *classe ouvrière* (0.40), followed by *cahier des charges*, *chemin de fer*, *conseil communal* and *suffrage universel* (0.13–0.15). A left speaker is predicted to use *classe ouvrière* 219 times per 100,000 phrases, compared with 58 for other parties (Section 5.2).

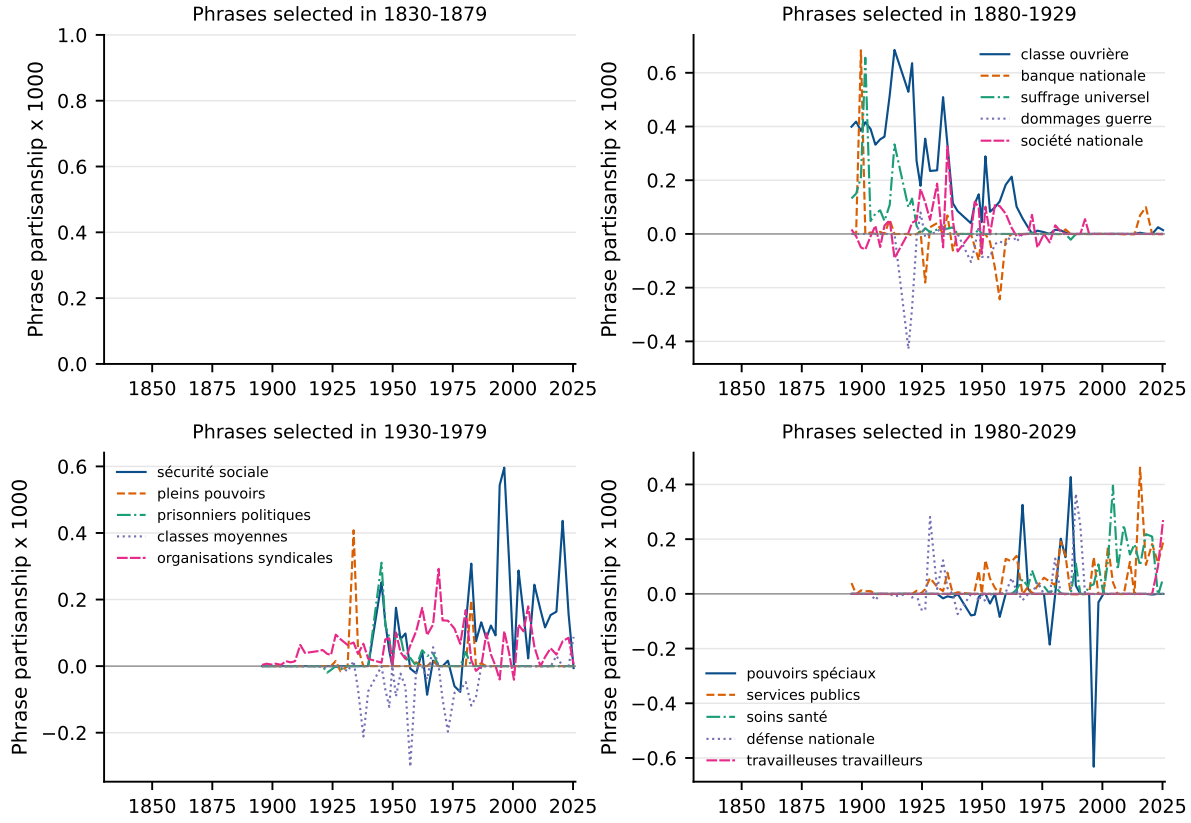


Figure A14: Left–others partisanship ζ_{jt} of selected phrases

Notes: As Figure A13; positive values: phrases of the left.

Figure A15 plots monarchy-related phrases as a share of all phrases alongside religious partisanship (top). The bottom panels show Catholic members' count and share of these phrases. The matches are spread over many phrases; the most frequent accounts for 6 per cent. They include institutions named after the Crown (*commissaire royal*, *bibliothèque royale*, *athénées royaux*), the monarchy itself (*famille royale*, *prérogative royale*, *question royale*) and fragments of royal-decree formulas (*royal pris*, *royal délibéré*), whose main phrase, *arrêté royal*, is no longer in the vocabulary (Section 3). Almost all are French. Their share peaks in 1949–1950, before the referendum on Leopold III's return. Catholics use 19 per cent of these phrases, below their shares of speakers (46 per cent) and speeches (28 per cent). Religious partisanship then exceeds its benchmark at a modest level (0.0006), and no period of 1950–1958 exceeds it (Sections 5.1.1 and 6).

Figure A16 plots language-policy phrases (top) and labour phrases (bottom) alongside the corresponding partisanship series. These compare Flemish and Walloon French speech before 1968, and the left with other parties. Language-policy phrases are most frequent in 1906–1908, 1921–1923 and 1929–1932, and around the state reforms of 1970 and 1988. Regional partisanship does not track them: the correlation is −0.01. Labour phrases reach 1.4–2.7 per cent in 1894–1914

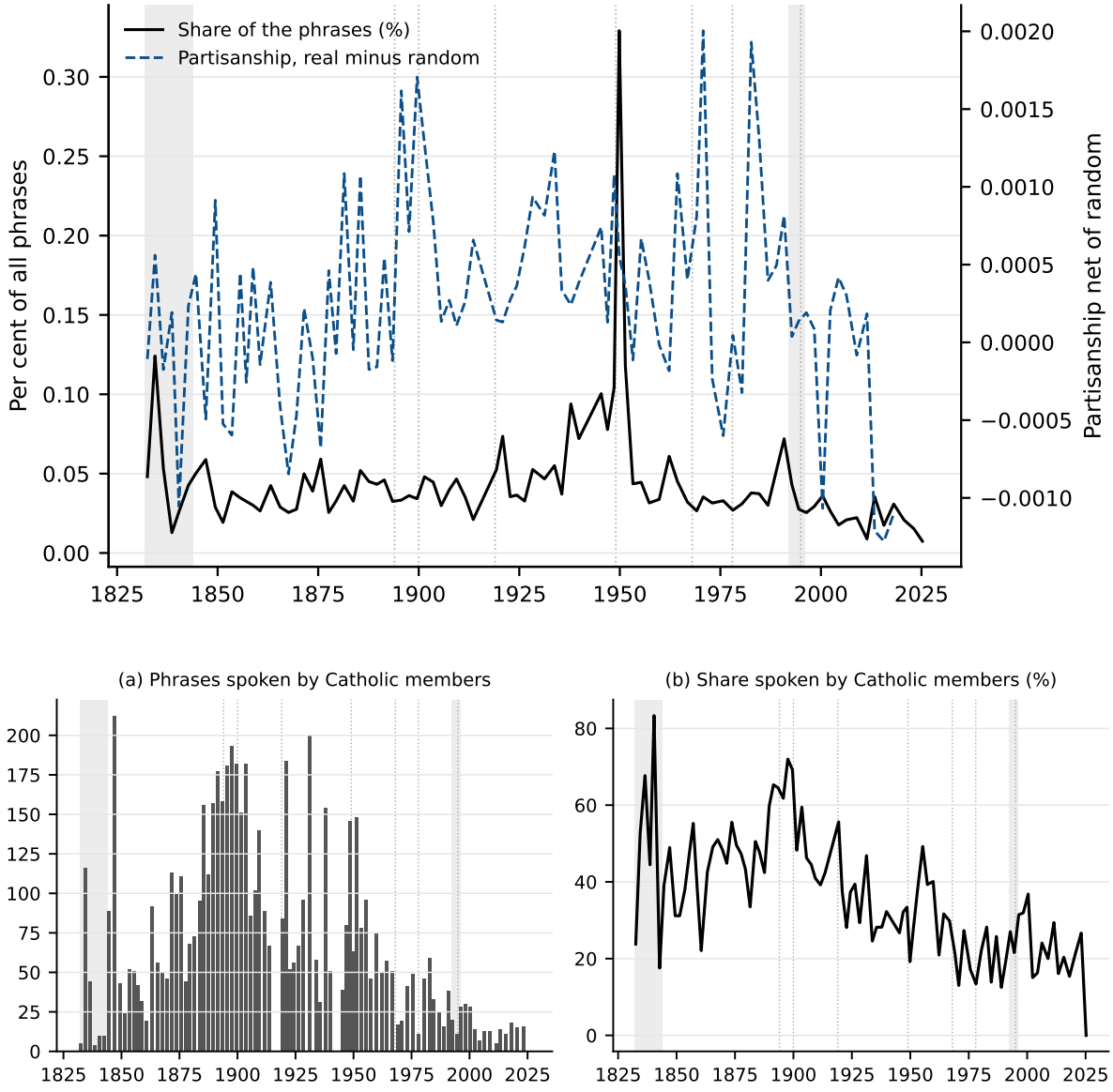


Figure A15: Phrases related to the Royal Question

after the Socialists enter and become less frequent after 1934. Left–others partisanship tracks them only loosely, with a correlation of 0.36.

The labour dictionary matches few Dutch phrases (2 per cent of matches, against 13 per cent of all phrases), so Dutch speech's growing share partly explains the later decline. In French alone, labour phrases still account for 0.7–1.3 per cent in 2016–2025. Left–others differences thus move loosely with labour attention; regional differences do not move with language-policy attention (Section 6).

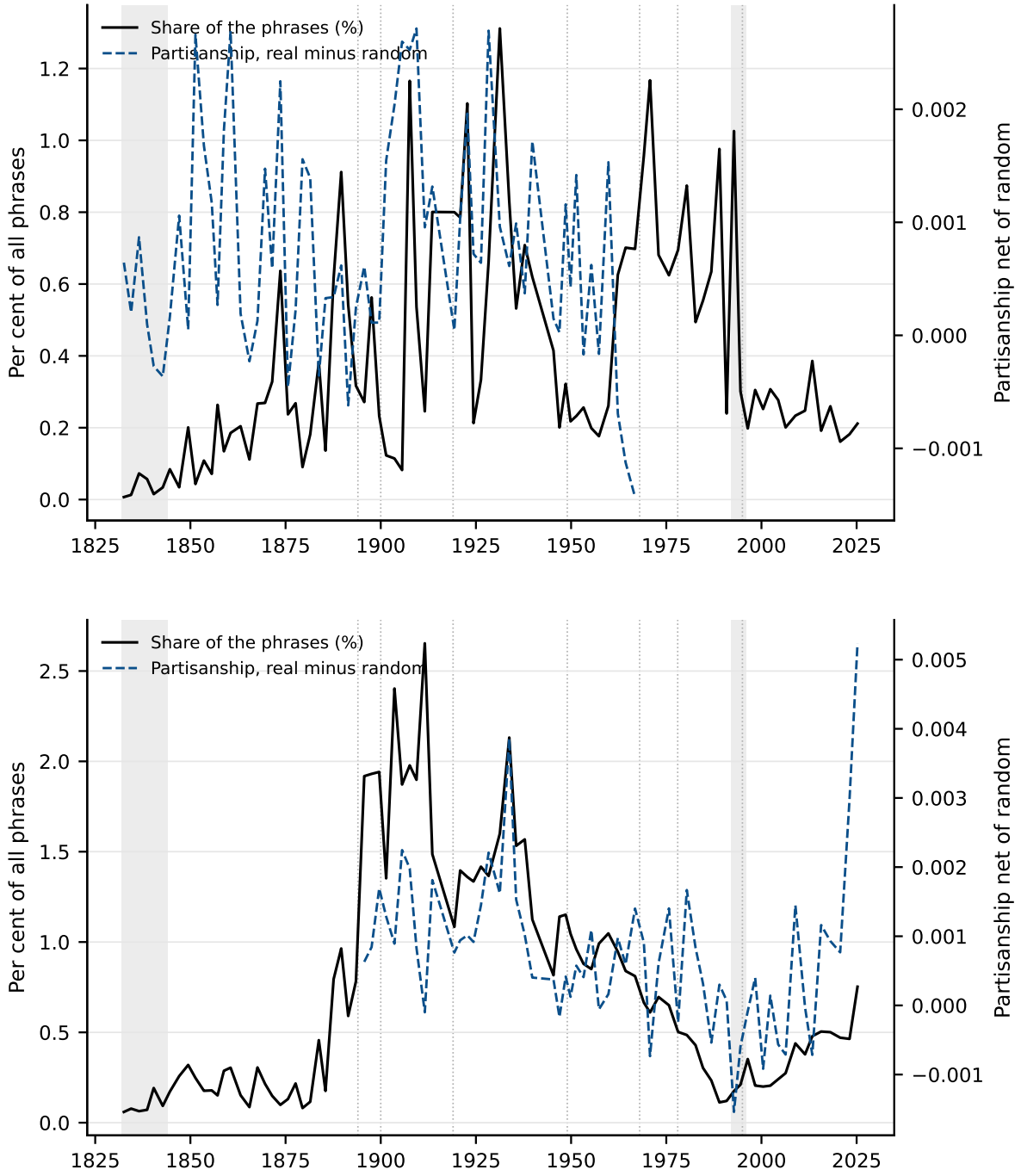


Figure A16: Phrases related to the language question (top) and to labour (bottom)

Roll-call votes. The dataset records the Chamber's roll-call votes, linked to members (Graziani, 2026). We keep yes and no votes. In 2,402 of the 31,615 roll calls, a member appears more than once, because several votes are recorded under one number, most of them in 1981–1983, 1995–1997 and 2019–2021; we leave these roll calls out. The division between two groups in a roll call is the index of party unlikeness of Rice (1925), the absolute difference between the groups' shares of yes among their yes and no votes. A roll call enters a comparison if each group casts at least five yes or no votes. The division of a period is the mean over its roll calls; periods with fewer than 20 roll calls and periods with low coverage of the sittings are left out of the correlations and the figures. Speech is net partisanship of the same groups in the same period. For the comparison of all Flemish with all Walloon members (Section 5.7), each region must cast at least ten yes or no votes, and we subtract from each roll call's index its expected value when the region labels are assigned at random among the members voting, computed exactly from the hypergeometric distribution. Within families, the division of a roll call is the mean over the families with at least five yes or no votes in each region, weighted by their votes. Subjects come from the title of the bill, taken from the legislative files of the dataset or, failing these, from the title printed in the report, and classified by keywords in French and Dutch. Produced by `code/python/28_votes.py` and `code/python/29_regions.py`.

Figure A17 compares the three main speech series with the division in votes between the same groups, and Table A1 reports their correlations by institutional period. Religious-group voting differences (panel a) are greatest under plural voting (0.50–0.81 in 1894–1914) and in 1954–1956 (0.64). Left–others voting differences (panel b) move with Socialist government status. They range from 0.45 to 0.63 in 1894–1914, fall to about 0.1 in 1934–1939 and 1961–1965, when the Socialists govern, and peak under the centre-right governments of 1981–1987 (0.61–0.73). Across periods, their correlation with the share of Socialist members whose party is in government is -0.82, against -0.30 for speech (54 periods). Votes therefore move with government–opposition divisions more closely than speech does (Section 5.4). Government–opposition voting differences (panel c) range from 0.18 to 0.81 and move little with speech (0.05).

Table A1: Correlation between division in roll calls and partisanship in speech, by institutional period

	All	1831–1893	1894–1918	1919–1944	1945–1970	1971–1994	1995–2025
Catholic vs other parties	0.19 (64)	−0.38 (12)	−0.63 (9)	0.47 (8)	−0.34 (14)	0.31 (9)	0.45 (12)
Left vs other parties	0.18 (54)	–	0.35 (9)	0.41 (8)	0.18 (14)	−0.11 (9)	0.02 (14)
Government vs opposition	0.05 (65)	−0.24 (12)	−0.86 (9)	0.15 (7)	−0.34 (14)	−0.46 (9)	−0.08 (14)

Notes: Correlation across periods between the division in roll calls (Figure A17) and net partisanship in speech; number of periods in parentheses. –: no periods. Correlations describe how the two measures move together; with 7 to 14 periods per institutional period, they are imprecise.

Figure A18 compares pairs of parties of the same language since 1979 (Section 5.5). Each point is a pair in a period: the division of the two parties in roll calls against their net partisanship in speech. A pair is on the same side when both parties are in government, or both in opposition, for at least half the period. Pairs on the same side vote alike and are harder to tell apart in speech. Panels (b) and (c) follow two pairs over time. The division between N-VA, with its predecessor the Volksunie, and Vlaams Belang is 0.10 to 0.30 in the periods in which both are in opposition and rises to 0.55 in 2024–2025, when N-VA governs for most of the period. CD&V and N-VA,

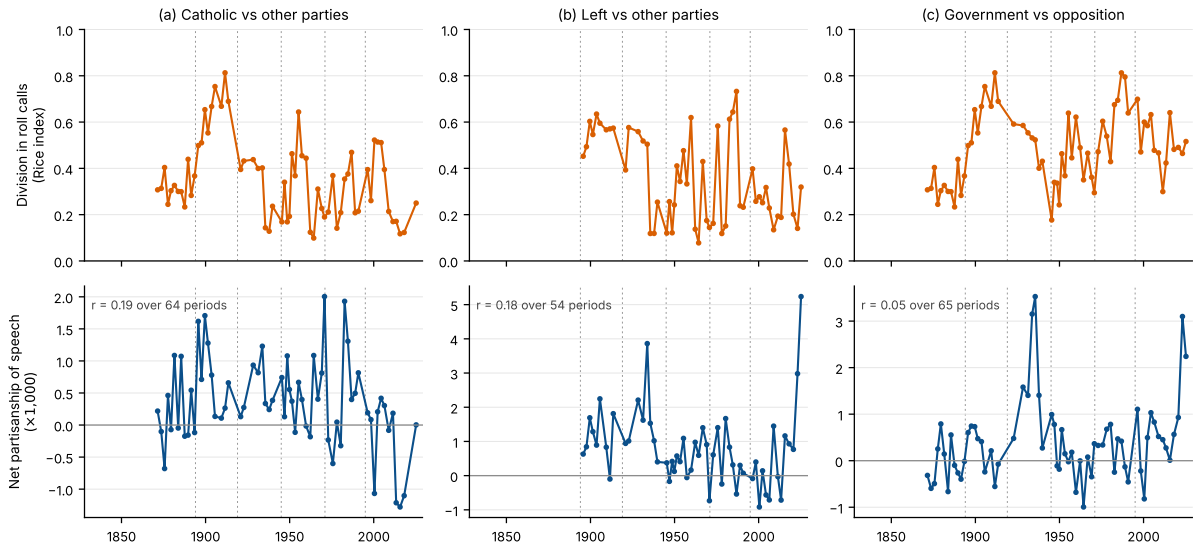


Figure A17: Partisanship in speech and division in roll-call votes

Notes: Top: division in roll calls (Rice, 1925), mean over the roll calls of the period in which each group casts at least five yes or no votes. Bottom: net partisanship of the same groups (Figures 1, 2 and 4), multiplied by 1,000; *r*: correlation over the periods shown. Periods with fewer than 20 roll calls and periods with low coverage of the sittings are left out; dotted lines: start of the institutional periods. Produced by `code/python/28_votes.py`.

with their predecessors, vote alike when both govern (a division of at most 0.04) and less so otherwise. Over time, the two measures are correlated at 0.16 for the first pair (6 periods) and 0.47 for the second (14 periods).

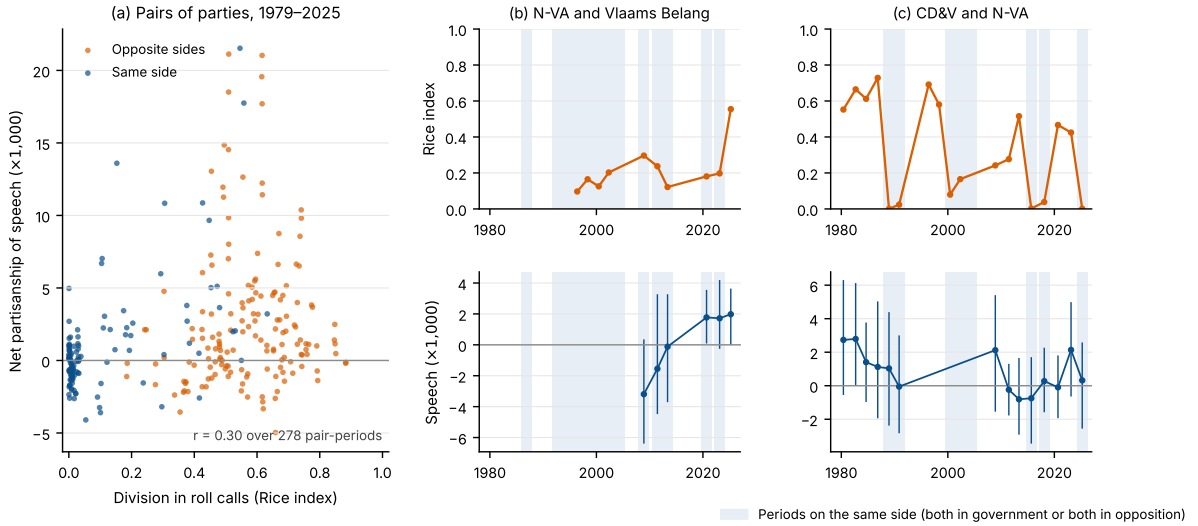


Figure A18: Pairs of parties since 1979: speech and roll-call votes

Notes: (a) Pairs of parties of the same language and period (Section 5.5): division in roll calls between the members of the two parties and net partisanship of the pair in speech, without covariates, multiplied by 1,000; same side: both parties in government, or both in opposition, for at least half the period; *r*: correlation over the pair-periods. (b) and (c) The division in roll calls (top) and net partisanship in speech with its 80 per cent interval (bottom) for two pairs; shaded: periods on the same side. The parties of the votes are the party units of Online Appendix B; N-VA includes the Volksunie. Produced by `code/python/28_votes.py`.

Figure A19 compares Flemish and Walloon members within the national parties before their splits. In each of the three parties, the members of the two regions seldom vote differently: the

division, pooled over the three parties, averages 0.04 and is at most 0.19, in the early 1920s. It is correlated at 0.17 with Flemish–Walloon partisanship in French speech (49 periods). Figure A20 reports the division between Flemish and Walloon members by subject of the bill (Section 5.7).

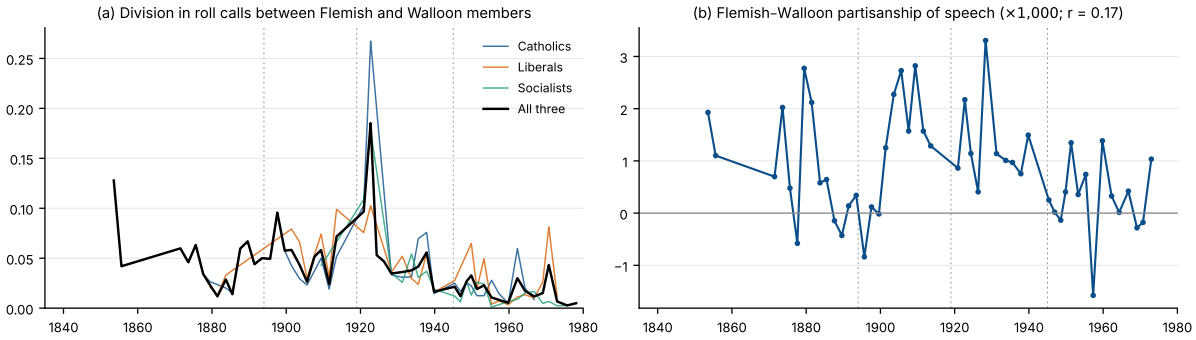


Figure A19: Flemish and Walloon members of the national parties: roll-call votes and speech

Notes: (a) Division in roll calls between the members of each national party elected in Flemish and in Walloon districts, before the party's split, and the three parties pooled (Brussels left out); not net of its random expectation. (b) Flemish–Walloon net partisanship in French speech (Figure 22 until 1967, then the families not yet split), multiplied by 1,000; *r*: correlation with the pooled division in roll calls. Produced by `code/python/28_votes.py`.



Figure A20: Flemish and Walloon members in roll-call votes, by subject of the bill

Notes: Net division in roll calls (Figure 8) by subject of the bill and institutional period: (a) all Flemish against all Walloon members; (b) members of the same party family. Subjects from keywords in French and Dutch in the title of the bill, the first matching class in the order shown; roll calls without a usable title are in *Other or unknown*. –: fewer than 10 roll calls. Periods with fewer than 20 roll calls and periods with low coverage of the sittings are left out. Produced by `code/python/29_regions.py`.

Figure A21 reports the experiment from Section 4. In panel (a), simulated character errors in born-digital reports from 2019–2025 steadily reduce left–others partisanship without covariates. At 5 per cent error, its distance above 0.5 falls to 72 per cent of the clean-text distance. With covariates, attenuation is smaller and less regular, retaining 81 per cent at 5 per cent error. The religious series is undefined in most of these years.

Panel (b) shows the share of tokens outside a lexicon of words occurring at least twice in each of at least ten periods. For French, this share is 6–12 per cent before 1843 and 1.1–3.0 per cent from 1845 to 2019, close to or below the 2.9 per cent of the clean reports used in the experiment.

Applied to the experiment, these rates suggest no attenuation from 1845. However, the lexicon also excludes new and rare words. These shares therefore do not yet measure historical OCR error rates (Section 5.10).

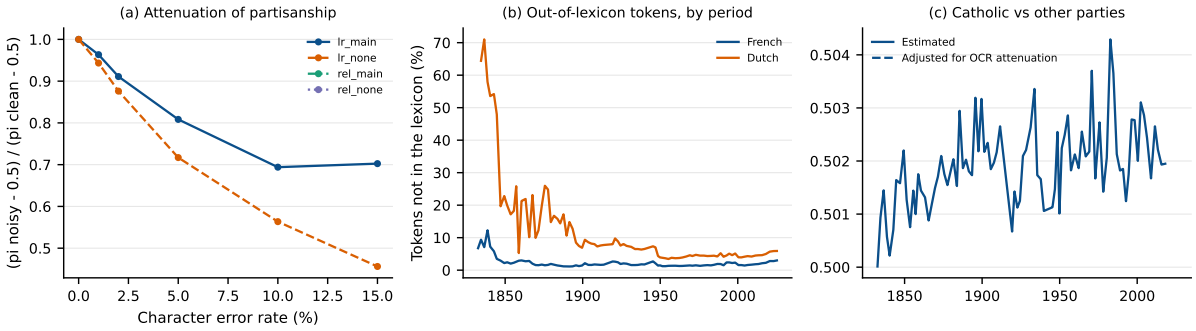


Figure A21: OCR noise and measured partisanship

Notes: (a) Attenuation of partisanship in the born-digital reports of 2019–2025 with simulated OCR errors; the religious series is not defined in these years. (b) Share of the tokens that are not in the reference lexicon (the words used at least twice in each of at least ten periods), by language. (c) Religious partisanship, real series; the series adjusted for the attenuation implied by panel (b) is not defined, because the religious series is not defined in the experiment.

Figure A22 applies Figure 26' decomposition to left–others partisanship. From 1894–1896 to 1912–1914, cumulative leave-out partisanship falls by 0.013, almost entirely through renewal; sitting-member changes sum to about zero. The First World War adds a further decline of 0.016, mainly among sitting members (-0.013). Afterwards, the within component varies more than renewal, with about one and a half times its standard deviation. The cumulative total stays negative from 1898–1900 to 2021–2024 and returns to about zero only in 2024–2025. Across the full sample, renewal subtracts 0.047 and sitting members add 0.048, as in the religious series; the change into 1999–2001 alone adds 0.042 to the within component. The components have opposite signs in three of five institutional periods (Table 10). The pattern in Section 5.8 is therefore weaker for left–others differences.

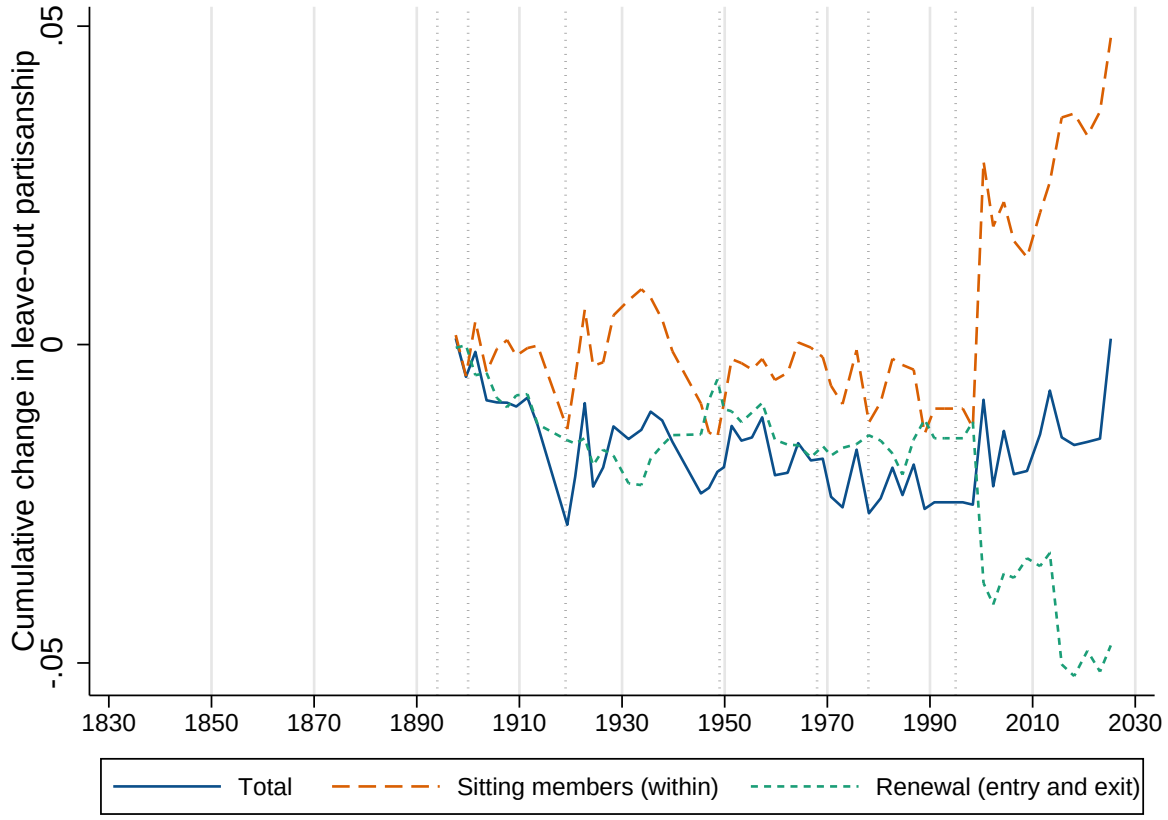


Figure A22: Left–others partisanship: renewal of the Chamber and adaptation of sitting members

Notes: As Figure 26.

Online Appendix B: Data and preprocessing

Sample selection. We start from the dataset's [nBlocks] blocks and select speech blocks from plenary sittings. Plenary reports are identified by document code from 1999 and by the 1985–1999 classification in `committee_scan.csv` for earlier reports. We exclude the chair's turns, retain French or Dutch speeches linked to a member through exact, fuzzy or manual matching or to a minister, and require Chamber membership on the sitting date. Table B1 gives block counts at each step. We retain speeches by members serving as ministers; Online Appendix Figure A10 excludes those delivered in a ministerial capacity.

Table B1: Sample selection

Step	Blocks
All blocks	[nBlocks]
Speech blocks	2,370,355
In plenary sittings	2,262,560
Not turns of the chair	1,826,697
In French or Dutch, linked to a member or minister	[nSpeechesSelected]
Delivered by members of the Chamber	[nSpeeches]

Preprocessing. We remove parenthetical text containing interruptions, applause and laughter, and rejoin words hyphenated at line breaks. In the PDF reports of 1 July 1999 to 14 July 2000, every accent is printed as a separate character after its letter (*pre´sident*); we join each accent to its letter. We spell out euro, dollar, per cent and paragraph signs, replace other non-alphanumeric characters with spaces, lowercase the text and split it into words. We remove each language's Snowball stopwords and single-character tokens, then apply the French and Dutch Snowball stemmers (Porter, 2001). For Dutch, we use the original algorithm, which retains the prefix *ge-*.

We form bigrams from consecutive stems. We remove those containing the surname of a member of the Chamber, including members who served as ministers, a party acronym, an address to the chair or colleagues, or a month. We retain surnames printed in lowercase at least half the time, treating them as common words. We also exclude listed procedural phrases and party names. Party adjectives such as *catholique* and *socialiste* remain because they can describe policy objects, as in *enseignement catholique*. The lists are in `code/python/textproc.py`.

Phrases without content. After the vocabulary is formed, we leave out phrases without content, with the rules of the project's website (`code/python/04b_filter_vocabulary.py`): procedural formulas (*projet loi*, *section centrale*, *ter tafel*), courtesies and titles (*premier ministre*, *eerste minister*), names of parties and parliamentary groups (*groupe socialiste*), amounts, dates and other numbers (a law with its year and an age remain, as in *67 ans*), person names that the surname filter misses, such as those of ministers who never sat in the Chamber (*général cousebant*, *van houtte*), fragments of words, pairs of filler words (*eh bien*, *ten eerste*) and the names of ministries, which serve as titles (*ministre finances*, *binnenlandse zaken*). French keeps 77,324 of 84,812 phrases, with 85.8 per cent of the occurrences, and Dutch 18,045 of 22,240, with 71.7 per cent. The phrases left out are listed, with their frequencies, in `output/checks/vocab_dropped_fr.csv` and `vocab_dropped_nl.csv`

of the replication package. Phrases that name a subject, such as *classe ouvrière*, remain, and so do a few phrases without content that the rules miss, among them some text of the voting procedure (Section 5.12).

Parties. Rules in `code/python/parties.py` map party labels to families, language communities and party units. We manually check these mappings in `party_crosswalk.csv` and use the dataset's family label when available. A party unit is the national party before its family's split—Catholics in 1968, Liberals in 1972 and Socialists in 1978—and the language-community party afterwards. Where a family has several parties at the same time, each party is a separate unit. Members of the joint parliamentary group of the two Green parties belong to the community of their district or, in Brussels, of the language they speak. A unit is in government on a day if one of its members holds ministerial office; ministers without a party place no unit in government. Unionist governments include both Catholics and Liberals. Table B2 lists every party label of the register, with its family and community, the years in which it appears, its speakers and speeches, its side in every comparison and its years in government. Table B3 lists the parties in government between consecutive changes of composition.

Table B2: Party labels, families and groups in each comparison

Party label	Community	Years	Leg.	Speakers	Speeches	Words (000s)	Side in the comparison: Rel.	L–O	L–R	R–O	Flemish–Walloon	Years in government	Multiparty family	Party of 2024
Catholic and Christian democrat														
Katholiek[r]	national	1831–1946	33	589	518,034	37,079	A	B	–	B	A, B	1831–1840, 1841–1847, 1855–1857, 1870–1878, 1884–1945	Catholic	CD&V, Les Engagés
CVP-PSC	national	1946–1968	7	191	75,713	7,161	A	B	–	B	A, B	1947–1954, 1958–1968	Catholic	CD&V, Les Engagés
Daensist	Dutch	1894–1898, 1900–1921	6	5	8,928	353	A	B	–	B	A	1894–1898, 1900–1921	Catholic	CD&V
CVP-PSC[r]	Dutch	1968–1971	2	52	3,436	618	A	B	–	B	–	1968–1971	Catholic	CD&V
CVP	Dutch	1971–2003	10	141	38,255	5,158	A	B	–	B	–	1971–1999	Catholic	CD&V
CD&V[r]	Dutch	2003–2025	7	62	12,688	1,609	A	B	–	B	–	2007–2025	Catholic	CD&V
CD&V/N-VA[s]	Dutch	2007–2024	4	2	294	46	A	B	–	B	–	2007–2021	Catholic	CD&V
RSCL	French	1954–1958	1	1	26	6	A	B	–	B	–	never	Catholic	Les Engagés
CVP-PSC	French	1968–1971	1	19	1,378	284	A	B	–	B	–	1968–1971	Catholic	Les Engagés
PSC	French	1971–2003	10	70	15,512	2,056	A	B	–	B	–	1971–1999	Catholic	Les Engagés
cdH[r]	French	2003–2024	6	29	7,246	1,106	A	B	–	B	–	2007–2014	Catholic	Les Engagés
Les Engagés	French	2024–2025	1	18	467	69	A	B	–	B	–	2025	Catholic	Les Engagés
Liberal														
Liberaal[r]	national	1831–1946	33	488	350,431	28,123	B	B	B	A	A, B	1831–1846, 1847–1870, 1878–1884, 1918–1925, 1925–1939, 1939–1946	Liberal	Open VLD, MR
LP	national	1946–1965	6	50	24,567	2,372	B	B	B	A	A, B	1946, 1946–1947, 1949–1950, 1954–1958, 1958–1961	Liberal	Open VLD, MR
PVV-PLP	national	1965–1971	2	54	6,334	908	B	B	B	A	A, B 1965–1968; ext. A, B 1968–1971	1966–1968	Liberal	Open VLD, MR
PVV[r]	Dutch	1971–1995	11	65	17,027	2,174	B	B	B	A	ext. A 1971–1974	1973–1977, 1980, 1981–1988	Liberal	Open VLD
VLD[r]	Dutch	1995–2007	6	49	8,595	1,078	B	B	B	A	–	1999–2007	Liberal	Open VLD
Open Vld[r]	Dutch	2007–2025	6	44	7,526	1,069	B	B	B	A	–	2007–2025	Liberal	Open VLD
PLP[r]	French	1971–1977	3	17	1,340	205	B	B	B	A	ext. B 1971–1974	1973–1977	Liberal	MR
PLDP	French	1974–1977	1	3	613	63	B	B	B	A	–	1974–1977	Liberal	MR
PL	French	1977–1981	2	2	516	71	B	B	B	A	–	1977, 1980	Liberal	MR

Table B2: Party labels, families and groups in each comparison (continued)

Party label	Community	Years	Leg.	Speakers	Speeches	Words (000s)	Side in the comparison: Rel.	L–O	L–R	R–O	Flemish–Walloon	Years in government	Multiparty family	Party of 2024
PRLW[r]	French	1977–1981	3	16	3,961	426	B	B	B	A	–	1977, 1980	Liberal	MR
PRL[r]	French	1981–2003	7	58	13,953	2,079	B	B	B	A	–	1981–1988, 1999–2003	Liberal	MR
PRL-FDF-MCC[s]	French	1995–2003	2	4	79	19	B	B	B	A	–	1999–2001	Liberal	MR
MCC	French	1999–2003	1	2	92	15	B	B	B	A	–	1999–2003	Liberal	MR
MR	French	2003–2025	6	70	9,519	1,497	B	B	B	A	–	2003–2025	Liberal	MR
Socialist														
Socialist[r]	national	1894–1946	14	176	285,158	14,108	B	A	A	B	A, B	1918–1921, 1925–1927, 1935–1939, 1939–1946	Socialist	Vooruit, PS
BSP-PSB[r]	national	1946–1971	9	164	83,210	6,698	B	A	A	B	A, B 1946–1968; ext. A, B 1968–1971	1946–1949, 1954–1958, 1961–1966, 1968–1971	Socialist	Vooruit, PS
BSP	national	1971–1978	3	49	7,922	808	B	A	A	B	ext. A 1971–1978	1971–1974, 1977–1978	Socialist	Vooruit, PS
BSP-Rode Leeuwen	national	1971–1974	1	1	13	5	B	A	A	B	–	1971–1974	Socialist	Vooruit, PS
PSB[r]	national	1971–1978	5	48	8,548	1,238	B	A	A	B	ext. B 1971–1978	1971–1974, 1977–1978	Socialist	Vooruit, PS
BSP[r]	Dutch	1978–1981	2	26	2,530	398	B	A	A	B	–	1979–1981	Socialist	Vooruit
SP[r]	Dutch	1981–2003	9	75	19,627	2,389	B	A	A	B	–	1981, 1988–2003	Socialist	Vooruit
Spirit	Dutch	2003–2007	1	7	457	60	B	A	A	B	–	2003–2007	Socialist	Vooruit
sp.a[r]	Dutch	2003–2024	6	45	8,103	965	B	A	A	B	–	2003–2007, 2011–2014, 2020–2024	Socialist	Vooruit
sp.a-Spirit[s]	Dutch	2003–2010	2	7	666	80	B	A	A	B	–	2003–2007	Socialist	Vooruit
Vooruit	Dutch	2024–2025	1	15	422	43	B	A	A	B	–	2024–2025	Socialist	Vooruit
PS	French	1978–2025	13	161	29,202	3,891	B	A	A	B	–	1979–1981, 1988–2014, 2020–2025	Socialist	PS
Communist and radical left														
Communist[r]	national	1925–1946	6	14	12,530	920	B	A	A	B	B	1944, 1945–1946	Communist	PVDA-PTB
KPB-PCB	national	1946–1985	13	33	19,826	2,415	B	A	A	B	A, B 1946–1968	1946, 1946–1947	Communist	PVDA-PTB
UGS	national	1965–1968	1	1	135	26	B	A	A	B	–	never	Communist	PVDA-PTB
PTB-go!	national	2014–2019	1	2	1,270	175	B	A	A	B	–	never	Communist	PVDA-PTB
PVDA/PTB	national	2019–2025	2	21	3,714	566	B	A	A	B	–	never	Communist	PVDA-PTB

Table B2: Party labels, families and groups in each comparison (continued)

Party label	Community	Years	Leg.	Speakers	Speeches	Words (000s)	Side in the comparison: Rel.	L–O	L–R	R–O	Flemish–Walloon	Years in government	Multiparty family	Party of 2024
Flemish nationalist, except N-VA														
Vlaams-nationalist	Dutch	1919–1946	7	29	15,453	1,089	B	B	–	B	A 1919–1940	never	Flemish nationalist	N-VA
VNV	Dutch	1939–1946	1	2	312	27	B	B	–	B	A 1939–1940	never	Flemish nationalist	N-VA
CVV	Dutch	1954–1958	1	1	112	15	B	B	–	B	A 1956–1958	never	Flemish nationalist	N-VA
Volksunie	Dutch	1958–1999	13	58	26,445	3,730	B	B	–	B	A 1961–1968	1977–1979, 1988–1991	Flemish nationalist	N-VA
VU-ID	Dutch	1999–2003	1	7	1,783	252	B	B	–	B	–	never	Flemish nationalist	N-VA
Flemish nationalist: N-VA														
N-VA	Dutch	2003–2025	6	76	11,686	1,734	B	B	B	A	–	2014–2018, 2025	Flemish nationalist	N-VA
Populist radical right														
Rexist	national	1936–1946	2	22	2,700	189	B	B	B	A	A, B	never	Radical right	Vlaams Belang
Vlaams Blok	Dutch	1978–2007	8	33	7,861	1,113	B	B	B	A	–	never	Radical right	Vlaams Belang
Vlaams Belang^{r}	Dutch	2007–2025	6	48	6,863	1,180	B	B	B	A	–	never	Radical right	Vlaams Belang
Front National	French	1991–2010	5	5	454	67	B	B	B	A	–	never	Radical right	–
Parti Populaire	French	2010–2019	2	2	571	74	B	B	B	A	–	never	Radical right	–
Francophone regionalist														
FDF	French	1965–1987, 1995–2003, 2010–2019	12	24	9,301	1,560	B	B	–	B	–	1977–1980	Francophone regionalist	DéFI
Front Wallon	French	1965–1968	1	1	226	34	B	B	–	B	B	never	Francophone regionalist	DéFI
PWT	French	1965–1968	1	1	319	34	B	B	–	B	B	never	Francophone regionalist	DéFI
RW	French	1968–1985	6	20	5,102	640	B	B	–	B	–	1974–1977	Francophone regionalist	DéFI
FDF/PPW	French	1988–1995	2	5	863	196	B	B	–	B	–	never	Francophone regionalist	DéFI

Table B2: Party labels, families and groups in each comparison (continued)

Party label	Community	Years	Leg.	Speakers	Speeches	Words (000s)	Side in the comparison: Rel.	L–O	L–R	R–O	Flemish–Walloon	Years in government	Multiparty family	Party of 2024
DéFI	French	2019–2025	2	2	1,012	222	B	B	–	B	–	never	Francophone regionalist	DéFI
Green														
Agalev	Dutch	1981–2003	6	24	4,788	758	B	B	–[a]	B	–	1999–2003	Green	Groen
Groen!	Dutch	2007–2014	2	6	1,526	260	B	B	–[a]	B	–	never	Green	Groen
Groen	Dutch	2014–2025	3	15	2,869	406	B	B	–[a]	B	–	2020–2025	Green	Groen
Ecolo	French	1981–2025	12	49	8,721	1,423	B	B	–[a]	B	–	1999–2003, 2020–2025	Green	Ecolo
Ecolo-Agalev[s]	Dutch, French	1991–2003	3	8	966	166	B	B	–[a]	B	–	1999–2003	Green	Groen, Ecolo
Ecolo-Groen[s]	Dutch, French	2007–2025	5	13	1,358	221	B	B	–[a]	B	–	2020–2025	Green	Groen, Ecolo
Other parties and independents														
Middenstanders	national	1919–1921	1	1	19	2	B	B	–	B	–	never	–	–
Oud-strijders	national	1919–1925, 1939–1946	3	3	1,027	81	B	B	–	B	–	never	–	–
Renaissance nationale	national	1919–1921	1	1	47	2	B	B	–	B	–	never	–	–
Onafhankelijk	national	1929–1932, 2019–2025	3	2	484	56	B	B	–	B	A 1929–1932	never	–	–
Onafhankelijke	national	1932–1939, 1988–1991	3	2	211	33	B	B	–	B	–	never	–	–
Technocraat	national	1939–1946	1	1	122	11	B	B	–	B	A	never	–	–
UDB	national	1946–1949	1	1	38	2	B	B	–	B	B	1946	–	–
PSI	national	1961–1965	1	1	85	8	B	B	–	B	B	never	–	–
RN	national	1961–1965	1	1	96	31	B	B	–	B	–	never	–	–
UDRT-RAD	national	1978–1987	3	3	660	110	B	B	–	B	–	never	–	–
ROSSEM	Dutch	1991–1995	1	3	165	19	B	B	–	B	–	never	–	–
LDD	Dutch	2007–2014	2	6	949	126	B	B	–	B	–	never	–	–

Notes: One row per party label as it appears in the register of members (78 rows), by family. Community: *national* before the family's split; *Dutch* and *French*, parties of one language community, after the split or from their foundation. Years: first and last year in which a member with the label sits in the Chamber; Leg.: legislatures. Speakers, speeches and words

come from the 14,487 speaker-periods of the analysis; the 84 without a family are not in the table, which counts 1,725,057 speeches and 146.3 million words. A family groups parties over time for comparison; it does not imply organizational continuity. A and B: the two sides of each comparison; –: left out of the comparison. Rel.: Catholic family against all other parties; L–O: Socialist and Communist families against all other parties; L–R: the left against the Liberal family, the populist radical right and N-VA ([a]: in a variant, the Greens join the left); R–O: the right against all other parties; Flemish–Walloon: members elected in Flemish (A) and in Walloon (B) districts, before the split, and in the extended series (ext.) the families not yet split. Government and opposition are defined by government status, not by family (Table B3). Multiparty family: the family in the multiparty index (Section 5.5). Party of 2024: the party elected in 2024 to which the label leads, as a grouping. Members whose party is unknown (341 of 16,793 member-periods) are left out of every comparison. [r] Residual uses, counted in the row: speakers who carry the label in a legislature in which most speakers of its lineage carry another name (legislature, speakers): Katholiek: 1946–1949 (1); CVP-PSC (Dutch): 1971–1974 (1); CD&V: 1999–2003 (1); cdH: 2024–2025 (1); Liberaal: 1946–1949 (2); PVV: 1999–2003 (1), 2003–2007 (1), 2007–2010 (1); VLD: 1991–1995 (2), 2007–2010 (2), 2010–2014 (1); Open Vld: 2003–2007 (1); PLP: 1977–1978 (1); PRLW: 1981–1985 (1); PRL: 2003–2007 (2); Socialist: 1946–1949 (5); BSP-PSB: 1971–1974 (1); PSB (national): 1978–1981 (2), 1981–1985 (1), speakers of the French community; BSP (Dutch): 1981–1985 (3); SP: 2003–2007 (2), 2019–2024 (1), 2024–2025 (1); sp.a: 1999–2003 (1); Communist: 1946–1949 (1); Vlaams Belang: 2003–2007 (1). [s] Label printed in the speeches, for members listed without a party in the register in that year: CD&V/N-VA, Ecolo-Agalev, Ecolo-Groen, PRL-FDF-MCC, sp.a-Spirit. Printed variants of a register label are counted with it: FDF-PPW with FDF/PPW; FN with Front National; PVDA-PTB with PVDA/PTB; VB with Vlaams Belang; VL.BLOK with Vlaams Blok; VLAAMS BLOK with Vlaams Blok. Produced by `code/python/27_paper_extras.py`.

Table B3: Parties in government, 1831–2025

First sitting	Last sitting	Sittings	Parties in government	Periods
14.09.1831	06.04.1840	474	Catholic, Liberal	1831–1833 to 1839–1841
24.04.1840	17.03.1841	20	Liberal	1839–1841
16.11.1841	25.03.1846	412	Catholic, Liberal	1841–1843 to 1845–1848
20.04.1846	10.05.1847	192	Catholic	1845–1848
10.11.1847	28.03.1855	1002	Liberal	1845–1848 to 1854–1856
24.04.1855	28.05.1857	264	Catholic, Liberal	1854–1856 to 1856–1857
15.12.1857	13.05.1870	1383	Liberal	1857–1859 to 1868–1870
09.08.1870	24.05.1878	828	Catholic	1870–1872 to 1876–1878
24.07.1878	17.05.1884	604	Liberal	1878–1880 to 1882–1884
22.07.1884	04.08.1914	3427	Catholic	1882–1884 to 1912–1914
28.11.1918	21.10.1921	287	Catholic, Liberal, Socialist	1918–1919 to 1919–1921
06.12.1921	12.05.1925	354	Catholic, Liberal	1921–1923 to 1925–1927
19.05.1925	22.05.1925	3	Catholic	1925–1927
23.06.1925	15.11.1927	215	Catholic, Liberal, Socialist	1925–1927 to 1927–1929
29.11.1927	14.03.1935	638	Catholic, Liberal	1927–1929 to 1934–1936
29.03.1935	22.02.1939	288	Catholic, Liberal, Socialist	1934–1936 to 1936–1939
23.02.1939	13.04.1939	4	Catholic, Socialist	1936–1939 to 1939–1940
18.04.1939	18.04.1939	1	Catholic, Liberal, Socialist	1939–1940
21.04.1939	30.06.1939	27	Catholic, Liberal	1939–1940
05.09.1939	20.09.1944	56	Catholic, Liberal, Socialist	1939–1940 to 1944–1946
03.10.1944	14.11.1944	4	Catholic, Liberal, Socialist, Communist	1944–1946
21.11.1944	07.02.1945	8	Catholic, Liberal, Socialist	1944–1946
14.02.1945	26.07.1945	26	Catholic, Liberal, Socialist, Communist	1944–1946
07.08.1945	08.03.1946	45	Liberal, Socialist, Communist, UDB	1944–1946 to 1946–1947
19.03.1946	26.03.1946	3	Socialist	1946–1947
03.04.1946	11.03.1947	97	Liberal, Socialist, Communist	1946–1947
25.03.1947	26.07.1949	204	Catholic, Socialist	1946–1947 to 1949–1950
16.08.1949	09.03.1950	42	Catholic, Liberal	1949–1950
20.06.1950	12.03.1954	356	Catholic	1950–1952 to 1952–1954
27.04.1954	18.06.1958	314	Liberal, Socialist	1954–1956 to 1958–1961
01.07.1958	11.10.1958	9	Catholic	1958–1961
18.11.1958	18.04.1961	195	Catholic, Liberal	1958–1961 to 1961–1963
27.04.1961	08.02.1966	302	Catholic, Socialist	1961–1963 to 1965–1968
23.03.1966	28.02.1968	155	Catholic, Liberal	1965–1968
23.04.1968	23.04.1968	1	CVP, PSC, Liberal	1968–1969
25.06.1968	07.12.1972	325	CVP, PSC, Socialist	1968–1969 to 1971–1974
30.01.1973	17.01.1974	79	CVP, PSC, Liberal, Socialist	1971–1974
28.03.1974	28.03.1974	1	CVP, PSC, PVV, PLP, Socialist	1974–1977
30.04.1974	06.06.1974	14	CVP, PSC, PVV, PLP	1974–1977
12.06.1974	05.05.1977	235	CVP, PSC, PVV, PLP/PRLW, RW	1974–1977 to 1977–1978
07.06.1977	14.11.1978	93	CVP, PSC, Socialist, Volksunie, FDF	1977–1978
11.01.1979	03.04.1979	2	CVP, PSC, BSP, PS, Volksunie, FDF	1979–1981
05.04.1979	16.01.1980	48	CVP, PSC, BSP/SP, PS, FDF	1979–1981
17.01.1980	03.04.1980	23	CVP, PSC, SP, PS	1979–1981
20.05.1980	14.10.1980	32	CVP, PSC, PVV, PRL, SP, PS	1979–1981
24.10.1980	15.12.1981	69	CVP, PSC, SP, PS	1979–1981 to 1981–1983
18.12.1981	21.04.1988	365	CVP, PSC, PVV, PRL	1981–1983 to 1988–1989
10.05.1988	13.07.1991	170	CVP, PSC, SP, PS, Volksunie	1988–1989 to 1989–1991
02.10.1991	01.07.1999	413	CVP, PSC, SP, PS	1989–1991 to 1999–2001

Table B3: Parties in government, 1831–2025 (continued)

First sitting	Last sitting	Sittings	Parties in government	Periods
14.07.1999	08.04.2003	234	VLD, PRL/MR, SP/sp.a, PS, Agalev, Ecolo	1999–2001 to 2001–2003
05.06.2003	26.06.2003	2	VLD, MR, sp.a, PS, Agalev	2003–2005
14.07.2003	20.12.2007	182	VLD/Open Vld, MR, sp.a, PS	2003–2005 to 2007–2010
21.12.2007	21.12.2007	1	CD&V, cdH, Open Vld, MR, sp.a, PS	2007–2010
23.12.2007	01.12.2011	155	CD&V, cdH, Open Vld, MR, PS	2007–2010 to 2010–2012
07.12.2011	07.10.2014	126	CD&V, cdH, Open Vld, MR, sp.a, PS	2010–2012 to 2014–2016
14.10.2014	06.12.2018	202	CD&V, Open Vld, MR, N-VA	2014–2016 to 2016–2019
12.12.2018	24.09.2020	67	CD&V, Open Vld, MR	2016–2019 to 2019–2021
01.10.2020	30.01.2025	197	CD&V, Open Vld, MR, sp.a/Vooruit, PS, Groen, Ecolo	2019–2021 to 2024–2025
04.02.2025	18.12.2025	44	CD&V, Les Engagés, MR, Vooruit, N-VA	2024–2025

Notes: Spells of unchanged composition of the parties in government, by sitting day: a party is in government on a day if one of its members holds ministerial office. Before their family's split (Catholics 1968, Liberals 1972, Socialists 1978), parties are named by family; afterwards, by the name of each community's party, with both names when a party was renamed during the spell. Sittings: days with plenary sittings in the spell; periods: the periods of Section 3 that the spell covers. Produced by `code/python/27_paper_extras.py` from `output/data/government_units.csv`.

Districts. We map electoral districts to provinces and the three regions in `code/python/districts.py`, treating Brussels, Flemish Brabant and Walloon Brabant separately. The arrondissement of Brussels before 1995 and the constituency of Brussels-Halle-Vilvoorde are coded as Brussels.

Seats of the governing parties. Panel (c) of Figure 8 counts the seats of all members, whether or not they speak. From 1919, the seats of a legislature are the members of its table in the register, as many as the seats of the Chamber. Before 1919, half of the Chamber was renewed every two years, while the register keeps one table for each legislature. We therefore reconstruct the members sitting on each day from their speeches and roll-call votes. A member counts from his first speech or vote in the legislature, or from its start when that day comes no later than the tenth day with a speech or vote in the legislature or 90 days after the first such day. A member of the table counts until the end of the legislature, and a member who left it until his last speech or vote. Members of the table who never speak or vote count throughout; members without a party in the register, a few who left, are not counted. The members found give each region's shares of governing and other members, which we multiply by the region's seats in the table; before the partial elections that enlarged the Chamber in the middle of a legislature, in 1847, 1859, 1866 and 1902, by its seats in the previous legislature's table. A party is in government on a day as in Section 3 (Online Appendix Table B3); a national party counts with the government when one of its community parties is in it, and the other way round. As in the rest of Section 5.7, the members elected in Brussels, including Brussels-Halle-Vilvoorde, are left out. Counted by language community instead, with Brussels, the governments of 1999–2007 and of 2020–2024 had about as many or more French-speaking than Dutch-speaking seats, while by district they had slightly more seats from Flanders than from Wallonia. The composition of a government is its seats elected in Flanders minus those

elected in Wallonia, as a share of both; panel (c) shows its mean over the sitting days of each period. Produced by `code/python/29_regions.py`, which also writes the seats of each spell of unchanged composition to `output/series/regions_government_seats.csv`.

Groups of phrases. School, Royal Question, language-policy and labour phrases are vocabulary phrases whose stems match the patterns in `code/python/08_dictionary.py`. Matches are listed in `output/phrases/`.

Language of the speeches. Each block starts from the dataset's language label. Before counting phrases, we cut each block at a line that introduces a translation (*Traduction :*, *Vertaling :*) and remove the roll-call formulas (*Début du vote*, *Résultat du vote*, *Uitslag van de stemming*, *Tout le monde a-t-il voté*). We relabel a block when it contains at least two function words of the other language and more than twice as many as of its labeled language, or at least one and none of its labeled language. Within a block, sentences that are clearly in the other language by the same rule are counted as speech in that language when they contain at least five words in all (`code/python/textproc.py`). Function words are, for example, *het*, *een*, *ik* and *mijnheer* for Dutch, and *le*, *que*, *dans* and *monsieur* for French. The lists leave out words that the two languages share, such as *de*, *en*, *je* and *pas*.

We check the remaining text in the other language by counting phrases containing its function words (`code/python/19_language_check.py`). The lists of this check also leave out words that occur in proper names, such as *van*, and the check misses other-language phrases without a function word. After the corrections, flagged phrases account for at most 0.25 per cent of French phrases in every period and, from 1919, for at most 0.21 per cent of Dutch phrases. Before 1919, Dutch counts are small: flagged phrases are one of the 38 Dutch phrases of 1892–1894 and at most 0.12 per cent from 1894. The French vocabulary contains 16 flagged phrases and the Dutch vocabulary 8.

Three sources accounted for most text in the other language before the corrections. First, the record includes translated formulas. From the interwar years to the 1950s, these concern submission of reports and bills (*De heer X legt ter tafel het verslag over het wetsontwerp*; *sur le bureau de la Chambre le rapport*). From the 1990s, they also close interpellations (*het incident is gesloten*, *l'incident est clos*). Second, some speeches use both languages. Third, some speeches have the wrong language label.

The effect was largest when such text concentrated in one group. In an earlier estimation without the corrections, phrases with a Dutch function word accounted for 1.6–1.7 per cent of Catholics' French phrases in 1950–1954, compared with 0.2–0.4 per cent for Liberals and Socialists, and the shares reversed in 1954–1958. After 2010, many Flemish nationalist and Vlaams Belang speeches labeled French were Dutch, and French passages accounted for 41–50 per cent of Communist members' Dutch phrases in 1969–1981. After the corrections, flagged phrases account for at most 0.03 per cent of the French phrases of Catholics, Liberals, Socialists and Communists in 1950–1958, and for at most 0.1 per cent of Communist members' Dutch phrases in 1969–1981.

Leave-out estimates without text in the other language. The same script recomputes the three leave-out series without the phrases that contain a function word from the other language (`output/series/lo_without_other_language.csv`). In an earlier estimation without the corrections, these phrases accounted for 20 to 26 per cent of religious, left–others and government–opposition partisanship in 1949–1958 and raised several government–opposition estimates after 1995. With the corrections, they account for at most 3 per cent of net leave-out partisanship in 1949–1958 and change no estimate after 1995 by more than 0.0003. The largest changes are in 1965–1968 and 1974–1977, up to 0.0026.

The language rule. After the Catholic, Liberal and Socialist families split, their members enter only their community's language estimation. From 1968, the rule also covers the parties associated with one community: Flemish nationalists, Vlaams Blok and Vlaams Belang, Francophone regionalists and the two Green parties (`code/python/designs.py`). Members of national parties, PVDA-PTB among them, enter both languages, as do all members before 1968. Flemish National League members, for example, did speak French in the 1930s.

Without the rule, the problem is larger in pairwise comparisons: a party represented in one language by a few mislabeled interventions can produce a component where the groups share almost no phrases. The pairs of the multiparty index (Section 5.5) are estimated with the rule: from 1968, pairs involving Flemish nationalists or the radical right have only a Dutch component, and pairs involving Francophone regionalists only a French one.

Procedural text. In 1993–1995 and 1997–2007, many speech blocks end with roll-call formulas: *Début du vote*, *Fin du vote*, *Résultat du vote*, and the Dutch equivalents *Begin van de stemming*, *Einde van de stemming*, *Uitslag van de stemming*. They concentrate among members speaking just before votes; in an earlier estimation, they reached 1,100–1,400 per 100,000 French phrases of Christian-democratic members in 1999–2005. We remove them before counting phrases. Other text of the voting procedure is not removed: the vote tallies printed in both languages (*oui nee*, *abstentions totaal*) and the formula that applies the result of a previous vote (*uitslag vorige*, *stemming gelden*) are among the most distinctive phrases of some lists, mostly Dutch lists of 1988–2010 (Online Appendix C).

Tables and companion modules. `code/python/20_main_tables.py` produces Tables 1 and 2 from the estimated series. `code/python/27_paper_extras.py` produces Tables B2, B3 and 3, Figures 3 and 7, and the macros in `tables/extras_macros.tex`. It reads `output/data`, the left–right and right–others series in `output/series` (including their leave-out series) and the party comparisons in `output/dyads`. The emotions module's run of 14 October 2026 supplies the data for Figures 13, 14 and 15, and Section 7 (`code/emotions`; Online Appendix E). `code/python/21_emotions_link.py` reads these outputs and draws the figures. It writes decade correlations to `output/series/emotions_corr.csv` and text macros to `tables/emotions_macros.tex`.

`code/python/22_economy.py` supplies Figure 10 and the growth results in Section 5.9. It reads the Maddison Project Database (`external/mpd2023_web.xlsx`), writes correlations and slopes to `output/series/economy_corr.csv`, and writes text macros to `tables/economy_macros.tex`.

The topic module (`code/topics`) labels every speech; its labels are in `topics/step5/topics.csv`. `code/python/23_topic_areas.py` estimates the series of Section 6 by topic and policy area, for periods and legislatures, with the random-label series, and writes them to `output/series/topic_partisanship.csv`. `code/python/26_topic_figures.py` draws Figure 12, writes Table 6, and writes text macros to `tables/topic_macros.tex`.

Token files. Token files for all [nPeriods] periods supply the very-long-speech counts from 1850 in Figure 21, panel b (`code/python/14_stata_inputs.py`), and the out-of-lexicon rates in Figure A21, panel b (`code/python/10_ocr_noise.py`).

Online Appendix C: Most partisan phrases

The tables list each group's five most partisan phrases by $\hat{\zeta}_{jt}$ and their predicted counts per 100,000 phrases for a speaker in each group, following Gentzkow et al. (2019). Religious, left–others and government–opposition comparisons cover, for French and for Dutch, every period in which that language's estimate is defined, with one table per decade. The Flemish–Walloon comparison uses French speech; Table C94 selects one period every ten years, and Tables C95–C108 list every period. Phrases are stemmed bigrams displayed in their most frequent unstemmed form. The vocabulary leaves out phrases without content (Online Appendix B), so the lists contain few formulas for submitting bills and reports, titles, or names of parties and persons. Some remain, such as *honneur proposer*, *général chazal* and *liberale groep*, as do some pairs of common words, such as *non seulement*. A list contains only phrases that favour its group, so some lists, mostly Dutch lists before 1914, contain fewer than five phrases, and a few are empty. Where a group lacks strongly partisan phrases, its list contains phrases with nearly equal predicted counts, as for both groups in 1831–1833 and in several Dutch lists before 1914. Some phrases are text in the other language or text of the voting procedure (Online Appendix B). Counts in this appendix cover all printed lists, including periods with low coverage of the sittings.

Tables C1–C20 (French) and C21–C33 (Dutch) show the religious comparison. Most phrases name subjects of legislation and debate. Education phrases appear from 1841–1843 to 1985–1987. They appear on both sides during the first school war (1878–1886). In 1950–1958, they are mostly Catholic phrases, both when the Catholics governed alone (1950–1954) and when they were in opposition during the second school war (1954–1958). *classe ouvrière* identifies other parties in 19 of 21 periods of 1894–1940. Catholic lists include family allowances (*allocations familiales*) in five periods from 1929 to 1958, the Congo in 1908–1912 and 1958–1963 (*comité spécial*, *union minière*, *force publique*), development aid in every period of 1961–1971 (*assistance technique*, *coopération développement*), the German-speaking community in most periods of 1971–1991 (*langue allemande*, *communauté germanophone*) and end-of-life care in 2001–2003 (*soins palliatifs*, *fin vie*). Other parties' lists include the Royal Question in 1949–1950 (*consultation populaire*, *question royale*) and the School Pact in 1958–1961 (*pacte scolaire*).

In Dutch, Catholic lists include large families (*kroostrijke gezinnen*) in every period from 1923–1925 to 1947–1949 except 1939–1940, technical education (*technisch onderwijs*) in every period from 1932–1934 to 1956–1958 except 1949–1950, and unborn life in 1989–1991 (*ongeboren leven*). Other parties' lists include veterans (*oud strijders*) in most periods between the wars, the Franco-Belgian military agreement (*fransch belgisch*, *militair akkoord*) in every period of 1925–1936, the Flemish people (*vlaamsche volk*) in every period of 1929–1940 and national education (*nationale opvoeding*) in six of the seven periods of 1961–1977. Record artefacts include headings of the printed reports in French lists of 1977–1993 (*annales parlementaires*, *belgique kamer*) and text of the voting procedure in Dutch lists. Three of the five Catholic phrases of 2001–2003 come from vote tallies printed in both languages (*oui nee*, *abstentions totaal*), and four of those of 2003–2005 from the formula that applies the result of a previous vote (*uitslag vorige*, *stemming gelden*).

The left–others lists appear in Tables C34–C47 and C48–C60. *classe ouvrière* identifies the French-speaking left in every period from 1894 to 1940. Before 1914 it appears alongside universal suffrage, the eight-hour day and labour courts (*prud hommes*); in 1932–1934, alongside

pleins pouvoirs. Social security and trade unions follow. In the federal era, the left's lists also include the index jump (*saut index*), pension age (*67 ans*) and tax havens (*paradis fiscaux*). Special powers (*pleins pouvoirs*, *pouvoirs spéciaux*) appear in the left's lists in 1932–1934, 1965–1968 and 1981–1987, mostly under governments without the Socialists, and in the remaining parties' lists in 1944–1947, 1956–1958, 1977–1978 and 1995–1997, when the Socialists governed. The other parties' lists include the middle classes (*classes moyennes*) in ten periods from 1934 to 1985 and the Congo in 1950–1954 and 1958–1961.

In Dutch, the left uses *werkende klas* in every period from 1919 to 1934 and in 14 of the 19 periods from 1902 to 1950; it returns in 2021–2025. In 2019–2025, the left's lists include *sterke overheid* in every period, *gewone mensen* in 2021–2024 and *werkende mensen* in 2024–2025, and the remaining parties' lists include guaranteed service (*gegarandeerde dienstverlening*) in every period from 2014–2016. Between the wars, the remaining parties' lists include veterans (*oud strijders*) in seven of ten periods. Record artefacts include Dutch text in French lists (*ik citeer* in 1969–1971 and *handelingen plenaire*, which heads the remaining parties' list of 1989–1991), headings of the reports in 1991–1995 (*représentants belgique*, *kamer volksvertegenwoordigers*), the French formula that closes interpellations in a Dutch list of 2014–2016 (*clos incident*) and the French pairing formula of votes in Dutch lists of 2005–2010 (*pairé avec*).

The government–opposition lists appear in Tables C61–C80 and C81–C93. Under governments formed by a single family, before 1914 and in 1950–1954, the French lists share most of their phrases with the religious lists, with the sides exchanged under Liberal governments. Special powers (*pleins pouvoirs*, *pouvoirs spéciaux*) are opposition phrases in 10 of the 11 French periods in which they appear. Several current issues identify the opposition: *suffrage universel* in 9 of the 10 periods in which it appears before 1914, *classe ouvrière* in every listed period from 1894–1896 to 1936–1939, the Royal Question in 1949–1950, *loi unique* in 1958–1963, and taxes in every period from 2007 to 2021 (*intérêts notionnels*, *fraude fiscale*, *tax shift*, *paradis fiscaux*). Governing members speak of the government agreement (*accord gouvernement*) in 2010–2016 and 2024–2025. Double forms such as *travailleurs travailleuses* and *celles ceux* are government phrases in 2021–2024 and opposition phrases in 2024–2025.

In Dutch, phrases of the Flemish movement (*vlaamsche volk*, *vlaamsche soldaten*) appear in the opposition's list in every listed period from 1921 to 1940, as does the Franco-Belgian military agreement in 1927–1936. From the 1960s, phrases on the Flemish community and its institutions (*vlaamse gemeenschap*, *vlaams parlement*, *halle vilvoorde*) are mostly opposition phrases, while governing members more often speak of Europe in 1981–2010 (*europese gemeenschap*, *europese unie*) and of the social partners (*sociale partners*) in most periods from 2001. Record artefacts include the vote tallies printed in both languages in the French opposition list of 2021–2024 (*oui nee*, *abstentions totaal*).

Table C94 suggests that regional economic interests often distinguish Flemish and Walloon members within the same families. Flemish lists include sugar and Ostend's port in 1839–1841; linen in 1848–1850; Ghent and West Flanders in 1888–1890; devastated regions in 1918–1919; the Congo in 1958–1961; and national product and public debt in 1965–1968. Walloon lists include the Société Générale in 1839–1841; the Charleroi canal and Meuse in 1859–1861; working hours and labour courts in 1908–1910; the national coal council in 1949–1950; and the Walloon

motorway in 1965–1968.

Language policy is absent from the periods of Table C94 apart from *pays flamand* in 1888–1890 and cultural councils in 1939–1940. In the full lists, it also appears on the Flemish side through *langue flamande* in 1856–1857 and 1861–1864, *pays flamand* in 1892–1894 and 1906–1908, Ghent's Flemish university in 1921–1923 and 1925–1927, and *problème linguistique* in 1961–1963, and on the Walloon side through *problème wallon* in 1944–1946.

Table C1: Most partisan phrases, Catholic and Other parties, French speech: periods beginning in the 1830s

Catholic			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1831–1833					
flandre orientale	105	80	pommes terre	71	88
province luxembourg	4	4	cour cassation	405	405
los renten	5	5	cour appel	263	263
grand duché	6	6	ville gand	51	51
maîtres forges	11	11	savoir si	141	141
1833–1835					
conseil communal	266	139	chemin fer	139	276
commissaire police	124	34	route fer	62	103
loi communale	188	127	cour comptes	81	118
pouvoir législatif	107	52	opinion projet	6	28
collège bourgmestre	69	27	péages canaux	10	21
1835–1837					
los renten	461	65	caisse épargne	15	54
ministres cultes	100	25	non activité	23	59
société générale	88	42	mines houille	7	26
chemin fer	138	97	plus voisin	6	25
substitut procureur	56	15	corps judiciaires	18	35
1837–1839					
sous officiers	73	16	chemin fer	380	459
état major	66	14	vote secret	20	82
sous armes	51	13	fils laine	6	21
réduction proposée	40	11	usages locaux	13	24
non activité	33	7	travaux publics	226	226
1839–1841					
société générale	256	135	sucre exotique	105	185
adjudication publique	45	41	tribunal commerce	51	104
chemin fer	513	513	chemins vicinaux	49	73
travaux publics	266	266	caisses prévoyance	9	20
plus grande	145	144			

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Catholic: positive; Other parties: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 1; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C2: Most partisan phrases, Catholic and Other parties, French speech: periods beginning in the 1840s

Catholic			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1841–1843					
chemin fer	609	514	écoles normales	190	308
sucre indigène	139	50	cours normaux	25	80
sucre exotique	112	44	écoles primaires	78	121
ministres cultes	53	24	loi communale	74	104
canalisation campine	45	17	conseil communal	134	162
1843–1845					
chemin fer	1089	694	droits différentiels	141	193
bulletin officiel	54	7	pavillon national	35	74
cahier charges	82	46	cour comptes	43	81
juge paix	103	69	pavillon étranger	26	60
st trond	56	26	droit sortie	19	50
1845–1848					
sucre indigène	111	35	chemin fer	419	545
sucre exotique	86	29	travaux publics	308	425
cour comptes	152	110	garde civique	37	104
dette flottante	53	24	conseil communal	56	97
tribunal commerce	33	8	ponts chaussées	22	59
1848–1850					
chemin fer	611	307	garde civique	49	84
enseignement religieux	131	49	société générale	23	57
enseignement moyen	159	86	banque nationale	18	45
conseil mines	89	28	lettre voiture	8	35
canal charleroy	93	42	code pénal	18	43

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Catholic: positive; Other parties: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 1; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C3: Most partisan phrases, Catholic and Other parties, French speech: periods beginning in the 1850s

Catholic			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1850–1852					
chemin fer	828	606	flandre occidentale	23	70
trésor public	118	55	code civil	41	80
droits différentiels	99	43	peine mort	8	31
travaux publics	370	324	code pénal	19	41
dépôts mendicité	62	27	cour comptes	59	81
1852–1854					
commissaire arrondissement	78	24	flandre occidentale	41	97
garde civique	301	269	demande concession	30	48
prohibition sortie	50	19	banque nationale	13	27
flandre orientale	69	43	écoles moyennes	7	20
chemins vicinaux	63	39	pommes terre	32	44
1854–1856					
libre sortie	71	29	chemin fer	844	979
prohibition sortie	79	38	travaux publics	336	450
fer fort	45	7	département travaux	45	87
honneur proposer	77	41	dendre waes	40	82
luttre denderleeuw	94	58	code pénal	39	73
1856–1857					
chemin fer	416	237	examen écrit	39	138
civil élémentaire	63	19	personnification civile	40	84
travaux publics	213	170	administrateurs spéciaux	149	176
prohibition sortie	74	31	opinion libérale	15	32
universités etat	119	79	examen oral	27	42
1857–1859					
ministres cultes	234	134	prud hommes	135	302
critique censure	115	29	conseils prud	80	224
chemin fer	394	331	contrainte corps	25	106
administrateurs spéciaux	63	17	ville anvers	59	103
propose renvoi	91	49	enseignement obligatoire	12	45
1859–1861					
vol domestique	130	24	garde civique	10	88
passe port	99	20	bruxelles louvain	43	85
fonds communal	147	83	sucre betterave	18	48
cours légal	85	44	cour appel	42	72
communes octroi	119	78	cours certificat	10	39

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Catholic: positive; Other parties: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 1; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C4: Most partisan phrases, Catholic and Other parties, French speech: periods beginning in the 1860s

Catholic			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1861–1864					
ministres cultes	83	35	garde civique	22	105
loi 1842	73	34	servitudes militaires	27	102
personnes civiles	56	23	citadelle nord	21	67
réforme communale	37	7	travaux publics	245	273
instruction primaire	83	56	ville anvers	59	86
1864–1866					
station intérieure	55	9	chemin fer	441	522
responsabilité ministérielle	39	13	écoles normales	5	28
loi communale	65	39	jonction directe	4	26
armée permanente	56	30	banque nationale	17	38
général chazal	47	22	centrale examiné	10	28
1866–1868					
conseil communal	161	94	écoles adultes	77	150
communal enghien	56	9	loi 1842	98	168
suffrage universel	145	113	enseignement primaire	49	93
peine mort	148	125	cour assises	38	77
corps électoral	86	63	code pénal	89	125
1868–1870					
listes électorales	100	41	sociétés coopératives	4	76
conseil communal	109	75	assemblée générale	42	97
service militaire	90	56	sociétés anonymes	85	137
visites domiciliaires	50	19	écoles normales	11	57
police judiciaire	35	9	travail enfants	3	32

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Catholic: positive; Other parties: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 1; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C5: Most partisan phrases, Catholic and Other parties, French speech: periods beginning in the 1870s

Catholic			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1870–1872					
chemin fer	807	599	écoles moyennes	101	189
bassins houillers	245	159	banque nationale	71	148
compagnie bassins	101	28	enseignement primaire	85	131
arrondissement charleroi	37	7	suffrage universel	32	77
conseil communal	121	96	écoles communales	22	64
1872–1874					
juge instruction	80	28	écoles moyennes	52	187
partie civile	70	28	servitudes militaires	48	151
détention préventive	62	29	ville anvers	88	129
compagnie bassins	41	8	flandre occidentale	39	78
code instruction	55	24	sociétés coopératives	17	54
1874–1876					
juge paix	128	51	garde civique	77	280
peine mort	49	5	chambre commerce	177	305
code pénal	50	14	chemin fer	476	587
caisses prévoyance	40	9	travaux publics	306	359
loi ventôse	54	25	commerce ostende	2	31
1876–1878					
cours eau	188	137	écoles moyennes	35	124
bassins houillers	66	22	port ostende	9	58
code civil	83	48	écoles normales	22	67
lit cours	29	5	chevaux mixtes	27	70
question propriété	29	7	ville gand	14	46
1878–1880					
instruction publique	279	119	chemin fer	349	455
corps électoral	117	42	écoles communales	81	133
valeur locative	76	18	marques fabrique	19	50
enseignement libre	80	28	département travaux	35	64
contribution personnelle	60	18	voies navigables	26	52

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Catholic: positive; Other parties: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 1; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C6: Most partisan phrases, Catholic and Other parties, French speech: periods beginning in the 1880s

Catholic			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1880–1882					
instruction publique	210	92	chemin fer	304	397
cardinal nina	83	9	écoles communales	54	118
conseil communal	196	124	droit pêche	21	74
ministres cultes	117	52	cours eau	68	94
corps électoral	99	52	travaux publics	251	277
1882–1884					
instruction publique	285	102	suffrage universel	38	74
général major	61	13	droit électoral	35	68
conseil communal	116	73	carte géologique	14	37
major gratry	52	12	eaux vie	17	37
députation permanente	157	126	écoles moyennes	57	77
1884–1886					
chemin fer	312	212	écoles communales	168	389
droit auteur	129	37	pères famille	96	191
cuve matière	58	3	écoles adoptées	107	181
écoles moyennes	60	32	instruction publique	93	160
oeuvres littéraires	31	5	conseil communal	150	206
1886–1888					
concordat préventif	68	9	écoles communales	79	194
procureur général	71	28	conseil communal	115	166
tabac indigène	36	4	écoles adoptées	47	89
juge instruction	53	23	pères famille	42	83
partie civile	45	21	général major	12	49
1888–1890					
chemin fer	400	299	sûreté publique	21	105
prud hommes	245	161	agents provocateurs	4	51
conseils prud	172	106	ville gand	17	63
pays flamand	90	46	écoles spéciales	63	103
programme candidature	23	3	conseil communal	44	82

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Catholic: positive; Other parties: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 1; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C7: Most partisan phrases, Catholic and Other parties, French speech: periods beginning in the 1890s

Catholic			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1890–1892					
fonds communes	117	44	lieutenant général	29	105
dépôts mendicité	42	13	traitement attente	71	145
service hospitalier	32	5	conseil communal	58	124
conseil supérieur	47	20	suffrage universel	72	128
loi 1876	52	26	service personnel	48	98
1892–1894					
représentation proportionnelle	199	117	suffrage universel	176	310
juge paix	63	23	chemin fer	267	383
minimum salaire	37	7	droit électoral	67	129
traverses métalliques	35	6	députation permanente	56	107
deux langues	32	5	loi électorale	77	117
1894–1896					
droit licence	90	15	conseil communal	65	178
chemin fer	405	339	suffrage universel	43	126
droit accise	77	35	classe ouvrière	67	145
id id	42	2	députation permanente	52	99
tabac indigène	49	12	loi scolaire	11	52
1896–1898					
garde civique	251	173	classe ouvrière	44	149
unions professionnelles	278	221	suffrage universel	28	92
pourrait être	106	81	représentation proportionnelle	43	77
faire commerce	41	17	vote plural	9	32
sociétés secours	70	48	service personnel	54	74
1898–1900					
juge paix	106	43	banque nationale	137	358
chemin fer	308	270	classe ouvrière	37	132
fer vicinaux	58	25	tramways bruxellois	32	118
cour militaire	41	9	suffrage universel	52	130
conseil guerre	89	61	conseil communal	48	96

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Catholic: positive; Other parties: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 1; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C8: Most partisan phrases, Catholic and Other parties, French speech: periods beginning in the 1900s

Catholic			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1900–1902					
chemin fer	405	282	suffrage universel	225	372
jeux hasard	118	29	classe ouvrière	40	120
cercles privés	83	27	représentation proportionnelle	65	129
maisons jeux	79	24	conseil communal	90	144
gare centrale	65	23	vote plural	50	95
1902–1904					
chemin fer	519	329	classe ouvrière	49	132
chef entreprise	64	19	lieutenant général	31	96
faute lourde	75	35	conseil communal	66	120
assurance obligatoire	100	68	heures travail	16	38
droit proportionnel	35	5	conseils prud	16	37
1904–1906					
repos dominical	94	55	lieutenant général	91	173
ligne défense	42	11	classe ouvrière	42	110
société belge	44	15	navire école	26	69
droit patente	42	15	chemin fer	299	340
code civil	87	61	heures travail	37	65
1906–1908					
langue flamande	107	52	classe ouvrière	28	110
pays flamand	72	45	lieutenant général	19	95
flamande pays	45	22	conseil communal	59	98
domaine national	37	15	heures travail	50	85
enseignement flamand	42	21	fer postes	39	68
1908–1910					
comité spécial	113	29	chemin fer	333	420
loi 1902	84	23	classe ouvrière	35	105
temps service	93	50	fer postes	34	93
effectif paix	70	36	écoles normales	40	94
union minière	48	18	postes télégraphes	37	90

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Catholic: positive; Other parties: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 1; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C9: Most partisan phrases, Catholic and Other parties, French speech: periods beginning in the 1910s

Catholic			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1910–1912					
caisses prévoyance	148	40	classe ouvrière	48	156
chemin fer	464	381	lieutenant général	19	109
comité spécial	47	8	conseil communal	60	127
caisse retraite	68	32	députation permanente	65	103
gouverneur général	52	21	chef cabinet	23	60
1912–1914					
assemblée générale	45	17	classe ouvrière	46	159
sous officiers	37	14	écoles communales	87	200
habitations bon	40	18	écoles adoptées	99	157
société nationale	75	56	suffrage universel	75	125
examiné projet	27	8	loi scolaire	38	84
1918–1919					
dommages guerre	204	142	classe ouvrière	61	115
oeuvre nationale	43	8	comité national	50	93
tribunaux dommages	95	62	avant guerre	143	174
écoles normales	76	45	chemin fer	149	177
orphelins guerre	39	12	pendant guerre	159	179
1919–1921					
administration centrale	46	16	classe ouvrière	93	190
victimes civiles	35	10	temps service	19	68
deux langues	46	23	avant guerre	155	202
partie flamande	29	9	huit heures	37	83
traité versailles	43	24	pendant guerre	118	148

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Catholic: positive; Other parties: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 1; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C10: Most partisan phrases, Catholic and Other parties, French speech: periods beginning in the 1920s

Catholic			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1921–1923					
gardes champêtres	157	54	chemin fer	260	339
pays flamand	70	31	écoles normales	34	84
extrême gauche	83	49	écoles moyennes	19	46
police rurale	45	13	classe ouvrière	83	108
locataire principal	32	11	avant guerre	136	158
1923–1925					
chemin fer	266	228	dommages guerre	43	98
sous officiers	44	7	service sauvetage	7	52
examiné projet	46	20	classe ouvrière	35	70
prix maxima	48	23	droits spécifiques	13	49
tarif minimum	37	13	ad valorem	6	38
1925–1927					
neuf ans	85	35	chemin fer	275	369
code civil	63	16	classe ouvrière	47	135
familles nombreuses	61	31	avant guerre	84	149
partie immeuble	35	5	huit heures	35	91
prud hommes	66	36	coût vie	18	48
1927–1929					
prud hommes	78	36	défense nationale	106	189
conseil supérieur	73	31	classe ouvrière	49	106
conseils prud	74	32	avant guerre	85	128
ville anvers	48	9	personnel enseignant	24	66
prison école	64	29	service militaire	18	46
1929–1932					
pensions vieillesse	95	56	société nationale	114	166
allocations familiales	73	41	classe ouvrière	75	127
main oeuvre	82	52	sciences arts	61	102
produits agricoles	41	14	écoles moyennes	19	56
responsabilité limitée	31	5	écoles communales	19	47

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Catholic: positive; Other parties: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 1; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C11: Most partisan phrases, Catholic and Other parties, French speech: periods beginning in the 1930s

Catholic			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1932–1934					
allocations familiales	141	65	pleins pouvoirs	40	154
pensions vieillesse	141	90	classe ouvrière	130	220
caisse nationale	73	29	défense nationale	68	144
législation sociale	58	20	télégraphes téléphones	32	78
majoration gratuite	42	6	commerce intérieur	25	67
1934–1936					
gouverneur général	53	2	arrêtés lois	56	130
prix revient	86	37	classe ouvrière	54	125
pouvoirs spéciaux	117	79	défense nationale	99	165
prix vente	45	9	société nationale	149	205
nombre députés	53	20	chemin fer	182	230
1936–1939					
taxe crise	81	27	non intervention	20	42
taxe professionnelle	49	20	nationalistes flamands	14	35
loi cadre	36	11	gouvernement espagnol	4	24
section législation	34	10	pommes terre	26	44
conseil mines	25	3	éducation physique	14	29
1939–1940					
chemin fer	157	80	personnel enseignant	26	69
pommes terre	85	18	nationalistes flamands	12	46
assurance chômage	53	18	politique indépendance	11	37
allocations familiales	91	58	sous officiers	25	48
caisses chômage	42	9	politique étrangère	12	25

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Catholic: positive; Other parties: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 1; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C12: Most partisan phrases, Catholic and Other parties, French speech: periods beginning in the 1940s

Catholic			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1944–1946					
dommages guerre	139	72	prisonniers politiques	103	196
impôt exceptionnel	72	7	secours civil	11	60
impôt capital	90	37	personnel enseignant	10	34
emprunt forcé	61	10	nations unies	36	55
application impôt	56	9	sécurité sociale	50	68
1946–1947					
allocations familiales	159	50	assistance publique	22	88
sous officiers	70	10	prisonniers politiques	80	131
pouvoirs spéciaux	54	7	personnel enseignant	15	56
banque nationale	74	42	collaboration économique	23	42
balance comptes	38	7	réparation intégrale	29	46
1947–1949					
juge paix	117	34	prisonniers politiques	37	84
banque nationale	141	58	organisations syndicales	11	54
neuf ans	84	8	classe ouvrière	27	64
arrêté loi	100	28	coût vie	29	63
conseil administration	85	37	mobilisation civile	5	39
1949–1950					
assurés libres	73	17	consultation populaire	201	406
travailleurs indépendants	66	15	question royale	105	148
conseil entreprise	65	20	stations épuration	9	31
pouvoirs représentants	53	9	sociétés habitations	14	31
central provincial	45	8	eaux usées	27	43

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Catholic: positive; Other parties: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 1; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C13: Most partisan phrases, Catholic and Other parties, French speech: periods beginning in the 1950s

Catholic			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1950–1952					
reviseurs entreprises	46	14	personnel enseignant	24	96
experts comptables	42	10	quatre mois	29	93
fonds communes	56	25	classe ouvrière	26	86
enseignement public	62	32	coût vie	43	94
plan décennal	47	19	etats unis	64	114
1952–1954					
enseignement public	99	44	plan schuman	21	138
enseignement technique	104	55	assurés libres	20	61
experts comptables	50	5	victimes civiles	34	71
enseignement moyen	123	81	office séquestres	1	35
droit civil	59	18	etats unis	65	96
1954–1956					
instruction publique	184	92	etats unis	42	117
enseignement moyen	185	98	travailleurs indépendants	38	90
enseignement public	103	36	haute autorité	19	66
enseignement technique	125	69	personnel enseignant	25	64
fonds routes	69	13	société nationale	119	157
1956–1958					
enseignement libre	90	30	conseil national	58	148
pouvoirs spéciaux	69	8	oeuvre nationale	20	53
allocations familiales	126	74	national charbonnages	29	62
moyen libre	31	4	conseil administration	25	57
quatre ans	62	37	secteur privé	42	71
1958–1961					
politique scientifique	53	3	classe ouvrière	26	74
budget ordinaire	59	26	loi unique	62	109
gouverneur général	61	31	haute autorité	27	72
force publique	62	31	personnel enseignant	9	54
économique régional	39	10	pacte scolaire	33	78

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Catholic: positive; Other parties: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 1; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C14: Most partisan phrases, Catholic and Other parties, French speech: periods beginning in the 1960s

Catholic			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1961–1963					
assistance technique	118	21	education nationale	57	102
chemin fer	178	119	classe ouvrière	12	53
général gaulle	49	7	maladie invalidité	24	53
pays voie	55	18	assistance publique	65	93
affaires africaines	44	9	classes moyennes	49	75
1963–1965					
assistance technique	177	28	corps médical	87	149
pays voie	77	8	maladie invalidité	119	177
voie développement	72	8	victimes guerre	17	65
outre mer	51	8	etats unis	59	106
sous traitants	51	12	hausse prix	15	45
1965–1968					
assistance technique	91	12	pouvoirs spéciaux	68	134
cour comptes	96	25	maladie invalidité	55	100
chemin fer	126	65	assurance maladie	73	102
coopération développement	47	7	corps médical	19	43
pays voie	50	12	gaz naturel	8	30
1968–1969					
coopération développement	85	15	personnes âgées	22	75
registre national	67	3	organisations syndicales	36	75
pays voie	71	11	conseil économique	117	148
education nationale	131	72	assistance publique	28	54
nations unies	71	13	travailleurs indépendants	13	31
1969–1971					
pays voie	282	26	police judiciaire	13	94
voie développement	262	27	gaz naturel	11	54
coopération développement	136	15	santé publique	72	112
parlement européen	127	40	travailleurs indépendants	16	50
gouvernement belge	121	40	personnes âgées	12	45

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Catholic: positive; Other parties: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 1; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C15: Most partisan phrases, Catholic and Other parties, French speech: periods beginning in the 1970s

Catholic			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1971–1974					
classes moyennes	179	72	expansion économique	52	97
santé publique	165	89	organisations syndicales	48	77
société nationale	90	35	conseil économique	41	65
langue allemande	61	20	fraude fiscale	8	28
exposé général	65	24	télégraphes téléphones	12	27
1974–1977					
arrondissement mons	79	5	aide sociale	23	102
économique sociale	136	76	travailleurs indépendants	15	79
langue allemande	82	23	centres publics	11	44
catégories sociales	57	9	publics aide	13	43
sous régions	54	12	autant plus	77	101
1977–1978					
finances publiques	198	75	travailleurs indépendants	25	67
langue allemande	135	22	plan secteur	13	49
maîtrise finances	83	3	fonds communes	20	43
politique industrielle	129	58	sociétés développement	11	17
education nationale	138	71	parlementaires parlementaire	190	192
1979–1981					
gouvernement belge	128	46	organisations syndicales	26	56
communauté germanophone	97	23	travailleurs indépendants	15	44
coopération développement	84	29	région bruxelles	32	56
parlement européen	68	17	communauté flamande	26	46
education nationale	113	72	communautés régions	41	61

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Catholic: positive; Other parties: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 1; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C16: Most partisan phrases, Catholic and Other parties, French speech: periods beginning in the 1980s

Catholic			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1981–1983					
troisième circuit	160	15	pouvoirs spéciaux	409	600
temps travail	135	29	cockerill sambre	8	76
circuit travail	109	12	fraude fiscale	12	77
conseil communal	116	36	pleins pouvoirs	13	50
crédits provisoires	114	40	utilisation rationnelle	10	46
1983–1985					
temps travail	180	37	pouvoirs spéciaux	80	128
politique scientifique	126	19	union soviétique	11	44
police communale	93	17	enseignement etat	23	54
langue allemande	62	4	education nationale	128	158
fonction publique	136	80	arriéré judiciaire	5	31
1985–1987					
région bruxelloise	375	182	education nationale	64	155
secteur textile	152	8	pouvoirs spéciaux	259	290
cockerill sambre	94	16	enseignement rénové	17	47
secteurs nationaux	105	38	fraude fiscale	16	46
service militaire	108	49	travailleurs indépendants	13	33
1988–1989					
communauté germanophone	142	15	culturelles nationales	20	57
société canal	102	8	scientifiques culturelles	21	56
accord coopération	109	32	publics institutions	26	58
mouscron comines	82	21	services publics	84	98
autorité nationale	70	13	annales parlementaires	219	221
1989–1991					
juge instruction	162	72	culturelles nationales	16	69
services police	161	80	scientifiques culturelles	17	69
précompte mobilier	103	36	loi financement	26	72
agents change	69	7	publics institutions	24	65
communauté germanophone	64	12	etat major	10	33

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Catholic: positive; Other parties: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 1; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C17: Most partisan phrases, Catholic and Other parties, French speech: periods beginning in the 1990s

Catholic			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1991–1993					
travaux intérêt	63	4	réforme etat	103	169
commissariat royal	57	8	gecontroleerd oui	8	41
acquisition nationalité	44	9	région bruxelles	51	84
nom groupe	48	12	conseil communauté	131	141
établissements pénitentiaires	45	10	belgique kamer	684	687
1993–1995					
êtres humains	194	60	sécurité sociale	252	325
traite êtres	167	47	entreprises publiques	52	100
juge instruction	117	16	dépenses électorales	48	92
remise dette	87	9	campagne information	20	51
procureur général	107	29	union européenne	143	170
1995–1997					
union monétaire	163	49	pouvoirs spéciaux	62	308
union européenne	140	72	sécurité sociale	535	621
dette publique	112	50	matière sécurité	25	55
contributions directes	68	7	services police	44	66
etats membres	109	48	police judiciaire	25	43
1997–1999					
union européenne	196	77	juge instruction	119	170
banque nationale	96	14	procureur général	40	84
banque centrale	98	17	nouveau belgique	3	45
extrême droite	107	38	organisations criminelles	7	45
administration fiscale	90	22	vie privée	42	70
1999–2001					
coopération développement	151	43	services police	32	92
arc ciel	76	31	criminalité organisée	14	55
principe précaution	77	35	union européenne	129	169
nations unies	66	29	réduction temps	18	55
soumis sanction	53	17	police locale	36	71

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Catholic: positive; Other parties: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 1; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C18: Most partisan phrases, Catholic and Other parties, French speech: periods beginning in the 2000s

Catholic			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
2001–2003					
soins palliatifs	146	33	police locale	37	91
réforme polices	136	68	criminalité organisée	13	57
arc ciel	107	44	police fédérale	29	70
fin vie	65	14	union européenne	113	152
protection jeunesse	66	17	accords bilatéraux	9	48
2003–2005					
titres services	126	20	peine travail	5	33
entités fédérées	123	52	tax shelter	2	30
communautés régions	92	30	police fédérale	15	38
nouvelle fois	89	28	arriéré judiciaire	21	44
insécurité juridique	67	10	coopération développement	41	50
2007–2010					
partenaires sociaux	243	117	intérêts notionnels	32	73
demandeurs emploi	111	29	fraude fiscale	69	102
plan relance	113	57	demandeurs asile	20	52
titres services	107	54	aide financière	12	34
coopération développement	109	56	médecins généralistes	53	69

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Catholic: positive; Other parties: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 1; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C19: Most partisan phrases, Catholic and Other parties, French speech: periods beginning in the 2010s

Catholic			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
2014–2016					
aide juridique	121	8	sécurité sociale	102	175
contre terrorisme	175	84	saut index	165	232
cour assises	102	19	doel doel	22	89
partenaires sociaux	236	171	union européenne	199	264
très clairement	145	85	enveloppe bien	7	60
2016–2019					
juge instruction	125	45	sécurité sociale	153	201
secret professionnel	84	29	saut index	10	53
aide juridique	68	16	hommes femmes	21	49
toute série	125	74	droits femmes	21	45
arabie saoudite	88	44	droits humains	17	41

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Catholic: positive; Other parties: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 1; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C20: Most partisan phrases, Catholic and Other parties, French speech: periods beginning in the 2020s

Catholic			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
2024–2025					
union européenne	459	178	flexi jobs	40	115
coopération développement	127	33	malades longue	17	86
norme croissance	119	48	travailleurs travailleuses	14	73
consommation drogues	47	6	travailleuses travailleurs	11	62
spf justice	39	4	tableaux budgétaires	10	56

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Catholic: positive; Other parties: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 1; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C21: Most partisan phrases, Catholic and Other parties, Dutch speech: periods beginning in the 1900s

Catholic			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1900–1902					
vlaamsche volk	633	315	algemeen stemrecht	565	566
			enkele woorden	383	384
			vlaamsche land	213	214
			onderlingen bijstand	279	280
			werkende klas	384	385
1902–1904					
10 jaar	1	1			
sommige vragen	1	1			
sommigen wel	1	1			
sommige zaken	1	1			
sommigen zeggen	1	1			
1904–1906					
			10 jaar	23	23
			sommige vragen	2	2
			sommigen wel	1	1
			sommige zaken	1	1
			sommigen zeggen	1	1
1906–1908					
mager vee	443	438	nationale bank	161	644
jaar dienst	451	447	bedienden werklieden	141	404
vlaamsche volk	410	406	private nijverheid	30	108
west vlaanderen	353	350			
vlaamsche taal	360	357			
1908–1910					
kleine bedienden	298	91	ambtenaar burgerlijke	34	196
spoorwegen posterijen	550	549	burgerlijke stand	69	230
vlaamsche volk	315	315	land waas	33	65
posterijen telegrafen	384	383	werkende klas	440	440
vlaamsche taal	329	329	67 jaar	57	57

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Catholic: positive; Other parties: negative) in Dutch speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 1; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C22: Most partisan phrases, Catholic and Other parties, Dutch speech: periods beginning in the 1910s

Catholic			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1910–1912					
onderlingen bijstand	488	486	private nijverheid	21	347
kleine bedienden	355	354			
werkende klas	342	341			
vlaamsche land	360	359			
twintig jaar	358	357			
1912–1914					
algemeen stemrecht	870	297	lagere scholen	125	342
vlaamsche soldaten	136	136	tweede taal	74	266
officieele scholen	326	326	lager onderwijs	121	303
twintig jaar	259	258	vlaamsche volk	281	360
jaar dienst	267	267	werkende klas	329	329
1918–1919					
			10 jaar	3	3
			sommige vragen	3	3
			sommigen wel	3	3
			sommige zaken	2	2
			sommigen zeggen	2	2
1919–1921					
landelijke politie	361	15	werkende klas	201	533
syndicale vrijheid	154	32	oud strijders	472	608
land waes	88	27	politieke gevangenen	65	133
lager onderwijs	91	39	nationale maatschappij	134	164
onze nationale	95	71	vlaamsche volk	219	220

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Catholic: positive; Other parties: negative) in Dutch speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 1; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C23: Most partisan phrases, Catholic and Other parties, Dutch speech: periods beginning in the 1920s

Catholic			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1921–1923					
middelbaar onderwijs	469	194	oud strijders	109	288
lager onderwijs	368	202	werkende klas	67	208
vierde graad	166	40	vlaamsche waalsche	83	203
landelijke politie	229	114	nationaal werk	48	149
militair statuut	122	59	vlaamsche soldaten	58	153
1923–1925					
vierde graad	187	34	vóór oorlog	344	346
kroostrijke gezinnen	154	14	nationale maatschappij	312	313
fransche taal	204	87	oud strijders	283	284
			kosteloos pensioen	265	267
			65 jaar	268	269
1925–1927					
onze soldaten	175	45	oud strijders	143	279
militaire overheid	152	48	nationale bank	107	189
tusschen beide	124	51	fransch belgisch	50	105
land waes	113	42	belgische regering	108	130
kroostrijke gezinnen	102	34	militair akkoord	15	35
1927–1929					
kroostrijke gezinnen	338	104	oud strijders	209	426
openbare werken	440	229	fransch belgisch	38	165
speciaal fonds	94	15	militair akkoord	20	110
één oplossing	135	61	veertien maanden	35	104
onderwijzend personeel	123	68	werkende klas	66	128
1929–1932					
kroostrijke gezinnen	256	71	vlaamsche volk	131	297
ambachten neringen	136	12	fransch belgisch	12	143
openbare werken	451	340	belgische regering	66	188
kamers ambachten	103	14	militair akkoord	4	119
kadastraal inkomen	127	40	belgische staat	56	153

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Catholic: positive; Other parties: negative) in Dutch speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 1; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C24: Most partisan phrases, Catholic and Other parties, Dutch speech: periods beginning in the 1930s

Catholic			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1932–1934					
nationale maatschappij	460	169	vlaamsche volk	133	236
maatschappij buurtspoorwegen	301	17	oud strijders	93	143
wet 1930	212	14	militair akkoord	10	57
technisch onderwijs	202	22	fransch belgisch	26	52
kroostrijke gezinnen	243	106	openbare werken	334	338
1934–1936					
kroostrijke gezinnen	260	69	vlaamsche volk	84	341
brusselsche agglomeratie	192	52	oud strijders	57	238
kadastraal inkomen	213	77	belgische staat	55	128
landen tuinbouw	106	13	fransch belgisch	103	168
technisch onderwijs	106	29	militair akkoord	62	119
1936–1939					
nationale bank	340	112	officieel onderwijs	77	144
kroostrijke gezinnen	198	31	vlaamsche volk	180	244
gewijzigd wet	106	22	jaar dienst	67	115
wet 1930	112	33	belgische staat	55	102
technisch onderwijs	129	55	laatste woord	49	83
1939–1940					
technisch onderwijs	511	125	vlaamsche volk	66	138
nationale maatschappij	523	256	vlaamsche soldaten	31	100
onze nationale	157	80	nationaal werk	76	133
ambachten neringen	112	47	vierde graad	15	51
bevoorrading land	83	22	openbaar onderwijs	308	311

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Catholic: positive; Other parties: negative) in Dutch speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 1; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C25: Most partisan phrases, Catholic and Other parties, Dutch speech: periods beginning in the 1940s

Catholic			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1944–1946					
dienst pensioenen	373	91	gedurende oorlog	234	236
kroostrijke gezinnen	223	33	arbeid sociale	161	163
belasting kapitaal	237	81	tijdens bezetting	203	205
technisch onderwijs	151	105	tijdens oorlog	259	261
			moeten betalen	140	141
1946–1947					
technisch onderwijs	339	106	politieke gevangenen	92	281
maatschappelijke zekerheid	248	122	dienst pensioenen	50	186
ambachten neringen	140	28	economische unie	110	229
kroostrijke gezinnen	89	17	economische problemen	30	77
sociale voorzorg	239	171	werkende klas	56	81
1947–1949					
nationale maatschappij	370	228	grote steden	38	119
technisch onderwijs	223	87	centraal bestuur	46	117
kroostrijke gezinnen	155	37	fonds gemeenten	41	98
nationale bank	164	64	werkende klas	29	83
volksgezondheid gezin	171	82	sint niklaas	53	107
1949–1950					
rechterlijke orde	199	63	algemene paritaire	36	139
burgerlijke partij	167	57	sociale voorzorg	168	168
			arbeid sociale	170	170
			vrij verzekerden	183	183
			staan vóór	144	144

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Catholic: positive; Other parties: negative) in Dutch speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 1; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C26: Most partisan phrases, Catholic and Other parties, Dutch speech: periods beginning in the 1950s

Catholic			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1950–1952					
maatschappelijke zekerheid	297	138	twintig maanden	52	199
belgisch congo	176	54	openbare werken	179	288
technisch onderwijs	194	76	gehuwde vrouw	40	111
landen tuinbouw	141	39	sociale zekerheid	93	156
lager onderwijs	118	44	sociale voorzorg	232	292
1952–1954					
technisch onderwijs	366	74	heilige stoel	52	310
hoge autoriteit	193	61	achttien maanden	36	246
belgisch congo	260	132	twintig maanden	75	277
zelfstandige arbeiders	326	210	militaire diensttijd	9	79
gemeenschappelijke markt	114	35	officieel onderwijs	72	141
1954–1956					
technisch onderwijs	282	18	vier jaar	124	288
openbaar onderwijs	324	159	vorige regering	150	267
vrij verzekerden	218	77	jonge mannen	12	78
sociale voorzorg	323	188	openbare werken	198	257
dienst pensioenen	148	16	rijksmiddelbare school	32	86
1956–1958					
sociale voorzorg	277	140	jonge mannen	9	171
technisch onderwijs	170	44	openbare werken	253	315
arbeid sociale	257	158	nationaal inkomen	55	108
belgisch congo	224	130	twintig maanden	21	70
kroostrijke gezinnen	97	29	commissaris generaal	22	71
1958–1961					
openbare werken	302	216	nationale bank	14	151
ronde tafelconferentie	116	51	vorige regering	58	153
volksgezondheid gezin	111	47	openbare schuld	28	100
regionale economie	80	17	nationaal inkomen	56	127
nationaal werk	108	46	belgisch congo	138	201

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Catholic: positive; Other parties: negative) in Dutch speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 1; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C27: Most partisan phrases, Catholic and Other parties, Dutch speech: periods beginning in the 1960s

Catholic			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1961–1963					
dienst pensioenen	220	50	fiscale hervorming	215	357
verenigde staten	154	52	arrondissement leper	9	138
kadastraal inkomen	235	147	economische expansie	134	207
openbare schuld	105	33	nationale opvoeding	188	254
haven antwerpen	126	57	vitaal belang	11	64
1963–1965					
europese school	148	33	nationale opvoeding	155	287
nationale maatschappij	141	74	tweede taal	74	170
openbare werken	272	218	nationale kas	17	109
technisch onderwijs	79	31	vlaamse gemeenschap	34	109
maatschappelijke zekerheid	70	31	twee gemeenschappen	19	84
1965–1968					
openbare werken	338	170	nationale opvoeding	202	345
wetenschappelijk onderzoek	135	63	vrije universiteiten	17	113
departement openbare	71	21	generaal toerisme	6	52
verzekering geven	73	29	huidige regering	68	112
maatschappelijke zekerheid	73	39	vlaamse gemeenschap	26	68
1968–1969					
openbare onderstand	198	76	economische expansie	89	133
gewaarborgd inkomen	166	44	vrije universiteiten	31	75
openbare werken	363	245	regionale economische	56	92
collectieve arbeidsovereenkomsten	83	6	onze landbouwers	21	57
sociale voorzorg	211	149	socialistische mutualiteiten	31	60
1969–1971					
hoger onderwijs	226	71	nationale opvoeding	139	204
nationale maatschappij	223	93	lager onderwijs	47	102
openbare werken	375	281	economische hogeschool	13	62
ruimtelijke ordening	139	74	universitaire expansie	21	70
gewaarborgd inkomen	81	17	vlaamse gemeenschap	29	70

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Catholic: positive; Other parties: negative) in Dutch speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 1; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C28: Most partisan phrases, Catholic and Other parties, Dutch speech: periods beginning in the 1970s

Catholic			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1971–1974					
openbare werken	430	326	sociale voorzorg	75	183
landen tuinbouw	89	15	vorige regering	102	157
buitengewoon onderwijs	102	35	traditionele partijen	17	60
departement openbare	106	44	nationale opvoeding	115	156
middelgrote ondernemingen	97	40	belgische staat	27	65
1974–1977					
sociale voorzorg	180	91	nationale opvoeding	140	263
hoger onderwijs	167	87	openbaar ambt	91	143
sociale zekerheid	125	66	vlaamse aangelegenheden	42	88
minder validen	128	72	openbare werken	259	303
maatschappelijke zekerheid	61	19	vlaamse gemeenschap	14	56
1977–1978					
halle vilvoorde	143	64	openbaar vervoer	47	151
vlaamse gemeenschap	151	82	nationale investeringsmaatschappij	28	107
sociale voorzorg	109	55	vorige regering	81	155
maatschappelijke zekerheid	60	10	openbare holding	7	69
hoofdstedelijk gebied	59	27	toegevoegd eerste	6	52
1979–1981					
hoofdstedelijk gebied	173	23	openbaar vervoer	59	106
beide gemeenschappen	85	20	studiemeester opvoeder	3	50
sociale zekerheid	425	362	wijze waarop	77	120
europees parlement	83	34	centrale regering	7	41
franstalige gemeenschap	71	25	vorige regering	57	91

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Catholic: positive; Other parties: negative) in Dutch speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 1; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C29: Most partisan phrases, Catholic and Other parties, Dutch speech: periods beginning in the 1980s

Catholic			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1981–1983					
cockerill sambre	314	150	centrale regering	23	87
europese commissie	177	40	vlaamse gemeenschap	75	135
europese gemeenschap	216	95	fiscale druk	11	69
vrije beroepen	117	40	openbaar vervoer	84	137
hoog comité	75	10	openbare ziekenhuizen	26	76
1983–1985					
europese gemeenschap	217	113	fiscale druk	27	91
nationale sectoren	204	114	centrale regering	9	57
cockerill sambre	160	89	verenigde staten	206	249
vijf nationale	80	13	sociale zekerheid	256	299
stand gebracht	85	34	fiscale fraude	8	49
1985–1987					
europese gemeenschap	295	137	wetenschappelijk onderzoek	65	168
sociale zekerheid	306	197	politieke vluchtelingen	45	129
sociale gesprekspartners	90	10	centrale regering	8	69
politieke samenwerking	84	4	openbare financiën	88	142
nationale overheid	86	28	voorlopige kredieten	67	118
1988–1989					
roerende voorheffing	200	79	openbare financiën	87	171
nationale bank	241	132	wetenschappelijke culturele	18	94
europees parlement	118	49	neen geantwoord	61	132
nationaal vlak	100	44	nationale wetenschappelijke	17	86
economische monetaire	151	99	diensten nationale	25	94
1989–1991					
europese gemeenschap	390	210	diensten nationale	30	98
europees parlement	185	62	nationale wetenschappelijke	31	98
menselijk leven	80	9	midden oosten	66	132
ongeboren leven	97	27	wetenschappelijke culturele	31	97
politieke unie	108	39	culturele instellingen	31	96

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Catholic: positive; Other parties: negative) in Dutch speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 1; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C30: Most partisan phrases, Catholic and Other parties, Dutch speech: periods beginning in the 1990s

Catholic			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1991–1993					
bijzondere wet	430	150	sociale zekerheid	615	665
hoofdstedelijk gewest	113	26	halle vilvoorde	60	108
gemeenschappen gewesten	175	98	stemming gelden	1	33
monetaire unie	101	26	uitslag vorige	4	36
fiscale druk	114	58	franstalige brusselaars	9	36
1993–1995					
globaal plan	406	282	sociale zekerheid	794	988
georganiseerde criminaliteit	74	8	sociaal pact	114	198
algemene uitgavenbegroting	64	11	objectieve aansprakelijkheid	58	134
wou stemmen	88	43	sociale programmawet	46	108
ontwerp verklaring	35	4	stemming gelden	12	68
1995–1997					
procureur generaal	436	275	sociale zekerheid	888	1196
monetaire unie	145	54	sociale partners	203	249
onbemande camera	144	54	medisch dossier	21	63
college procureurs	125	48	politieke cultuur	39	76
strafrechtelijk beleid	94	27	vlaams parlement	32	69
1997–1999					
voorwaardelijke invrijheidstelling	189	54	sociale zekerheid	103	208
procureur konings	190	105	vlaams parlement	87	192
procureur generaal	289	220	politieke partijen	122	203
lokale politie	99	33	medisch dossier	7	58
college procureurs	121	59	fiscale discriminatie	19	61
1999–2001					
federale politie	235	89	halle vilvoorde	30	156
sociale partners	200	102	europese unie	117	188
actieve welvaartsstaat	186	100	gerechtelijk arrondissement	32	92
georganiseerde criminaliteit	80	9	voorlopige hechtenis	13	62
belgische nationaliteit	174	106	stemming gelden	4	48

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Catholic: positive; Other parties: negative) in Dutch speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 1; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C31: Most partisan phrases, Catholic and Other parties, Dutch speech: periods beginning in the 2000s

Catholic			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
2001–2003					
veiligheid staat	247	100	europese unie	102	224
ontwerp verklaring	138	18	toegevoegde rechters	19	98
oui nee	148	39	lambermont akkoord	57	132
abstentions totaal	141	41	verenigde staten	79	141
total bijgevolg	128	38	georganiseerde misdaad	10	69
2003–2005					
stemming gelden	142	1	europese unie	149	340
uitslag vorige	141	1	halle vilvoorde	114	222
federale overheid	221	85	sociale partners	81	181
vorige stemming	143	10	europees parlement	56	148
gelden stemming	136	4	net zoals	66	134
2005–2007					
pairé avec	78	8	europese unie	89	325
inspectie financiën	77	16	sociale partners	108	212
volgende week	188	130	europese grondwet	26	91
onze vraag	75	29	europees parlement	26	80
nieuwe belastingen	80	36	europese commissie	117	169
2007–2010					
sociale partners	376	249	nationale bank	79	196
nmbs groep	106	8	halle vilvoorde	65	167
financiële crisis	189	105	notionele intrestaftrek	20	98
bnp paribas	87	27	vlaams parlement	49	112
voorlopige hechtenis	75	25	fiscale fraude	143	190

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Catholic: positive; Other parties: negative) in Dutch speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 1; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C32: Most partisan phrases, Catholic and Other parties, Dutch speech: periods beginning in the 2010s

Catholic			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
2010–2012					
sociale partners	336	108	minnelijke schikking	49	200
nmbs groep	184	53	notionele intrestaftrek	10	102
nationale bank	285	191	fiscale fraude	92	135
bilaterale akkoorden	99	21	halle vilvoorde	41	74
begeleide minderjarigen	85	20	lopende zaken	462	468
2012–2014					
sociale partners	497	203	halle vilvoorde	44	164
voorliggend ontwerp	290	96	europese unie	354	463
europese commissie	346	211	gemiste kans	59	118
erg belangrijk	105	25	vlaamse partijen	40	95
ambtenaar burgerlijke	103	25	civiele lijst	20	74
2014–2016					
nationale bank	312	115	gegarandeerde dienstverlening	26	95
sociale partners	519	336	europese unie	225	294
volgend jaar	253	125	verenigde staten	69	134
sociale bescherming	188	75	erkende vluchtelingen	30	85
beschikbaar inkomen	115	12	bijzonder ernstig	11	29
2016–2019					
sociale zekerheid	389	272	gegarandeerde dienstverlening	14	57
europese commissie	330	219	openbare orde	21	60
nationale bank	291	183	regering michel	18	54
beschikbaar inkomen	102	12	vlaams parlement	40	73
minnelijke schikking	209	134	jaar gewerkt	18	47

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Catholic: positive; Other parties: negative) in Dutch speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 1; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C33: Most partisan phrases, Catholic and Other parties, Dutch speech: periods beginning in the 2020s

Catholic			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
2024–2025					
fiscale hervorming	392	83	sociale zekerheid	378	385
nationale loterij	216	61	voorlopige twaalfden	227	232
zodat we	244	110	langdurig zieken	254	258
sociaal tarief	174	43	twee jaar	197	201
binnen europa	130	21	vorige regering	191	195

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Catholic: positive; Other parties: negative) in Dutch speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 1; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C34: Most partisan phrases, Left and Other parties, French speech: periods beginning in the 1890s

Left			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1894–1896					
classe ouvrière	219	58	droit licence	6	87
cahier charges	93	33	droits entrée	49	101
chemin fer	424	368	distillateurs agricoles	7	57
conseil communal	149	94	droit accise	28	73
suffrage universel	112	58	tabac indigène	10	46
1896–1898					
classe ouvrière	207	39	garde civique	145	261
suffrage universel	94	34	fabricants sucre	7	40
sociétés coopératives	78	42	tribunal liége	1	28
grand central	119	83	pourrait être	80	103
syndicats professionnels	38	9	observations présentées	20	43
1898–1900					
banque nationale	414	139	juge paix	27	113
classe ouvrière	186	31	conseil guerre	42	96
suffrage universel	143	57	députation permanente	39	88
caisse épargne	155	71	cour militaire	5	50
tramways bruxellois	111	36	cour appel	15	51

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Left: positive; Other parties: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 2; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C35: Most partisan phrases, Left and Other parties, French speech: periods beginning in the 1900s

Left			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1900–1902					
suffrage universel	502	239	chemin fer	244	366
classe ouvrière	202	34	jeux hasard	10	88
industrie travail	101	44	cercles privés	14	69
général major	129	83	gare centrale	7	55
comités patronage	36	9	distilleries agricoles	10	57
1902–1904					
classe ouvrière	203	46	chemin fer	349	449
heures travail	62	13	conseil communal	57	109
prix viande	79	35	faute lourde	27	68
industrie travail	109	74	chef entreprise	21	50
augmentation prix	49	16	ville bruxelles	12	42
1904–1906					
classe ouvrière	176	42	code civil	34	88
lieutenant général	211	110	repos dominical	36	88
heures travail	113	30	grande coupure	44	96
industrie travail	185	106	navire école	8	52
etat indépendant	93	36	conseil communal	56	91
1906–1908					
classe ouvrière	176	35	langue flamande	23	97
industrie travail	216	101	seconde ligne	4	37
huit heures	172	66	pays flamand	35	65
heures travail	130	47	enseignement moyen	16	45
chemin fer	533	455	ligne défense	7	35
1908–1910					
classe ouvrière	180	34	comité spécial	6	85
chemin fer	477	342	union minière	4	45
prud hommes	299	168	loi 1902	26	61
conseils prud	200	121	budget extraordinaire	33	63
huit heures	120	46	effectif paix	31	60

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Left: positive; Other parties: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 2; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C36: Most partisan phrases, Left and Other parties, French speech: periods beginning in the 1910s

Left			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1910–1912					
classe ouvrière	258	50	caisses prévoyance	50	107
industrie travail	163	101	juge paix	32	77
lieutenant général	112	53	cour comptes	8	48
heures travail	74	25	députation permanente	57	93
suffrage universel	64	20	loi 1900	22	54
1912–1914					
classe ouvrière	314	38	bons trésor	9	69
suffrage universel	204	70	écoles communales	110	159
industrie travail	105	56	société nationale	37	74
quelque chose	102	54	enseignement public	17	52
heures travail	53	14	budget extraordinaire	16	50
1918–1919					
classe ouvrière	258	46	dommages guerre	43	214
comité national	133	55	tribunaux dommages	13	99
huit heures	50	10	écoles normales	22	73
suffrage universel	90	50	juge unique	40	77
suffrage femmes	69	37	langue flamande	10	44
1919–1921					
classe ouvrière	325	69	dommages guerre	56	166
passage eau	55	1	tribunaux dommages	23	61
suffrage universel	81	28	commissaires etat	15	52
avant guerre	216	167	chemin fer	191	228
huit heures	97	51	régions dévastées	60	85

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Left: positive; Other parties: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 2; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C37: Most partisan phrases, Left and Other parties, French speech: periods beginning in the 1920s

Left			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1921–1923					
classe ouvrière	169	60	chemin fer	286	336
industrie travail	113	56	gardes champêtres	69	113
sciences arts	147	95	extrême gauche	41	73
quatrième degré	54	13	grand duché	21	53
administration communale	117	76	pays flamand	25	56
1923–1925					
service sauvetage	98	8	chemin fer	184	291
classe ouvrière	103	31	union douanière	13	50
société nationale	113	46	pro deo	11	39
industrie travail	152	97	prix maxima	16	43
sciences arts	80	40	examiné projet	15	40
1925–1927					
classe ouvrière	183	40	banque nationale	52	125
chemin fer	392	287	dette flottante	19	75
huit heures	112	37	général kestens	19	70
avant guerre	161	93	neuf ans	29	77
société nationale	103	55	etats unis	53	97
1927–1929					
défense nationale	211	98	prison école	17	69
classe ouvrière	137	42	juge paix	10	56
personnel enseignant	75	22	grands lacs	5	47
huit heures	71	26	condamnation conditionnelle	10	49
industrie travail	144	101	ville anvers	10	40
1929–1932					
industrie travail	180	77	sociétés privées	6	33
classe ouvrière	164	69	langue flamande	14	42
société nationale	191	116	libre échange	24	52
huit heures	63	21	responsabilité limitée	2	27
assurances sociales	61	24	deux langues	20	43

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Left: positive; Other parties: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 2; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C38: Most partisan phrases, Left and Other parties, French speech: periods beginning in the 1930s

Left			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1932–1934					
classe ouvrière	308	102	caisse nationale	11	76
pleins pouvoirs	204	39	juge paix	27	85
liste arrondissement	76	14	allocations familiales	63	119
allocations chômage	97	37	tenir compte	103	159
défense nationale	142	93	agents change	13	60
1934–1936					
classe ouvrière	176	42	prix revient	31	77
société nationale	260	129	gouverneur général	5	35
arrêtés lois	156	57	classes moyennes	32	59
chemin fer	265	175	prix vente	8	33
nationale chemins	125	58	revenu cadastral	10	34
1936–1939					
travaux publics	145	75	classes moyennes	57	142
classe ouvrière	129	83	grands magasins	16	68
non intervention	58	15	prix revient	38	66
administration communale	72	29	juge paix	9	35
caisses chômage	52	22	banque nationale	108	133
1939–1940					
personnel enseignant	102	23	santé publique	80	116
instruction publique	156	105	budget extraordinaire	21	51
classe ouvrière	81	47	classes moyennes	33	63
union soviétique	29	2	défense nationale	239	269
masses laborieuses	25	1	fournitures bureau	3	32

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Left: positive; Other parties: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 2; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C39: Most partisan phrases, Left and Other parties, French speech: periods beginning in the 1940s

Left			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1944–1946					
prisonniers politiques	236	112	impôt exceptionnel	5	48
sécurité sociale	116	15	dommages guerre	65	107
secours civil	94	9	prix revient	22	62
ouvriers mineurs	54	7	pouvoirs extraordinaires	26	59
pensions vieillesse	60	19	pouvoirs spéciaux	66	98
1946–1947					
sécurité sociale	103	50	allocations familiales	57	111
prisonniers politiques	140	91	juge paix	38	89
société nationale	85	37	sous officiers	13	46
classe ouvrière	103	57	prix revient	78	109
personnel enseignant	64	20	pouvoirs spéciaux	7	37
1947–1949					
plan marshall	174	100	classes moyennes	88	137
classe ouvrière	85	26	neuf ans	6	53
mobilisation civile	57	5	commerce extérieur	45	90
allocations compensatoires	72	28	juge paix	36	78
société nationale	131	92	banque nationale	62	102
1949–1950					
membres conseil	149	70	eaux usées	10	66
consultation populaire	400	323	dommages guerre	29	63
libre belgique	62	16	société nationale	53	83
question royale	162	122	collège électoral	18	47
vrai oui	46	11	organisations ouvrières	15	44

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Left: positive; Other parties: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 2; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C40: Most partisan phrases, Left and Other parties, French speech: periods beginning in the 1950s

Left			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1950–1952					
classe ouvrière	133	17	classes moyennes	46	83
personnel enseignant	115	38	dommages guerre	42	77
sécurité sociale	138	68	enseignement moyen	41	69
chemin fer	105	55	congo belge	18	44
services publics	81	31	loi 1947	7	30
1952–1954					
plan schuman	142	73	dommages guerre	22	56
victimes civiles	89	32	après ordres	2	30
assurés libres	74	21	experts comptables	4	30
personnel enseignant	84	32	commerce extérieur	19	44
santé publique	83	44	congo belge	24	49
1954–1956					
personnel enseignant	92	19	banque nationale	27	77
travailleurs indépendants	106	47	fonds routes	11	55
enseignement libre	211	154	taxe transmission	18	56
enseignement officiel	114	58	instruction publique	108	142
santé publique	129	77	enseignement public	38	72
1956–1958					
oeuvre nationale	77	10	classes moyennes	69	190
conseil administration	87	22	banque nationale	22	120
santé publique	104	50	dette publique	11	65
classe ouvrière	78	29	haute conjoncture	21	69
société nationale	124	82	pouvoirs spéciaux	10	44
1958–1961					
classe ouvrière	94	20	congo belge	70	172
loi unique	120	58	table ronde	99	141
pacte scolaire	89	33	politique scientifique	3	40
services publics	103	52	ruanda urundi	73	107
personnel enseignant	60	12	marché commun	71	100

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Left: positive; Other parties: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 2; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C41: Most partisan phrases, Left and Other parties, French speech: periods beginning in the 1960s

Left			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1961–1963					
classe ouvrière	89	4	assistance technique	26	67
organisations syndicales	97	27	marché commun	107	141
services publics	106	58	pays voie	11	42
santé publique	124	81	besoins vitaux	12	42
loi unique	108	75	voie développement	10	40
1963–1965					
corps médical	168	91	assistance technique	35	114
services publics	142	86	pays voie	8	48
etats unis	117	64	voie développement	8	46
maladie invalidité	184	132	outre mer	5	41
assurance maladie	190	142	sécurité sociale	50	85
1965–1968					
pouvoirs spéciaux	194	63	cour comptes	10	75
assurance maladie	147	54	assistance technique	6	54
maladie invalidité	135	52	économique wallon	15	53
corps médical	69	14	lois linguistiques	18	48
ouvriers mineurs	60	8	autoroute wallonie	21	51
1968–1969					
organisations syndicales	132	15	conseil économique	102	190
maladies professionnelles	95	38	code judiciaire	6	53
ouvriers mineurs	64	18	économique wallon	30	71
marché commun	129	89	région flamande	17	57
union soviétique	38	6	minimum vital	11	48
1969–1971					
santé publique	153	75	conseil culturel	34	106
distribution eau	70	9	police judiciaire	42	99
organisations syndicales	114	59	ik citeer	9	62
etats unis	119	75	conseil économique	41	89
marché commun	105	67	six communes	19	66

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Left: positive; Other parties: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 2; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C42: Most partisan phrases, Left and Other parties, French speech: periods beginning in the 1970s

Left			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1971–1974					
travailleurs frontaliers	68	14	classes moyennes	36	115
organisations syndicales	109	54	santé publique	57	131
marche famenne	35	2	conseil culturel	52	103
fonds communes	54	23	moyennes entreprises	14	48
régie postes	49	20	aménagement territoire	32	64
1974–1977					
aide sociale	166	16	économique sociale	67	109
centres publics	77	5	langue allemande	16	57
publics aide	71	6	communes facilités	3	43
conseil communal	127	68	région bruxelles	5	42
assistance publique	71	17	classes moyennes	67	100
1977–1978					
santé publique	85	39	finances publiques	30	126
conseil administration	69	38	pouvoirs spéciaux	67	142
famille libérale	48	17	pacte egmont	14	62
groupes financiers	35	6	conseil régional	21	64
etats unis	87	59	moyennes entreprises	83	120
1979–1981					
loi programme	184	103	région bruxelles	15	66
organisations syndicales	96	28	communauté germanophone	5	56
etats unis	94	37	finances publiques	37	86
défense nationale	126	74	région wallonne	106	149
sécurité sociale	194	144	matières personnalisables	14	50

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Left: positive; Other parties: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 2; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C43: Most partisan phrases, Left and Other parties, French speech: periods beginning in the 1980s

Left			Other parties		
Phrase	$\hat{\eta}^A$	$\hat{\eta}^B$	Phrase	$\hat{\eta}^A$	$\hat{\eta}^B$
1981–1983					
sécurité sociale	244	120	troisième circuit	14	67
cockerill sambre	116	8	classes moyennes	26	74
pouvoirs spéciaux	619	539	temps travail	29	70
pleins pouvoirs	91	11	circuit travail	11	49
services publics	137	61	conseil communal	31	67
1983–1985					
services publics	131	75	temps travail	28	103
région wallonne	103	47	classes moyennes	25	62
pouvoirs spéciaux	151	97	année scolaire	22	51
secteurs nationaux	94	44	langue allemande	2	29
poser question	151	109	travaux publics	48	72
1985–1987					
pouvoirs spéciaux	373	202	droits succession	134	230
publicité commerciale	88	16	communauté européenne	27	64
sécurité sociale	196	143	finances publiques	75	112
education nationale	166	115	secteur textile	8	44
politique scientifique	63	24	juge instruction	12	44
1988–1989					
défense nationale	217	70	pacte scolaire	22	131
cour comptes	113	8	etat central	117	177
biologie clinique	75	17	région flamande	21	71
loi programme	110	51	cour arbitrage	52	88
budget général	61	5	conseil communal	77	112
1989–1991					
défense nationale	191	92	handelingen plenaire	250	352
etat major	90	10	loi financement	27	76
office contrôle	88	31	communauté européenne	73	110
santé publique	84	33	cour arbitrage	14	46
sécurité sociale	154	105	précompte professionnel	13	35

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Left: positive; Other parties: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 2; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C44: Most partisan phrases, Left and Other parties, French speech: periods beginning in the 1990s

Left			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1991–1993					
chemin fer	122	15	communauté germanophone	33	172
partis politiques	131	32	plan convergence	24	102
conseil régional	164	104	loyauté fédérale	25	87
biologie clinique	63	8	congé politique	18	64
conseil administration	68	15	francophones périphérie	14	59
1993–1995					
sécurité sociale	487	270	temps travail	46	76
congé politique	127	22	organisations syndicales	3	19
voulais voter	109	34	santé publique	56	64
organismes assureurs	83	9	représentants belgique	635	639
clauses sociales	102	53	annales handelingen	588	592
1995–1997					
sécurité sociale	786	548	pouvoirs spéciaux	91	344
services publics	88	33	fiscale parafiscale	29	90
maladies chroniques	51	8	pression fiscale	48	107
organisations syndicales	53	11	union monétaire	33	80
sociales fiscales	58	16	huit ans	22	54
1997–1999					
sécurité sociale	199	70	cour comptes	24	129
santé publique	104	46	juge instruction	124	177
etats membres	86	42	nouveau belgique	2	49
soins santé	108	65	supérieur contrôle	17	57
socialiste votera	32	3	purement simplement	77	110
1999–2001					
conséquence proposition	57	4	organisations syndicales	33	49
proposition modification	54	8	poser question	98	114
parole déclaration	75	33	non seulement	169	170
temps travail	95	55	union européenne	160	161
défense européenne	43	10	celle ci	154	155

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Left: positive; Other parties: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 2; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C45: Most partisan phrases, Left and Other parties, French speech: periods beginning in the 2000s

Left			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
2001–2003					
sécurité sociale	183	68	cour arbitrage	25	81
santé publique	199	90	arc ciel	27	69
services publics	174	104	accords bilatéraux	8	50
zones police	151	87	poser question	80	100
organisations syndicales	83	33	cour comptes	21	38
2003–2005					
soins santé	370	212	consultation populaire	29	73
service universel	103	19	gouvernement fédéral	37	73
sécurité sociale	218	155	vote automatisé	6	31
établissements pénitentiaires	66	10	poser question	101	119
organisations syndicales	53	11	pécule vacances	13	27
2005–2007					
organisations syndicales	84	13	dossier confidentiel	32	90
service universel	74	33	sécurité routière	33	72
santé publique	92	52	cour comptes	19	49
soins santé	126	91	adopter enfant	5	26
établissements pénitentiaires	40	6	gouvernement fédéral	51	67
2007–2010					
santé publique	182	77	titres services	29	75
soins santé	183	82	coopération développement	48	76
sécurité sociale	193	95	jeux hasard	27	51
crise financière	159	84	droits enfant	20	40
paradis fiscaux	82	24	état fédéral	31	49

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Left: positive; Other parties: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 2; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C46: Most partisan phrases, Left and Other parties, French speech: periods beginning in the 2010s

Left			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
2010–2012					
groupe sncb	154	53	gouvernement affaires	29	67
santé publique	131	55	assistance avocat	10	31
sécurité sociale	138	70	union européenne	284	287
observatoire prix	71	6	celle ci	143	144
soins santé	122	64	doivent être	135	136
2012–2014					
partenaires sociaux	200	125	arrondissement judiciaire	40	82
pouvoir achat	143	69	supérieur justice	9	34
soins santé	138	65	juge paix	16	36
gouvernement fédéral	126	53	allocations chômage	37	53
états membres	109	55	chefs corps	20	35
2014–2016					
services publics	244	58	partenaires sociaux	149	203
saut index	339	155	contre terrorisme	78	111
tax shift	201	125	gouvernement fédéral	52	83
plus riches	79	9	arabie saoudite	24	55
sécurité sociale	203	141	génocide arménien	17	47
2016–2019					
saut index	111	13	extrêmement important	28	67
soins santé	195	108	pacte énergétique	18	56
sécurité sociale	238	172	santé publique	43	80
67 ans	71	7	entités fédérées	80	118
tax shift	148	86	prendre compte	14	48
2019–2021					
sécurité sociale	236	61	droit passerelle	62	181
paradis fiscaux	105	19	entités fédérées	61	117
soins santé	237	152	niveaux pouvoir	35	76
services publics	136	63	état islamique	4	44
conditions travail	123	56	gouvernement fédéral	40	77

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Left: positive; Other parties: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 2; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C47: Most partisan phrases, Left and Other parties, French speech: periods beginning in the 2020s

Left			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
2021–2024					
sécurité sociale	145	81	ose espérer	9	47
conditions travail	104	44	cour constitutionnelle	53	91
gaz électricité	76	19	partis majorité	21	58
cessez feu	72	18	droits patient	13	48
services publics	95	50	raison laquelle	127	155
2024–2025					
travailleuses travailleurs	119	10	énergies fossiles	8	58
67 ans	106	6	transition énergétique	6	32
travailleurs travailleuses	125	28	aidants proches	12	34
pouvoir achat	205	123	santé publique	36	56
services publics	161	85	travailleurs sociaux	29	49

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Left: positive; Other parties: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 2; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C48: Most partisan phrases, Left and Other parties, Dutch speech: periods beginning in the 1900s

Left			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1900–1902					
10 jaar	2	2			
sommige vragen	2	2			
sommigen wel	2	2			
sommige zaken	2	2			
sommigen zeggen	2	2			
1902–1904					
werkende klas	231	26	mager vee	1347	1349
			vlaamsche taal	673	674
			vlaamsche land	479	480
			ten gevolge	416	417
			één grote	290	291
1906–1908					
werkende klas	555	126	nationale bank	188	516
mager vee	437	437	jaar dienst	448	449
christen democratie	67	67	kleine bedienden	273	274
uitsluitend fransch	168	168	werklieden bedienden	177	178
katholieke regeering	153	153	dertig jaar	185	185
1908–1910					
werkende klas	524	265	kleine bedienden	76	266
grote mogendheden	138	22	spoorwegen posterijen	553	554
officieele scholen	127	63	jaar dienst	333	334
jaar 1962	1	1	vlaamsche taal	331	332
			vlaamsche volk	313	314

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Left: positive; Other parties: negative) in Dutch speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 2; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C49: Most partisan phrases, Left and Other parties, Dutch speech: periods beginning in the 1910s

Left			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1910–1912					
private nijverheid	657	22	onderlingen bijstand	483	489
kleine bedienden	745	281	algemeen stemrecht	324	329
			werkende klas	334	338
			twintig jaar	352	356
			vlaamsche land	356	360
1912–1914					
uren dagen	561	157	vrije scholen	277	320
ten honderd	441	69	algemeen stemrecht	708	716
14 jaar	493	176	officieele scholen	321	325
landen tuinbouw	103	19	vlaamsche volk	303	306
lager onderwijs	202	165	één grote	280	283
1919–1921					
werkende klas	614	84	oud strijders	206	698
voort brengen	97	26	landelijke politie	69	210
nationale maatschappij	193	127	syndicale vrijheid	50	117
kleine eigenaars	65	19	gewestelijke indeeling	64	104
arbeidende klas	102	90	verwoeste streken	204	204

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Left: positive; Other parties: negative) in Dutch speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 2; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C50: Most partisan phrases, Left and Other parties, Dutch speech: periods beginning in the 1920s

Left			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1921–1923					
werkende klas	226	23	middelbaar onderwijs	172	436
nationaal werk	167	58	gewestelijke indeeling	64	192
zoo lang	153	61	vierde graad	30	157
pensioen ontvangen	108	26	lager onderwijs	229	326
oud strijders	243	172	vlaamsche land	105	173
1923–1925					
werkende klas	263	48	vierde graad	34	144
werk kinderwelzijn	187	60	oud strijders	229	325
arbeidende klas	72	59	65 jaar	269	269
kanaal terneuzen	35	35	grote markt	178	179
sluis terneuzen	1	1	volgende bepaling	132	133
1925–1927					
openbare werken	808	374	oud strijders	108	308
drie jaar	601	274	zeer zeker	41	75
negen jaar	397	111	vóór oorlog	292	296
landbouw openbare	240	132	zes maanden	263	267
werkende klas	90	19	twee jaar	237	239
1927–1929					
gewestelijke indeeling	316	52	oud strijders	183	531
zes maanden	560	313	fransch belgisch	37	204
werkende klas	160	5	militair akkoord	9	142
maanden dienst	199	46	nationale maatschappij	395	514
departement landsverdediging	166	19	openbare werken	229	337
1929–1932					
nijverheid arbeid	471	256	oud strijders	134	356
openbare werken	510	329	vlaamsche land	90	239
werkende klas	118	13	vlaamsche volk	153	269
officieele scholen	137	42	fransch belgisch	16	130
dien zin	167	81	katholieke vlaamsche	22	100

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Left: positive; Other parties: negative) in Dutch speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 2; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C51: Most partisan phrases, Left and Other parties, Dutch speech: periods beginning in the 1930s

Left			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1932–1934					
openbaren onderstand	293	32	wet 1930	27	104
sint truiden	190	64	oud strijders	80	139
openbare werken	427	306	nationale maatschappij	283	285
vrije scholen	127	24	ten gevolge	208	209
werkende klas	124	51	sociale voorzorg	184	185
1934–1936					
openbare werken	677	416	oud strijders	41	162
west vlaanderen	328	142	fransch belgisch	43	157
provincie west	168	35	kroostrijke gezinnen	92	189
officieele scholen	125	38	militair akkoord	20	106
toestand landbouw	112	27	landen tuinbouw	24	71
1936–1939					
officieel onderwijs	307	62	nationale bank	56	258
jaar dienst	186	69	vlaamsche land	142	260
vrije scholen	145	35	verkeerswezen posterijen	46	141
officieele scholen	126	30	buitenlandsche politiek	52	140
terugwerkende kracht	86	21	kroostrijke gezinnen	30	114
1939–1940					
vierde graad	148	9	landen tuinbouw	24	74
werkende klas	93	3	nationale maatschappij	366	368
culturele autonomie	101	32	openbaar onderwijs	306	308
openbaren onderstand	151	88	technisch onderwijs	295	297
openbare werken	320	257	bespreking begroting	232	233

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Left: positive; Other parties: negative) in Dutch speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 2; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C52: Most partisan phrases, Left and Other parties, Dutch speech: periods beginning in the 1940s

Left			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1944–1946					
universitair hoger	1	1			
10 jaar	0	0			
sommige sprekers	19	19			
sommige streken	56	56			
sommige vragen	11	11			
1946–1947					
economische unie	271	92	technisch onderwijs	144	297
politieke gevangenen	272	112	maatschappelijke zekerheid	120	239
werkende klas	100	14	onderwijzend personeel	41	148
economische problemen	103	19	ambachten neringen	35	130
internationale organisatie	96	16	sociale voorzorg	182	224
1947–1949					
fonds gemeenten	123	28	technisch onderwijs	98	208
steden gemeenten	102	21	nationale bank	66	162
centraal bestuur	123	43	volksgezondheid gezin	76	166
grote steden	116	41	nationale maatschappij	256	343
werkende klas	77	6	kroostrijke gezinnen	57	132
1949–1950					
algemene paritaire	153	34	rechterlijke orde	51	183
werkende klas	73	29	burgerlijke partij	61	139
gemengde militaire	53	23	rode kruis	60	80
arrondissement kortrijk	71	71	stad antwerpen	57	57
west vlaanderen	66	66	eenheid land	158	158

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Left: positive; Other parties: negative) in Dutch speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 2; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C53: Most partisan phrases, Left and Other parties, Dutch speech: periods beginning in the 1950s

Left			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1950–1952					
twintig maanden	217	49	maatschappelijke zekerheid	152	278
openbare werken	315	169	landen tuinbouw	40	135
officieel onderwijs	192	91	gebrekkigen verminkten	41	127
gehuwde vrouw	121	38	technisch onderwijs	94	174
burgerlijke oorlogsslachtoffers	79	18	belgisch congo	73	151
1952–1954					
heilige stoel	330	48	zelfstandige arbeiders	114	397
achttien maanden	274	30	technisch onderwijs	95	330
twintig maanden	307	70	hoge autoriteit	56	195
officieel onderwijs	168	61	belgisch congo	153	240
buitengewone begroting	140	56	gemeenschap kolen	15	90
1954–1956					
vier jaar	310	131	technisch onderwijs	30	256
rijksmiddelbare school	130	27	vrij onderwijs	74	196
vorige regering	253	159	lager onderwijs	41	152
openbare werken	284	195	vrij verzekerden	95	199
sint niklaas	139	50	openbaar onderwijs	197	299
1956–1958					
jonge mannen	201	12	sociale voorzorg	168	256
buitenlandse handel	144	68	technisch onderwijs	69	151
twintig maanden	92	18	zelfstandige arbeiders	20	100
nationaal inkomen	123	54	arbeid sociale	178	244
medisch schooltoezicht	64	10	openbare schuld	29	95
1958–1961					
nationale bank	196	15	vlaamse land	65	222
vorige regering	170	63	regionale economische	24	87
belgisch congo	231	138	ten overstaan	166	223
openbare schuld	114	32	rode kruis	8	50
belgische spoorwegen	124	59	economische politiek	105	146

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Left: positive; Other parties: negative) in Dutch speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 2; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C54: Most partisan phrases, Left and Other parties, Dutch speech: periods beginning in the 1960s

Left			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1961–1963					
arrondissement leper	171	5	fiscale hervorming	193	316
vitaal belang	127	8	kleine middelgrote	20	133
zaken energie	127	31	kadastraal inkomen	110	215
nationaal inkomen	157	69	vlaamse waalse	34	128
burgerlijke oorlogsslachtoffers	89	9	vlaamse land	56	144
1963–1965					
economische expansie	221	98	europese school	9	113
sociale partners	152	31	tweede taal	60	148
oud strijders	128	11	vlaamse land	98	176
orde geneesheren	119	35	vlaamse gemeenschap	16	87
burgerlijke oorlogsslachtoffers	94	15	middelbaar onderwijs	68	132
1965–1968					
onze landbouwers	99	23	openbare werken	152	268
bijzondere machten	92	25	universitaire expansie	33	104
lid staten	101	37	nationale partijen	30	100
vrije verzekering	69	6	wetenschappelijk onderzoek	45	110
nieuwe belastingen	106	44	jaar 1967	16	67
1968–1969					
openbare besturen	149	57	technische ingenieurs	33	175
collectieve overeenkomsten	96	12	nationale opvoeding	163	276
onze landbouwers	107	26	openbare werken	214	318
sluiting ondernemingen	89	18	gewaarborgd inkomen	29	110
openbaar ambt	105	45	sociale voorzorg	119	190
1969–1971					
verdovende middelen	142	11	hoger onderwijs	30	166
verenigde staten	163	61	secundair onderwijs	42	133
gerechtelijke politie	96	15	buitengewoon onderwijs	15	71
buitenlandse handel	123	52	universitair onderwijs	22	66
openbaar ambt	130	65	vlaamse gemeenschap	20	61

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Left: positive; Other parties: negative) in Dutch speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 2; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C55: Most partisan phrases, Left and Other parties, Dutch speech: periods beginning in the 1970s

Left			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1971–1974					
minder validen	329	140	openbare werken	292	389
arrondissement aalst	144	29	openbare onderstand	51	107
sociale voorzorg	220	116	academisch ziekenhuis	4	61
gewaarborgd inkomen	170	98	kleine middelgrote	56	110
grote steden	88	33	middelgrote ondernemingen	28	74
1974–1977					
openbare sector	166	62	hoger onderwijs	50	138
openbare werken	360	262	kleine middelgrote	32	118
openbare holding	102	13	sociale voorzorg	61	138
privé sector	155	80	middelgrote ondernemingen	17	93
hoog comité	81	26	lager onderwijs	32	89
1977–1978					
buitengewoon onderwijs	110	6	vlaamse gemeenschap	20	141
openbare werken	311	210	buitenlandse handel	60	136
nationale opvoeding	199	125	halle vilvoorde	57	106
rekening gehouden	162	110	hoog comité	8	48
secundair onderwijs	76	29	belgische dienst	13	53
1979–1981					
openbare sector	196	46	vlaamse gemeenschap	48	273
privé sector	133	60	gemeenschappen gewesten	70	160
el salvador	77	10	hoofdstedelijk gebied	15	86
rekening houdend	173	113	vlaanderen wallonië	36	95
bestuur postchecks	78	20	sociale zekerheid	344	394

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Left: positive; Other parties: negative) in Dutch speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 2; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C56: Most partisan phrases, Left and Other parties, Dutch speech: periods beginning in the 1980s

Left			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1981–1983					
bijzondere machten	487	358	cockerill sambre	109	236
sociale zekerheid	358	238	vlaamse gemeenschap	49	142
privé sector	141	51	openbare werken	67	149
openbare sector	128	53	gemeenschappen gewesten	43	121
gesubsidieerd onderwijs	85	33	openbaar vervoer	82	143
1983–1985					
deeltijdse arbeid	103	42	nationale sectoren	76	173
fiscale fraude	74	18	cockerill sambre	55	141
openbare diensten	102	46	vlaamse gemeenschap	34	106
inrichtende machten	72	22	openbare werken	57	126
sociale promotie	92	42	gemeenschappen gewesten	34	87
1985–1987					
openbaar ambt	213	47	europese gemeenschap	122	209
fiscale uitgaven	158	36	gemeenschappen gewesten	36	111
openbare diensten	103	29	vlaamse gemeenschap	52	123
vijf jaar	167	103	nationale regering	108	152
buitengewoon onderwijs	45	2	financieren saldo	71	109
1988–1989					
vaste leden	112	3	neen geantwoord	16	126
oudere werklozen	126	37	nationale bank	88	169
wetenschappelijk onderzoek	158	71	roerende voorheffing	60	116
nucleaire sector	81	15	wetenschappelijke culturele	29	84
sociale partners	95	33	vlaamse gemeenschap	58	98
1989–1991					
gewaarborgd inkomen	180	47	derde fase	61	177
18 jaar	126	23	wetenschappelijke culturele	35	90
mannen vrouwen	133	49	openbare diensten	60	112
inkomen bejaarden	100	24	ongeboren leven	6	54
politie inlichtingendiensten	80	20	roerende voorheffing	69	114

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Left: positive; Other parties: negative) in Dutch speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 2; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C57: Most partisan phrases, Left and Other parties, Dutch speech: periods beginning in the 1990s

Left			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1991–1993					
europese gemeenschap	299	110	sint michielsakkoord	172	413
rijbewijs punten	121	29	halle vilvoorde	67	94
18 jaar	99	20	kamer volksvertegenwoordigers	1486	1491
sociaal verzekerden	57	20	volksvertegenwoordigers belgië	1429	1434
belgië parlementaire	45	10	représentants belgique	1420	1425
1993–1995					
verenigde staten	196	94	sociale zekerheid	714	984
problemen rijzen	77	16	nationale bank	56	156
intergouvernementele conferentie	70	15	politieke partijen	105	143
buitenlands beleid	97	43	hoog comité	25	57
radioactief afval	50	11	comité toezicht	28	34
1995–1997					
sociale zekerheid	1311	1088	nationale bank	27	68
sociaal overleg	252	71	politieke benoemingen	19	53
medisch dossier	145	37	fiscale druk	45	69
modernisering sociale	171	76	vlaams parlement	41	64
deeltijdse arbeid	143	76	parlementaire onderzoekscommissie	37	58
1997–1999					
medisch dossier	193	14	hoog comité	18	143
politieke partijen	314	153	comité toezicht	14	115
chronische zieken	134	20	vlaams parlement	101	184
mannen vrouwen	124	30	belgische nationaliteit	20	91
europese unie	220	144	nationale bank	32	81
1999–2001					
actieve welvaartsstaat	369	96	belgische nationaliteit	36	142
sociaal overleg	245	48	halle vilvoorde	34	132
chronische zieken	167	24	snel belg	11	25
sociale partners	253	126	belg wet	13	27
sociale gesprekspartners	118	10	sociale zekerheid	228	231

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Left: positive; Other parties: negative) in Dutch speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 2; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C58: Most partisan phrases, Left and Other parties, Dutch speech: periods beginning in the 2000s

Left			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
2001–2003					
binnen kader	130	19	vlaams parlement	76	241
sociale partners	171	68	veiligheid staat	32	165
bepaald ogenblik	164	63	halle vilvoorde	46	135
denk goed	86	22	snel belg	10	87
vrije beroepen	103	40	belg wet	11	85
2003–2005					
sociale partners	321	105	snel belg	14	155
wet major	82	4	belg wet	16	154
denk goed	89	31	fiscale amnestie	18	155
denk inderdaad	74	22	halle vilvoorde	95	205
economische inspectie	53	3	stemming bijgevolg	12	53
2005–2007					
sociale zekerheid	332	138	snel belg	7	84
sociale partners	305	134	belg wet	8	84
huiselijk geweld	112	22	belgische nationaliteit	24	90
collectieve schuldenregeling	97	30	nationale bank	36	90
zwarte lijst	68	6	wet houdende	24	68
2007–2010					
sociale partners	528	209	parlementaire onderzoekscommissie	61	154
sociale zekerheid	456	224	nationale bank	112	188
sociaal overleg	173	50	halle vilvoorde	89	164
pairé avec	99	8	vlaams parlement	47	112
oudere werknemers	88	6	snel belgwet	12	61

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Left: positive; Other parties: negative) in Dutch speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 2; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C59: Most partisan phrases, Left and Other parties, Dutch speech: periods beginning in the 2010s

Left			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
2010–2012					
nucleaire centrales	109	23	fiscale fraude	79	135
hoofdelijke aansprakelijkheid	107	23	halle vilvoorde	38	74
interprofessioneel akkoord	132	57	sociale partners	139	155
nederland duitsland	98	23	lopende zaken	463	468
sociaal overleg	156	87	vorig jaar	233	236
2012–2014					
magistrale bereidingen	151	10	halle vilvoorde	74	147
fiscale paradijzen	96	14	gemiste kans	67	114
tijden crisis	97	17	sociale partners	220	253
heel europa	70	17	roerende voorheffing	44	69
europese commissie	280	228	federale overheid	253	255
2014–2016					
tax shift	561	187	gegarandeerde dienstverlening	43	532
vorige regering	513	217	zware beroepen	68	178
grote vermogens	184	25	europese unie	237	294
fiscale fraude	214	97	clos incident	173	222
sociaal overleg	375	267	intrafamiliaal geweld	14	58
2016–2019					
tweede pijler	144	21	europese unie	145	296
45 jaar	182	66	saudi arabië	106	198
alle duidelijkheid	207	95	sociale zekerheid	245	309
minnelijke schikking	230	127	gegarandeerde dienstverlening	5	59
europese commissie	321	224	riziv nummers	15	58
2019–2021					
sterke overheid	157	1	gegarandeerde dienstverlening	0	432
sociale bescherming	131	16	europese unie	289	380
intrafamiliaal geweld	158	88	nieuwe belastingen	27	76
sociale zekerheid	321	258	ter stemming	56	103
internationaal recht	85	26	regering michel	22	58

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Left: positive; Other parties: negative) in Dutch speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 2; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C60: Most partisan phrases, Left and Other parties, Dutch speech: periods beginning in the 2020s

Left			Other parties		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
2021–2024					
sociale bescherming	218	39	gegarandeerde dienstverlening	0	792
sterke overheid	172	1	nationale loterij	14	124
gewone mensen	143	7	nationale bank	51	120
werkende klas	126	8	state the	27	87
flexi jobs	146	40	europese commissie	219	278
2024–2025					
sterke overheid	282	2	gegarandeerde dienstverlening	0	1581
werkende mensen	311	56	flexi jobs	55	174
sociale zekerheid	563	314	werkloosheidsuitkeringen tijd	29	116
werkende klas	217	3	nationale bank	76	140
sociaal akkoord	200	26	zweedse regering	9	71

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Left: positive; Other parties: negative) in Dutch speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 2; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C61: Most partisan phrases, Government and Opposition, French speech: periods beginning in the 1830s

Government			Opposition		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1839–1841					
chemin fer	515	512	société générale	112	382
travaux publics	266	265	fil main	9	63
sucre exotique	134	133	avantages commerce	6	59
plus grande	145	144	fil mécanique	14	59
bons trésor	126	125	liste civile	17	57

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Government: positive; Opposition: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 4; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C62: Most partisan phrases, Government and Opposition, French speech: periods beginning in the 1840s

Government			Opposition		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1845–1848					
cour comptes	149	50	travaux publics	356	401
chemin fer	487	446	juge paix	40	80
garde civique	76	46	industrie linière	41	80
sucre indigène	75	47	école vétérinaire	9	32
sucre exotique	59	36	province liège	25	48
1848–1850					
société générale	58	17	chemin fer	308	611
garde civique	83	49	enseignement religieux	49	131
banque nationale	45	18	enseignement moyen	86	161
lettre voiture	35	8	conseil mines	27	89
code pénal	43	18	canal charleroy	40	99

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Government: positive; Opposition: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 4; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C63: Most partisan phrases, Government and Opposition, French speech: periods beginning in the 1850s

Government			Opposition		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1850–1852					
flandre occidentale	72	22	chemin fer	608	827
code civil	81	39	trésor public	56	115
ville wavre	33	6	droits différentiels	43	99
peine mort	32	8	travaux publics	322	372
cour comptes	82	58	dépôts mendicité	27	61
1852–1854					
flandre occidentale	99	40	commissaire arrondissement	24	78
demande concession	48	30	garde civique	265	304
banque nationale	27	13	flandre orientale	43	69
écoles moyennes	20	6	chemins vicinaux	39	63
expropriation forcée	14	2	prohibition sortie	21	43
1857–1859					
prud hommes	299	141	critique censure	29	115
conseils prud	221	87	ministres cultes	140	216
contrainte corps	107	23	chemin fer	331	393
ville anvers	100	66	administrateurs spéciaux	15	69
enseignement obligatoire	45	12	propose renvoi	49	91
1859–1861					
garde civique	87	9	vol domestique	24	129
bruxelles louvain	85	43	passe port	20	99
sucre betterave	49	17	fonds communal	82	149
cours certificat	41	9	cours légal	43	87
direct bruxelles	54	24	communes octroi	78	119

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Government: positive; Opposition: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 4; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C64: Most partisan phrases, Government and Opposition, French speech: periods beginning in the 1860s

Government			Opposition		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1861–1864					
garde civique	104	21	loi 1842	34	73
servitudes militaires	101	27	ministres cultes	39	76
citadelle nord	67	21	personnes civiles	24	54
travaux publics	271	246	réforme communale	7	35
ville anvers	83	62	instruction primaire	56	84
1864–1866					
chemin fer	523	441	station intérieure	9	55
jonction directe	26	4	loi électorale	38	66
banque nationale	38	17	responsabilité ministérielle	13	40
écoles normales	26	7	armée permanente	30	56
centrale examiné	28	10	loi communale	40	65
1866–1868					
écoles adultes	150	77	conseil communal	94	161
loi 1842	168	98	communal enghien	9	58
cour assises	78	38	suffrage universel	114	144
enseignement primaire	91	52	peine mort	124	150
code pénal	125	89	corps électoral	63	86
1868–1870					
sociétés coopératives	75	6	listes électorales	37	103
assemblée générale	97	42	conseil communal	75	109
sociétés anonymes	138	84	service militaire	56	90
écoles normales	54	13	visites domiciliaires	19	51
travail enfants	32	3	police judiciaire	7	37

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Government: positive; Opposition: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 4; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C65: Most partisan phrases, Government and Opposition, French speech: periods beginning in the 1870s

Government			Opposition		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1870–1872					
chemin fer	805	599	écoles moyennes	101	187
bassins houillers	245	159	banque nationale	71	152
compagnie bassins	102	28	écoles communales	22	65
arrondissement charleroi	36	7	suffrage universel	33	76
conseil communal	121	95	enseignement moyen	48	90
1872–1874					
juge instruction	80	28	écoles moyennes	52	185
partie civile	70	28	servitudes militaires	48	151
détention préventive	64	28	flandre occidentale	38	82
compagnie bassins	42	8	cour comptes	14	52
code instruction	54	25	sociétés coopératives	18	53
1874–1876					
juge paix	128	51	garde civique	77	277
peine mort	50	5	chambre commerce	175	309
code pénal	50	14	chemin fer	476	586
loi ventôse	54	25	travaux publics	308	357
caisses prévoyance	38	10	aucune espèce	27	57
1876–1878					
cours eau	192	130	écoles moyennes	35	123
bassins houillers	66	22	port ostende	9	58
code civil	84	46	chevaux mixtes	27	70
lit cours	30	5	listes électorales	33	74
question propriété	30	6	écoles normales	24	63
1878–1880					
chemin fer	453	349	instruction publique	114	279
écoles communales	135	81	corps électoral	42	117
marques fabrique	50	19	valeur locative	19	76
parti clérical	31	3	enseignement libre	28	80
voies navigables	52	26	contribution personnelle	18	60

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Government: positive; Opposition: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 4; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C66: Most partisan phrases, Government and Opposition, French speech: periods beginning in the 1880s

Government			Opposition		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1880–1882					
chemin fer	397	305	instruction publique	88	212
écoles communales	119	53	cardinal nina	9	83
droit pêche	73	21	conseil communal	123	196
cours eau	96	66	ministres cultes	57	110
enseignement moyen	68	43	corps électoral	52	99
1882–1884					
suffrage universel	73	39	instruction publique	99	285
droit électoral	68	35	général major	13	61
carte géologique	37	15	conseil communal	73	116
eaux vie	37	16	major gratry	12	52
droits pilotage	24	4	députation permanente	125	157
1884–1886					
chemin fer	312	212	écoles communales	167	393
droit auteur	130	36	pères famille	94	193
cuve matière	59	3	instruction publique	86	167
écoles moyennes	61	32	écoles adoptées	107	181
vaine pâture	46	19	école mixte	18	78
1886–1888					
concordat préventif	68	9	écoles communales	78	197
procureur général	71	28	conseil communal	115	166
tabac indigène	35	4	pères famille	40	86
juge instruction	53	23	écoles adoptées	47	89
partie civile	45	21	écoles libres	64	101
1888–1890					
chemin fer	401	297	sûreté publique	20	108
prud hommes	245	162	ville gand	17	66
conseils prud	172	108	agents provocateurs	4	51
pays flamand	90	45	écoles spéciales	63	102
fer vicinaux	64	36	conseil communal	44	81

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Government: positive; Opposition: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 4; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C67: Most partisan phrases, Government and Opposition, French speech: periods beginning in the 1890s

Government			Opposition		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1890–1892					
fonds communes	116	44	lieutenant général	29	105
dépôts mendicité	42	13	traitement attente	71	145
conseil supérieur	47	19	conseil communal	58	123
service hospitalier	32	5	suffrage universel	73	127
loi 1876	53	26	service personnel	48	99
1892–1894					
représentation proportionnelle	199	118	suffrage universel	178	308
juge paix	63	23	chemin fer	269	381
minimum salaire	37	6	droit électoral	67	129
traverses métalliques	35	6	députation permanente	56	107
deux langues	33	4	loi électorale	76	118
1894–1896					
droit licence	90	15	conseil communal	65	177
chemin fer	406	340	suffrage universel	44	125
droit accise	77	35	classe ouvrière	67	146
id id	42	2	députation permanente	52	99
tabac indigène	49	12	loi scolaire	11	52
1896–1898					
garde civique	254	171	classe ouvrière	43	150
unions professionnelles	278	221	suffrage universel	28	90
pourrait être	106	81	représentation proportionnelle	44	74
faire commerce	41	17	vote plural	9	31
sociétés secours	70	48	service personnel	54	74
1898–1900					
juge paix	106	43	banque nationale	136	365
fer vicinaux	62	22	classe ouvrière	36	133
chemin fer	308	270	tramways bruxellois	32	118
cour militaire	41	9	suffrage universel	52	129
conseil guerre	90	60	conseil communal	48	96

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Government: positive; Opposition: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 4; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C68: Most partisan phrases, Government and Opposition, French speech: periods beginning in the 1900s

Government			Opposition		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1900–1902					
chemin fer	405	282	suffrage universel	227	370
jeux hasard	115	31	classe ouvrière	39	121
maisons jeux	81	23	représentation proportionnelle	67	127
cercles privés	80	29	conseil communal	91	144
gare centrale	65	23	vote plural	50	95
1902–1904					
chemin fer	520	329	classe ouvrière	48	133
chef entreprise	63	19	lieutenant général	31	96
faute lourde	76	35	conseil communal	66	119
droit proportionnel	36	5	instruction publique	42	67
assurance obligatoire	98	70	heures travail	16	39
1904–1906					
repos dominical	94	54	lieutenant général	91	173
ligne défense	41	12	classe ouvrière	42	111
société belge	44	15	navire école	26	69
code civil	88	60	chemin fer	299	339
ponts chaussées	44	16	heures travail	37	65
1906–1908					
langue flamande	107	52	classe ouvrière	27	111
pays flamand	72	45	lieutenant général	19	95
code civil	47	25	conseil communal	59	98
enseignement moyen	47	25	heures travail	49	85
flamande pays	45	23	fer postes	39	68
1908–1910					
comité spécial	114	28	chemin fer	333	420
loi 1902	85	23	classe ouvrière	34	106
temps service	94	50	fer postes	34	93
effectif paix	70	35	postes télégraphes	38	89
union minière	48	18	écoles normales	41	92

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Government: positive; Opposition: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 4; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C69: Most partisan phrases, Government and Opposition, French speech: periods beginning in the 1910s

Government			Opposition		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1910–1912					
caisses prévoyance	146	41	classe ouvrière	47	157
chemin fer	463	381	lieutenant général	19	109
comité spécial	48	7	conseil communal	60	127
caisse retraite	68	32	députation permanente	65	103
gouverneur général	51	21	chef cabinet	23	60
1912–1914					
assemblée générale	45	17	classe ouvrière	45	161
bon marché	64	39	écoles communales	87	201
sous officiers	37	14	écoles adoptées	99	157
société nationale	78	55	suffrage universel	75	125
habitations bon	40	19	loi scolaire	38	84

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Government: positive; Opposition: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 4; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C70: Most partisan phrases, Government and Opposition, French speech: periods beginning in the 1920s

Government			Opposition		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1921–1923					
chemin fer	347	262	classe ouvrière	64	147
gardes champêtres	111	66	sciences arts	92	152
extrême gauche	76	38	industrie travail	59	104
grand duché	55	22	quatrième degré	14	49
pays flamand	51	29	écoles libres	35	66
1923–1925					
chemin fer	292	170	service sauvetage	9	85
union douanière	52	14	classe ouvrière	33	91
prix maxima	43	16	sciences arts	34	90
sous officiers	31	6	société nationale	47	102
point arrêt	29	4	industrie travail	99	142
1927–1929					
prud hommes	81	19	défense nationale	104	214
conseils prud	74	18	classe ouvrière	47	128
prison école	67	18	personnel enseignant	20	83
juge paix	51	11	service militaire	16	57
condamnation conditionnelle	46	9	six mois	89	123
1929–1932					
sociétés privées	34	6	industrie travail	77	171
libre échange	51	23	classe ouvrière	71	149
seconde langue	70	45	société nationale	117	180
deux langues	46	21	écoles communales	18	57
responsabilité limitée	26	2	sciences arts	69	107

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Government: positive; Opposition: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 4; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C71: Most partisan phrases, Government and Opposition, French speech: periods beginning in the 1930s

Government			Opposition		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1932–1934					
caisse nationale	77	11	classe ouvrière	101	275
allocations familiales	123	60	pleins pouvoirs	39	202
tenir compte	163	102	liste arrondissement	14	81
juge paix	87	30	allocations chômage	37	92
agents change	56	15	défense nationale	93	142
1934–1936					
juge paix	48	7	classe ouvrière	56	313
nord midi	47	13	arrêtés lois	66	305
société nations	43	23	union nationale	20	194
croix feu	27	8	gouvernement union	7	138
gouverneur général	25	5	petite épargne	55	170
1936–1939					
instruction publique	84	43	banque nationale	89	254
travaux publics	104	76	classes moyennes	86	154
santé publique	104	76	classe ouvrière	85	136
éducation physique	31	8	quarante heures	23	74
beaux arts	23	6	crédit industrie	2	38
1939–1940					
instruction publique	142	94	charges sociales	16	66
santé publique	107	61	sous officiers	32	69
société nationale	112	78	masses laborieuses	1	37
beaux arts	39	14	heures travail	19	54
écoles moyennes	39	16	allocations familiales	58	90

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Government: positive; Opposition: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 4; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C72: Most partisan phrases, Government and Opposition, French speech: periods beginning in the 1940s

Government			Opposition		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1944–1946					
prisonniers politiques	190	95	impôt exceptionnel	6	91
secours civil	59	5	emprunt forcé	9	74
sécurité sociale	72	33	impôt spécial	22	86
nations unies	56	29	amnistie fiscale	19	77
théâtre national	46	26	impôt capital	40	93
1946–1947					
assistance publique	98	18	allocations familiales	47	146
personnel enseignant	62	13	banque emission	8	78
main oeuvre	135	93	banque nationale	27	96
prisonniers politiques	131	88	sous officiers	11	58
santé publique	101	64	pouvoirs spéciaux	6	46
1947–1949					
société nationale	135	56	prisonniers politiques	48	112
chemin fer	109	46	allocations compensatoires	30	86
juge paix	76	20	coût vie	37	88
plan marshall	145	103	syndicats libéraux	7	57
arrêté loi	58	30	services publics	36	73
1949–1950					
eaux usées	54	22	question royale	97	177
cours eau	52	26	membres conseil	84	136
prince régent	52	26	libre belgique	17	62
pouvoirs représentants	32	7	vrai oui	9	48
après ordres	29	4	oui non	51	71

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Government: positive; Opposition: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 4; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C73: Most partisan phrases, Government and Opposition, French speech: periods beginning in the 1950s

Government			Opposition		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1950–1952					
experts comptables	42	9	personnel enseignant	25	96
reviseurs entreprises	46	14	classe ouvrière	25	87
fonds communes	56	25	quatre mois	33	91
enseignement public	62	32	etats unis	62	115
neuf ans	42	14	coût vie	43	95
1952–1954					
enseignement public	99	44	plan schuman	30	134
enseignement technique	102	56	classes moyennes	62	98
experts comptables	50	5	office séquestres	1	35
enseignement moyen	125	80	etats unis	63	96
pouvoirs spéciaux	56	13	victimes civiles	38	71
1954–1956					
haute autorité	73	20	instruction publique	100	159
société nationale	165	117	enseignement public	40	84
gouvernement précédent	123	79	maladie invalidité	40	82
après ordres	35	1	enseignement moyen	112	153
grande bretagne	50	20	fonds routes	15	56
1956–1958					
conseil national	160	60	allocations familiales	55	141
marché commun	164	114	pouvoirs spéciaux	5	62
secteur privé	80	37	enseignement libre	35	73
national charbonnages	69	27	travail prévoyance	47	85
oeuvre nationale	56	22	ouvriers mineurs	43	79
1958–1961					
congo belge	175	74	loi unique	61	124
table ronde	146	99	classe ouvrière	23	86
politique scientifique	40	3	personnel enseignant	11	61
ruanda urundi	104	76	pacte scolaire	35	84
gouverneur général	53	30	instruction publique	42	87

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Government: positive; Opposition: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 4; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C74: Most partisan phrases, Government and Opposition, French speech: periods beginning in the 1960s

Government			Opposition		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1961–1963					
assistance technique	66	14	réforme fiscale	92	172
nations unies	73	22	loi unique	64	144
chemin fer	156	107	assistance publique	56	136
allocations familiales	90	53	classe ouvrière	23	72
santé publique	111	79	taxe transmission	15	62
1963–1965					
assistance technique	97	19	indice prix	12	73
corps médical	147	85	prix détail	25	84
allocations familiales	93	31	hausse prix	20	66
organismes assureurs	57	26	dépenses militaires	10	52
sécurité sociale	76	45	frais production	7	44
1965–1968					
cour comptes	78	11	pouvoirs spéciaux	60	176
assistance technique	62	5	maladie invalidité	57	117
budget ordinaire	49	17	assurance maladie	66	121
coopération développement	33	2	corps médical	15	58
défense nationale	106	78	education nationale	88	122
1968–1969					
etats unis	131	70	personnes âgées	40	92
centres santé	45	4	conseil économique	119	170
question posée	133	93	revenu garanti	22	58
nations unies	39	8	ouvriers mineurs	28	58
gouvernement belge	63	34	économique wallon	35	62
1969–1971					
pays voie	132	15	police judiciaire	32	124
voie développement	130	13	classes moyennes	22	77
parlement européen	97	17	conseil culturel	56	95
coopération développement	65	10	région flamande	23	52
secteur public	72	27	chemin fer	25	53

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Government: positive; Opposition: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 4; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C75: Most partisan phrases, Government and Opposition, French speech: periods beginning in the 1970s

Government			Opposition		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1971–1974					
chemin fer	95	40	conseil culturel	64	99
travailleurs indépendants	87	34	emplois nouveaux	12	39
société nationale	64	23	partis traditionnels	3	23
pension nationale	35	2	fraude fiscale	16	35
promotion sociale	40	8	bruxelles capitale	11	30
1974–1977					
classes moyennes	137	60	aide sociale	25	117
langue allemande	64	15	conseil communal	69	121
catégories sociales	43	4	education nationale	52	104
partenaires sociaux	54	16	centres publics	7	55
économique sociale	110	74	publics aide	8	52
1977–1978					
langue allemande	68	9	pouvoirs spéciaux	49	192
régie postes	68	15	classes moyennes	55	165
education nationale	96	67	moyennes entreprises	72	145
eau heure	31	3	petites moyennes	88	156
industrie aéronautique	44	16	secteur public	66	105
1979–1981					
etats unis	92	38	taux intérêt	29	79
gouvernement belge	90	41	communauté flamande	15	59
education nationale	101	64	région bruxelles	24	63
pacte scolaire	53	16	communautés régions	35	72
coopération développement	59	24	secteur public	35	69

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Government: positive; Opposition: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 4; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C76: Most partisan phrases, Government and Opposition, French speech: periods beginning in the 1980s

Government			Opposition		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1981–1983					
classes moyennes	96	32	pouvoirs spéciaux	417	657
troisième circuit	71	17	cockerill sambre	7	98
education nationale	210	159	fraude fiscale	23	92
temps travail	76	32	pleins pouvoirs	8	63
circuit travail	52	13	sécurité sociale	143	196
1983–1985					
temps travail	102	41	pouvoirs spéciaux	69	153
politique scientifique	61	20	modération salariale	48	89
education nationale	173	136	région wallonne	54	88
chefs établissements	56	21	sécurité sociale	98	131
année scolaire	55	24	pouvoir achat	20	53
1985–1987					
secteur textile	63	6	pouvoirs spéciaux	182	335
juge instruction	62	13	libre belgique	21	68
candidats réfugiés	49	14	francophones majorité	6	50
police judiciaire	43	10	publicité commerciale	27	66
procureur général	49	17	education nationale	115	150
1988–1989					
cour comptes	65	10	pacte scolaire	56	135
communauté germanophone	66	17	conseil communal	54	121
charges intérêts	52	13	culturelles nationales	13	74
rigueur budgétaire	40	4	etat central	126	184
crédits provisoires	45	10	publics institutions	19	74
1989–1991					
juge instruction	143	50	loi financement	25	90
nations unies	106	51	culturelles nationales	28	86
procureur général	91	39	scientifiques culturelles	31	82
police judiciaire	93	43	publics institutions	32	79
etat major	47	15	institutions scientifiques	33	75

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Government: positive; Opposition: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 4; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C77: Most partisan phrases, Government and Opposition, French speech: periods beginning in the 1990s

Government			Opposition		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1991–1993					
aide sociale	76	34	loyauté fédérale	24	96
travailleurs sociaux	46	11	réforme etat	113	178
chemin fer	58	23	francophones périphérie	9	68
comité avis	47	11	communauté germanophone	103	162
partis politiques	69	38	etat fédéral	132	183
1993–1995					
êtres humains	127	61	campagne information	19	64
traite êtres	110	46	allocations familiales	40	75
congé politique	77	23	dépenses électorales	63	97
juge instruction	63	17	électeurs inscrits	17	46
remise dette	47	8	campagne électorale	19	48
1995–1997					
sécurité sociale	669	558	pouvoirs spéciaux	72	399
dette publique	105	30	pression fiscale	55	113
etats membres	91	34	fiscale parafiscale	41	93
union européenne	116	61	peines incompressibles	22	70
monnaie unique	86	38	libération conditionnelle	45	90
1997–1999					
etats membres	97	17	cour comptes	46	147
sécurité sociale	139	75	nouveau belgique	2	93
consommation drogues	42	3	juge instruction	118	194
union européenne	117	81	purement simplement	67	123
extrême droite	68	34	comité supérieur	18	69
1999–2001					
police locale	79	25	coopération développement	44	144
services police	88	41	nations unies	21	84
criminalité organisée	55	15	principe précaution	32	83
organisations syndicales	58	23	purement simplement	74	121
temps travail	70	49	pénale internationale	13	51

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Government: positive; Opposition: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 4; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C78: Most partisan phrases, Government and Opposition, French speech: periods beginning in the 2000s

Government			Opposition		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
2001–2003					
police locale	99	24	cour arbitrage	25	183
criminalité organisée	60	12	soins palliatifs	32	140
accords bilatéraux	51	9	réforme polices	68	131
organisations syndicales	55	14	bruxelles hal	7	51
nations unies	67	30	arc ciel	48	86
2003–2005					
service universel	71	20	entités fédérées	40	109
santé publique	130	82	gouvernement fédéral	40	95
procureur général	59	22	titres services	16	68
peine travail	39	8	créances alimentaires	54	93
établissements pénitentiaires	39	13	communautés régions	29	68
2005–2007					
cour arbitrage	144	92	one shot	7	55
services police	72	29	cour comptes	21	66
juge instruction	109	69	demandeurs asile	21	63
organisations syndicales	45	14	nouvelle fois	22	63
loi 2003	35	10	centres fermés	32	72
2007–2010					
santé publique	129	47	sécurité sociale	107	146
pouvoir achat	155	80	fonds vieillissement	14	49
partenaires sociaux	151	110	entités fédérées	30	59
plan relance	76	45	intérêts notionnels	58	81
maladies chroniques	48	19	organisation travaux	5	28

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Government: positive; Opposition: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 4; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C79: Most partisan phrases, Government and Opposition, French speech: periods beginning in the 2010s

Government			Opposition		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
2010–2012					
accord gouvernement	105	58	intérêts notionnels	34	107
abus sexuels	74	31	temps partiel	8	57
coopération développement	61	18	crédit temps	8	42
nations unies	60	29	autorité nationale	3	31
nom groupe	44	25	efficacité énergétique	5	31
2012–2014					
accord gouvernement	121	43	intérêts notionnels	66	285
santé publique	82	36	région bruxelloise	32	109
partenaires sociaux	160	115	fraude fiscale	63	125
coopération développement	68	24	ajustement budgétaire	22	69
compétitivité entreprises	54	14	système fiscal	20	67
2014–2016					
accord gouvernement	230	118	saut index	94	256
réforme pensions	110	37	sécurité sociale	103	179
âge légal	77	38	services publics	67	143
entités fédérées	94	56	fraude fiscale	64	139
63 ans	43	6	tax shift	94	167
2016–2019					
pouvoir achat	141	45	sécurité sociale	136	211
extrêmement important	120	30	voitures société	18	67
entités fédérées	151	89	saut index	16	55
pension complémentaire	72	12	fraude fiscale	55	94
commission européenne	154	102	paradis fiscaux	6	44
2019–2021					
droit passerelle	225	79	comptes titres	19	101
union européenne	223	149	paradis fiscaux	14	90
travailleurs indépendants	73	23	sécurité sociale	77	152
transition énergétique	53	5	pouvoirs spéciaux	33	98
niveau européen	95	53	taxe comptes	13	63

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Government: positive; Opposition: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 4; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C80: Most partisan phrases, Government and Opposition, French speech: periods beginning in the 2020s

Government			Opposition		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
2021–2024					
travailleurs travailleuses	70	11	courtes peines	9	56
femmes hommes	72	19	police judiciaire	46	88
celles ceux	64	22	ose espérer	11	52
toutes tous	49	12	oui nee	1	41
nations unies	63	27	abstentions totaal	1	41
2024–2025					
accord gouvernement	359	260	travailleurs travailleuses	10	87
union européenne	291	193	67 ans	5	74
transition énergétique	50	5	tableaux budgétaires	14	80
travailleurs sociaux	68	28	travailleuses travailleurs	14	80
entités fédérées	83	43	celles ceux	111	175

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Government: positive; Opposition: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 4; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C81: Most partisan phrases, Government and Opposition, Dutch speech: periods beginning in the 1900s

Government			Opposition		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1900–1902					
vlaamsche volk	1133	96	algemeen stemrecht	562	568
			enkele woorden	380	385
			vlaamsche land	210	213
			onderlingen bijstand	278	280
			werkende klas	370	373
1902–1904					
10 jaar	1	1			
sommige vragen	1	1			
sommigen wel	1	1			
sommige zaken	1	1			
sommigen zeggen	1	1			
1904–1906					
alleen kwestie	1	1			
belgische nationale	1	1			
10 jaar	23	23			
speciaal aandacht	1	1			
spaaren lijfrentekas	1	1			
1906–1908					
vlaamsche volk	493	310	nationale bank	185	619
mager vee	442	439	bedienden werklieden	129	419
jaar dienst	451	449			
west vlaanderen	352	350			
werkende klas	352	350			
1908–1910					
kleine bedienden	299	91	burgerlijke stand	42	255
spoorwegen posterijen	550	548	ambtenaar burgerlijke	50	182
posterijen telegrafen	384	383	vlaamsche volk	264	367
vlaamsche taal	329	328	werkende klas	423	423
jaar dienst	336	335	67 jaar	57	57

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Government: positive; Opposition: negative) in Dutch speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 4; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C82: Most partisan phrases, Government and Opposition, Dutch speech: periods beginning in the 1910s

Government			Opposition		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1910–1912					
onderlingen bijstand	488	486	private nijverheid	54	278
kleine bedienden	355	354			
vlaamsche land	359	358			
werkende klas	338	337			
twintig jaar	358	357			
1912–1914					
algemeen stemrecht	868	306	lagere scholen	61	517
officieele scholen	325	323	vlaamsche volk	220	480
vrije scholen	315	313	lager onderwijs	110	337
één grote	284	282	ten honderd	87	208
jaar dienst	268	266	tweede taal	96	217

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Government: positive; Opposition: negative) in Dutch speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 4; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C83: Most partisan phrases, Government and Opposition, Dutch speech: periods beginning in the 1920s

Government			Opposition		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1921–1923					
middelbaar onderwijs	520	116	oud strijders	118	301
lager onderwijs	359	202	vlaamsche soldaten	54	166
vierde graad	160	40	werkende klas	82	192
vlaamsche land	172	127	pensioen ontvangen	24	101
landen tuinbouw	61	27	pensioen toegekend	23	87
1923–1925					
vierde graad	154	46	vlaamsche volk	99	177
kroostrijke gezinnen	103	32	vóór oorlog	345	346
openbare werken	294	240	kosteloos pensioen	266	266
lagere scholen	185	169	nationale maatschappij	310	311
			65 jaar	269	269
1927–1929					
openbare werken	476	195	oud strijders	184	459
kroostrijke gezinnen	274	114	zes maanden	323	474
onderwijzend personeel	154	50	fransch belgisch	39	176
speciaal fonds	86	13	gewestelijke indeeling	72	204
één oplossing	125	61	vlaamsche volk	76	177
1929–1932					
kroostrijke gezinnen	233	72	vlaamsche volk	111	314
ambachten neringen	134	7	fransch belgisch	13	149
kadastraal inkomen	149	26	militair akkoord	2	126
openbare werken	446	335	belgische staat	49	162
kamers ambachten	95	16	belgische regering	87	180

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Government: positive; Opposition: negative) in Dutch speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 4; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C84: Most partisan phrases, Government and Opposition, Dutch speech: periods beginning in the 1930s

Government			Opposition		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1932–1934					
nationale maatschappij	431	163	vlaamsche volk	94	281
maatschappij buurtspoorwegen	238	32	besluit wet	124	231
technisch onderwijs	174	26	vlaamsche katholieken	12	106
wet 1930	165	19	openbaren onderstand	36	124
sociale voorzorg	262	130	militair akkoord	5	69
1934–1936					
openbare werken	511	356	vlaamsche volk	75	471
kroostrijke gezinnen	204	96	oud strijders	59	333
kadastraal inkomen	171	94	belgische staat	36	208
technisch onderwijs	90	21	fransch belgisch	90	228
onze nationale	117	52	vlaamsche land	130	265
1936–1939					
openbare werken	271	138	vlaamsche volk	163	326
officieel onderwijs	153	38	vlaamsche land	181	336
sociale voorzorg	187	96	belgische staat	49	160
nationale maatschappij	332	243	buitenlandsche politiek	88	179
middelbare scholen	106	38	wij vlaamsche	27	106
1939–1940					
technisch onderwijs	446	72	vlaamsche volk	51	184
nationale maatschappij	446	265	vlaamsche soldaten	26	135
buitengewone begroting	134	33	vlaamsche land	102	178
openbare werken	301	230	uitsluitend fransch	23	69
onze nationale	135	83	wij vlaamsche	19	59

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Government: positive; Opposition: negative) in Dutch speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 4; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C85: Most partisan phrases, Government and Opposition, Dutch speech: periods beginning in the 1940s

Government			Opposition		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1944–1946					
ambachten neringen	184	34	belasting kapitaal	60	324
arbeid sociale	162	161	dienst pensioenen	130	365
tijdens bezetting	204	203	onze jeugd	107	107
gedurende oorlog	235	234	voldoende reden	1	1
moeten betalen	138	137	rol overheid	1	1
1946–1947					
economische unie	236	94	technisch onderwijs	107	345
politieke gevangenen	265	132	dienst pensioenen	44	197
economische problemen	83	28	ambachten neringen	13	159
internationale organisatie	69	22	onderwijzend personeel	74	141
sint niklaas	73	45	landen tuinbouw	43	103
1947–1949					
nationale maatschappij	327	186	werkende klas	23	256
technisch onderwijs	177	53	verenigde staten	130	324
middelbaar onderwijs	154	95	sovjet unie	11	184
lager onderwijs	87	37	vlaamse volk	24	137
kroostrijke gezinnen	106	74	onze hoop	8	107
1949–1950					
vrij verzekerden	183	183	algemene paritaire	29	156
eenheid land	158	158			
arbeid sociale	170	170			
wetgevende macht	155	154			
sociale voorzorg	169	168			

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Government: positive; Opposition: negative) in Dutch speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 4; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C86: Most partisan phrases, Government and Opposition, Dutch speech: periods beginning in the 1950s

Government			Opposition		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1950–1952					
maatschappelijke zekerheid	276	159	twintig maanden	55	195
technisch onderwijs	192	76	openbare werken	173	290
landen tuinbouw	136	40	gehuwde vrouw	40	111
gebrekkigen verminkten	135	40	burgerlijke oorlogsslachtoffers	13	78
belgisch congo	162	67	sociale voorzorg	231	292
1952–1954					
technisch onderwijs	361	74	heilige stoel	48	317
zelfstandige arbeiders	334	201	achttien maanden	31	253
hoge autoriteit	190	68	twintig maanden	78	272
belgisch congo	248	143	buitengewone begroting	42	154
gemeenschappelijke markt	116	31	militaire diensttijd	10	77
1954–1956					
vier jaar	309	118	technisch onderwijs	19	277
vorige regering	259	156	openbaar onderwijs	172	317
openbare werken	268	192	sociale voorzorg	182	323
jonge mannen	82	12	vrij onderwijs	70	206
rijksmiddelbare school	94	29	vrij verzekerden	80	215
1956–1958					
jonge mannen	183	11	sociale voorzorg	136	276
openbare werken	315	252	technisch onderwijs	48	166
gemeenschappelijke markt	164	105	arbeid sociale	171	250
twintig maanden	71	22	belgisch congo	139	218
verenigde staten	140	95	kroostrijke gezinnen	35	91
1958–1961					
openbare werken	291	207	nationale bank	17	174
haven antwerpen	77	36	vorige regering	64	166
regionale economische	79	38	belgisch congo	137	217
liberale groep	55	15	hoger onderwijs	42	120
economische politiek	147	110	openbare schuld	34	108

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Government: positive; Opposition: negative) in Dutch speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 4; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C87: Most partisan phrases, Government and Opposition, Dutch speech: periods beginning in the 1960s

Government			Opposition		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1961–1963					
openbare werken	198	86	fiscale hervorming	214	488
dienst pensioenen	145	46	arbeiders bedienden	115	250
burgerlijke oorlogsslachtoffers	42	3	nationale opvoeding	194	309
haven antwerpen	97	59	kleine middelgrote	79	174
vitaal belang	50	19	vlaamse land	94	189
1963–1965					
openbare werken	286	130	tweede taal	64	264
gewestelijke investeringsmaatschappijen	55	6	vlaamse gemeenschap	21	201
nationale maatschappij	122	73	nationale opvoeding	175	337
burgerlijke oorlogsslachtoffers	48	3	hoger onderwijs	80	237
technisch onderwijs	67	22	vlaamse land	127	257
1965–1968					
openbare werken	278	197	vrije universiteiten	36	127
jaar 1966	106	36	vlaamse gemeenschap	19	94
wetenschappelijk onderzoek	121	59	nationale opvoeding	254	319
jaar 1967	77	23	universitaire expansie	60	121
gemengde loopbaan	80	32	generaal toerisme	6	64
1968–1969					
openbare onderstand	163	69	technische ingenieurs	74	259
gewaarborgd inkomen	118	60	nationale opvoeding	206	333
paritaire comités	59	18	hoger onderwijs	78	165
collectieve arbeidsovereenkomsten	51	10	parlementaire vraag	59	143
technische bijstand	53	15	culturele autonomie	102	177
1969–1971					
openbare werken	361	257	nationale opvoeding	135	242
nationale maatschappij	184	97	lager onderwijs	40	138
hoger onderwijs	171	95	universitaire expansie	13	109
minder validen	94	21	vlaamse gemeenschap	20	98
verzekering geven	70	21	economische hogeschool	11	85

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Government: positive; Opposition: negative) in Dutch speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 4; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C88: Most partisan phrases, Government and Opposition, Dutch speech: periods beginning in the 1970s

Government			Opposition		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1971–1974					
minder validen	213	119	traditionele partijen	7	114
buitengewoon onderwijs	82	17	academisch ziekenhuis	13	115
openbare schuld	78	15	bevoegdheid cultuurraden	10	68
verzekering geven	87	26	militaire technische	7	63
openbare werken	384	336	syndicale vrijheid	4	59
1974–1977					
hoger onderwijs	181	54	openbare werken	233	336
buitengewoon onderwijs	107	35	nationale opvoeding	183	263
sociale zekerheid	120	54	vlaamse aangelegenheden	33	103
lager onderwijs	106	45	langstlevende echtgenoot	17	77
sociale voorzorg	143	96	vlaamse gemeenschap	11	66
1977–1978					
openbare werken	276	84	nationale investeringsmaatschappij	29	245
nationale opvoeding	168	67	openbare holding	15	174
vlaamse gemeenschap	134	36	zuid afrika	26	138
openbaar vervoer	127	65	toegevoegd eerste	15	121
sociale voorzorg	88	43	vlaamse aangelegenheden	41	127
1979–1981					
hoofdstedelijk gebied	122	27	vlaamse gemeenschap	151	292
openbare sector	105	46	studiemeester opvoeder	10	72
privé sector	99	49	vlaanderen wallonië	52	113
sociale gesprekspartners	68	23	gemeenschappen gewesten	115	170
rekening houdend	146	101	centrale regering	5	58

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Government: positive; Opposition: negative) in Dutch speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 4; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C89: Most partisan phrases, Government and Opposition, Dutch speech: periods beginning in the 1980s

Government			Opposition		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1981–1983					
europese gemeenschap	189	90	vlaamse gemeenschap	62	153
europese commissie	133	43	centrale regering	22	98
cockerill sambre	242	165	openbare ziekenhuizen	29	80
vrije beroepen	104	34	sociale zekerheid	252	294
sovjet unie	98	32	openbare werken	100	137
1983–1985					
europese gemeenschap	206	101	vlaamse gemeenschap	48	109
stand gebracht	85	24	centrale regering	9	65
nationale sectoren	170	118	onze mening	14	56
vijf nationale	60	13	fiscale amnestie	8	49
westeuropese unie	47	6	sociale zekerheid	263	304
1985–1987					
europese gemeenschap	309	102	politieke vluchtelingen	48	140
sociale zekerheid	282	194	centrale regering	6	82
tijdens komende	96	13	openbaar ambt	66	124
netto financieren	136	55	vier jaar	118	168
financieren saldo	135	68	vlaamse meerderheid	2	47
1988–1989					
gemeenschappen gewesten	352	168	wetenschappelijke culturele	24	134
netto financieren	156	80	diensten nationale	28	132
financieren saldo	150	83	openbare financiën	100	197
sociale zekerheid	441	385	culturele instellingen	26	118
wij wensen	111	58	nationale wetenschappelijke	28	117
1989–1991					
europese gemeenschap	330	190	derde fase	114	211
europees parlement	159	30	vergaderingen vrijdag	64	149
gemeenschappen gewesten	137	65	openbaar vervoer	71	143
politieke unie	90	31	wetenschappelijke culturele	50	117
netto financieren	75	16	diensten nationale	50	117

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Government: positive; Opposition: negative) in Dutch speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 4; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C90: Most partisan phrases, Government and Opposition, Dutch speech: periods beginning in the 1990s

Government			Opposition		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1991–1993					
bijzondere wet	292	179	sint michielsakkoord	281	438
europese gemeenschap	172	107	halle vilvoorde	49	119
hoofdstedelijk gewest	78	32	vlaanderen wallonië	48	110
voorstel bijzondere	44	14	sociale zekerheid	622	672
vlaams parlement	99	75	vlaamse beweging	34	75
1993–1995					
globaal plan	403	250	sociale zekerheid	737	1092
wou stemmen	102	22	gelden stemming	12	84
onbemande camera	66	7	sociale programmawet	56	120
verdrag maastricht	98	52	sociaal pact	141	202
intergouvernementele conferentie	47	9	vlaamse regering	25	84
1995–1997					
procureur generaal	376	272	traditionele partijen	9	83
gerechtelijk onderzoek	128	48	vlaams parlement	17	88
gerechtelijke politie	121	49	sociale zekerheid	1081	1149
onbemande camera	111	40	politieke cultuur	32	90
sociaal overleg	139	71	politieke benoemingen	15	69
1997–1999					
procureur konings	179	84	vlaams parlement	76	231
voorwaardelijke invrijheidstelling	136	50	hoog comité	67	155
europese unie	198	128	sociale zekerheid	133	219
college procureurs	105	50	politieke verantwoordelijkheid	41	109
medisch dossier	76	21	comité toezicht	60	121
1999–2001					
europese unie	229	129	federale politie	43	190
actieve welvaartsstaat	187	96	halle vilvoorde	31	163
oudere werknemers	71	23	vlaams parlement	72	142
ten volle	88	40	politieke cultuur	34	97
chronische zieken	70	23	vlaamse partijen	4	65

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Government: positive; Opposition: negative) in Dutch speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 4; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C91: Most partisan phrases, Government and Opposition, Dutch speech: periods beginning in the 2000s

Government			Opposition		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
2001–2003					
europese unie	267	134	vlaams parlement	60	300
duurzame ontwikkeling	138	59	veiligheid staat	50	202
sociale partners	118	58	snel belg	13	123
verenigde staten	152	102	belg wet	13	120
binnen kader	60	14	lambermont akkoord	38	136
2003–2005					
europese unie	354	206	snel belg	6	220
federale politie	154	27	belg wet	6	217
sociale partners	213	95	halle vilvoorde	108	233
europees parlement	181	71	fiscale amnestie	54	177
oudere werknemers	119	63	vlaams parlement	125	221
2005–2007					
europese unie	344	145	snel belg	11	123
sociale partners	228	126	belg wet	11	122
europees parlement	105	25	nieuwe belastingen	4	87
sociale zekerheid	219	145	federaal agentschap	15	84
buitenlandse werknemers	55	13	pairé avec	3	64
2007–2010					
europese unie	226	132	halle vilvoorde	58	186
opel antwerpen	126	39	vlaams parlement	46	125
federale politie	92	19	hebt gedaan	31	97
voorlopige hechtenis	67	20	vlaamse partijen	9	67
nmbs groep	55	11	binnen regering	48	103

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Government: positive; Opposition: negative) in Dutch speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 4; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C92: Most partisan phrases, Government and Opposition, Dutch speech: periods beginning in the 2010s

Government			Opposition		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
2010–2012					
sociale partners	200	123	seksueel misbruik	91	173
financiële instellingen	87	38	notionele intrestaftrek	36	115
neemt minuut	52	3	halle vilvoorde	30	89
stilte acht	52	3	onze amendementen	11	61
financiële markten	77	29	vlaamse partijen	19	67
2012–2014					
europese commissie	321	178	halle vilvoorde	52	216
sociale partners	311	206	gemiste kans	28	164
sociaal overleg	135	67	vlaamse partijen	18	137
elektronisch toezicht	146	83	constitutieve autonomie	10	74
erg belangrijk	67	19	politieke benoemingen	7	63
2014–2016					
sociale partners	434	295	tax shift	181	402
economische groei	176	67	vorige regering	238	352
gegarandeerde dienstverlening	107	5	grote vermogens	25	118
onze economie	117	49	saoedi arabië	31	122
nationale bank	179	119	europese unie	245	329
2016–2019					
sociale zekerheid	354	198	saudi arabië	131	252
gegarandeerde dienstverlening	68	4	zware beroepen	97	170
riziv nummers	69	6	gemiste kans	27	84
flexi jobs	69	30	tweede pijler	30	86
verenigd koninkrijk	89	53	fiscale rechtvaardigheid	18	66
2019–2021					
tijdelijke werkloosheid	295	155	europese unie	285	417
sociale partners	205	100	nieuwe belastingen	8	103
komende weken	133	40	sociale zekerheid	218	304
seksueel geweld	101	29	intrafamiliaal geweld	65	132
hernieuwbare energie	104	43	onze patiënten	14	69

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Government: positive; Opposition: negative) in Dutch speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 4; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C93: Most partisan phrases, Government and Opposition, Dutch speech: periods beginning in the 2020s

Government			Opposition		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
2021–2024					
sociaal tarief	446	217	europese commissie	203	323
hernieuwbare energie	201	59	nationale loterij	59	144
vorige regering	186	79	gemiste kans	10	89
stappen gezet	115	34	sociale zekerheid	184	252
onze samenleving	151	76	tijdens bespreking	60	121
2024–2025					
sterke overheid	109	3	flexi jobs	52	182
eerlijke bijdrage	110	18	nieuwe belastingen	19	138
werken we	111	22	moeten betalen	103	190
mensen werken	188	101	sterkste schouders	154	218
gaan werken	148	61	asiel migratie	64	128

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Government: positive; Opposition: negative) in Dutch speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 4; covariates: female, province. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C94: Most partisan phrases, Flemish districts and Walloon districts, French speech: one period every ten years

Flemish districts			Walloon districts		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1839–1841					
sucre indigène	150	49	société générale	156	248
sucre exotique	149	95	tribunal commerce	18	104
fil mécanique	49	7	publicité concurrence	78	141
gouvernement hollandais	69	44	ville gand	7	24
port ostende	23	4	juge paix	24	36
1848–1850					
chemin fer	531	307	conseil mines	29	67
enseignement moyen	143	77	juge paix	11	36
industrie linière	71	23	société générale	38	62
garde civique	88	46	code pénal	21	45
instruction primaire	73	32	cour comptes	33	56
1859–1861					
écoles normales	125	4	canal charleroi	40	262
conseil communal	124	52	canalisation meuse	7	103
élève universitaire	112	58	travaux publics	280	360
examen écrit	60	9	code pénal	101	168
flandre occidentale	57	11	communes octroi	68	128
1868–1870					
députation permanente	224	110	sociétés anonymes	66	186
listes électorales	109	41	assemblée générale	27	136
lieutenant général	96	30	sociétés commandite	35	101
conseil communal	124	76	code pénal	56	121
flandre orientale	55	10	chemin fer	289	348
1878–1880					
députation permanente	236	84	chemin fer	262	764
conseil communal	177	84	travaux publics	269	384
instruction publique	227	144	voies navigables	25	78
valeur locative	85	7	canal charleroi	8	57
administration communale	161	85	ministres cultes	59	107
1888–1890					
prud hommes	262	183	tribunal commerce	33	95
pays flamand	98	24	commis greffiers	6	56
conseils prud	183	117	général major	12	57
ville gand	65	9	sûreté publique	22	63
flandre occidentale	45	8	caisses prévoyance	15	52
1898–1900					
représentation proportionnelle	345	262	banque nationale	165	235
banque etat	73	30	chemin fer	266	315
fer vicinaux	61	25	députation permanente	68	93
société nationale	44	9	contrat travail	111	133
banque émission	54	24	suffrage universel	64	86
1908–1910					
travaux publics	117	52	prud hommes	129	251
administration communale	117	58	conseils prud	83	186
effectif paix	83	26	neuf heures	7	90
temps service	92	41	écoles normales	31	110
collège échevinal	66	16	huit heures	38	94
1918–1919					
régions dévastées	87	19	traité paix	26	95
flandre occidentale	66	9	juge paix	23	66
administration communale	123	78	chemin fer	139	176
députation permanente	56	18	prix revient	11	46
commissaire royal	28	4	tribunaux dommages	63	95
1929–1932					
enseignement moyen	88	25	chemin fer	152	241
sciences arts	108	58	société nationale	128	174
écoles libres	72	26	impôt foncier	20	54
application loi	106	62	défense nationale	81	107
allocations familiales	89	46	droits douane	19	43
1939–1940					
crédit anversois	155	18	sous officiers	11	45
conseil culturel	115	9	prix revient	38	67
santé publique	104	42	bénéfices exceptionnels	15	43
supérieur contrôle	63	2	charges sociales	16	36
pleins pouvoirs	69	10	pommes terre	24	44

Table C94: Most partisan phrases, Flemish districts and Walloon districts, French speech: one period every ten years (continued)

Flemish districts			Walloon districts		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1949–1950					
société nationale	143	39	travailleurs indépendants	6	53
bureaux dépouillement	76	16	prisonniers guerre	40	80
code électoral	75	26	conseil national	31	71
nationale habitations	62	18	national charbonnages	11	50
logements bon	55	12	prix revient	47	85
1958–1961					
table ronde	209	59	allocations familiales	45	120
gouverneur général	116	19	conseil national	32	103
congo belge	142	55	haute autorité	31	81
ruanda urundi	128	47	national charbonnages	20	62
force publique	97	21	travailleurs indépendants	26	66
1965–1968					
etats unis	207	108	autoroute wallonie	12	73
dette publique	115	19	plan restructuration	0	17
national brut	121	30	santé publique	66	82
travailleurs indépendants	125	35	province luxembourg	15	30
produit national	123	40	ouvriers mineurs	41	51

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Flemish districts: positive; Walloon districts: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 22; covariates: family, female. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C95: Most partisan phrases, Flemish districts and Walloon districts, French speech: periods beginning in the 1830s

Flemish districts			Walloon districts		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1831–1833					
flandre orientale	121	34	conseil guerre	49	125
flandre occidentale	75	12	gouvernement provisoire	97	155
ville gand	68	10	excès pouvoir	13	69
près cour	80	32	pommes terre	80	117
tribunal commerce	38	18	cour cassation	386	386
1833–1835					
prix céréales	45	4	sociétés anonymes	24	70
bas prix	40	17	ministres cultes	30	68
session ordinaire	27	6	cour comptes	69	101
loi électorale	34	14	opinion projet	7	37
députation conseil	33	14	élection directe	29	55
1835–1837					
chemin fer	228	72	los renten	114	286
société générale	103	45	bois communaux	9	72
chambre représentans	68	29	cour bruxelles	26	70
contribution foncière	75	38	loi 1810	33	72
caissier général	36	4	mis circulation	13	47
1837–1839					
sous officiers	77	14	chemin fer	280	489
flandre orientale	66	22	droit timbre	26	68
traité paix	56	20	grand duché	16	32
port ostende	40	5	plus grande	226	226
non activité	42	8	travaux publics	226	226
1839–1841					
sucre indigène	150	49	société générale	156	248
sucre exotique	149	95	tribunal commerce	18	104
fil mécanique	49	7	publicité concurrence	78	141
gouvernement hollandais	69	44	ville gand	7	24
port ostende	23	4	juge paix	24	36

Notes: The five phrases with the largest partisanship $\hat{\xi}_{jt}$ for each group (Flemish districts: positive; Walloon districts: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 22; covariates: family, female. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C96: Most partisan phrases, Flemish districts and Walloon districts, French speech: periods beginning in the 1840s

Flemish districts			Walloon districts		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1841–1843					
travaux publics	247	190	instruction primaire	45	109
canalisation campine	65	14	deux navires	6	44
canal zelzaete	51	10	an xii	13	43
sucre indigène	120	85	ancienne majorité	4	32
industrie linière	57	27	morale religion	17	40
1843–1845					
cour appel	123	52	chemin fer	694	1093
domicile secours	59	13	cahier charges	37	90
flandre occidentale	58	13	trésor public	40	84
ville gand	50	6	entre sambre	25	57
avocat général	52	9	canal latéral	21	50
1845–1848					
cour comptes	164	82	chemin fer	404	683
juge paix	60	24	travaux publics	330	449
bons trésor	70	35	ponts chaussées	22	63
flandre occidentale	52	18	santo thomas	7	37
dépôts mendicité	51	20	trésor public	37	63
1848–1850					
chemin fer	531	307	conseil mines	29	67
enseignement moyen	143	77	juge paix	11	36
industrie linière	71	23	société générale	38	62
garde civique	88	46	code pénal	21	45
instruction primaire	73	32	cour comptes	33	56

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Flemish districts: positive; Walloon districts: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 22; covariates: family, female. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C97: Most partisan phrases, Flemish districts and Walloon districts, French speech: periods beginning in the 1850s

Flemish districts			Walloon districts		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1850–1852					
chemin fer	917	558	lettres gage	66	172
travaux publics	411	322	crédit foncier	71	172
flandre occidentale	70	16	code civil	28	90
ville gand	54	1	administration forestière	3	50
trésor public	101	60	hypothèque judiciaire	11	55
1852–1854					
chemin fer	991	738	conseil communal	74	162
travaux publics	543	419	droit commun	16	54
flandre occidentale	107	22	conseils milice	5	40
flandre orientale	97	15	administration forestière	2	35
cours eau	122	43	province namur	12	43
1854–1856					
canal schipdonck	55	16	chemin fer	877	1124
flandre orientale	56	19	minerai fer	19	131
cours eau	44	11	luttre denderleeuw	45	111
contich lierre	46	15	code pénal	35	97
ville anvers	38	7	fer fort	3	64
1856–1857					
examen écrit	113	48	chemin fer	321	458
balance romaine	53	11	travaux publics	204	247
écoles moyennes	57	20	province namur	9	47
élève universitaire	112	76	civil élémentaire	29	63
langue flamande	50	16	libre entrée	54	82
1857–1859					
prud hommes	267	185	ministres cultes	102	310
conseils prud	193	136	code pénal	84	286
chemin fer	376	327	critique censure	21	122
flandre occidentale	59	14	canal charleroi	27	94
chambre commerce	81	38	ville anvers	95	133
1859–1861					
écoles normales	125	4	canal charleroi	40	262
conseil communal	124	52	canalisation meuse	7	103
élève universitaire	112	58	travaux publics	280	360
examen écrit	60	9	code pénal	101	168
flandre occidentale	57	11	communes octroi	68	128

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Flemish districts: positive; Walloon districts: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 22; covariates: family, female. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C98: Most partisan phrases, Flemish districts and Walloon districts, French speech: periods beginning in the 1860s

Flemish districts			Walloon districts		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1861–1864					
servitudes militaires	92	30	canal charleroi	20	84
chemin fer	423	376	juge paix	67	121
peinture murale	57	10	réforme communale	2	49
flandre occidentale	47	11	travaux publics	247	288
langue flamande	42	7	bourses études	32	63
1864–1866					
général chazal	55	14	travaux publics	310	414
flandre orientale	45	6	code pénal	41	94
flandre occidentale	43	5	dépenses électorales	12	47
armée permanente	61	24	bourses études	32	64
lieutenant général	43	15	banque nationale	17	48
1866–1868					
suffrage universel	165	77	code pénal	62	164
conseil communal	167	93	chemin fer	210	294
lieutenant général	65	19	province namur	10	59
écoles gardiennes	40	11	peine mort	117	159
tribunal commerce	96	69	ministres cultes	28	63
1868–1870					
députation permanente	224	110	sociétés anonymes	66	186
listes électorales	109	41	assemblée générale	27	136
lieutenant général	96	30	sociétés commandite	35	101
conseil communal	124	76	code pénal	56	121
flandre orientale	55	10	chemin fer	289	348

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Flemish districts: positive; Walloon districts: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 22; covariates: family, female. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C99: Most partisan phrases, Flemish districts and Walloon districts, French speech: periods beginning in the 1870s

Flemish districts			Walloon districts		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1870–1872					
écoles normales	213	99	chemin fer	480	947
conseil communal	170	78	bassins houillers	166	288
écoles primaires	119	43	compagnie bassins	14	117
ville gand	64	5	travaux publics	366	431
écoles communales	78	24	société générale	30	84
1872–1874					
ville gand	99	9	chemin fer	416	905
ville anvers	144	63	travaux publics	389	448
garde civique	111	55	fer athus	0	43
juge instruction	86	31	grand luxembourg	7	47
flandre occidentale	70	19	commissaire arrondissement	30	69
1874–1876					
domicile secours	152	40	chemin fer	433	677
juge paix	111	52	garde civique	97	213
fonds communes	71	14	travaux publics	310	385
commerce ostende	44	1	pommes terre	34	93
flandre occidentale	49	7	exercice profession	21	54
1876–1878					
port ostende	76	1	chemin fer	319	456
cours eau	211	162	ministres cultes	67	112
députation permanente	193	161	travaux publics	258	297
flandre occidentale	51	21	chevaux mixtes	30	66
ville gand	36	11	fer athus	0	33
1878–1880					
députation permanente	236	84	chemin fer	262	764
conseil communal	177	84	travaux publics	269	384
instruction publique	227	144	voies navigables	25	78
valeur locative	85	7	canal charleroi	8	57
administration communale	161	85	ministres cultes	59	107

Notes: The five phrases with the largest partisanship $\hat{\xi}_{jt}$ for each group (Flemish districts: positive; Walloon districts: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 22; covariates: family, female. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C100: Most partisan phrases, Flemish districts and Walloon districts, French speech: periods beginning in the 1880s

Flemish districts			Walloon districts		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1880–1882					
commissaire spécial	310	38	chemin fer	269	529
députation permanente	431	172	travaux publics	238	330
administration communale	242	101	province namur	21	65
conseil communal	199	119	écoles communales	67	97
bureaux bienfaisance	124	47	première communion	13	40
1882–1884					
instruction publique	229	106	chemin fer	362	566
administration communale	183	106	ministres cultes	37	74
députation permanente	185	111	droit commun	65	101
conseil communal	118	66	travaux publics	176	212
carte géologique	52	3	concordat préventif	11	47
1884–1886					
écoles communales	295	200	commissaire arrondissement	70	147
ville gand	73	9	chemin fer	236	310
cuve matière	48	3	instruction publique	107	156
écoles normales	107	66	province namur	12	54
écoles adoptées	153	119	droit auteur	47	89
1886–1888					
service personnel	167	68	chemin fer	480	562
procureur général	80	23	ville bruxelles	21	65
port ostende	56	1	fer vicinaux	60	99
juge instruction	69	21	champs expérience	5	43
conseil communal	135	95	général major	9	46
1888–1890					
prud hommes	262	183	tribunal commerce	33	95
pays flamand	98	24	commis greffiers	6	56
conseils prud	183	117	général major	12	57
ville gand	65	9	sûreté publique	22	63
flandre occidentale	45	8	caisses prévoyance	15	52

Notes: The five phrases with the largest partisanship $\hat{\xi}_{jt}$ for each group (Flemish districts: positive; Walloon districts: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 22; covariates: family, female. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C101: Most partisan phrases, Flemish districts and Walloon districts, French speech: periods beginning in the 1890s

Flemish districts			Walloon districts		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1890–1892					
fonds communes	170	30	chemin fer	621	757
flandre occidentale	73	15	lieutenant général	21	86
loi 1876	64	19	conseil communal	58	110
domicile secours	112	68	cahier charges	26	68
dépôts mendicité	58	14	écoles normales	22	63
1892–1894					
représentation proportionnelle	236	107	chemin fer	268	397
deux langues	39	4	corps électoral	155	224
pays flamand	40	7	loi électorale	67	121
enseignement moyen	81	50	triple vote	18	58
général brassine	59	31	suffrage universel	172	211
1894–1896					
écoles adoptées	70	17	chemin fer	319	399
id id	36	1	conseil communal	80	140
administration communale	97	71	etat congo	8	60
sociétés coopératives	34	9	distilleries agricoles	53	91
flandre occidentale	33	10	loi communale	12	43
1896–1898					
unions professionnelles	345	157	grand central	47	102
garde civique	282	171	caisses prévoyance	14	47
ateliers chômage	64	21	chemin fer	232	263
administration communale	81	51	petite propriété	33	61
faire commerce	42	15	douaniers wallons	3	25
1898–1900					
représentation proportionnelle	345	262	banque nationale	165	235
banque etat	73	30	chemin fer	266	315
fer vicinaux	61	25	députation permanente	68	93
société nationale	44	9	contrat travail	111	133
banque émission	54	24	suffrage universel	64	86

Notes: The five phrases with the largest partisanship $\hat{\xi}_{jt}$ for each group (Flemish districts: positive; Walloon districts: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 22; covariates: family, female. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C102: Most partisan phrases, Flemish districts and Walloon districts, French speech: periods beginning in the 1900s

Flemish districts			Walloon districts		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1900–1902					
chemin fer	357	272	jeux hasard	41	151
ville anvers	53	10	maisons jeux	30	96
flandre occidentale	47	6	industrie travail	40	85
cahier charges	47	12	code pénal	23	55
compagnie générale	36	1	général major	63	93
1902–1904					
gare nord	44	5	caisses prévoyance	7	127
flandre occidentale	48	10	accidents travail	79	150
caisse épargne	101	64	faute lourde	35	92
ville bruxelles	41	7	ouvriers mineurs	6	52
bruxelles anvers	42	9	industrie travail	69	115
1904–1906					
grande coupure	165	19	caisses prévoyance	3	100
ville anvers	83	12	industrie travail	108	164
lieutenant général	143	86	ouvriers mineurs	7	54
installations maritimes	55	14	repos dominical	47	94
société belge	51	14	administration mines	4	47
1906–1908					
lieutenant général	109	34	conseil mines	15	74
pays flamand	90	29	ouvriers mineurs	14	68
seconde ligne	61	7	bon secours	5	54
ligne défense	57	9	effet utile	11	56
flandre occidentale	58	13	huit heures	77	120
1908–1910					
travaux publics	117	52	prud hommes	129	251
administration communale	117	58	conseils prud	83	186
effectif paix	83	26	neuf heures	7	90
temps service	92	41	écoles normales	31	110
collège échevinal	66	16	huit heures	38	94

Notes: The five phrases with the largest partisanship $\hat{\xi}_{jt}$ for each group (Flemish districts: positive; Walloon districts: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 22; covariates: family, female. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C103: Most partisan phrases, Flemish districts and Walloon districts, French speech: periods beginning in the 1910s

Flemish districts			Walloon districts		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1910–1912					
cour comptes	67	11	caisses prévoyance	45	165
syndicats chrétiens	59	11	ouvriers mineurs	32	92
unions professionnelles	62	15	60 ans	14	65
lock out	43	5	55 ans	17	67
écoles communales	64	28	industrie travail	111	158
1912–1914					
écoles communales	203	107	chemin fer	229	307
conseil régional	72	20	industrie travail	54	94
écoles adoptées	169	118	huit heures	9	40
écoles libres	95	47	caisses prévoyance	12	38
loi scolaire	70	32	garde civique	8	30
1918–1919					
régions dévastées	87	19	traité paix	26	95
flandre occidentale	66	9	juge paix	23	66
administration communale	123	78	chemin fer	139	176
députation permanente	56	18	prix revient	11	46
commissaire royal	28	4	tribunaux dommages	63	95
1919–1921					
travaux publics	114	46	huit heures	56	98
cour comptes	80	27	temps service	27	67
régions dévastées	100	49	défense nationale	84	121
dommages guerre	153	104	comité national	11	38
commissaires etat	59	21	prix revient	21	48

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Flemish districts: positive; Walloon districts: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 22; covariates: family, female. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C104: Most partisan phrases, Flemish districts and Walloon districts, French speech: periods beginning in the 1920s

Flemish districts			Walloon districts		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1921–1923					
université gand	183	60	défense nationale	138	207
écoles normales	129	50	gardes champêtres	76	128
huit heures	141	67	grand duché	35	78
sciences arts	155	85	union économique	7	35
université flamande	126	56	six mois	41	70
1923–1925					
coopératives locataires	68	4	ouvriers mineurs	31	80
service sauvetage	53	1	défense nationale	71	111
prix maxima	63	18	chemin fer	216	254
écoles moyennes	57	19	société nations	17	51
droits spécifiques	58	20	fonds communes	12	45
1925–1927					
chemin fer	386	302	neuf ans	27	98
sciences arts	106	46	eupen malmédy	12	66
société nationale	107	52	conseil communal	40	80
université gand	48	5	bail ferme	26	66
partie mobile	48	13	société nations	39	79
1927–1929					
sciences arts	201	69	prison école	6	99
société nationale	210	105	défense nationale	134	210
écoles libres	112	27	neuf ans	36	96
comité sélection	99	14	bail neuf	16	53
chemin fer	279	211	eupen malmédy	9	45
1929–1932					
enseignement moyen	88	25	chemin fer	152	241
sciences arts	108	58	société nationale	128	174
écoles libres	72	26	impôt foncier	20	54
application loi	106	62	défense nationale	81	107
allocations familiales	89	46	droits douane	19	43

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Flemish districts: positive; Walloon districts: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 22; covariates: family, female. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C105: Most partisan phrases, Flemish districts and Walloon districts, French speech: periods beginning in the 1930s

Flemish districts			Walloon districts		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1932–1934					
allocations familiales	188	62	juge paix	16	95
caisse nationale	106	25	agents change	5	56
conventions collectives	78	5	défense nationale	93	140
liste arrondissement	88	16	chemin fer	116	152
fonds communes	87	17	taxe crise	26	56
1934–1936					
gouverneur général	78	7	défense nationale	86	161
ouvriers frontaliers	63	12	arrêté loi	42	110
droit sortie	58	10	arrêtés lois	37	98
voies hydrauliques	46	2	société nations	18	73
général voies	39	1	grand duché	11	57
1936–1939					
banque nationale	234	63	allocations familiales	28	71
congo belge	63	3	congés payés	42	78
ruanda urundi	59	2	quarante heures	8	42
lieutenant général	131	78	prix revient	32	63
grands magasins	70	36	état major	15	43
1939–1940					
crédit anversois	155	18	sous officiers	11	45
conseil culturel	115	9	prix revient	38	67
santé publique	104	42	bénéfices exceptionnels	15	43
supérieur contrôle	63	2	charges sociales	16	36
pleins pouvoirs	69	10	pommes terre	24	44

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Flemish districts: positive; Walloon districts: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 22; covariates: family, female. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C106: Most partisan phrases, Flemish districts and Walloon districts, French speech: periods beginning in the 1940s

Flemish districts			Walloon districts		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1944–1946					
sous occupation	139	49	prisonniers politiques	38	127
arrêté loi	90	35	prix revient	17	66
création conseil	48	3	pommes terre	34	69
services incendie	44	13	conseils professionnels	17	51
pouvoirs extraordinaires	51	26	problème wallon	5	33
1946–1947					
auditeur général	76	12	juge paix	37	106
prisonniers politiques	169	109	travaux publics	57	118
banque emission	95	40	sous officiers	5	52
collaboration économique	80	26	réparation intégrale	29	64
augmentation salaire	52	12	travaux communaux	14	42
1947–1949					
banque emission	88	14	juge paix	18	108
syndicats libéraux	66	4	prisonniers politiques	10	85
banque nationale	136	79	travaux publics	64	123
organisations syndicales	101	46	arrêté loi	30	65
industrie textile	71	19	conseils professionnels	7	35
1949–1950					
société nationale	143	39	travailleurs indépendants	6	53
bureaux dépouillement	76	16	prisonniers guerre	40	80
code électoral	75	26	conseil national	31	71
nationale habitations	62	18	national charbonnages	11	50
logements bon	55	12	prix revient	47	85

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Flemish districts: positive; Walloon districts: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 22; covariates: family, female. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C107: Most partisan phrases, Flemish districts and Walloon districts, French speech: periods beginning in the 1950s

Flemish districts			Walloon districts		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1950–1952					
etats unis	153	43	allocations familiales	47	83
coût vie	139	67	prisonniers politiques	25	59
bénéfices exceptionnels	99	49	enseignement moyen	13	45
vingt quatre	113	64	prisonniers guerre	14	43
européenne paiements	55	10	conseil administration	28	52
1952–1954					
plan schuman	207	35	allocations familiales	27	110
haute autorité	69	15	travailleurs indépendants	36	118
etats unis	108	54	victimes civiles	16	93
dette publique	68	15	prisonniers politiques	17	67
procureur général	68	19	civiles guerre	19	52
1954–1956					
tirer parti	109	6	enseignement technique	36	130
banque nationale	114	39	enseignement moyen	53	142
bancs socialistes	89	32	travailleurs indépendants	24	89
gouvernement précédent	150	93	enseignement public	27	80
impôt complémentaire	59	10	enseignement libre	180	231
1956–1958					
banque nationale	217	55	conseil national	32	155
dette publique	155	26	national charbonnages	10	104
allocations familiales	221	121	ouvriers mineurs	39	121
combats boxe	73	12	industrie charbonnière	21	83
court terme	65	10	santé publique	51	90
1958–1961					
table ronde	209	59	allocations familiales	45	120
gouverneur général	116	19	conseil national	32	103
congo belge	142	55	haute autorité	31	81
ruanda urundi	128	47	national charbonnages	20	62
force publique	97	21	travailleurs indépendants	26	66

Notes: The five phrases with the largest partisanship $\hat{\xi}_{jt}$ for each group (Flemish districts: positive; Walloon districts: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 22; covariates: family, female. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Table C108: Most partisan phrases, Flemish districts and Walloon districts, French speech: periods beginning in the 1960s

Flemish districts			Walloon districts		
Phrase	$\hat{q}^A$	$\hat{q}^B$	Phrase	$\hat{q}^A$	$\hat{q}^B$
1961–1963					
flandre wallonie	115	11	assistance publique	33	84
problème linguistique	74	8	allocations familiales	56	104
région mouscron	49	2	autoroute wallonie	1	30
assistance technique	89	45	santé publique	111	135
grande bretagne	66	27	pacte scolaire	20	43
1963–1965					
politique énergétique	55	9	santé publique	84	158
pays tiers	57	12	pacte scolaire	12	72
prix détail	62	18	allocations familiales	44	97
plein emploi	51	9	autoroute wallonie	8	53
chambre syndicale	52	17	travailleurs indépendants	68	108
1965–1968					
etats unis	207	108	autoroute wallonie	12	73
dette publique	115	19	plan restructuration	0	17
national brut	121	30	santé publique	66	82
travailleurs indépendants	125	35	province luxembourg	15	30
produit national	123	40	ouvriers mineurs	41	51

Notes: The five phrases with the largest partisanship $\hat{\zeta}_{jt}$ for each group (Flemish districts: positive; Walloon districts: negative) in French speech, and their predicted counts per 100,000 phrases for a speaker of each group. Specification: Figure 22; covariates: family, female. Phrases are shown in their most frequent unstemmed form. The vocabulary leaves out procedural phrases, forms of address and titles, names of parties and parliamentary groups, amounts, dates, names of persons, fragments of words and pairs of filler words (`code/python/04b_filter_vocabulary.py`).

Online Appendix D: Descriptive figures and tables

This appendix describes Chamber composition and the speech volume behind the estimates. A period's members are all those listed for its legislature, whether or not they speak. This includes replacements and, before 1919, members entering or leaving through partial elections during the legislature. Speeches follow the sample in Section 3, except that the tables and Figures D1, D2 and D5 leave out 75 speaker-periods (4,910 speeches) whose speakers are missing from the list of members. Here, government status means the member's party held office on at least half the period's sitting days. Unlike the main estimates, it is not based on the timing of that member's speeches.

Table D1 selects one period every twenty years plus the final period. It reports each family's shares of members and speeches, and speeches per member. Catholics and Liberals share the Chamber until 1894. Socialists account for 19 per cent of members in 1898–1900. Flemish nationalists and Communists enter between the wars. In 2019–2021, no family has more than a quarter of members; in 2024–2025, the largest, the Flemish nationalist family, which in this table includes the Vlaams Belang, has 26 per cent.

Opposition families usually deliver more speeches per member than governing families. Socialists deliver 406 versus Catholics' 169 in 1898–1900; Communists deliver 652 in 1958–1961; and Christian democrats deliver 81 versus Socialists' 20 in 1999–2001. In the last two rows, 7–8 per cent of members lack a family label because the register lacks their party. Nearly all who speak nevertheless have a party in the speaker data used for estimation.

Table D1: Share of members and speeches by party family

Period	Party family	Share of members	Share of speeches	Speeches per member
1839–1841	Catholic / Christian democrat	0.50	0.49	4.7
	Liberal	0.49	0.51	5.0
	Unknown	0.02	0.00	0.0
1859–1861	Liberal	0.64	0.63	60.8
	Catholic / Christian democrat	0.35	0.37	64.2
	Unknown	0.01	0.00	1.5
1878–1880	Liberal	0.56	0.52	74.2
	Catholic / Christian democrat	0.43	0.48	87.9
	Unknown	0.01	0.00	12.0
1898–1900	Catholic / Christian democrat	0.72	0.56	169.0
	Socialist	0.19	0.36	406.4
	Liberal	0.08	0.08	201.5
	Unknown	0.01	0.00	85.0
1918–1919	Catholic / Christian democrat	0.52	0.39	45.9
	Liberal	0.25	0.31	75.6
	Socialist	0.21	0.30	86.9
	Unknown	0.03	0.00	0.0
1939–1940	Catholic / Christian democrat	0.34	0.30	42.9
	Socialist	0.33	0.30	45.0
	Liberal	0.16	0.16	49.3
	Flemish nationalist	0.08	0.11	67.8
	Communist / radical left	0.04	0.09	107.6
	Unknown	0.03	0.01	9.1
	Other	0.03	0.03	61.2
1958–1961	Catholic / Christian democrat	0.48	0.33	96.9

Table D1: Share of members and speeches by party family (continued)

Period	Party family	Share of members	Share of speeches	Speeches per member
	Socialist	0.39	0.54	192.8
	Liberal	0.11	0.08	105.6
	Communist / radical left	0.01	0.04	651.5
	Unknown	0.01	0.00	0.0
	Flemish nationalist	0.00	0.01	204.0
1979–1981	Catholic / Christian democrat	0.38	0.26	67.7
	Socialist	0.27	0.19	68.0
	Liberal	0.17	0.19	116.3
	Francophone regionalist	0.08	0.12	159.9
	Flemish nationalist	0.07	0.16	227.9
	Communist / radical left	0.02	0.07	411.5
	Unknown	0.01	0.00	6.0
	Other	0.00	0.00	63.0
1999–2001	Liberal	0.27	0.21	31.6
	Socialist	0.24	0.12	19.7
	Catholic / Christian democrat	0.19	0.37	80.6
	Green	0.13	0.08	24.7
	Flemish nationalist	0.13	0.20	65.3
	Unknown	0.03	0.01	15.0
	Francophone regionalist	0.01	0.00	9.0
	Other	0.01	0.00	18.0
2019–2021	Flemish nationalist	0.25	0.27	64.0
	Socialist	0.19	0.15	44.8
	Liberal	0.18	0.16	55.2
	Green	0.14	0.08	34.6
	Catholic / Christian democrat	0.10	0.13	78.0
	Communist / radical left	0.07	0.11	95.8
	Unknown	0.07	0.05	45.6
	Francophone regionalist	0.01	0.03	175.5
	Other	0.01	0.02	176.0
2024–2025	Flemish nationalist	0.26	0.25	40.1
	Socialist	0.18	0.17	38.5
	Catholic / Christian democrat	0.16	0.11	28.3
	Liberal	0.16	0.17	44.7
	Communist / radical left	0.08	0.11	53.8
	Unknown	0.08	0.04	16.9
	Green	0.06	0.11	76.9
	Other	0.01	0.01	57.0
	Francophone regionalist	0.01	0.03	206.0

Notes: One period every twenty years and the last period.

Table D2 reports, every ten years, the member, speaker and speech shares for Catholics, the left, women and governing parties. Since most members speak, speaker shares differ from member shares by at most 0.05. Speech shares differ more. Excluding low-coverage years, governing parties deliver a smaller speech share than member share in 53 of 55 periods since 1894 with at least ten opposition members, compared with 10 of 21 earlier periods. In 1949–1950, governing Catholics hold 0.50 of membership but deliver 0.27 of speeches, while the left accounts for 0.37 and 0.57. Women's member share stays below a tenth until 1995 and exceeds two fifths from 2019 (Section 5.5). Their speech share is below their seat share in 40 of 42 adequately covered periods since 1929.

Table D3 reports member and speech shares for Flemish, Walloon and Brussels districts. Flemish districts elect 0.45–0.52 of members until 2010 and 0.55–0.57 from 2019. Walloon districts

Table D2: Summary statistics: Catholics, left parties, women and government parties

	Catholics			Left parties			Women			Government		
Period	MPs	Speakers	Speeches	MPs	Speakers	Speeches	MPs	Speakers	Speeches	MPs	Speakers	Speeches
1839–1841	0.50	0.47	0.49	0.00	0.00	0.00	0.00	0.00	0.00	0.49	0.53	0.51
1848–1850	0.28	0.24	0.28	0.00	0.00	0.00	0.00	0.00	0.00	0.70	0.75	0.72
1859–1861	0.35	0.38	0.37	0.00	0.00	0.00	0.00	0.00	0.00	0.64	0.61	0.63
1868–1870	0.41	0.40	0.40	0.00	0.00	0.00	0.00	0.00	0.00	0.59	0.60	0.60
1878–1880	0.43	0.45	0.48	0.00	0.00	0.00	0.00	0.00	0.00	0.56	0.55	0.52
1888–1890	0.70	0.69	0.53	0.00	0.00	0.00	0.00	0.00	0.00	0.70	0.69	0.53
1898–1900	0.72	0.71	0.56	0.19	0.20	0.36	0.00	0.00	0.00	0.72	0.71	0.56
1908–1910	0.53	0.52	0.42	0.20	0.21	0.33	0.00	0.00	0.00	0.53	0.52	0.42
1918–1919	0.52	0.54	0.39	0.21	0.23	0.30	0.00	0.00	0.00	0.97	1.00	1.00
1929–1932	0.41	0.39	0.37	0.37	0.39	0.43	0.01	0.01	0.00	0.56	0.55	0.49
1939–1940	0.34	0.36	0.30	0.37	0.35	0.40	0.01	0.01	0.02	0.83	0.84	0.76
1949–1950	0.50	0.46	0.27	0.37	0.39	0.57	0.03	0.02	0.01	0.63	0.61	0.43
1958–1961	0.48	0.50	0.33	0.40	0.39	0.58	0.05	0.04	0.01	0.59	0.60	0.42
1969–1971	0.32	0.34	0.26	0.31	0.29	0.28	0.04	0.03	0.01	0.55	0.56	0.44
1979–1981	0.38	0.38	0.26	0.29	0.29	0.26	0.07	0.07	0.03	0.65	0.64	0.44
1989–1991	0.28	0.28	0.25	0.35	0.32	0.15	0.08	0.10	0.08	0.71	0.69	0.48
1999–2001	0.19	0.20	0.37	0.24	0.23	0.12	0.21	0.22	0.17	0.64	0.62	0.41
2007–2010	0.22	0.23	0.20	0.20	0.22	0.23	0.36	0.36	0.28	0.58	0.61	0.46
2019–2021	0.10	0.11	0.13	0.26	0.24	0.25	0.43	0.43	0.39	0.58	0.57	0.46
2024–2025	0.16	0.17	0.11	0.27	0.27	0.28	0.41	0.40	0.36	0.51	0.53	0.40

Notes: Share of the members, of the speakers (members with at least one speech) and of the speeches that belong to each category; one period every ten years and the last period.

elect 0.38–0.43 until 1940 and about a third from 1969. Brussels accounts for 0.10–0.17 from 1848.

Speech shares differ from seat shares. In every displayed period from 1908 to 1981, Brussels members' speech share is larger than their seat share, and Flemish members' is smaller except in 1958–1961; Brussels delivers 0.28 of speeches against 0.14 of seats in 1929–1932. In the displayed rows from 1989, Flemish members' speech share exceeds their seat share. Brussels members deliver 0.18–0.29 of speeches in the 1908–1991 rows and are excluded from Section 5.6's regional comparison. The low-coverage 1839–1841 row uses 598 speeches, and 4 per cent of members lack a region.

Table D3: Summary statistics: regions

	Flanders		Wallonia		Brussels	
Period	MPs	Speeches	MPs	Speeches	MPs	Speeches
1839–1841	0.50	0.30	0.40	0.55	0.06	0.14
1848–1850	0.50	0.49	0.40	0.44	0.10	0.07
1859–1861	0.45	0.50	0.42	0.37	0.13	0.14
1868–1870	0.47	0.37	0.43	0.52	0.10	0.10
1878–1880	0.45	0.53	0.43	0.36	0.12	0.11
1888–1890	0.47	0.48	0.39	0.43	0.14	0.08
1898–1900	0.46	0.42	0.42	0.51	0.12	0.06
1908–1910	0.48	0.37	0.38	0.41	0.14	0.21
1918–1919	0.45	0.32	0.42	0.45	0.14	0.23
1929–1932	0.47	0.37	0.39	0.36	0.14	0.28
1939–1940	0.45	0.36	0.40	0.43	0.15	0.21
1949–1950	0.49	0.38	0.36	0.35	0.15	0.26
1958–1961	0.50	0.50	0.36	0.32	0.15	0.18
1969–1971	0.50	0.43	0.34	0.29	0.15	0.29
1979–1981	0.50	0.48	0.33	0.27	0.17	0.26
1989–1991	0.51	0.54	0.32	0.25	0.17	0.21
1999–2001	0.52	0.63	0.34	0.26	0.14	0.12
2007–2010	0.51	0.52	0.34	0.26	0.15	0.21
2019–2021	0.55	0.59	0.34	0.27	0.11	0.14
2024–2025	0.57	0.58	0.32	0.30	0.11	0.12

Notes: Share of the members elected in, and of the speeches by members from, each region; one period every ten years and the last period.

Figure D1 shows the member shares of Catholic, Liberal and left (Socialist and Communist) families. Catholics and Liberals share the Chamber until 1894. Liberals reach 70 per cent in 1848–1852; Catholics hold 60–72 per cent in 1886–1900. Socialists enter with 17 per cent in 1894. In 1919–1968, Catholic and left shares are 31–51 and 34–45 per cent, while Liberals hold 9–22 per cent. After the party splits, the Catholic share falls most, reaching 10 per cent in 2019–2024. Liberals reach their highest post-1894 share, 27 per cent, in 1999–2007. Catholics, group A in the religious comparison, thus form a majority under Catholic governments in 1884–1914 and a small federal-era minority. In 2019–2024, they are too few for the religious series to be defined (Section 5.1).

Figure D2 reports, among members with a known party, the share whose parties govern on at least half the period's sitting days. The governing share is 100 per cent in 1831–1839, 1841–1848 and 1854–1857. These periods have unionist or mixed cabinets, or successive cabinets of both parties in 1845–1848. The share is 85–100 per cent under the national-union governments of 1918–1921, 1925–1927, 1934–1936 and 1939–1940, and 77 per cent in 1936–1939. Most such periods are absent from Figure 4.

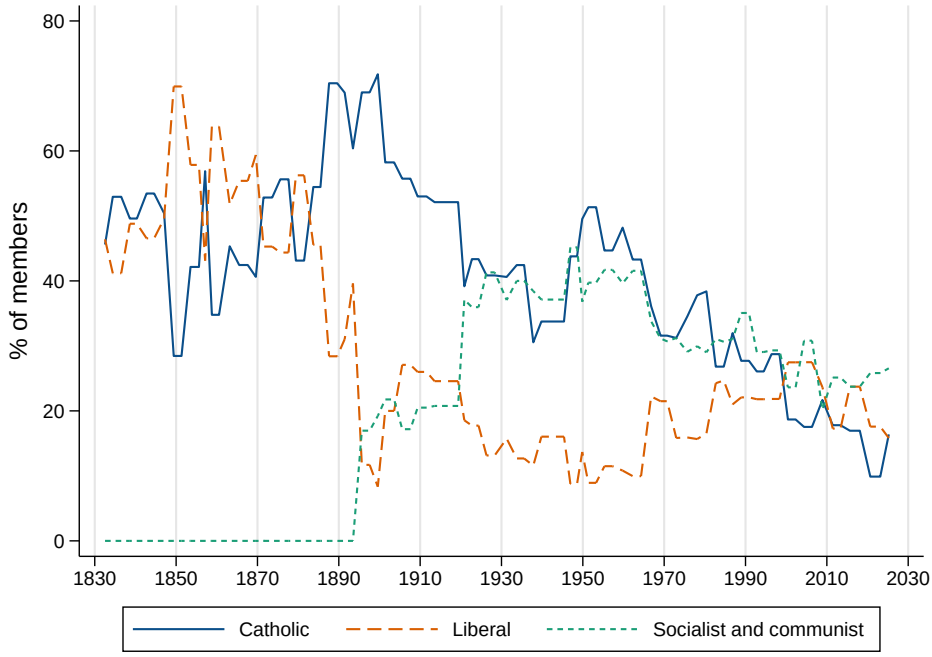


Figure D1: Members of the Catholic, Liberal and left families

Under single-party governments in 1847–1914, the governing family holds 53–72 per cent of members. The exception is 1882–1884 at 46 per cent, which includes members elected in 1884. After 1919, excluding national-union governments, the share ranges from 51–53 per cent in 1950–1958 and 1981–1987 to 82–85 per cent in 1961–1965 and 1977–1978. Federal-era shares are 56–66 per cent. The narrow majorities of 1950–1958 coincide with modest partisanship: net values are at most 0.0007 in the religious and government–opposition series, and only the latter exceeds its random benchmark, in 1950–1952. In an earlier estimation on all phrases, both series were high in these years (Section 5.12).

Figure D3 reports the decade share of plenary speech blocks linked to a member or minister, excluding the chair's turns. Linkage is 74 per cent in the 1830s, when sources are digitized years of the *Moniteur belge*, 93 per cent in the 1840s and 96 per cent in the 1850s. In every later decade it lies between 97.8 per cent (2000s) and 99.5 per cent (2010s). Overall, 98.5 per cent of 1,826,697 blocks are linked (Table B1). The lowest rates occur in the 1830s and 1840s, which include the low-coverage years excluded from averages. From the 1850s, at least 96 per cent of the blocks in the available reports are linked in every decade.

Figure D4 shows sample speech counts (a) and phrases per speaker (b). Speech counts are at most 3,500 per period before 1843, 33,000–57,000 in 1894–1914, and 69,371 in 1929–1932, a period lasting almost three years. From 1988, they range from 6,100 to 15,400. Panel (b) counts the phrases of the vocabulary before the phrases without content are left out. Phrases per speaker are highest in 1845–1848 (7,400) and 1894–1896 (6,600). They average about 4,700 in 1894–1918 and 2,300 in 1995–2025, while the mean number of days with plenary sittings falls from about 215 to about 105. With the vocabulary used for estimation, the two averages are about 4,100 and 1,800. The lowest values occur in the poorly covered years 1831–1843 and 1991–1995 and in short periods such as 1949–1950. Since penalization retains coefficients only when the data

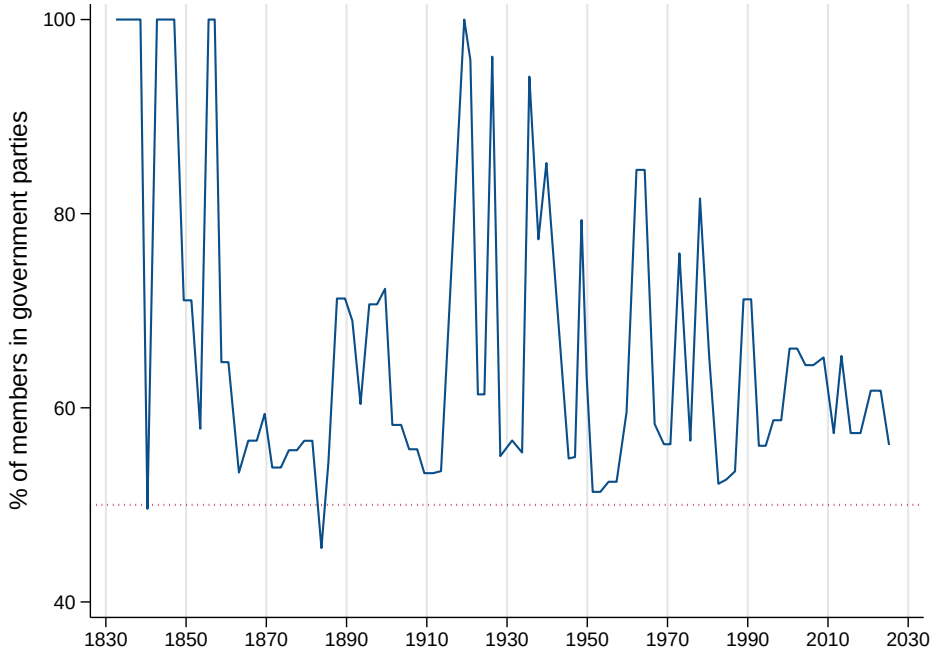


Figure D2: Members of the parties in government

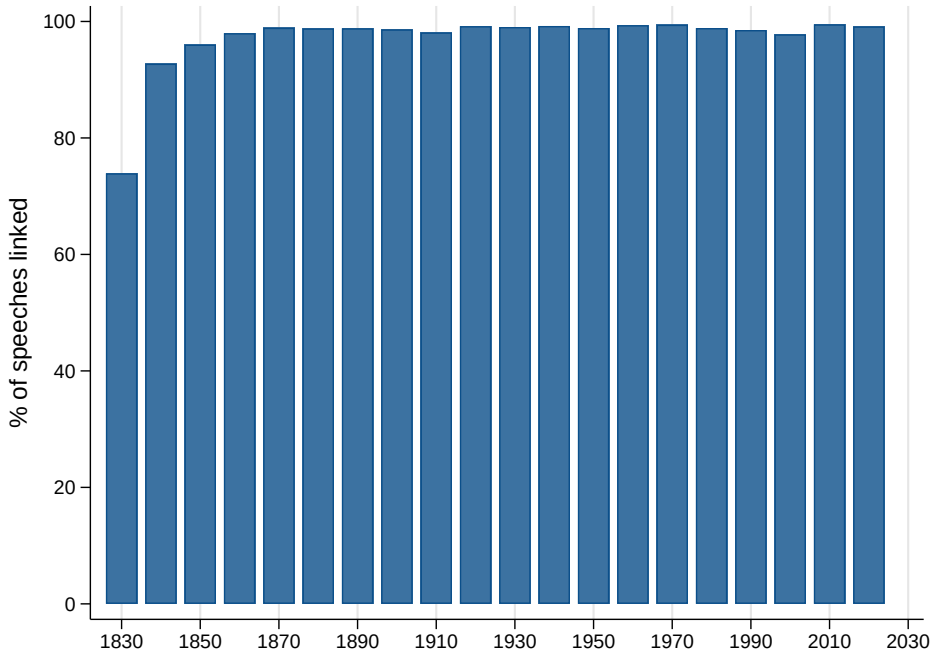


Figure D3: Speeches linked to a member or minister, by decade

Notes: See Graziani (2026) for the evaluation of the segmentation.

support them, declining speech volume complicates comparisons across eras (Section 5.12).

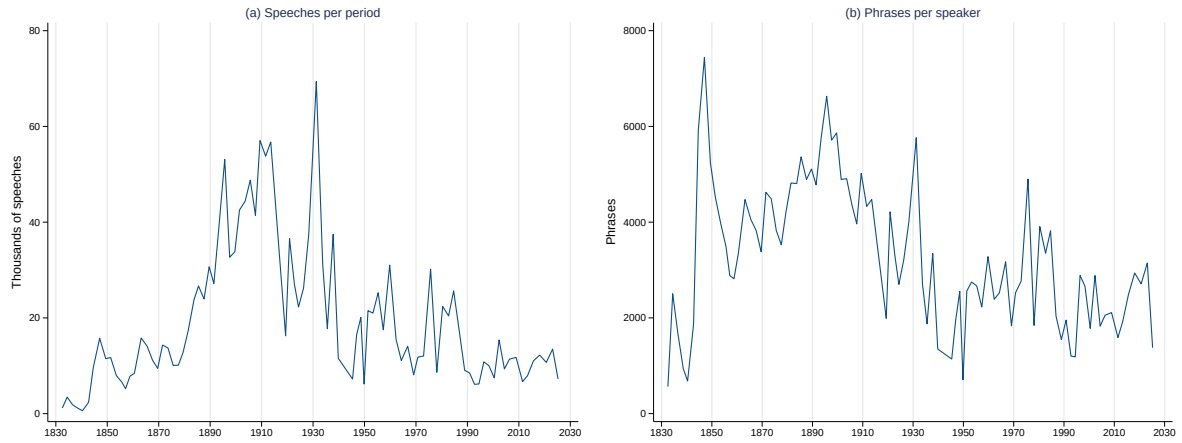


Figure D4: Amount of speech

Notes: (a) Speeches of the sample in each period. (b) Phrases of the vocabulary before the phrases without content are left out (`output/data/phrases_per_period.csv`), divided by the number of speakers.

Figure D5 shows the share of governing and opposition members who speak at least once per period. Both shares usually range from 80 to 100 per cent. Excluding low-coverage periods and periods with fewer than ten opposition members, opposition participation is higher in 37 of 55 periods since 1894 and 11 of 21 earlier periods, usually by a few points.

The lowest opposition shares, 59 per cent in 1884–1886 and 58 per cent in 1944–1946, include members who never speak. In 1884–1886, these are 34 Liberals who lose their seats in June 1884. In 1944–1946, they are all 18 Flemish nationalist members of the 1939 legislature, 24 Catholics and two Rexists. The opposition's higher speech share in Table D2 mainly reflects speaking more often. Since 1894, opposition speakers deliver a median of 1.7 times as many speeches as governing speakers, and more in 53 of 55 periods.

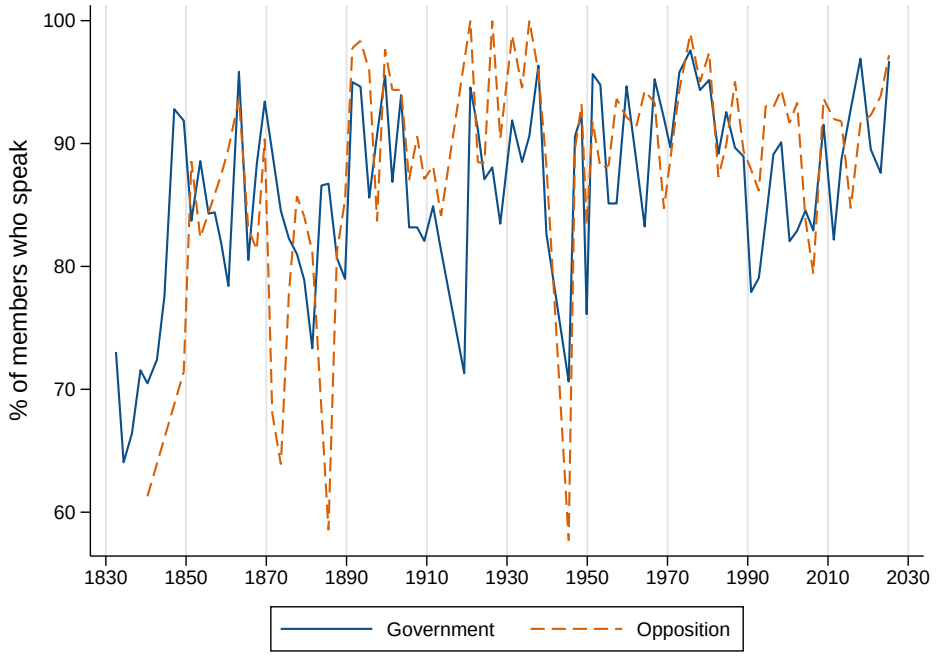


Figure D5: Share of government and opposition members who speak

Online Appendix E: Reactions, references and emotions

Section 7's measures come from a companion module (`code/emotions`). Its first steps cover the entire period without a language model. They detect stenographers' stage directions, the chair's interventions and references between members. Later steps classify emotions and tone using a language model and a classifier, as described below. The reported results come from the module's run of 14 October 2026. The later steps have not yet been run on the corpus, so Section 7 reports no emotion estimates. Reactions and references do not depend on these steps. Table E1 gives the detected counts.

Table E1: Reactions, calls to order and references found by the module

	Number	Attributed to a member or minister (%)
Stage directions (pieces of parentheses) in the speeches and procedural blocks		
Disorder	139,379	
Approval	140,385	
Laughter	80,753	
Stir	1,230	
Other (no measure)	959,805	
Turns of the chair		
Calls to order	1,223	
Censures	89	
References in the speeches		
By name	1,219,442	78.2
By office (ministers)	654,770	68.8
To the rapporteur	29,045	46.6
To the previous speaker	8,862	85.7
By constituency	4,096	42.9
Generic courtesy (no one designated)	102,555	

Notes: Plenary sittings, September 1831–2025: 2,260,972 speeches before the split of merged interventions (see text), 26,092 collective interjections and 639,824 procedural blocks in 15,557 sittings. Stage directions: pieces of a parenthesis cut at its dashes and semicolons; a piece counts once in every category it records; the counts include the reactions attributed to the public galleries, which the measures leave out. References: in the 1,833,156 speeches other than the turns of the chair, after the split. Run of `code/emotions` of 14 October 2026 (steps 0–2).

Sample. The module uses the version of the dataset used for the partisanship analysis (Graziani, 2026). It selects speeches, collective interjections and procedural blocks from plenary sittings between September 1831 and the end of 2025. As in the partisanship analysis, it excludes committee and Central Commission reports identified in `committee_scan.csv` for 1985–1999, and later documents with committee codes. Chair turns are identified from printed roles (*président*, *voorzitter* and variants), dataset flags or links.

The resulting sample contains 1,833,156 non-chair speeches by members and ministers, a small share unlinked to a person. Unlike the partisanship selection (Table B1), it requires neither a link to a member or minister nor Chamber membership on the sitting date, and it splits the interventions that the reports from 1999 merge with the preceding block (below). After removing parenthetical text, the speeches contain 304 million words. We measure reactions and references in these speeches. Each legislature is split at the midpoint between its first and last sitting, producing 112 half-legislatures for Figures 13 and 15.

From 1999, some numbered interventions remain inside the preceding block of the dataset, often a chair turn, sometimes with a summary in the other language. The module splits these blocks at the printed labels of the interventions (`criv.py`). It cuts 4,058 blocks at 10,569 labels and recovers 7,294 interventions, 98 per cent of them linked to a member or minister. It removes 1,348 summaries in the other language, moves 1.46 million words out of chair turns and 0.23 million out of other members' blocks, and separates 1,141 chair turns from members' blocks. In the 61 sittings of 1 July 1999 to 14 July 2000, whose PDF reports print every accent as a separate character, the module joins each accent to its letter, as in the partisanship analysis (Online Appendix B), and leaves the blocks as they are: there the dataset separates neither the chair's turns nor most members' interventions, which are not numbered. Levels after 1999 nevertheless remain not comparable with earlier ones (Section 7).

Reactions of the Chamber. Stenographers record reactions within speech parentheses, for example *(Interruptions.)*, *(Applaudissements à gauche. — Protestations à droite.)* and *(Gelach.)*. The module reads parentheses of at most 25 words in speeches and procedural blocks, splitting them at dashes and semicolons. It normalizes each piece to lowercase without accents or punctuation and matches patterns for one or more categories: disorder (interruptions, noise, murmurs, protests, exclamations, uproar, the chair's gavel and Dutch equivalents), approval (applause, *très bien*, *bravo*, assent), laughter, and stir (*mouvement*, *sensation*). Patterns match word beginnings to detect plurals and OCR-split forms such as *applau dissements*.

A single *(Non.)* or *(Neen.)* usually answers the chair's request for objections. Yes or no counts as approval or disorder only when repeated, as in *Non ! non !*, or attributed to a side of the House. Other pieces—petition or naturalization place names, document references, article numbers and footnotes—enter an unused residual category.

A reaction is assigned to the speech containing it. If printed in procedural text or a chair turn, it is assigned to the preceding non-chair speech in that sitting. We exclude reactions from public galleries. Disorder per 100,000 words is the count of disorder pieces assigned to a group's speeches divided by those speeches' word count. The hostile share is disorder divided by disorder plus approval. Decades follow sitting dates.

The chair. We detect calls to order in chair turns using patterns such as *je vous rappelle à l'ordre*, *à l'ordre !* and *ik roep u tot de orde*. Censures use *censure*, *inscription au procès-verbal* and *met vermelding in de notulen*. A turn containing both counts once in each category. We exclude agenda and business-order reminders: *je rappelle que … à l'ordre du jour*, *passer à l'ordre du jour*, *à l'ordre des travaux* and *tot de orde van de dag*. In a check of every detected call on an earlier version of the dataset, these exclusions removed 26 of 1,227 calls. We report calls and censures per 100 sittings. The module also records chair warnings, sitting suspensions and collective interjections, where shouts are attributed to several members. Section 7 does not use these.

Government, party family and language community. A speaker is on the governing side on the sitting date if they are a minister or belong to a party with a minister in office. Unlinked speeches have unknown status. We match parties by label, otherwise by family and party language, or by family alone when no language is specified, as before national-party splits. For

Flemish nationalists, Francophone regionalists and the residual category, family and language are not enough: the party itself, or its predecessor, must hold office.

Some ministerial terms lack end dates, including 81 of 95 prime-minister terms. We end these the day before the next term in the same portfolio or the next prime minister's appointment. Where neither follows, the minister is treated as currently in office. Independent and extra-parliamentary ministers count only in cabinets consisting entirely of such ministers. Most members therefore belong to the governing side under the Catholic–Liberal unionist cabinets of 1831–1847: 95 per cent of 1830s speeches with known status.

Party family comes from the member register or, if absent, from the party label. Language community follows party language for parties associated with one community: Flemish nationalists, Francophone regionalists, Greens and other community-specific parties throughout, and Catholics, Liberals and Socialists after their 1968, 1972 and 1978 splits. Otherwise it follows constituency region, assigning Flanders to Dutch and Wallonia to French. For Brussels members from 1919, we instead use the language of at least 70 per cent of their legislature's speeches, provided they deliver at least 20. Community is known for 94 per cent of member-legislatures. Almost every member speaks French before the 1890s, so early community labels describe constituency region rather than spoken language.

References between members. We detect references in non-chair speeches after removing parenthetical text. Six pattern types identify: names following titles (*M. Woeste*, *l'honorable M. de Theux*, *MM. Beernaert et Woeste*, *de heer Van Rompuy*, *Mijnheer Tindemans*, *collega De Wever*, *minister Vandenbroucke*); constituency-based references (*l'honorable membre de Gand*); ministerial offices (*M. le ministre des Finances*, *de eerste minister*); rapporteurs; previous speakers (*l'honorable préopinant*, *de vorige spreker*); and generic courtesy without a specific target (*l'honorable membre*). When matches overlap, we keep the most specific.

We match names against members of the legislature and ministers in office on the sitting date. First, we try exact surnames, including leading words of the printed name. Next, we try surnames without particles (*Theux* for *de Theux*), then leading words of longer surnames (*de Theux* for *de Theux de Meylandt*). Finally, names of at least six letters can match members' surnames by similarity to allow for OCR errors. We require a unique best match scoring at least 88 out of 100. Homonyms are resolved by first initial or, failing that, by who speaks in the same sitting. Otherwise they remain unresolved.

An office-based reference targets the minister holding that portfolio. If several qualify, we select the one speaking in the same debate. A minister mentioned without a portfolio is resolved only when a single minister speaks in that debate. Portfolios are unrecognized for 20 per cent of ministerial terms, leaving those ministers identifiable only by name. A rapporteur reference targets the most recent preceding speaker with that printed role in the same debate. A previous-speaker reference targets the last other member to speak in the debate. We exclude self-references.

Among name references, 72 per cent match surnames exactly. Another 6 per cent are resolved through particles, leading words, similarity, initials or sitting participation. One per cent remain ambiguous between homonyms, and 21 per cent match neither a legislature member nor a

current minister. These include OCR errors, outsiders and members or ministers from other years. Section 7 uses resolved name references, following Peeters and Lefever (2026)'s name-only counts. A reference crosses families when both family labels are known and differ; the residual small-party category counts as a family. It crosses the language border when both communities are known and differ.

Random pairing. Group sizes affect cross-group reference shares. When one family dominates membership, it receives many references even if speakers address opponents. Figure 15's benchmark randomly pairs each reference's speaker with a target from those referenced in the same half-legislature. The expected cross-group share is $1 - \sum_g s_g t_g$, where s_g is group g's share of references made and t_g its share received. We calculate observed and expected shares by half-legislature. Decade averages include half-legislatures whose midpoint falls in that decade, weighted by reference counts.

Validation. Peeters and Lefever (2026, Appendix 1) report counts for 53 investiture, State of the Union and confidence debates in 1961–2023. They count member and minister speeches, excluding chair turns, and name references within and across language groups from Chamber reports. We manually identify each debate's sitting dates using agenda headings, chair announcements and cabinet dates, and list them in the module. For those dates, we select speeches outside headings for other business: questions, interpellations, bills, budgets, motions, petitions, naturalizations and personal statements. Next, we count name references between members with known language communities.

The module finds speeches for 49 of 53 debates. Investiture sittings from 1992, 1995 and 2003 are missing, and the 1994 debate is unidentified. For 1987 only the government declaration is available. Speech counts differ from those of Peeters and Lefever (2026) by more than 15 per cent in 12 debates: the module finds fewer in 1999, 2006 (only one of the debate days is in the dataset), 2009, 2014, 2018 and 2021–2023, and more in 1977, 1987, 1997 and 2008, when our days include both debates of the first Leterme government. Section 7 reports the comparison.

Emotions and tone (in preparation). We remove parenthetical text and divide members' speeches into passages of at most 120 words, producing 4.2 million passages. One language model labels both French and Dutch: OpenAI's `gpt-6-luna`, with low reasoning effort and answers constrained to the codebook's JSON schema. In sampled passages, it rates eight emotions from Widmann and Wich (2023): anger, fear, disgust, sadness, joy, enthusiasm, pride and hope. Ratings range from 0 (absent) to 2 (strong).

In sampled references, the model labels the speaker's stance towards the target as supportive, neutral or critical. It also detects uncivil attacks—insults, mockery, attacks on character or motives, and accusations of lying—and denial of the target's legitimacy as a political actor (Muddiman, 2017). Tone is classified as no attack, civil attack (critical without incivility) or uncivil attack.

The emotion sample contains 8,000 training passages and 2,400 test passages. A quarter of training passages are drawn from text with emotion cues, including exclamations, indignation, pride, hope or recorded disorder, to represent rare emotions. Test passages are allocated equally

across institutional periods, languages and the 24 topics of Section 6, within the limits of each cell. The reference sample contains 5,000 training references, half crossing groups, and 2,000 test references, allocated equally across periods and topics. The training samples were drawn first, from an earlier version of the dataset; the test samples come from the version used here. The 74 test passages and 7 test references that also appear in the training samples stay in the gold standard but are left out of the classifiers' test metrics. The language model has labelled all four samples.

Two human coders independently label 300 test passages and 250 test references. We measure their agreement using Cohen's kappa and, for emotions, Krippendorff's alpha. We compare each with the language model using Cohen's kappa. For each task, a single multilingual XLM-R classifier (Conneau et al., 2020) is fine-tuned on language-model labels. Using one classifier for both languages avoids differences in calibration entering trends as Dutch speech grows from almost zero to about 60 per cent. Probabilities are recalibrated on validation labels and averaged by speech, weighting passages by word count. We correct group-period averages for classifier errors with the design-based supervised learning estimator of Egami et al. (2023). Its gold standard is the test sample drawn with known probabilities, using either language-model labels or the two coders' labels.

Online Appendix F: Event studies

Section 8's estimates come from the event-study module of the replication package (`code/events`, launched by `run_events.sh`). It reads the speech counts, speaker-periods, roll calls, topics and reactions used in the rest of the paper and writes every estimate, with the same columns, to `output/events/summary.csv`. For the events of Table F1, the file has 12,489 rows, 6,531 of them with an estimate (the others are terms that cannot be estimated, such as event times without observations). Table F1 lists the events and comparisons of the catalogue, fixed before estimation, that this paper reports, the designs used for each and the number of estimates; the entries marked as not estimated need data that the project does not have, have no sittings before the event, or were not studied in this paper.

Table F1: Catalogue of the event studies

Event	Designs	Estimates
Plural and compulsory voting (revision of 1893; election of 1894)	R	169
Proportional representation (1899; election of 1900)	R	243
Equal male suffrage (1919)	R	312
Women's suffrage (1948; election of 1949)	R	295
Gender quotas: law of 1994 (applied 1999) and parity of the lists of 2002 (from 2007)	R, 3	650
Provincial districts and 5% threshold (2002; election of 2003); cartels	R, 1	298
Federal Chamber of 150 (1993; election of 1995) and regional parliaments elected apart	R, 2, 3	280
Equality law (De Vriendt-Coremans, 1898)	3	9
Language laws of 1932 and 1935	2	34
Language border (laws of 1962 and 1963)	2, 6	53
Splits of the national parties: Catholics 1968, Liberals 1972, Socialists 1978	4, 5	99
State reforms: 1970, 1980, 1988-89, 1993, 2001, 2012-14	2	137
BHV and 541 days without a government (2010-2011)	6, 5	41
Every change of coalition, 1979-2025	1, 3, 5	2,621
Governments of national union: 1918-21, 1926, 1935, 1944-45	1, 3	48
Purple coalition (1999): Christian democrats in opposition	3, 1	30
N-VA in government (October 2014) and out (December 2018)	1, 3	45
Cordon sanitaire (1989; all parties after the vote of 1991)	6	46
Vivaldi (2020) and Arizona (2025)	1, 3	43
Elections and falls of governments	6	350

Table F1: Catalogue of the event studies (continued)

Event	Designs	Estimates
First school war (1879-84)	2	20
Royal question (1950)	2, 6	68
Second school war (1954-58) and School Pact (1958)	2	56
Independence of the Congo (30 June 1960)	2	28
Strike against the Unitary Law (1960-61)	2, 6	47
Special powers and austerity (1982-85); Global Plan (1993)	2	55
Dutroux affair and White March (1996)	2, 6	47
Great Depression (1929-35)	2	12
Oil shocks (1973 and 1979)	2	34
European integration: 1951, 1957, Maastricht 1992, euro 1999-2002	2	99
Financial crisis (2008) and euro crisis (2010-12)	1, 2, 6	98
Attacks (2001, 2015-16) and refugee crisis (2015)	1, 2	166
COVID-19 (March 2020)	3, 6, 2	69
Russian invasion of Ukraine (2022)	2	47
Published series of other parliaments (US, Finland, UK, Norway, Sweden) *(not estimated: needs published series of other countries)*		0
ParlaMint corpora (17 parliaments, Belgium 2015-2020) *(not estimated: needs the ParlaMint corpora)*		0
Proportional representation in Europe (staggered, 1899-1921) *(not estimated: needs digitized debates of other countries)*		0
Senate as an intensity control (1894, 1900, 1921) *(not estimated: needs the Senate debates after 1900)*		0
Exposure by arrondissement (electoral statistics) *(not estimated: needs electoral statistics by district)*		0
Types of speech (interpellations and questions against debates on bills) *(not estimated: not studied in this paper)*		0
Members around the unions of 1918 and 1944 *(not estimated: no sittings in the half-years before, during the world wars)*		0

Notes: Designs: 1, pairs of parties (coalitions); 2, policy areas; 3, members; 4, families around the splits; 5, roll calls; 6, sittings. Estimates: rows of `output/events/summary.csv` with an estimate for the entry. An estimate that concerns several entries, such as a change of coalition that is also a single event, is counted in each, so the column sums to more than the total.

Series. For 82 divisions, the module estimates net partisanship by period without covariates, with the penalized estimator of Section 4 and the leave-out estimator of Gentzkow et al. (2019), on the paper's vocabulary and on the two placebo vocabularies. The divisions are the religious,

left–others, government–opposition, left–right and right–others groupings; Flemish against Walloon members, overall and within the Catholic, Liberal and Socialist families; women against men; each of the eight families against all the others; the 28 pairs of families of the multiparty index; and, from 1979, the 36 pairs of parties of the same language of Section 5.5. The languages are combined as in Section 4. Each series is also estimated after permuting the group labels among the speakers of each period, 10 times for the penalized estimator and 20 times for the leave-out estimator, and net partisanship is the real series minus the mean of the permuted series.

Speech scores. For designs whose unit is a member, a half-year, a month or a policy area, each speech s of member i in period t receives the posterior probability on group A implied by its phrases, $a_s = m_s^{-1} \sum_j c_{sj} \rho_{jt}$. In the leave-out version, ρ_{jt} uses the frequencies of the two groups in the period without the member. In the cross-fitted version, the members are split into five folds, and ρ_{jt} comes from the penalized model, without covariates, fitted on the other four folds. In the leave-party-out version, the frequencies leave out every member of the speaker's party unit, in both groups; a speech is kept only when each group keeps at least five speakers outside the party, so periods of one-party governments drop out (67 per cent of the member-periods with a leave-out score keep one). Each version is computed with the real groups and with the permuted groups. The module stores the sums of m_s and $m_s a_s$ by member, period, language and month, so that every aggregate is a sum of these rows. With weight one half for each group and equal weights for the members of a group, the leave-out scores reproduce the leave-out series of the paper to within 10^{-15}, in each language and combined; the speech counts sum to the counts of the main estimation.

Placebo vocabularies. Among the phrases that the content filter leaves out of the vocabulary (Section 3), the numbers, amounts and dates (for example *000 frank*) form one placebo vocabulary and the pairs of filler words (for example *ten eerste*) the other: 2,189 and 1,326 French phrases, 852 and 1,392 Dutch phrases, against 77,324 and 18,045 in the paper's vocabulary. The same vocabulary thresholds apply. These vocabularies carry little political content, but they are not neutral: governments speak of budgets, and the members of a party share speech habits. A change that appears equally in the paper's vocabulary and in a placebo vocabulary is therefore read as a change in who speaks, how much and in what role, rather than in political vocabulary.

Results of Section 8: coalitions and roll calls. The pair panels use the 36 pairs of parties of the same language from 1979 (N-VA with the Volksunie, PVDA-PTB in both languages). The roll-call panels use the 25 pairs with roll-call data. Two parties are partners in a period or half-year when both are in government for at least half of it; being on the same side is measured as one minus the absolute difference between the shares of the period in which each party governs, which equals one when both govern or both oppose. Table F2 reports, for every panel, the two-way fixed-effects coefficients with the wild cluster bootstrap and the placebo distribution from shifting each pair's government path in time, and the estimator of de Chaisemartin and D'Haultfœuille (2020) for the first change of each pair. The panels by period use the series of the pairs; the panels by half-year use the leave-out speech scores; the dyads are the estimates of

Section 5.5. Rows marked "minus" use the difference between the paper's vocabulary and a placebo vocabulary, pair by pair and period by period. The estimator of de Chaisemartin and D'Haultfœuille (2020), which uses only the first change of each pair, gives a positive change in the penalized series of the pair (4.0 per thousand), but also a placebo of 2.4 in the period before ($p = 0.085$) and a positive change on both placebo vocabularies, so we do not read it as a change in political wording; these estimates rest on the six pairs that switch. Table F3 reports the single changes of coalition and the shocks that concern pairs of parties.

Table F2: Coalitions: pairs of parties of the same language, 1979–2025

	Partners			Same side			dCDH	
Panel	Coef.	p_{wild}	p_{dates}	Coef.	p_{wild}	p_{dates}	Effect	Placebo
Pairs, penalized (dyads, Section 5.5)	0.10	0.412	0.586	−0.51	<0.001	0.006	3.10 (0.004)	1.77 (0.160)
Pairs by period, penalized								
paper's vocabulary	0.12	0.471	0.560	−0.47	<0.001	0.034	4.05 (<0.001)	2.38 (0.085)
numbers and dates	1.21	0.032	0.018	0.48	0.509	0.412	5.80 (0.068)	5.94 (0.032)
filler pairs	−1.47	0.012	0.002	−2.18	<0.001	0.002	2.24 (0.139)	1.71 (0.022)
paper's minus numbers and dates	−1.04	0.069	0.044	−0.91	0.214	0.090	−1.71 (0.512)	−3.52 (0.050)
paper's minus filler pairs	1.57	0.008	0.002	1.69	0.001	0.002	1.83 (0.200)	0.68 (0.565)
Pairs by period, leave-out								
paper's vocabulary	−10.81	<0.001	0.002	−11.56	<0.001	0.002	8.43 (0.340)	−6.49 (0.414)
numbers and dates	−5.94	0.168	0.070	−7.26	0.067	0.022	−4.96 (0.763)	−4.57 (0.786)
filler pairs	−8.27	0.011	0.002	−9.62	<0.001	0.002	−1.21 (0.922)	2.47 (0.807)
paper's minus numbers and dates	−5.00	0.350	0.120	−4.49	0.323	0.114	13.31 (0.321)	−1.87 (0.914)
paper's minus filler pairs	−2.59	0.157	0.162	−2.08	0.188	0.294	9.33 (0.448)	−8.79 (0.227)
Pairs by half-year, leave-out								
paper's vocabulary	−10.00	<0.001	0.002	−9.96	<0.001	0.002	0.02 (0.998)	1.80 (0.689)
numbers and dates	−4.39	0.162	0.060	−5.04	0.081	0.012	−26.71 (0.207)	5.38 (0.403)
filler pairs	−10.14	<0.001	0.002	−9.69	<0.001	0.002	−9.12 (0.248)	−2.65 (0.687)
paper's minus numbers and dates	−5.79	0.101	0.024	−5.22	0.079	0.008	24.37 (0.129)	−5.82 (0.426)
paper's minus filler pairs	0.04	0.976	0.996	−0.41	0.705	0.648	5.71 (0.655)	1.60 (0.792)
Roll-call agreement by period	0.46	<0.001	0.002	0.34	<0.001	0.002	0.43 (<0.001)	−0.02 (0.474)
Roll-call agreement by half-year	0.47	<0.001	0.002	0.34	<0.001	0.002	0.59 (<0.001)	0.05 (0.275)

Notes: Partners: both parties in government; same side: one minus the absolute difference between the two parties' shares of the period in government (one when both govern or both oppose). Coefficients with pair and time fixed effects; weights: $1/(1/n_A + 1/n_B)$, with n the numbers of speakers of the two parties, for the series and half-year panels, the square root of the number of roll calls for roll-call agreement, and the weights of Section 5.5 for the dyads; net partisanship multiplied by 1,000; roll-call agreement on a 0–1 scale. p_{wild}: wild cluster bootstrap by pair with the null imposed (999 draws); p_{dates}: share of 499 placebo coefficients, from random circular shifts of each pair's treatment path, at least as far from their mean as the estimate (counting the estimate). dCDH: effect in the first cell after the change and placebo in the cell before, with bootstrap p-values in parentheses (de Chaisemartin and D'Haultfœuille, 2020, 2026).

Results of Section 8: sittings after elections. The sitting panels stack all elections since 1831, including the partial elections before 1919, and all changes of government outside the four months after an election, with bins of three months from twelve months before to eleven months after, event and month-of-year fixed effects and the wild cluster bootstrap by event (Table F4). The outcomes are the reactions recorded in the sittings per 100 speeches (Section 7), the shares of references by name and of adjacent turns that cross party families or language communities, and the government–opposition partisanship of the speeches of each month, from leave-out scores. Of the 20 tests in this family (months 0–2 after elections and after changes of government), 5 have $p \leq 0.05$. Besides disorder, approval and laughter, these are the chair's warnings, calls to order and censures after elections (0.10 per 100 speeches, $p = 0.022$) and the share of adjacent turns that cross party families (2.7 percentage points, $p = 0.040$); neither keeps a false discovery rate below 10 per cent ($q = 0.11$ and 0.16). In months 0–2 after changes of government outside

Table F3: Single changes of coalition and shocks, pairs of parties by half-year

Event	Words		Fillers		Words − fillers		Votes	
	Coef.	p_{dates}	Coef.	p_{dates}	Coef.	p_{dates}	Coef.	p_{dates}
N-VA enters government (2014): new partners	−6.09	0.667	−2.29	0.794	−3.97	0.810	0.57	0.015
N-VA enters government (2014): former partners	31.94	0.014	20.03	0.027	12.26	0.123	−0.49	0.014
N-VA leaves government (2018)	−9.06	0.492	−2.58	0.857	−6.40	0.492	−0.23	0.106
Vivaldi (2020): new partners	−18.87	0.027	−14.12	0.041	−4.63	0.479	0.34	0.029
Arizona (2025): new partners	−17.45	0.134	−21.17	0.098	3.71	0.793	–	–
Arizona (2025): former partners	9.34	0.207	12.54	0.171	−2.94	0.671	–	–
Purple coalition (1999): new partners	−1.64	0.712	1.17	0.808	−3.53	0.562	0.39	0.014
Purple coalition (1999): former partners	2.05	0.836	5.54	0.562	−4.37	0.671	−0.43	0.086
Financial crisis (2008): pairs on opposite sides	9.16	0.068	4.17	0.356	5.31	0.479	−0.32	0.029
Euro crisis (2010): pairs on opposite sides	−6.35	0.411	0.75	0.795	−7.06	0.479	0.11	0.086
Attacks of 2001: pairs with VB or N-VA	8.66	0.288	−2.96	0.671	11.72	0.151	0.02	0.729
Refugees (2015): pairs with VB or N-VA	−13.24	0.110	−4.06	0.479	−9.33	0.356	0.20	0.014
Brussels attacks (2016): pairs with VB or N-VA	−8.73	0.178	−1.81	0.740	−7.06	0.534	0.07	0.300
COVID-19 (2020): pairs on opposite sides	−5.60	0.370	−1.72	0.671	−3.98	0.479	0.19	0.071

Notes: Difference-in-differences by half-year, four half-years before and four after the event, pair and half-year fixed effects; treated pairs: as described in each row; controls: for changes of coalition, pairs whose status does not change in the window; for shocks, all other pairs. Words: leave-out net partisanship of the pair multiplied by 1,000; fillers: the same on pairs of filler words; votes: roll-call agreement, over all pairs of parties, including pairs of different languages. p_{dates}: share of the same estimates at the other half-years with a window that does not overlap the event, at least as far from their mean as the estimate (counting the estimate). The half-year of the event is the first of the four after it.

election periods, no outcome has $p \leq 0.05$. Of the 20 comparisons for the three months before the event, 2 have $p \leq 0.05$, against 1 expected by chance: disorder is lower before changes of government than in months 12 to 10 before them (−0.81 per 100 speeches, $p = 0.008$), and adjacent turns that cross language groups are more frequent before elections (1.8 percentage points, $p = 0.036$). These comparisons before an event are not part of the families of Table F12.

Table F4: Sittings around elections and changes of government

	Disorder		Approval		Laughter		Chair	
Events	0–2 m.	−3 to −1	0–2 m.	−3 to −1	0–2 m.	−3 to −1	0–2 m.	−3 to −1
Elections (73)	3.42	0.44	3.04	−0.58	0.83	−0.19	0.10	0.00
	(<0.001)	(0.329)	(<0.001)	(0.190)	(0.004)	(0.439)	(0.022)	(0.944)
without the first 30 days	3.28	0.47	2.66	−0.58	0.82	−0.19	0.11	0.00
	(0.002)	(0.301)	(<0.001)	(0.186)	(0.020)	(0.425)	(0.036)	(0.924)
without 14 days after the new government	2.86	0.48	2.65	−0.47	0.48	−0.16	0.11	0.00
	(0.002)	(0.297)	(<0.001)	(0.275)	(0.134)	(0.507)	(0.020)	(0.938)
without both	2.86	0.49	2.38	−0.50	0.52	−0.18	0.11	0.00
	(0.004)	(0.289)	(<0.001)	(0.234)	(0.150)	(0.463)	(0.028)	(0.912)
Changes of government (111)	−0.22	−0.81	0.11	0.49	0.20	−0.07	0.01	0.01
	(0.479)	(0.008)	(0.882)	(0.593)	(0.295)	(0.808)	(0.555)	(0.397)
Mean, months −12 to −10	6.12		6.40		3.87		0.23	

Notes: Reactions recorded in the sittings per 100 speeches (Section 7); chair: warnings, calls to order and censures. Coefficients for months 0–2 after the event and for the three months before, against months 12–10 before; event and month-of-year fixed effects; wild cluster bootstrap p by event in parentheses. Number of events in parentheses; the rows without some days use the same elections. Without the first 30 days: sittings of the 30 days after the election left out; without 14 days after the new government: sittings of the 14 days after the start of the first government formed after the election left out.

Results of Section 8: electoral reforms. For each reform, the change J in a series from the period before the first election under the new rules to the period after it is compared with the same change at the elections without a reform since 1831 (Figure 17). The change is measured

relative to the mean change at those elections, and the p-value is the share of those elections, counting the reform itself, whose change is at least as far from that mean. The module also restricts the comparison to the elections within 40 years of the reform and to the elections without another shock (1914 and 1939, followed by war; 1936, when Rex and the Flemish nationalists entered in strength; 1946, the first post-war election; 1968 and 1978, the Catholic and Socialist splits; 1981, when the Greens entered; and 1991, the breakthrough of the Vlaams Blok); the religious change of 1894 keeps a p-value of at most 0.03 with the penalized estimator, but some p-values move across 0.05: with the leave-out estimator, the religious change has $p = 0.032$ against all elections and 0.055 without the shock elections. Three further controls follow the reforms. The first is the mean change of the divisions that a reform should change minus that of the divisions that it should not (the lists are in `code/events/ev_events.py`). The second is the slope of the change of the three to six families (each against all the others) or pairs of families whose series is defined at the reform on the absolute change in their share of speeches, compared with the same slope at the other elections and with a permutation of the exposures. The third is the change on the placebo vocabularies. Table F5 reports the first comparison and Table F6 the other three. The module also follows the members present in the periods before and after each reform, and the series by half-year around the election.

Event studies without a robust change. The other designs find no change that stands out from comparable dates once placebo dates, placebo vocabularies and the number of tests are taken into account. The rest of this appendix summarizes them.

Members and government. The member panels contain one row per member and period (or half-year), with the member's scores averaged over the two languages by phrases. Table F7 reports, for every score, the estimator of de Chaisemartin and D'Haultfœuille (2020, 2026) for a member's party entering or leaving government and the estimators of Callaway and Sant'Anna (2021), Sun and Abraham (2021) and Borusyak et al. (2024) for a member's first appointment as minister. Figure F1 shows the first group. The scores of members whose party enters or leaves government are closer to the government side while their party governs: by 28.2 per thousand with leave-out scores and 1.6 with cross-fitted penalized scores, with placebos near zero in the period before. This difference, however, also appears on phrases without political content: 15.1 per thousand on filler pairs and 26.6 on numbers and dates. A member's leave-out score uses the speech of his party colleagues, who join the government side with him, so whatever distinguishes the party's members, including their filler words, moves with the party. Leaving the member's whole party out of the group frequencies removes this channel but reverses the sign (-91.5 per thousand, and -55.6 on filler pairs): under the leave-out frequencies, the posterior is pulled towards the group with more speech when phrases are rare in the other group, and leaving a whole party out shrinks the side to which the party belongs, while in the permuted benchmark the party's members are spread over both sides. On average over member-periods, the leave-party-out score of members of governing parties is lower than that of opposition members, while their leave-out score is higher. The member-level scores therefore mostly reflect the party composition of the two sides rather than a change in what members say, and we do not interpret them; the pair comparisons of Section 8.2 hold the two parties fixed

Table F5: Change at the first election after each reform, against the other elections

		Penalized			Leave-out			
Reform	Division	Jump	Pct.	p	Jump	Pct.	p	Elections
Plural voting (1894)	Religious	2.06	100	0.016	16.17	98	0.032	62
	Government–opposition	2.02	98	0.058	16.05	98	0.038	51
	Flemish–Walloon	2.22	100	0.037	13.02	96	0.074	53
Proportional representation (1900)	Religious	−0.73	8	0.175	−7.87	10	0.222	62
	Left–others	−0.77	21	0.400	5.51	82	0.400	34
	Government–opposition	−0.95	2	0.096	−6.33	8	0.250	51
	Flemish–Walloon	0.57	77	0.444	−3.92	34	0.667	53
Equal male suffrage (1919)	Religious	0.42	79	0.397	0.89	53	0.921	62
	Left–others	−0.19	38	0.800	7.84	88	0.200	34
	Flemish–Walloon	1.32	89	0.222	−0.82	49	0.907	53
	Left–right	−0.13	40	0.806	1.55	70	0.839	30
	Right–others	0.61	77	0.452	−4.63	23	0.484	30
Women's suffrage (1949)	Religious	−0.41	18	0.397	−10.03	5	0.111	62
	Left–others	−0.90	18	0.343	13.60	100	0.057	34
	Government–opposition	0.31	73	0.519	9.99	98	0.077	51
	Flemish–Walloon	−0.24	38	0.778	18.98	100	0.019	53
Federal Chamber of 150 (1995)	Religious	0.40	79	0.429	6.86	87	0.254	62
	Left–others	0.43	68	0.657	5.83	82	0.400	34
	Government–opposition	1.37	96	0.077	4.16	78	0.481	51
	Left–right	1.23	87	0.258	13.86	87	0.194	30
	Right–others	1.08	87	0.258	8.75	90	0.290	30
Gender quotas (1999)	Religious	−0.17	40	0.778	7.51	89	0.222	62
	Left–others	−1.04	12	0.229	16.51	100	0.029	34
	Government–opposition	−0.24	29	0.654	4.90	78	0.404	51
	Left–right	0.16	60	0.806	3.61	73	0.677	30
	Right–others	−0.12	53	1.000	−7.11	17	0.355	30
	Women–men	2.05	100	0.200	18.07	100	0.200	4
Provincial districts and threshold (2003)	Religious	0.40	79	0.429	2.84	73	0.651	62
	Left–others	0.36	65	0.714	7.44	88	0.257	34
	Government–opposition	0.68	94	0.154	−3.37	29	0.558	51
	Left–right	0.17	60	0.806	1.54	70	0.839	30
	Right–others	0.74	83	0.290	−3.00	30	0.742	30
	Women–men	1.17	75	0.600	−3.00	25	0.600	4
Parity of the lists (2007)	Religious	−0.24	32	0.651	−4.15	27	0.540	62
	Left–others	0.36	65	0.714	3.08	68	0.686	34
	Government–opposition	−0.13	35	0.788	−0.67	49	0.904	51
	Left–right	0.20	63	0.742	3.72	73	0.677	30
	Right–others	0.52	73	0.548	3.44	63	0.645	30
	Women–men	1.28	75	0.600	5.73	100	0.400	4

Notes: Jump: change in net partisanship (without covariates; real minus the mean of the permuted series) from the period before the first election under the reform to the period after it, minus the mean change at the elections without a reform at which the series is defined (column Elections), multiplied by 1,000. Pct.: percentile of the reform's change among the changes at those elections; p: share of those elections, counting the reform, whose change is at least as far from their mean. Penalized: 10 permutations; leave-out: 20.

Table F6: Electoral reforms: divisions, exposure and placebo vocabularies

	Targets − placebos		Exposure slope		Rel.: main − numbers		Rel.: main − fillers	
Reform	Pen.	Leave-out	Pen.	Leave-out	Pen.	Leave-out	Pen.	Leave-out
Plural voting (1894)	0.54	7.06	–	–	−1.65	2.90	1.72	16.35
	(0.65)	(0.46)			(0.40)	(0.90)	(0.02)	(0.10)
Proportional representation (1900)	−1.07	−20.90	0.00	−1.22	1.09	7.32	−0.76	−3.36
	(0.28)	(0.08)	(0.97)	(0.43)	(0.62)	(0.65)	(0.27)	(0.68)
Equal male suffrage (1919)	−0.75	11.50	−0.11	1.82	0.52	−1.57	0.35	−3.25
	(0.35)	(0.13)	(0.71)	(0.49)	(0.78)	(0.92)	(0.63)	(0.70)
Women's suffrage (1949)	0.17	−25.84	0.42	−3.31	1.01	−28.96	−0.29	−0.95
	(0.83)	(0.02)	(0.20)	(0.11)	(0.68)	(0.10)	(0.70)	(0.92)
Federal Chamber of 150 (1995)	–	–	−0.03	−0.92	−2.03	−7.49	−1.40	1.59
			(0.89)	(0.51)	(0.29)	(0.65)	(0.05)	(0.87)
Gender quotas (1999)	2.74	14.16	0.19	1.25	1.24	26.03	−0.24	14.87
	(0.20)	(0.20)	(0.49)	(0.63)	(0.56)	(0.14)	(0.78)	(0.11)
Provincial districts and threshold (2003)	–	–	0.08	1.72	1.16	−2.69	1.36	10.79
			(0.71)	(0.54)	(0.57)	(0.90)	(0.06)	(0.25)
Parity of the lists (2007)	1.51	10.16	−0.05	0.19	2.39	−3.26	−2.41	−15.71
	(0.40)	(0.40)	(0.83)	(0.89)	(0.22)	(0.89)	(0.02)	(0.10)

Notes: p-values, in parentheses below the estimates, compare the reform with the other elections. Targets − placebos: mean change of the divisions that the reform should change (for example the religious and government–opposition groupings in 1894) minus that of the divisions that it should not (`code/events/ev_events.py`), relative to the same difference at the elections without a reform, multiplied by 1,000. Exposure slope: the reform's own slope of the changes of the families (each against all the others) whose series is defined at the reform, three or four, on the absolute change in their share of speeches at the election, in per thousand per percentage point; p from the same slope at the other elections. Rel.: religious grouping, change on the paper's vocabulary minus change on the placebo vocabulary, relative to the same difference at the elections without a reform, multiplied by 1,000.

and avoid this problem. The scores of members appointed minister for the first time change in opposite directions across estimators and vocabularies. Table F8 reports the single changes of government.

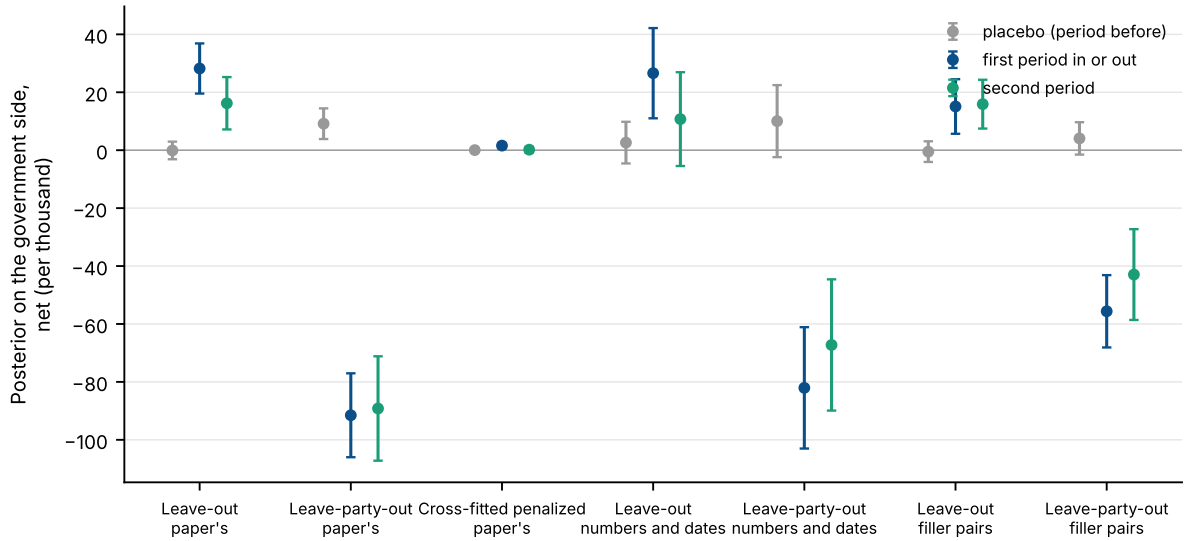


Figure F1: Members whose party enters or leaves government: scores on the government side

Notes: Estimator of de Chaisemartin and D'Haultfœuille (2020, 2026): change in a member's posterior on the government side (net of the permuted groups, multiplied by 1,000) from the period before the member's first change of status to the first and second periods after it, against members not yet switched, signed so that a positive value means more governmental speech while in government; placebo: the period before. Bars: ±1.96 bootstrap standard errors (199 draws, resampling members). Leave-out: frequencies without the member; leave-party-out: without the member's party; cross-fitted: penalized model without the member's fold.

Table F7: Members: party entering or leaving government and first appointment as minister

	Party enters or leaves government (dCDH)			First appointment as minister		
Outcome	Effect, 1st	Effect, 2nd	Placebo	CS	SA ($k = 1$)	BJS
Government side, leave-out	28.2	16.2	−0.1	4.1	3.3	1.0
	(<0.001)	(<0.001)	(0.954)	(0.113)	(0.414)	(0.677)
Government side, leave-party-out	−91.5	−89.2	9.1	−10.0	−21.3	−11.5
	(<0.001)	(<0.001)	(<0.001)	(0.144)	(0.010)	(0.041)
Government side, cross-fitted	1.6	0.2	0.0	−0.7	−1.2	−0.6
	(<0.001)	(0.716)	(0.871)	(0.026)	(0.015)	(0.031)
leave-out, numbers and dates	26.6	10.7	2.6	−9.4	−6.6	−12.3
	(<0.001)	(0.194)	(0.476)	(0.061)	(0.465)	(0.002)
leave-party-out, numbers and dates	−82.1	−67.2	10.0	−22.1	−15.5	−25.0
	(<0.001)	(<0.001)	(0.114)	(0.023)	(0.179)	(<0.001)
leave-out, filler pairs	15.1	15.9	−0.5	−0.8	3.4	−1.4
	(0.002)	(<0.001)	(0.787)	(0.788)	(0.413)	(0.508)
leave-party-out, filler pairs	−55.6	−42.9	4.1	−11.4	−5.4	−12.7
	(<0.001)	(<0.001)	(0.153)	(0.022)	(0.348)	(0.002)
Own family vs others, leave-out	−13.0	−7.3	1.0	−6.6	2.5	−5.8
	(0.015)	(0.280)	(0.685)	(0.085)	(0.694)	(0.067)
Own family vs others, cross-fitted	−0.3	0.0	−0.4	−0.5	0.4	−0.4
	(0.731)	(0.979)	(0.203)	(0.242)	(0.540)	(0.310)
Own religious group, leave-out	−12.9	−16.3	−2.2	–	–	–
	(<0.001)	(<0.001)	(0.161)			
Own religious group, cross-fitted	0.0	0.1	0.1	–	–	–
	(0.913)	(0.755)	(0.512)			
Own left–others group, leave-out	−13.2	−7.9	5.8	–	–	–
	(0.010)	(0.116)	(0.016)			
Own left–others group, cross-fitted	0.9	−0.4	0.0	–	–	–
	(0.118)	(0.563)	(0.907)			

Notes: Scores net of the permuted groups, multiplied by 1,000; p-values in parentheses. dCDH: first and second period after the member's first change of status, and placebo in the period before. First appointment as minister: for CS and BJS, average effect over the period of the appointment and the three after it; for SA, effect in the first period after it ($k = 1$); never- or not-yet-appointed members as controls (CS: Callaway and Sant'Anna, 2021; SA: Sun and Abraham, 2021; BJS: imputation, Borusyak et al., 2024).

Table F8: Members: single changes of government by half-year

Event	Leave-out		Leave-party-out		Cross-fitted		Leave-party-out, fillers	
	Coef.	p_{dates}	Coef.	p_{dates}	Coef.	p_{dates}	Coef.	p_{dates}
Purple coalition (1999), parties in	43.7	0.012	−65.8	0.049	1.3	0.415	−21.9	0.085
Purple coalition (1999), parties out	−52.0	0.011	28.4	0.287	−1.1	0.191	3.3	0.872
Michel I (2014), parties in	97.2	0.026	−2.2	0.939	3.9	0.092	17.2	0.452
Michel I (2014), parties out	−70.5	0.011	−49.0	0.181	−2.7	0.021	−29.3	0.170
N-VA leaves (2018)	−47.6	0.143	−6.1	0.852	−2.3	0.461	−15.6	0.574
Vivaldi (2020), parties in	32.8	0.103	−0.3	1.000	2.2	0.077	12.0	0.551
Arizona (2025), parties in	51.5	0.101	−3.4	0.915	3.7	0.107	20.2	0.331
Arizona (2025), parties out	−62.7	0.010	−12.3	0.588	−4.9	0.010	−26.0	0.196
National union (1925–26), parties in	6.9	0.239	−223.0	0.032	1.3	0.186	−70.7	0.301
National union (1935), parties in	21.1	0.157	−200.9	0.056	1.3	0.201	−130.2	0.097

Notes: Members of the parties that enter or leave government at the date against the members of the other parties, six half-years before and five after, member and half-year fixed effects; posterior on the government side net of the permuted groups, multiplied by 1,000. p_{dates}: as in Table F3.

Splits of the national parties. For the splits of 1968, 1972 and 1978, the outcomes are, by family and year from 1945 to 2000, the shares of references by name and of adjacent turns between members of the same family that cross the two language groups, and the roll-call agreement of the family's Flemish and Walloon members (Table F9). The roll-call agreement of a family's Flemish and Walloon members did not fall after the split (2.3 percentage points, $p = 0.233$ with

robust standard errors). After the split, the share of references by name within a family that cross the language groups changes by -5.0 percentage points ($p = 0.081$ with robust standard errors) and the share of adjacent turns by -4.5 ($p = 0.180$), but changes of this size are common when the three split years are moved together by up to ten years in either direction ($p = 0.238$ and 0.286). With three splits, there are only six ways of assigning the split years to the families, so that permutation cannot give a p-value below 0.17. With a bootstrap that rests on the three families, the estimators of Callaway and Sant'Anna (2021) and Borusyak et al. (2024) give 2.0 and -2.8 points for references and -7.8 and -2.2 for turns ($p = 0.064$ and 0.707). From 1968 to 1971, the reactions data give every member of the joint CVP-PSC label the language of the label, Dutch, Walloon members included; the module assigns these members the language of most of their speeches in that legislature (ties go to the region of the district; one minister without speeches then gets the language given to him under another party label). Without the language correction, the Catholic shares fall sharply in 1968–1971 and return to their earlier level in 1972. Turns are counted only when the two speeches are adjacent. The same coding of the joint label enters the references between language communities of Section 7 in those years.

Table F9: Splits of the national parties

Outcome	TWFE	$p_{\text{assign.}}$	p_{dates}	Callaway–Sant'Anna	Imputation
Mentions across language groups (same family)	-5.0 (0.081)	0.667	0.238	2.0 (0.755)	-2.8 (0.682)
Adjacent turns across language groups (same family)	-4.5 (0.180)	0.667	0.286	-7.8 (0.064)	-2.2 (0.707)
Roll-call agreement of the Flemish and Walloon wings	2.3 (0.233)	1.000	0.857	2.1 (0.195)	1.6 (0.301)

Notes: Families: Catholic (split 1968), Liberal (1972) and Socialist (1978), by year 1945–2000; percentage points. TWFE: family and year fixed effects, weights equal to the square root of the number of references, turns or roll calls, heteroskedasticity-robust p with the degrees of freedom of the fixed effects (with three families, clustered standard errors are not reliable); $p_{\text{assign.}}$: permutation of the three split years across the families (six assignments); p_{dates}: the three split years moved together by -10 to $+10$ years (20 placebo dates; the smallest possible value is $1/21$). Callaway–Sant'Anna and imputation: average effect after the split, with p from a bootstrap over the three families. From 1968 to 1971, members of the joint CVP-PSC label are assigned the language of most of their speeches.

Single events in the sittings. Single events compare the mean of the three or six months after with the months before, against the same contrast at every other month of the surrounding twenty years, away from elections; 28 of these 299 comparisons have $p \leq 0.05$, about twice as many as chance alone would produce, but none keeps a false discovery rate below 10 per cent. The largest rise in disorder follows the popular consultation on the Royal Question in March 1950: in the six months after it, disorder rises by 9.7 per 100 speeches ($p = 0.020$ against the same change at other dates). The Chamber did not sit between March and the election of June 1950, however, so the sittings in this window are those of June to August 1950, after the election and around the King's return in July. Table F10 lists the comparisons with $p \leq 0.05$. Most of their windows contain an election, while the placebo months are taken away from elections, so part of these changes may come from the elections themselves (Table F4); several also rest on few sittings. The comparisons for hostile collective interjections, cries recorded without a named speaker that contain words of protest such as *non*, *la clôture* or *à l'ordre*, rest on very few of them. After the King's return in July 1950, their rate falls (-0.26 per 100 speeches in three months, $p = 0.007$), but this compares three interjections in the five sittings of June 1950, just after the election, one of them a report title that the module reads as an interjection, with two

in the seventeen sittings of August to October. After the start of the strike against the Unitary Law in December 1960, it rises (0.20, $p = 0.022$), from none in the seven sittings of September to November to twelve in the twenty-one sittings of January to March 1961 (ten of them cries such as *Non ! non !* or *C'est faux !*). After the election of Black Sunday in November 1991, disorder and laughter become rarer (-11.9 and -1.6 per 100 speeches in three months, $p = 0.034$ and 0.010), but the panel has only three sittings in these three months, at the start of the years of incomplete coverage. After the government fell over Brussels-Halle-Vilvoorde in April 2010 and after the euro crisis of May 2010, references by name and adjacent turns that cross party families become rarer (in the three months after the euro crisis, -24 and -21 percentage points, $p = 0.014$ and 0.024); these windows contain the election of June 2010, and the three months after the euro crisis contain only two sittings. The cordon sanitaire is studied as the Vlaams Blok against the other opposition parties, by half-year, for the disorder during its speeches, the calls to order of its members and the references to them; the p-values come from a permutation of the treated party among seven opposition parties, so the smallest possible value is $1/7$, and the smallest of the six is 0.286.

Table F10: Single events in the sittings with $p \leq 0.05$ against placebo dates

Event	Outcome	Months	Change	p_{dates}
Royal Question: consultation (12 Mar. 1950)	approval	3	9.4	0.035
	approval	6	6.8	0.047
	disorder	6	9.7	0.020
	hostile collective interjections	3	0.21	0.014
	government–opposition partisanship	3	39.9	0.022
	turns across families	6	−8	0.034
Return of the King (22 Jul. 1950)	chair	3	−0.35	0.042
	hostile collective interjections	3	−0.26	0.007
	hostile collective interjections	6	−0.23	0.007
	government–opposition partisanship	3	−54.2	0.007
	government–opposition partisanship	6	−26.9	0.014
Strike against the Unitary Law (20 Dec. 1960)	hostile collective interjections	3	0.20	0.022
	turns across families	3	13	0.028
	turns across families	6	9	0.041
Cordon sanitaire (10 May 1989)	turns across language groups	3	30	0.040
	turns across language groups	6	18	0.032
Election of Black Sunday (24 Nov. 1991)	disorder	3	−11.9	0.034
	laughter	3	−1.6	0.010
White March (20 Oct. 1996)	turns across families	3	17	0.025
	turns across families	6	7	0.036
Government falls over BHV (22 Apr. 2010)	government–opposition partisanship	6	−12.3	0.020
	turns across families	3	−12	0.043
	turns across families	6	−15	0.020
	references across families	6	−12	0.041
Euro crisis (2 May 2010)	government–opposition partisanship	6	−14.9	0.010
	turns across families	3	−21	0.024
	references across families	3	−24	0.014
	references across families	6	−11	0.046

Notes: Change: mean of the monthly values over the three or six months after the month of the event minus the same mean over the months before it; each month's value is the mean of its sittings weighted by their speeches, and months without sittings are skipped. Disorder, approval, laughter, the chair's warnings, calls to order and censures, and hostile collective interjections (cries recorded without a named speaker, attributed to members, to a side of the House or to its benches, that contain words of protest such as *non*, *la clôture* or *à l'ordre*): per 100 speeches; shares of references by name and of adjacent turns that cross party families or language groups: percentage points; government–opposition partisanship of the speeches of the month, from leave-out scores: per thousand. p_{dates}: share of the same contrast at the other months of the ten years before and after the event, at least as far from their mean; placebo months are more than twice the window away from the event and more than two months away from an election. None of these 28 comparisons, out of 299, keeps a false discovery rate below 10 per cent.

Policy areas. Conflicts and shocks that concern a policy area are studied with the partisanship of the topics of that area (Section 6) against the other areas, by period and by legislature, with the penalized series of `23_topic_areas.py` and with the leave-out speech scores within areas and topics. With eight substantive areas, a permutation of the treated area cannot give a p-value below one eighth; the module therefore also reports placebo dates, the wild bootstrap when two areas or more are treated, and synthetic difference-in-differences (Arkhangelsky et al., 2021). Of the 611 tests, 27 have $p \leq 0.05$ against placebo dates, about as many as chance alone would produce (31), and none keeps a false discovery rate below 10 per cent. Table F11 lists them; an estimate that serves two dates of one event is listed once. Among them, the four largest leave-out estimates follow the independence of the Congo in 1960. With leave-out scores by period, the partisanship of foreign affairs, defence and colonies rises against the other areas by 16.0 per thousand in the religious grouping and by 15.9 in the left–others grouping ($p = 0.027$ and 0.023), and that of its topic of foreign relations and colonial governance by 29.4 in the left–others grouping and by 27.7 between government and opposition ($p = 0.023$ and 0.030). By half-year, the rises are smaller and do not stand out from placebo dates ($p = 0.711$ and 0.637). On numbers and dates, the same comparison by period falls (-40.0 and -34.4, $p = 0.107$ and 0.136). Synthetic difference-in-differences, with the seven other substantive areas as donors (for the topic, the other topics), the three periods of 1952–1958 before and the four of 1961–1969 after (1958–1961 left out), gives 10.9 and 22.0 per thousand for the area and 29.1 and 26.1 for the topic, with standard errors of 6.2, 14.8, 13.9 and 15.3, from the placebo estimates that give the treatment to each donor area in turn (for the topic, to 50 random donor topics).

Other events. The governments of national union, COVID-19 and the fall of the government over Brussels-Halle-Vilvoorde in 2010 show no change that stands out from comparable dates after the correction for multiple testing, in the outcomes studied for them (Table F1 lists the designs). After the equality law of 1898, the Flemish–Walloon partisanship of speech in French, by half-year, rises more than at placebo dates with the cross-fitted scores: by 4.8 per thousand from the two half-years before the law to the two after it ($p = 0.013$), and by 3.5 with four half-years on each side ($p = 0.028$). The cross-fitted scores are computed only on the paper's vocabulary. With the leave-out scores, the rise is larger (7.5 and 8.5), but rises of this size are common at placebo dates ($p = 0.338$ and 0.181), and numbers and dates rise by about as much (6.0 and 9.1). This comparison is not part of the families of Table F12.

Multiple testing. Table F12 groups the tests into families that answer one question with one kind of inference and reports, for each, the tests with $p \leq 0.05$, the number expected by chance, and the tests with Benjamini and Hochberg (1995) q-values of at most 0.10 and 0.05. Estimates on a placebo vocabulary alone and the robustness variants are left out of the families; at the reforms, the differences between the paper's vocabulary and the placebo vocabularies form a family of their own. The estimates that keep a false discovery rate below 10 per cent are the roll-call results of Section 8.2, including those of single changes of coalition and of the two crises; the reactions after elections; the associations of wording with governing together or being on the same side, which the placebo vocabularies share; and two changes in the wording of single pairs of parties, matched by filler pairs. None of the comparisons of the electoral reforms

Table F11: Policy areas: estimates with $p \leq 0.05$ against placebo dates

Event	Division	Panel	Estimate	p_{dates}	q
School Pact (1958)	Government–opposition	l.-o., half-year, areas	11.6	0.045	0.93
School Pact (1958)	Government–opposition	pen., period, areas	−2.9	0.049	0.93
School Pact (1958)	Left–others	l.-o., period, areas	−14.5	0.048	0.93
Independence of the Congo (1960)	Government–opposition	l.-o., period, topics	27.7	0.030	0.93
Independence of the Congo (1960)	Left–others	l.-o., period, areas	15.9	0.023	0.93
Independence of the Congo (1960)	Left–others	l.-o., period, topics	29.4	0.023	0.93
Independence of the Congo (1960)	Religious	l.-o., period, areas	16.0	0.027	0.93
Oil shock (1973)	Left–others	l.-o., period, areas	7.9	0.026	0.93
Oil shock (1973)	Left–right	pen., legislature, areas	−2.7	0.045	0.93
State reform (1980)	Right–others	pen., period, areas	3.2	0.029	0.93
State reform (1988–89)	Left–others	pen., legislature, topics	−3.4	0.048	0.93
State reform (1988–89)	Religious	pen., legislature, topics	1.5	0.026	0.93
Maastricht Treaty (1992)	Government–opposition	pen., legislature, topics	−3.9	0.030	0.93
State reform (1993)	Left–others	pen., legislature, topics	5.6	0.048	0.93
State reform (1993)	Religious	pen., legislature, topics	2.6	0.026	0.93
Global Plan (1993)	Left–right	pen., legislature, topics	1.5	0.050	0.93
Global Plan (1993)	Left–right	pen., period, topics	16.5	0.034	0.93
Dutroux affair (1996)	Right–others	pen., period, topics	−2.6	0.040	0.93
Euro (1999)	Government–opposition	pen., legislature, topics	6.0	0.028	0.93
Euro (1999)	Left–others	l.-o., half-year, areas	9.6	0.027	0.93
State reform (2001)	Government–opposition	pen., period, topics	−3.8	0.018	0.93
Attacks of 11 September 2001	Government–opposition	pen., period, areas	1.2	0.018	0.93
Attacks of 11 September 2001	Government–opposition	pen., period, topics	3.0	0.019	0.93
Financial crisis (2008)	Government–opposition	pen., legislature, areas	1.9	0.048	0.93
Financial crisis (2008) and euro crisis (2010)	Left–right	pen., period, areas	3.6	0.050	0.93
State reform (2012–14)	Religious	pen., period, topics	4.3	0.014	0.93

Notes: Difference-in-differences of the partisanship of the area (or topics) touched by the event against the other areas (topics), with unit and time fixed effects, twelve years on each side unless the catalogue sets another window. By period and by half-year, the unit that contains the date is left out of the estimate at the event (not of those at placebo dates); by legislature, the years after the event begin with the first legislature that starts in a later calendar year. Net partisanship multiplied by 1,000. Panels: pen., penalized series within areas or topics (`code/python/23_topic_areas.py`), by period or legislature; l.-o., leave-out speech scores within areas or topics, by period or half-year. p_{dates}: share of the same estimate at other dates of the sample, with windows of the same length centred on dates more than one window length away from the event (shortened at the ends of the sample), at least as far from their mean; q: Benjamini and Hochberg (1995) within the family of policy areas (Table F12). An estimate that serves two dates of one event, both in the same period, is listed once. Rows in order of date.

with the other elections does, on the paper's vocabulary or in its differences with the placebo vocabularies.

Table F12: Multiple testing

Family of tests	Tests	$p \leq 0.05$	Expected	$q \leq 0.10$	$q \leq 0.05$
Coalitions, pairs of parties (two-way fixed effects)	12	10	0.6	10	10
Single events on pairs of parties	27	8	1.4	8	0
Policy areas	611	27	30.6	0	0
Single events on members	55	11	2.8	0	0
Sittings after elections and changes of government	20	5	1.0	3	3
Sittings, single events	299	28	15.0	0	0
Reforms against the other elections	174	15	8.7	0	0
Reforms: paper's minus placebo vocabularies	344	16	17.2	0	0

Notes: Tests: estimates of the family with a p-value; expected: 5 per cent of the tests. Families: coalitions, two-way fixed effects with the wild bootstrap (paper's vocabulary and roll calls); single events on pairs of parties and on members (placebo dates); policy areas (placebo dates); sittings, single events (placebo dates) and around elections and changes of government (wild bootstrap); reforms against the other elections (all divisions and both estimators, paper's vocabulary) and their differences with the placebo vocabularies.

Not estimable here. Comparisons with other parliaments (published series for the United States, Finland, the United Kingdom, Norway and Sweden; the ParlaMint corpora; the staggered adoption of proportional representation in Europe), the Senate as a comparison chamber for the reforms after 1900 and exposure by electoral district need data that the project does not have. The members around the governments of national union of 1918 and 1944 have no sittings in the half-years before, during the world wars, and the types of speech (interpellations and questions against debates on bills) were not studied in this paper.